

EEG-FM-Compass: Progress, Benchmarking, and Future Directions for EEG Foundation Models

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ABSTRACT

Electroencephalography (EEG) foundation models (FMs) have recently emerged as a promising paradigm for brain-computer interfaces, aiming to learn transferable neural representations from large-scale heterogeneous recordings. Despite rapid progress, a fair and comprehensive comparison of existing EEG FMs is still lacking, owing to inconsistent pre-training objectives, preprocessing choices, and downstream evaluation protocols. To fill this gap, we present EEG-FM-Compass. We first review 55 representative models and organize their design choices into a unified taxonomic framework including data standardization, model architectures, and self-supervised pre-training strategies. We then evaluate 12 open source FMs and competitive specialist baselines across 13 EEG datasets spanning nine brain-computer interface paradigms. Emphasizing real-world deployments, we consider both cross-subject generalization under a leave-one-subject-out protocol and rapid calibration under a within-subject few-shot setting. We further compare full-parameter fine-tuning with linear probing to assess the transferability of pre-trained representations, and examine the relationship between model scale and downstream performance. Our results indicate that: 1) linear probing is frequently insufficient; 2) specialist models trained from scratch remain competitive across many tasks; and 3) larger FMs do not necessarily yield better generalization performance under current data regimes and training practices.

Keywords: Brain-computer interface, EEG foundation model, self-supervised learning, general representation, benchmark

1. INTRODUCTION

Brain-computer interfaces (BCIs) establish a direct communication pathway between the brain and external devices, by decoding various brain signals [1]. They can be diagnostic and therapeutic tools for a wide range of neurological and psychiatric diseases, e.g., epilepsy [2], disorder of consciousness [3], and mood disorders [4], and can support communications and interactions for individuals with severe motor or speech impairments [5] caused by amyotrophic lateral sclerosis, brainstem stroke, high level spinal cord injury, etc.

BCI systems are commonly categorized into invasive, non-invasive, and partially invasive systems. This paper focuses on non-invasive BCIs, which do not require implanting sensors inside the brain. Electroencephalography (EEG), recorded by electrodes placed on the scalp, is

one of the most widely used signal modalities in non-invasive BCIs because of its millisecond-level temporal resolution, portability, and relatively low acquisition cost [6].

Classical EEG-based BCI paradigms include motor imagery (MI), steady state visual evoked potentials (SSVEP), event related potentials (ERP), epilepsy recognition, and so on. Recent years have witnessed some emerging paradigms that target higher level cognitive states, such as mental workload [7] and imagined speech [8,9]. Together, these developments underscore the potential of EEG as a versatile interface to motor, perceptual, cognitive, and clinical brain states.

Deep learning has driven substantial progress in EEG decoding over the past decade. Early convolutional architectures that remain widely used, such as ShallowConvNet and DeepConvNet [10] and EEGNet [11], employ compact spatio-temporal filters to learn discriminative

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EEG patterns from limited labeled data. Beyond these generic backbones, a large body of paradigm-specific decoders has been developed to encode task-dependent inductive biases. For motor imagery, representative models range from filter-bank and spatial-filter-inspired networks, such as MSFBCNN [12] and FBCNet [13], to convolutional-transformer hybrids such as EEG Conformer [14] that couple local temporal filtering with global dependency modeling. For ERP, P300, and RSVP target detection, architectures such as HDNN and CNN4EEG [15], EEG-Inception [16], and MOCNN [17] exploit hierarchical convolutions, inception-style temporal filters, and multi-scale feature extraction to improve single-trial event decoding. For emotion recognition and cognitive-state decoding, graph- and attention-based models, including DGCNN [18], RGNN [19], and LGGNet [20], capture functional relations among EEG channels and brain regions. In clinical EEG analysis, specialized architectures have been tailored to sleep staging, seizure detection, and abnormality assessment, with representative examples including DeepSleepNet [21] for sleep staging and SPaR-CNet [22] for seizure and rhythmic / periodic pattern classification. These methods generally outperform classical pipelines that rely on hand-crafted features.

Despite these advances, real-world deployment of deep learning models remains challenging. First, most approaches require large amounts of labeled data, whereas EEG acquisition and expert annotation are costly and time consuming. Second, EEG devices differ widely in channel counts and electrode layouts, and conventional architectures often fail to accommodate heterogeneous inputs [23,24]. Third, many existing models are trained for a single task with limited capacity and transferability, which restricts their generalization to new BCI paradigms.

Built on recent progress on large language models (LLMs) [25] and vision large models (VLMs) [26], EEG FMs [27], as illustrated in Fig. 1, have emerged as a promising direction for addressing these challenges. The core premise is that a model pre-trained on large-scale heterogeneous EEG data can learn general-purpose representations that transfer effectively to diverse downstream tasks with minimal task-specific adaptation. This paradigm offers a principled solution to the data scarcity problem, as self-supervised pre-training can leverage vast amounts of unlabeled recordings that

would otherwise remain unutilized. Furthermore, FMs can be designed to accommodate diverse electrode configurations, enabling a single pre-trained model to generalize across heterogeneous devices.

Numerous EEG FMs were proposed in the past two years, with diverse pre-training objectives, architectures, and target applications. Some models focus on general-purpose representation learning across multiple paradigms, whereas others focus on specific clinical or cognitive applications. Pre-training strategies range from masked signal reconstruction and contrastive learning to codebook-based discrete modeling and autoregressive sequence prediction. Architecturally, these models have evolved from Transformer-based encoders to Mamba-based designs that offer improved efficiency for long sequences. The scale of pre-training data has also increased substantially, with some recent models leveraging thousands of hours of EEG recordings from dozens of public datasets.

Unfortunately, different studies evaluated their proposed models on different datasets using inconsistent protocols, making direct comparison difficult. Moreover, several fundamental questions regarding the capabilities and limitations of these models have not been rigorously examined. These considerations motivate the following three research questions in this paper:

Q1. Can EEG FMs extract generalizable EEG representations, so that they can be easily adapted to various different downstream tasks?

Q2. Do EEG FMs consistently and significantly outperform traditional and deep learning methods trained from scratch using only the fine-tuning data?

Q3. Do EEG FMs exhibit behavior consistent with scaling laws? Specifically, do larger model sizes and greater volumes of pre-training data lead to better generalization performance on downstream BCI tasks?

Our main contributions are:

1) A comprehensive overview of existing EEG FMs.

- We survey 55 EEG FMs, constituting the most comprehensive collection to date.
- We provide a detailed and structured comparison of their technical designs, encompassing basic information, pre-training data scale, pre-processing pipelines, pre-training strategies, and architectural choices.
- We propose a unified taxonomic framework for EEG FMs that organizes existing work into a coherent design space.

2) Fair and comprehensive benchmarking

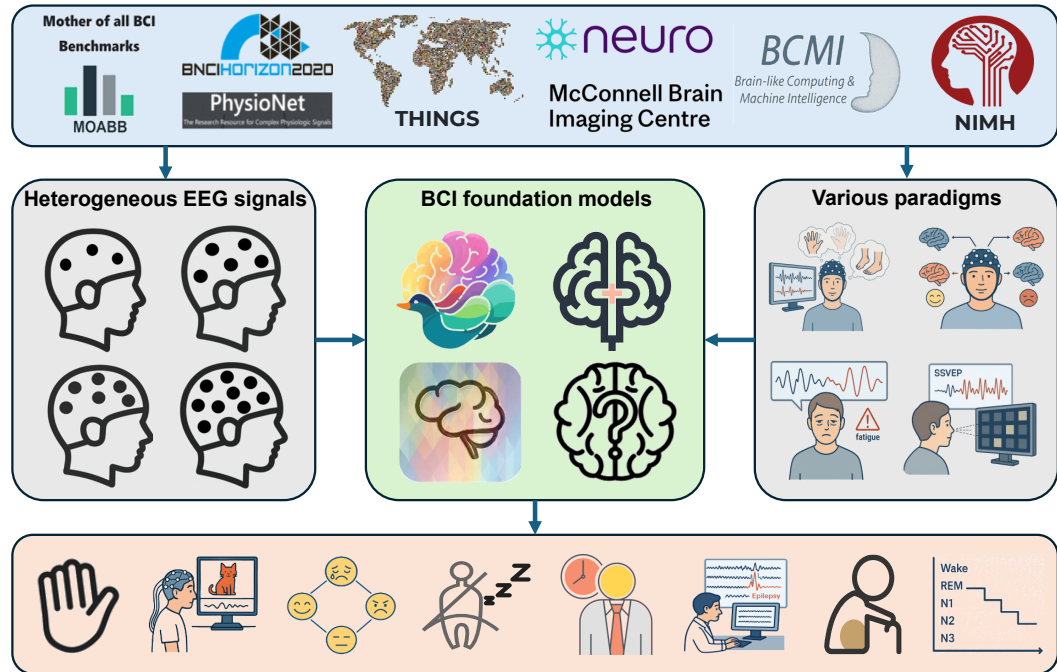


Figure 1. Overview of EEG FMs. Models are pre-trained on large scale heterogeneous EEG data collected from devices with diverse electrode configurations across various BCI paradigms. Through self-supervised pre-training, the learned representations may generalize to a wide range of downstream tasks.

for open source EEG FMs.

- We systematically compare “full-parameter fine-tuning” with “classification head fine-tuning” across various models and tasks to assess whether pre-trained encoders provide broadly transferable EEG representations. Beyond the commonly used leave-one-subject-out (LOSO) scenario, we introduce a within-subject few-shot adaptation scenario in which the fine-tuning data volume is approximately 1/20 ~ 1/100 of that typically used in LOSO protocols, as illustrated in Fig. 2b and c.
- We comprehensively compare traditional machine learning methods, CNN-based models, and Transformer-based models trained from scratch against fine-tuned EEG FMs to evaluate whether conventional approaches remain competitive.
- We evaluate EEG FMs of varying parameter sizes pre-trained on diverse datasets to investigate whether a larger model necessarily leads to better generalization performance.

The remainder of this paper is organized as follows. Section ‘Overview of EEG FMs’ reviews 55 different EEG FMs. Section ‘Benchmark of EEG FMs’ presents the benchmark. Section ‘Discussion’ points out future research directions. Finally, Section ‘Conclusion’ draws conclusions.

2. OVERVIEW OF EEG FMs

This section introduces the conceptual framework of EEG FMs, provides a comprehensive summary of existing approaches, and organizes prevalent pre-training strategies into a unified taxonomic framework, as shown in Fig. 2a.

2.1. Advances and Trends of EEG FMs

Fig. 3 presents an overview of the 55 existing EEG FMs. As shown in Fig. 3a, 16.4% of the surveyed studies were published in 2024 and 67.3% in 2025 or 2026, indicating a clear surge in research activity. This accelerated progress is accompanied by increasing diversity in model scope, signal modalities, backbone architectures, and training methodologies. Table 1 summarizes the 55 surveyed models in chronological order, reporting the affiliation of the first author, publication date, targeted modality, pre-training data scale, computational cost, and parameter size (bold represents open source).

Model scope has begun to bifurcate. As shown in Fig. 3b, while most studies aim to develop generalized EEG FMs, a nontrivial subset focuses on paradigm-specific FMs. In practical BCI deployment, the target paradigm is often known prior to downstream data collection. Motivated by this observation, paradigm-specific models are pre-trained exclusively on

Table 1. Comparative overview of EEG FMs. ‘—’ indicates that the corresponding information was not reported in the original paper.

Foundation Model	Author Affiliation	Signal Modalities	Pre-training Data Size	Computational Cost	Number of Parameters
BENDR [28]	UofT	EEG	1.5TB	—	4.0M
BrainBERT [29]	MIT	iEEG	43.7h	—	43.2M
MBrain [30]	ZJU	iEEG	470h	4 × 3090	—
BIOT [31]	MIT	EEG + ECG	58,021h	8 × A6000	3.2M
Brant [32]	ZJU	iEEG	2,528h	4 × A100, 67.2h	68M / 104M / 249M / 506M
LaBraM [33]	SJTU	EEG	2,500h	8 × A800	5.8M / 46M / 369M
Mentality [34]	UCLA	EEG	—	—	—
Neuro-GPT [35]	USC	EEG	5,656h	—	0.16M
MEET [36]	NPU	EEG (Emotion)	—	1 × 3090	30M / 61M / 215M
EEGFormer [37]	MSR	EEG	1.7TB	—	1.9M / 2.3M / 3.2M / 5.8M
BrainWave [38]	ZJU	EEG + iEEG	40,907h	4 × A100, 100h	115M / 204M / 459M / 1065M
NeuroLM [39]	SJTU	EEG	25,000h	8 × A100	254M / 500M / 1696M
Brant-X [40]	ZJU	EEG + EXG	4TB	2 × A100	>1B
FoME [41]	NPU	EEG + iEEG	26,000h	6 × 4090, 350h	476M / 745M
EEGPT [42]	HIT	EEG	—	8 × 3090	4.7M / 25M
BrainGPT [43]	CAS	EEG	37,500K trials	8 × A800, 20h	1.5M / 11.3M / 184M / 1.1B
GEFM [44]	UTokyo	EEG	—	—	—
CBraMod [45]	ZJU	EEG	27,062h	4 × A5000, 120h	4.0M
CEReBrO [46]	UZH	EEG	>20,000h	4 × 2080 Ti	3.58M / 39.95M / 85.15M
LEAD [47]	UNCC	EEG (AD)	730.48h	4 × A5000	—
FEMBA [48]	POLIMI	EEG	21,000h	—	7.9M / 47.7M / 77.8M / 389M
LCM [49]	Neuro Industry, Inc.	EEG	—	—	33.9M
TFM [50]	UIUC	EEG	≈1,900h	—	1.9M
Gram [51]	SJTU	EEG	7,291h	8 × A800, 240h	6.0M / 47.19M / 251.28M
ALFEE [52]	TJU (Tongji)	EEG	25,000h	8 × A100	16M / 44M / 120M / 300M / 540M
BrainOmni [53]	AI Lab	EEG + MEG	2,653h	16 × A100, 18h	8.4M / 33M
E3GT [54]	JHU	EEG	26,496h	—	96.4M
CodeBrain [55]	NUS	EEG	9,246h	A100	3.9M – 146.8M
DIVER-0 [56]	SNU	EEG	≈3,000h	4 × A100	—
UniMind [57]	AI Lab	EEG	929K trials	8 × A800, 21.78h	0.5B / 1.8B / 7B
CSBrain [58]	SHA AI Lab	EEG	9,000h	4 × A100, 101h	4.9M
DMAE-EEG [59]	NUDT	EEG	—	8 × 4090	—
EEGMamba [60]	ZJU	EEG	16,724h	1 × A5000, 120h	3.3M
MIRepNet [61]	HUST	EEG (MI)	50,355 trials	1 × 3090, 3h	5.2M
PSGFM [62]	JHUAPL	EEG + EXG (Sleep)	482,270 trials	—	97.1M
EEGDM [63]	XMUM	EEG	—	8 × 4090	12.8M
CoMET [64]	UCAS	EEG	>1,000K trials	4 × A100	5M / 19M / 151M
EpilepsyFM [65]	NPU	EEG (Epilepsy)	—	6 × 4090, 58h	6.3M
SingLEM [66]	TUAT	EEG	357,000h	4 × A100	3.3M
BrainPro [67]	NTU	EEG	2,180h	5 × A800	7.69M
Uni-NTFM [68]	UCAS	EEG	28,000h	32 × A100	57M / 427M / 912M / 1.9B
ELASTIQ [69]	NTU	EEG	1,153h	4 × H100	26.42M
BioCodec [70]	USC	EEG + EMG	>1,000h	4 × A100	0.3M – 2.6M
HEAR [71]	HKPU	EEG	8,782h	8 × A6000	3.1M / 6.0M
NeuroRVQ [72]	ICL	EEG	—	4 × V100	5.9M
NeurIPT [73]	XMUM	EEG	>2,100h	8 × 4090	73.5M
REVE [74]	LAB-STICC	EEG	61,415h	1 × A100, 260h	12M / 69M / 408M
mdJPT [75]	SUSTech	EEG (Emotion)	—	1 × 3090	1.0M
LUNA [76]	ETH	EEG	21,928h	8 × A100	7.0M / 43M / 311.4M
THD-BAR [77]	BUAA	EEG	2,123h	8 × L40S	124M / 354M / 1555M
EEG-X [78]	Emotiv	EEG	1,267h	—	—
SAMBA [79]	Emotiv	EEG	>1,000h	2 × A6000 Ada	1.0M
DeeperBrain [80]	ZJU	EEG	>17,200h	1 × A5000, 7h	—
EEGMoE [81]	UCAS	EEG	115h	A800	1.68M
Brain-OF [82]	FZJ	EEG + MEG + fMRI	5,870,984 trials	224 × A100, 144h	47.5M / 331M / 1.7B

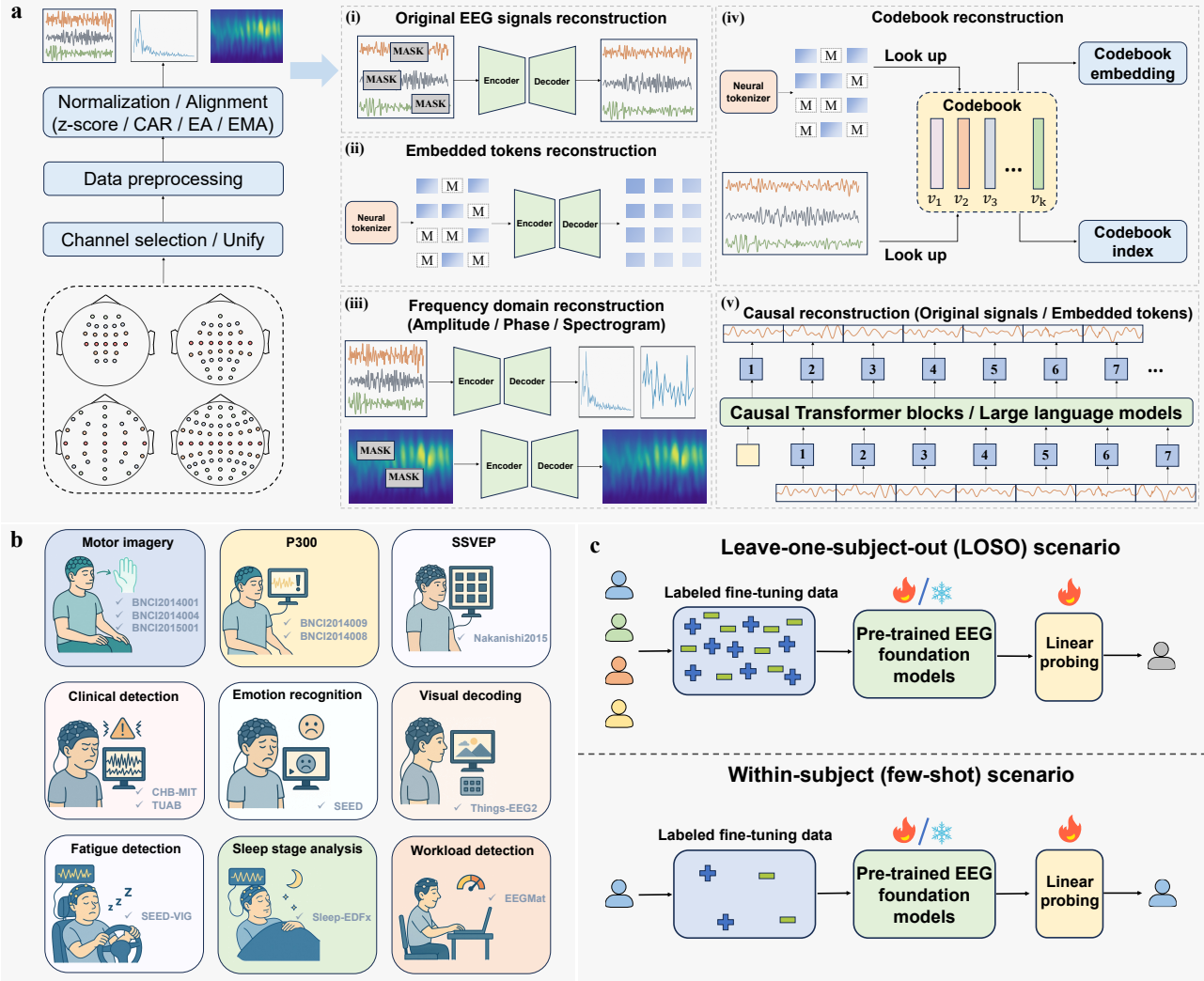


Figure 2. Overview of the EEG FM pre-training pipeline and benchmarking scenarios. (a) Raw EEG trials are first standardized through channel selection or unification, followed by dataset-dependent preprocessing and normalization/alignment. The standardized signals are then used for self-supervised pre-training with representative objectives: (i) masked reconstruction of raw EEG signals in the time domain; (ii) masked reconstruction of embedded tokens after tokenization; (iii) frequency-domain reconstruction, where the target can be the spectrogram, spectral amplitude, or phase-related representation; (iv) codebook-based reconstruction, where a tokenizer maps the signal to discrete codebook indices or embeddings and the model learns to predict the corresponding discrete units; and (v) autoregressive or causal reconstruction using causal masking, implemented with causal Transformer blocks or large language models. (b) The 13 downstream datasets spanning 9 representative BCI paradigms, including MI, P300, SSVEP, clinical detection, emotion recognition, visual decoding, fatigue detection, sleep stage analysis, and workload detection. (c) Illustration of the two evaluation scenarios: the LOSO scenario, which aggregates labeled data from multiple subjects for fine-tuning and evaluates on a held-out subject, and the within-subject few-shot scenario, which uses only a small amount of labeled data from the target subject for adaptation.

data from a single paradigm to prioritize domain-aligned representation learning, potentially at the expense of cross-paradigm generalizability.

The distribution of signal modalities further reflects this diversity. Fig. 3c shows that non-invasive scalp EEG remains the dominant modality, largely because it does not require surgical implantation and is substantially easier to collect at scale than invasive recordings. However, scalp EEG signals are attenuated and spatially blurred due to volume conduction through the scalp and skull, which limits both signal strength and spatial resolution [83]. To im-

prove robustness and enrich the supervisory signal, several studies incorporate auxiliary physiological modalities such as electrocardiogram (ECG) [31], electromyogram (EMG) [40], or magnetoencephalography (MEG) [53], suggesting that representation learning can benefit from correlated biosignals.

From an architectural perspective, Fig. 3d indicates that Transformer-based backbones dominate current EEG FMs. Model capacity and training resources, however, exhibit substantial variability rather than a monotonic scaling trend. Fig. 3e reveals a wide distribution of param-

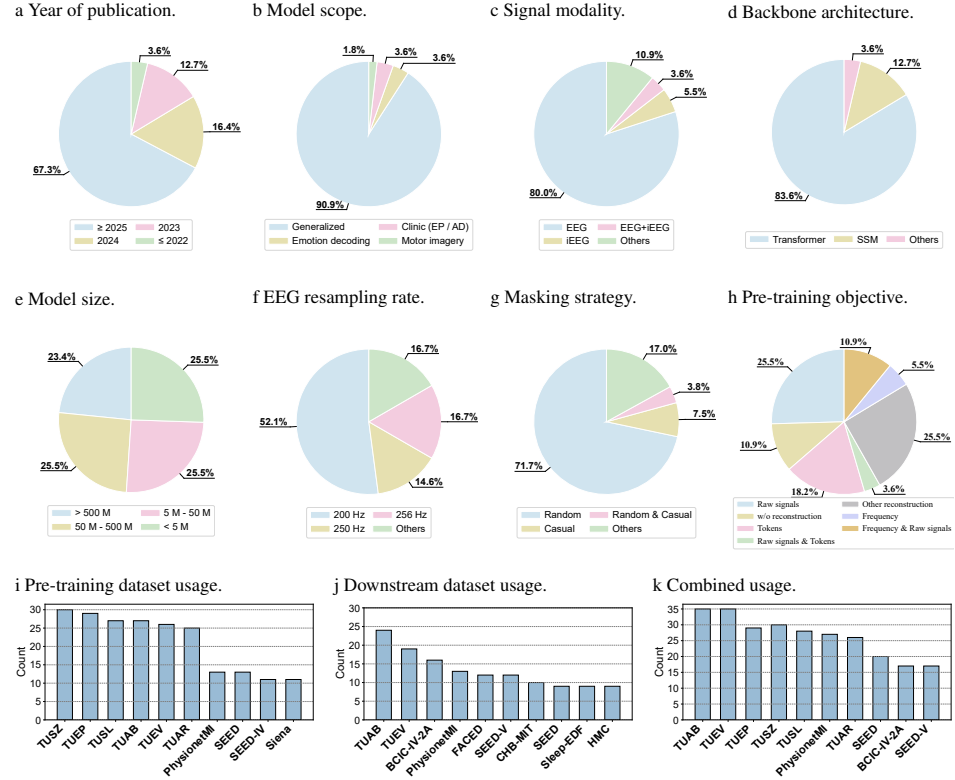


Figure 3. Statistical overview of 55 existing EEG FMs. (a-h) Distribution of key model characteristics: year of publication, model scope (generalized vs. paradigm-specific), signal modality, backbone architecture, model size, EEG resampling rate, masking strategy, and pre-training objective. (i-k) Dataset usage statistics: (i) frequency of datasets used during pre-training, (j) frequency of datasets used for downstream evaluation, and (k) combined frequency across both pre-training and downstream stages.

ter scales, and Table 1 confirms that model sizes range from fewer than one million parameters to several billion parameters.

In summary, EEG FMs have entered a phase of rapid exploration characterized by diverse model scopes, modalities, architectures, and pre-training strategies. However, existing models employed heterogeneous pre-training objectives and were evaluated under diverse downstream scenarios and fine-tuning protocols, which complicates the derivation of consistent conclusions on the factors that truly drive generalization. This observation motivates the unified framework illustrated in Fig. 2, which standardizes the major design axes of EEG FMs. The need for systematic comparison further calls for a comprehensive benchmark, which is presented in Section ‘Benchmark of EEG FMs’.

2.2. Definition of EEG FMs

EEG FMs aim to learn transferable and generalizable neural representations from large scale EEG data. In contrast to conventional BCI pipelines that are optimized for a single task and dataset, often through hand crafted features and

supervised training from scratch, EEG FMs are typically pre-trained on heterogeneous EEG data collected under different devices and paradigms. After pre-training, they can be adapted to downstream BCI tasks using fine-tuning or prompting, with the expectation of improved generalization and reduced dependence on task specific labels. Due to their versatility, FMs have become a prominent research direction in the BCI community.

A commonly used definition of an AI FM is [84]:

Definition 1 (FM) A foundation model is any model that is trained on broad data and can be adapted to a wide range of downstream tasks.

While this definition captures the essence of general purpose FMs, the design of EEG FMs exhibits several domain-specific characteristics:

- 1) EEG FMs are pre-trained on large scale EEG data, including both scalp EEG and intracranial EEG (iEEG). Additionally, other physiological signals such as electrocardiogram (ECG) and electromyogram (EMG) may serve as auxiliary data during pre-training.
- 2) General-purpose FMs are expected to han-

dle highly heterogeneous downstream tasks, such as semantic segmentation, object detection, long form question answering, and video generation [85]. In contrast, current EEG FMs primarily target classification tasks across various electrode configurations and BCI paradigms.

- 3) EEG data acquisition is resource-intensive, requiring participant recruitment, paradigm design, and stringent environmental control. As a result, EEG corpora are typically smaller than text or image corpora, and current EEG FMs are often trained with considerably fewer parameters and lower computational budgets.

Based on these domain specific considerations, we propose the following definition for EEG FMs:

Definition 2 (EEG FM) *An EEG foundation model is pre-trained on large scale electrophysiological data, and can be adapted through fine-tuning or prompting to heterogeneous EEG devices and downstream BCI tasks (categories).*

2.3. Problem Definition

Let the pre-training EEG corpus be $\mathcal{D}_{\text{pre}} = \{\mathcal{D}^{(m)}\}_{m=1}^M$, where $\mathcal{D}^{(m)} = \{X_i^{(m)}\}_{i=1}^{N_m}$, M is the number of EEG datasets, and N_m is the number of EEG trials in dataset m . Each raw EEG trial is a multi-channel time series $X_i^{(m)} \in \mathbb{R}^{C_m \times T_m}$, where C_m is the number of channels and T_m is the number of sampled time points.

Let the downstream task be $\mathcal{T} = \{\tau_j\}_{j=1}^J$, where each task τ_j specifies a paradigm, a device configuration, and a label space of size C_j . For each task τ_j , we define a labeled dataset:

$$\mathcal{D}(\tau_j) = \mathcal{D}_{\text{task}}^{(j)} = \{(X_k^{(j)}, y_k^{(j)})\}_{k=1}^{N_j},$$

$$X_k^{(j)} \in \mathbb{R}^{C_j \times T_j}, y_k^{(j)} \in \{1, 2, \dots, C_j\}. \quad (1)$$

We further denote the corpus of all downstream task datasets as

$$\mathcal{D}_{\text{down}} = \{\mathcal{D}_{\text{task}}^{(j)}\}_{j=1}^J. \quad (2)$$

An EEG FM is expected to be pre-trained on $\mathcal{D}_{\text{pre}}$ and then adapted to each downstream task in $\mathcal{T}$. Let f_{Θ} denote a pre-trained model with parameters Θ , which maps an input trial to a predictive distribution over C_j classes for task τ_j . Let $\mathcal{L}_{\text{pre}}$ denote a self-supervised pre-training objective defined on the pre-training corpus $\mathcal{D}_{\text{pre}}$. The pre-training stage estimates

$$\Theta^* = \arg \min_{\Theta} \sum_{m=1}^M \sum_{i=1}^{N_m} \mathcal{L}_{\text{pre}}(X_i^{(m)}; \Theta). \quad (3)$$

For each downstream task τ_j , we split its labeled dataset into a fine-tuning set and a test set:

$$\mathcal{D}_{\text{task}}^{(j)} = \mathcal{D}_{\text{ft}}^{(j)} \cup \mathcal{D}_{\text{te}}^{(j)}, \quad (4)$$

$$\mathcal{D}_{\text{ft}}^{(j)} \cap \mathcal{D}_{\text{te}}^{(j)} = \emptyset, \quad (5)$$

with

$$\mathcal{D}_{\text{ft}}^{(j)} = \{(X_k^{(j)}, y_k^{(j)})\}_{k=1}^{n_j}, \quad (6)$$

$$\mathcal{D}_{\text{te}}^{(j)} = \{(X_k^{(j)}, y_k^{(j)})\}_{k=n_j+1}^{N_j}. \quad (7)$$

We expect a pre-trained model could achieve strong generalization performance on $\mathcal{D}_{\text{te}}^{(j)}$ after fine-tuning on $\mathcal{D}_{\text{ft}}^{(j)}$.

Data collection and curation

Constructing EEG FMs begins with data collection, where the primary sources include public datasets and self-collected recordings. Fig. 3 presents the frequency of dataset usage in pre-training and downstream evaluation across existing EEG FMs. Most existing approaches aim to develop general-purpose models that accommodate various paradigms, and therefore aggregate large volumes of unlabeled data spanning diverse tasks for pre-training.

However, directly downloaded EEG data often exhibit inconsistent quality, including recordings from poor-performing subjects, corrupted channels, and various artifacts. Curating a large-scale, high-quality dataset is therefore critical for effective FM training [45,86]. Common curation strategies include subject-level screening to exclude participants with abnormally low task performance or excessive noise, and channel-level screening to remove channels exhibiting persistent artifacts or disconnections. Filtering out such noisy data facilitates more stable optimization and improves the quality of learned representations.

Data preprocessing

EEG signals exhibit substantial variability across subjects and devices. Variations in electrode placement, impedance, environmental noise, and physiological state can induce considerable distribution shifts, which may impair large scale pre-training and downstream transfer. Therefore, EEG FM pipelines incorporate preprocessing and normalization to reduce nuisance variability and stabilize optimization. Detailed preprocessing configurations are summarized in Section A1 of the supplementary information.

To maintain notational consistency throughout this section, we denote a raw EEG trial by

$X \in \mathbb{R}^{C \times T}$ and use the symbol $\tilde{X}$ for the model input after preprocessing. Specifically, we define a preprocessing operator

$$\tilde{X} = \mathcal{G}(X), \quad (8)$$

where $\mathcal{G}$ represents a specific alignment or normalization strategy.

Channel Unification. EEG signals exhibit substantial spatial heterogeneity because different devices adopt different electrode layouts, channel counts, reference schemes, and recording protocols. This heterogeneity makes it difficult to directly reuse conventional task-specific models across datasets. Transformer-based and sequence-based backbones can in principle process variable-length token sequences, but EEG channel tokens are not exchangeable: their meaning depends on electrode location, montage, and neurophysiological context. Among the surveyed EEG FMs, channel unification strategies can be summarized into the following categories.

- 1) *Fixed or common-montage modeling.* A straightforward strategy is to restrict pre-training to a fixed set of channels or to datasets that share a common montage, typically using standard 10–20 clinical EEG configurations. This simplifies cross-dataset alignment but discards recordings or channels that do not match the selected montage and limits transfer to unseen electrode layouts. Representative models include BENDR [28], Mentality [34], CBraMod [45], CodeBrain [55], and DIVER-0 [56].
- 2) *Canonical template or region-level remapping.* Another strategy maps heterogeneous channels into a predefined template space. This can be implemented by selecting channels that match a canonical montage, interpolating or aligning channels to a standard layout, or grouping electrodes into anatomically motivated brain regions. Compared with fixed-montage modeling, this strategy provides more explicit spatial normalization, but the resulting representation remains constrained by the chosen template or region definition. Representative models include LEAD [47], MIRepNet [61], ELASTIQ [69], DMAE-EEG [59], EpilepsyFM [65], CSBrain [58], BrainPro [67], and Uni-NTFM [68].
- 3) *Channel-wise or single-channel modeling.* Some models avoid strict montage unification by processing each channel independently or by tokenizing single-channel signals before cross-channel aggregation. This design

is naturally robust to missing channels and varying channel counts, because the model can operate on arbitrary subsets of electrodes. However, it may require additional mechanisms to recover inter-channel spatial interactions. Representative models include BIOT [31], BrainGPT [43], SingLEM [66], TFM [50], and BioCodec [70].

- 4) *Learnable channel embeddings and local spatial encodings.* Several EEG FMs inject channel identity or local spatial information through learnable channel embeddings, electrode IDs, local electrode convolutions, or positional encodings. This strategy preserves channel-specific information and is easy to integrate with Transformer backbones. However, if the embedding is tied to a fixed channel vocabulary or electrode index, its ability to generalize to unseen layouts is limited. Representative models include LaBraM [33], NeuroLM [39], EEGPT [42], CBraMod [45], and DIVER-0 [56].
- 5) *Coordinate-aware spatial encoding.* A more flexible strategy encodes electrode coordinates or sensor metadata directly. Instead of relying only on channel names or indices, these models use physical electrode positions, pairwise distances, sensor type, or 3D geometry to construct spatially aware representations. This design better supports heterogeneous and previously unseen electrode configurations. Representative models include HEAR [71], REVE [74], EEG-X [78], SAMBA [79], DeeperBrain [80], NeurIPT [73], LUNA [76], and BrainOmni [53].
- 6) *Projection to a unified latent or source space.* The most explicit form of channel unification maps heterogeneous channel sets into a fixed latent space before the main backbone. This can be achieved with learnable queries, cross-attention, sensor encoders, retrieval-based spatial modules, or channel encoders. The backbone then operates on a unified representation whose size is decoupled from the original number of electrodes. Representative models include LUNA [76], BrainOmni [53], HEAR [71], ALFEE [52], and BrainPro [67].

Resampling and Bandpass Filtering. Resampling and filtering are widely adopted to standardize temporal resolution and suppress nuisance components in EEG signals. Fig. 3f summarizes the resampling choices across the surveyed studies. Specifically, 50.0% of the studies resample signals to 200 Hz, while 34.1%

resample to 250 Hz or 256 Hz. Bandpass filtering is commonly applied to attenuate slow drifts and high-frequency noise, and notch filters at 50 Hz or 60 Hz are frequently employed to reduce power-line interference.

Normalization and Marginal Alignment. Below we describe several widely adopted normalization or data alignment approaches used in preprocessing.

- 1) *z-score Normalization.* *z*-score normalization rescales each channel to zero-mean and unit variance, ensuring comparable magnitude across channels. This technique is widely employed in the preprocessing stage of EEG FMs. For a trial X , the channel-wise statistics are defined as

$$\mu_c = \frac{1}{T} \sum_{t=1}^T X_{c,t}, \quad (9)$$

$$\sigma_c = \sqrt{\frac{1}{T} \sum_{t=1}^T (X_{c,t} - \mu_c)^2 + \epsilon}, \quad (10)$$

and the normalized signal is given by

$$\mathcal{G}_z(X)_{c,t} = \frac{X_{c,t} - \mu_c}{\sigma_c}, \quad (11)$$

where $\epsilon > 0$ is a small constant for numerical stability. Depending on the protocol, the statistics can be computed per trial, per session, or over the entire training set. Representative FMs that adopt *z*-score normalization include BrainBERT [29], Neuro-GPT [35], Brant-X [40], EEGPT [42], BrainGPT [43], LEAD [47], ALFEE [52], BrainOmni [53], UniMind [57], EpilepsyFM [65], BioCodec [70], REVE [74], and LUNA [76].

- 2) *Common Average Reference (CAR).* Another standard preprocessing technique is CAR, which suppresses common mode activity shared across all channels. Let $\mathbf{1}_C \in \mathbb{R}^C$ denote an all-ones vector. CAR transforms X as follows:

$$\mathcal{G}_{\text{car}}(X) = X - \frac{1}{C} \mathbf{1}_C \mathbf{1}_C^\top X. \quad (12)$$

The underlying assumption of CAR is that signals recorded at all electrodes contain a common noise component, such as reference electrode drift or environmental interference. By subtracting the instantaneous average across all electrodes from each individual electrode, CAR effectively attenuates this common mode component while preserving spatially localized neural activity.

Representative FMs employing CAR include EEGPT [42], E3GT [54], HEAR [71], and mdJPT [75].

- 3) *Euclidean Alignment (EA).* EA [87,88] performs subject-wise or session-wise whitening to reduce covariance shifts and improve cross-subject consistency. Assume a subject contains n trials $\{X_i\}_{i=1}^n$, where each $X_i \in \mathbb{R}^{C \times T}$. EA first computes the mean covariance matrix as

$$\bar{R} = \frac{1}{n} \sum_{i=1}^n X_i X_i^\top, \quad (13)$$

and then applies the whitening transformation

$$\mathcal{G}_{\text{ea}}(X_i) = \bar{R}^{-1/2} X_i. \quad (14)$$

After this transformation, the mean covariance of the aligned trials becomes the identity matrix, thereby reducing discrepancies in second-order statistics across subjects. Representative FMs adopting EA include EEGPT [42] and MIREPNet [61].

- 4) *Exponential Moving Average (EMA) Normalization.* To handle gradual drift in long recordings, some approaches adopt exponential moving average normalization, in which normalization statistics are updated sequentially. Let $x_t \in \mathbb{R}^C$ denote the multichannel sample at time t . EMA maintains exponentially decaying estimates of the first and second moments as follows:

$$m_t = \alpha m_{t-1} + (1 - \alpha) x_t, \quad (15)$$

$$s_t = \alpha s_{t-1} + (1 - \alpha) x_t \odot x_t, \quad (16)$$

where $\alpha \in (0, 1)$ is the decay factor and $\odot$ denotes the element-wise product. The variance estimate is given by $v_t = s_t - m_t \odot m_t$, and the normalized sample is computed as

$$\mathcal{G}_{\text{ema}}(x)_t = \frac{x_t - m_t}{\sqrt{v_t + \epsilon}}. \quad (17)$$

EMA normalization is particularly suitable for online or streaming settings, as it does not require precomputed global statistics and can adapt to non-stationarities in the signal. A representative FM employing EMA normalization is FoME [41].

- 5) *Summary.* *z*-score normalization, CAR, and EMA normalization are widely adopted as generic preprocessing components, whereas EA provides an explicit mechanism to reduce session-level covariance shifts. In practice, $\mathcal{G}$ can be instantiated as a composition of these operations. For notational simplicity, we use

the unified notation $\tilde{X} = \mathcal{G}(X)$ to denote the preprocessed model input throughout the remainder of this section.

Data Augmentation. Although data augmentation is widely adopted in supervised EEG learning, its role in EEG FM pre-training remains relatively under-explored. Among the surveyed models, only a few explicitly incorporate augmentation-like mechanisms. Specifically, BIOT [31], LEAD [47], and CoMET [64] adopt contrastive learning with perturbed views; PSGFM [62] enlarges the effective pre-training corpus through overlapping temporal windows; Uni-NTFM [68] applies signal-level perturbations including Gaussian noise, channel dropout, and temporal shifts; and REVE [74] injects Gaussian noise into electrode coordinates to improve robustness against spatial variability. These observations indicate that augmentation strategies for EEG FMs remain sparse and heterogeneous, and that the systematic design of physiologically grounded augmentation policies constitutes a promising direction for future research. Such policies could leverage neurophysiological priors specific to EEG signals, including oscillatory structure, spatial topography, and inter-trial variability, to introduce meaningful invariances while preserving task-relevant information.

Model pre-training

Most EEG FMs are pre-trained with self-supervised objectives that remove or corrupt part of the input and require the model to recover the masked information. Section A1 of the supplementary information summarizes the pre-training strategies of 55 FMs, and Fig. 3 highlights eight empirical trends across these models. Several insights can be drawn from this analysis. First, a Transformer-based backbone is adopted by approximately 83.6% of the models. Second, masked reconstruction constitutes the dominant pre-training paradigm; among the masking strategies, random masking is the most prevalent, accounting for approximately 71.7%, while causal masking and mixed masking together occupy a smaller portion. Third, regarding the reconstruction target, raw-signal reconstruction is the most frequent strategy, accounting for approximately 25.5%, while token reconstruction and hybrid approaches combining raw and token reconstruction together constitute a comparable fraction. Codebook-based and frequency-domain objectives appear less frequently as standalone targets, but are often employed as auxiliary supervision in multi-

target designs. Based on these observations, we organize the mainstream pre-training strategies into five categories: masked reconstruction of raw signals, masked reconstruction of embedded tokens, frequency-domain reconstruction, codebook-based objectives, and autoregressive pre-training.

Given a raw EEG trial $X \in \mathbb{R}^{C \times T}$, we denote the model input after preprocessing by $\tilde{X} = \mathcal{G}(X)$. Since EEG is typically sampled at a high temporal resolution, most EEG FMs further aggregate time steps into patches to reduce the sequence length and capture local temporal structure. Let the patch length be M and the stride be S , with overlap $O = M - S$. We segment $\tilde{X}$ along the temporal axis into N_p patches, where

$$N_p = \left\lfloor \frac{T - M}{S} \right\rfloor + 1, \quad (18)$$

and denote the resulting patch tensor by

$$P = \mathcal{S}(\tilde{X}) \in \mathbb{R}^{N_p \times C \times M}, \quad (19)$$

where $\mathcal{S}$ denotes the patching operator. The k -th patch of channel c is denoted by $p_{k,c} \in \mathbb{R}^M$.

For notational clarity, the superscripts used in Section A1 of the supplementary information indicate the target space: raw signals (*rs*), embedded tokens (*et* or *emb*), spectrograms (*spec*), spectral amplitude (*sa*), band power (*bp*), codebook indices (*ci*), denoised raw signals (*nrs*), and diffusion velocity (*v*).

Masked Reconstruction of Raw Signals. Raw signal reconstruction is the most prevalent pre-training strategy, employed by representative models such as Brant [32], FoME [41], CBraMod [45], CSBrain [58], EEGMamba [60], BrainPro [67], REVE [74], and EEG-X [78]. In this strategy, masking is applied directly to the patched raw signal,

$$P_{\text{msk}} = \mathcal{M}_x(P), \quad (20)$$

and the encoder consumes the masked patches. The model is trained to reconstruct the original patches using the mean squared error (MSE) loss:

$$\hat{\tilde{X}} = \mathcal{D}_\phi(\mathcal{E}_\theta(P_{\text{msk}})), \quad (21)$$

$$\mathcal{L}_{\text{mse}}^{\text{rs}} = \|\hat{\tilde{X}} - \tilde{X}\|_2^2. \quad (22)$$

This approach directly constrains the encoder to preserve the waveform structure and the cross-channel dependencies that are critical for event-related components and oscillatory bursts. However, non-invasive EEG typically exhibits a

low signal-to-noise ratio and contains substantial nuisance variability arising from artifacts, impedance fluctuations, and background activity. When the pre-training target is the waveform itself, a model with sufficient capacity may expend representational power on reconstructing idiosyncratic noise patterns that are not predictive for downstream tasks.

To mitigate this issue, several models replace or augment the $\mathcal{L}_{\text{mse}}^{\text{rs}}$ with more robust variants. For example, FEMBA employs the smooth ℓ_1 loss $\mathcal{L}_{s-l1}^{\text{rs}}$, BrainPro uses a weighted variant $\mathcal{L}_{w\text{-mse}}^{\text{rs}}$ together with a decomposition loss $\mathcal{L}_{\text{dec}}$, and REVE incorporates an auxiliary loss $\mathcal{L}_{\text{aux}}$. EEG-X further emphasizes denoising by reconstructing noise-removed signals via $\mathcal{L}_{\text{mse}}^{\text{nrs}}$ and incorporates teacher-student distillation $\mathcal{L}_{\text{kd}}^{\text{tea-stu}}$, which aligns with the practical need to suppress artifacts rather than reproduce them. Denoising variants like EEG-X replace the noisy target $p_{k,c}$ with a cleaned or artifact-suppressed target $p_{k,c}^{\text{den}}$:

$$\mathcal{L}_{\text{mse}}^{\text{nrs}} = \frac{1}{|\Omega|} \sum_{(k,c) \in \Omega} \|\hat{p}_{k,c} - p_{k,c}^{\text{den}}\|_2^2. \quad (23)$$

These robust and denoising variants are particularly relevant for non-invasive EEG, where direct waveform regression may encourage the model to reproduce artifacts and recording-specific noise rather than task-relevant neural activity.

In summary, raw-signal reconstruction serves as a strong baseline when the data quality is controlled and the masking is sufficiently challenging, but it may benefit from explicit regularization that discourages the memorization of noise.

Masked Reconstruction of Embedded Tokens. Instead of reconstructing waveforms, another group of EEG FMs reconstructs latent representations produced by a tokenizer, convolutional embedder, or patch embedding module. This strategy is adopted by BENDR [28], BIOT [31], GEFM [44], CEReBrO [46], LCM [49], LUNA [76], ELASTIQ [69], MIRepNet [61], and EEGMoE [81].

For token-based pre-training, each patch is first mapped to an embedding through a tokenizer:

$$Z = \mathcal{T}_{\psi}(P), \quad Z \in \mathbb{R}^{N_P \times d}, \quad (24)$$

where d denotes the embedding dimension. Masking is then applied in the token space:

$$Z_{\text{msk}} = \mathcal{M}_z(Z). \quad (25)$$

The model learns to predict the original token embeddings:

$$\hat{Z} = \mathcal{D}_{\phi}(\mathcal{E}_{\theta}(Z_{\text{msk}})), \quad \mathcal{L}_{\text{mse}}^{\text{et}} = \|\hat{Z} - Z\|_2^2. \quad (26)$$

This embedded-token reconstruction loss, denoted as $\mathcal{L}_{\text{mse}}^{\text{et}}$ or $\mathcal{L}_{\text{mse}}^{\text{emb}}$ in Section A1 of the supplementary information, is used by CEReBrO [46] for masked token reconstruction, by LCM [49] as a reconstruction term combined with contrastive learning, and by MIRepNet [61] as part of its hybrid pre-training objective together with a supervised MI classification loss. EEGMoE [81] adopts an ℓ_1 variant, denoted as $\mathcal{L}_{\ell_1}^{\text{et}}$, to learn domain-decoupled token representations with additional MoE regularization.

Compared to raw signal reconstruction, token-level objectives aim to reduce sensitivity to amplitude scaling and local waveform noise, as the tokenizer compresses the input into a representation that can be designed to emphasize spatio-temporal structure. This design often improves optimization stability for large encoders and naturally supports patch-based processing, which is consistent with the high prevalence of patching observed in Fig. 3.

Contrastive learning constitutes a complementary latent-space objective, in which a representation is contrasted against semantically related and unrelated counterparts, which is used by BENDR [28], BIOT [31], GEFM [44], LCM [49], Brant-X [40], LEAD [47], MBrain [30], and CoMET [64]. Given an anchor representation h_i , a positive representation h_i^+ , a set of negatives $\mathcal{N}_i$, a similarity function $\text{sim}(\cdot, \cdot)$, and a temperature τ , a generic InfoNCE-style loss is

$$s_i^+ = \frac{\text{sim}(h_i, h_i^+)}{\tau}, \quad s_{ij}^- = \frac{\text{sim}(h_i, h_j)}{\tau}. \quad (27)$$

$$\mathcal{L}_{\text{cl}} = -\frac{1}{B} \sum_{i=1}^B \log \frac{\exp(s_i^+)}{\exp(s_i^+) + \sum_{h_j \in \mathcal{N}_i} \exp(s_{ij}^-)}. \quad (28)$$

The construction of positive-negative pairs varies considerably across models. BENDR and GEFM apply contrastive prediction over latent EEG tokens, denoted $\mathcal{L}_{\text{cl}}^{\text{et}}$; BIOT contrasts channel-wise biosignal tokens; and CoMET augments masked reconstruction with a global contrastive term $\mathcal{L}_{\text{cl}}^{\text{glob}}$ to reinforce global discriminative structure. Brant-X performs two-level contrastive alignment, denoted $\mathcal{L}_{\text{cl}}^{\text{coarse}}$ and $\mathcal{L}_{\text{cl}}^{\text{fine}}$, to align EEG with auxiliary physiological signals at different semantic scales, whereas

MBrain combines multi-channel contrastive predictive coding with time-shift prediction to capture spatial and temporal dependencies. LEAD employs contrastive pre-training to learn AD-relevant EEG representations.

Compared with raw-signal reconstruction, latent-token reconstruction and contrastive learning reduce direct dependence on local waveform amplitude and can improve robustness to nuisance variability. However, their effectiveness depends strongly on the tokenizer, augmentation scheme, and positive-negative pair construction. If the tokenizer is too shallow, the latent reconstruction target may remain close to waveform regression; if it is too coarse, transient or task-relevant EEG patterns may be discarded.

Several models address this tension by combining token-level objectives with auxiliary terms. CEREBrO incorporates $\mathcal{L}_{\text{aux}}$ to enrich the learning signal. LCM combines contrastive learning $\mathcal{L}_{\text{cl}}^{\text{emb}}$ with a weighted reconstruction term $\lambda \mathcal{L}_{\text{mse}}^{\text{et}}$. MIRepNet integrates $\mathcal{L}_{\text{mse}}^{\text{et}}$ with a classification loss $\mathcal{L}_{\text{cls}}$ to bias the representation toward discriminative structure relevant to its target paradigm. These examples suggest that token reconstruction often benefits from complementary objectives that encourage global semantic learning rather than purely local fidelity.

Frequency-Domain Reconstruction. Frequency-domain objectives supervise EEG FMs with spectral targets rather than, or in addition to, time-domain waveforms. This strategy is used by BrainBERT and BrainWave for spectrogram or time-frequency reconstruction, Mentality for joint raw-signal and spectrogram reconstruction, EEGFormer for spectral-amplitude reconstruction, Uni-NTFM for frequency / band-power supervision, Gram for spectral mimic learning with codebook classification, E3GT and PSGFM for spectrogram-derived pseudo-label prediction, HEAR for codebook-embedding and frequency reconstruction, BioCodec for STFT-based codec reconstruction, and Brain-OF for joint raw-signal and spectral-amplitude reconstruction.

Let $\mathcal{F}_r$ denote a spectral transform and

$$S^r = \mathcal{F}_r(P) \quad (29)$$

be the spectral target of EEG patches P , where r denotes the target type. For regression-based spectral supervision, the generic masked spectral reconstruction loss is

$$\mathcal{L}_{\text{mse}}^r = \frac{1}{|\Omega|} \sum_{(k,c) \in \Omega} \left\| \hat{S}_{k,c}^r - S_{k,c}^r \right\|_2^2, \quad (30)$$

$$r \in \{\text{spec, sa, bp, freq}\}.$$

Here, $\mathcal{L}_{\text{mse}}^{\text{spec}}$ is used by BrainBERT and Mentality; $\mathcal{L}_{\text{mse}}^{\text{sa}}$ is used by EEGFormer; $\mathcal{L}_{\text{mse}}^{\text{bp}}$ is used by Uni-NTFM; and $\mathcal{L}_{\text{mse}}^{\text{freq}}$ is used by HEAR and SAMBA as a frequency-domain reconstruction term. Brain-OF uses a robust smooth- ℓ_1 variant for spectral-amplitude reconstruction, denoted by $\mathcal{L}_{s\text{-}l1}^{\text{sa}}$.

Some models use discrete spectral pseudo-labels instead of continuous spectral regression. E3GT and PSGFM follow a HuBERT-style design, where spectrogram-derived k -means labels are treated as prediction targets. If $q_{k,c}$ denotes the spectral k -means label of patch (k, c) and $\hat{\pi}_{k,c}$ denotes the predicted categorical distribution, the corresponding loss is

$$\mathcal{L}_{\text{cls}}^{\text{skl}} = -\frac{1}{|\Omega|} \sum_{(k,c) \in \Omega} \log \hat{\pi}_{k,c,q_{k,c}}, \quad (31)$$

where $\mathcal{L}_{\text{cls}}^{\text{skl}}$ refers to spectral pseudo-label classification, as used by E3GT and PSGFM.

Several models combine spectral supervision with other reconstruction or auxiliary terms. EEGFormer combines spectral-amplitude reconstruction with codebook-embedding reconstruction, denoted as $\mathcal{L}_{\text{mse}}^{\text{sa}} + \mathcal{L}_{\text{mse}}^{\text{cbe}}$. Gram combines spectral-amplitude mimic learning with codebook-index classification, denoted as $\mathcal{L}_{\text{mse}}^{\text{sa}} + \mathcal{L}_{\text{cls}}^{\text{ci}}$. HEAR combines codebook-embedding reconstruction and frequency-domain reconstruction, denoted as $\mathcal{L}_{\text{mse}}^{\text{ce}} + \mathcal{L}_{\text{mse}}^{\text{freq}}$, where the superscript ce refers to codebook embedding rather than cross-entropy. BioCodec combines a Huber-style raw-signal reconstruction term with multi-scale STFT losses and an auxiliary codec regularizer, summarized as $\mathcal{L}_{\text{huber}}^{\text{rs}} + \mathcal{L}_{\ell_1}^{\text{stft}} + \mathcal{L}_{\ell_2}^{\text{stft}} + \mathcal{L}_{\text{aux}}$. SAMBA combines time-domain and frequency-domain reconstruction, denoted as $\mathcal{L}_{\text{mae}}^{\text{os}} + \mathcal{L}_{\text{mse}}^{\text{freq}}$. ALFEE combines raw-signal reconstruction, PSD-enhanced reconstruction, and downstream-task prediction, denoted as $\mathcal{L}_{\text{mse}}^{\text{rs}} + \mathcal{L}_{\text{mse}}^{\text{rs} \oplus \text{PSD}} + \mathcal{L}_{\text{cls}}^{\text{dt}}$. These designs indicate that frequency-domain supervision is usually used to emphasize rhythmic EEG structure, while time-domain or auxiliary losses are often retained to preserve waveform fidelity and task-relevant information.

Codebook-Based Objectives. Codebook-based pre-training discretizes continuous EEG representations into neural tokens, which can then be predicted as code indices or reconstructed as codebook embeddings. This strategy is adopted by LaBraM [33], BrainOmni [53], CodeBrain [55], EpilepsyFM [65], NeuroRVQ [72], NeuroLM [39], and THD-BAR [77].

Let a quantizer map embeddings to discrete indices:

$$I = Q(Z), \quad I \in \{1, 2, \dots, K\}^L, \quad (32)$$

where K denotes the codebook size and L denotes the sequence length. For masked code prediction, the model estimates a categorical distribution $\hat{P} \in [0, 1]^{L \times K}$ over the codebook and minimizes the cross-entropy loss:

$$\mathcal{L}_{\text{cls}}^{ci} = \sum_{\ell=1}^L \mathcal{L}_{\text{ce}}(I_{\ell}, \hat{P}_{\ell}). \quad (33)$$

Autoregressive index modeling employs negative log-likelihood objectives such as $\mathcal{L}_{\text{nl}}^{ci}$, which is adopted by NeuroLM and THD-BAR. An alternative branch aligns predicted embeddings with codebook embeddings, represented by $\mathcal{L}_{\text{mse}}^{che}$ or related terms, as employed by HEAR [71].

The advantage of codebook-based objectives is that discretization provides a compact symbolic representation of EEG, which can reduce the influence of low-level waveform noise and enable large-scale sequence modeling. It also facilitates causal generation and prompt-style adaptation when combined with decoder-only architectures. However, codebook learning introduces additional design choices, including codebook size, commitment regularization, residual quantization depth, and update schedules. If these factors are not carefully controlled, the learned codebook may collapse or show imbalanced usage, thereby weakening the learned representations.

Autoregressive Pre-training. Autoregressive pre-training imposes a causal factorization over an ordered EEG token sequence. It is used by Neuro-GPT [35], BrainGPT [43], NeuroLM [39], and THD-BAR [77]. Given an ordered token sequence $u_{1:L}$, where u_{ℓ} can represent an EEG chunk, an embedded token, or a codebook index, the generic autoregressive loss is

$$\mathcal{L}_{\text{ar}} = - \sum_{\ell=1}^L \log p_{\Theta}(u_{\ell} \mid u_{1:\ell-1}). \quad (34)$$

For continuous EEG tokens, as in Neuro-GPT and BrainGPT, this objective is typically implemented through regression or next-token prediction over EEG chunks or embeddings. For discrete neural codes, as in NeuroLM, Eq. 34 corresponds to the negative log-likelihood objective $\mathcal{L}_{\text{nl}}^{ci}$.

Autoregressive modeling is attractive because it aligns naturally with decoder-only Transformer architectures and supports sequence continuation, generative modeling, and prompt-based adaptation; it also provides a principled means of modeling temporal dependencies in EEG. Nevertheless, causal objectives can be more difficult to optimize than bidirectional masked reconstruction, particularly when the EEG tokens are high-dimensional or when their ordering does not faithfully reflect the underlying neurophysiological structure. A purely temporal ordering may further bias the model toward local continuity rather than task-relevant global structure, whereas topology-aware designs, exemplified by THD-BAR, instead seek to jointly capture spatial hierarchy and temporal evolution.

Hybrid Objectives and Practical Selection. Several models adopt hybrid designs that combine two or more complementary targets. Examples include Mentality, which combines $\mathcal{L}_{\text{mse}}^{rs}$ and $\mathcal{L}_{\text{mse}}^{spec}$; EEGPT [42], which combines $\mathcal{L}_{\text{mse}}^{rs}$ with an embedding reconstruction term $\mathcal{L}_{\text{mse}}^{emb}$; DMAE-EEG [59], which combines $\mathcal{L}_{\text{mse}}^{rs}$ and $\mathcal{L}_{\text{mse}}^{et}$; and SAMBA [79], which combines time-domain and frequency-domain reconstruction. These hybrid formulations should be interpreted as deliberate design choices to constrain the representation from complementary perspectives, rather than an indiscriminate aggregation of all available losses.

In practice, the choice of pre-training objective should be guided by the intended deployment setting. When robustness to cross-dataset heterogeneity and low signal-to-noise ratio is critical, token-level or codebook-based objectives can provide stronger invariance than direct waveform regression. When rhythmic neural activity is central to the downstream task, frequency-domain supervision can be beneficial, especially when combined with a time-domain constraint. When causal generation, prompt-based adaptation, or LLM-style modeling is desired, autoregressive objectives become a natural choice, but require careful tokenization and ordering design.

Summary. Existing EEG FMs largely adhere to a masked prediction paradigm, but differ substantially in their target space and implied inductive biases. Table 2 of the supplementary information summarizes these design choices through the reconstruction objective and loss function columns, encompassing $\mathcal{L}_{\text{mse}}^{rs}$ and its robust variants, $\mathcal{L}_{\text{mse}}^{et}$ and $\mathcal{L}_{\text{cl}}^{et}$, spectral losses

such as $\mathcal{L}_{mse}^{spec}$ and $\mathcal{L}_{mse}^{sa}$, codebook index losses such as $\mathcal{L}_{cls}^{ci}$ and $\mathcal{L}_{nll}^{ci}$, as well as additional auxiliary terms that refine the learning signal. This taxonomy provides a consistent framework for comparing pre-training strategies under heterogeneous EEG settings and clarifies why different approaches may be preferable for specific BCI paradigms and deployment constraints. Detailed experimental analysis is presented in the following section.

Downstream generalization

After pre-training on large-scale EEG corpora, downstream evaluation is required to assess whether the learned representations transfer effectively to practical BCI tasks. Fig. 3j summarizes the most frequently used downstream datasets. These datasets span multiple representative paradigms. For example, TUAB, TUEV, and CHB-MIT are clinical EEG datasets; BCIC-IV-2A and PhysioNetMI are MI datasets; FACED, SEED, and SEED-V are emotion recognition datasets; and Sleep-EDF is a sleep-related dataset. This distribution indicates that downstream evaluation in existing work is largely concentrated on clinical applications, MI, affective decoding, and sleep analysis.

Following the problem definition, each downstream task τ_j is associated with a dataset $\mathcal{D}_{task}^{(j)}$, which is partitioned into a fine-tuning set $\mathcal{D}_{ft}^{(j)}$ and a held-out test set $\mathcal{D}_{te}^{(j)}$. Given a pre-trained model f_{Θ^*} , task-specific fine-tuning estimates

$$\Theta_j^* = \arg \min_{\Theta} \sum_{(X,y) \in \mathcal{D}_{ft}^{(j)}} \mathcal{L}_{cls}^{(j)}(y, f_{\Theta}(X)), \quad (35)$$

where $\mathcal{L}_{cls}^{(j)}$ denotes a supervised loss for task τ_j . The fine-tuned model $f_{\Theta_j^*}$ is then evaluated on the held-out test set. Taking classification accuracy as an example, the evaluation is formulated as follows:

$$\hat{y} = \arg \max_{c \in \{1,2,\dots,C_j\}} \left[f_{\Theta_j^*}(X) \right]_c, \quad (36)$$

$$\text{Acc}(\tau_j) = \frac{1}{|\mathcal{D}_{te}^{(j)}|} \sum_{(X,y) \in \mathcal{D}_{te}^{(j)}} \mathbf{1}(\hat{y} = y), \quad (37)$$

where $\mathbf{1}(\cdot)$ denotes the indicator function.

Regarding evaluation scenarios, most existing studies adopt a leave-one-subject-out (LOSO) setting or perform subject-disjoint splits into training, validation, and test sets. Such protocols are valuable for assessing cross-subject generalization, as the test subject remains unseen during fine-tuning. However, these protocols typically require a substantial amount of labeled data from

the same device and paradigm for fine-tuning, since data from multiple training subjects are aggregated for adaptation. This requirement can be restrictive in practical deployment scenarios where only limited calibration data may be available for a new user.

In principle, a FM is expected to reduce dependence on task-matched labeled data and enable effective adaptation with minimal calibration. This motivates the need for a more comprehensive evaluation suite that encompasses both data-rich cross-subject transfer and data-limited calibration regimes under consistent protocols. Section ‘Benchmark of EEG FMs’ presents our benchmark design, which is constructed to address these complementary requirements.

3. BENCHMARK OF EEG FMs

This section presents a comprehensive benchmark for EEG FMs. We evaluate 12 open source FMs and 8 specialist baselines, including conventional machine learning and deep learning methods, on 13 datasets spanning 9 representative BCI paradigms. To assess model generalization under realistic deployment constraints, we design a set of comprehensive and fair evaluation scenarios. An overview of the datasets and evaluation scenarios is illustrated in Fig. 2.

3.1. Datasets

Fig. 3j summarizes the downstream datasets most frequently adopted in prior studies, where clinical EEG, MI, emotion recognition, and sleep staging emerge as the dominant evaluation paradigms. However, this concentration on a limited set of paradigms may not fully reflect the generalization capability of FMs across diverse BCI applications. To address this limitation, we selected 13 datasets spanning 9 BCI paradigms for downstream evaluation, providing broader coverage of representative BCI scenarios. Specifically, the selected datasets include BNCI2014001 [89], BNCI2014004 [90], BNCI2015001 [91], BNCI2014009 [92], BNCI2014008 [92], CHB-MIT [93], TUAB [94], Sleep-EDFx [95], SEED [96], Nakanishi2015 [97], EEGMat [98], Things-EEG2 [99], and SEED-VIG [100]. For paradigms with widespread practical applications, such as MI, P300, and clinical diagnosis, we included two to three datasets to ensure comprehensive evaluation. The dataset characteristics are summarized in Section A3 of the supplementary information.

3.2. Evaluation Scenarios

Most existing EEG FM studies evaluated downstream transfer under a LOSO scenario or subject-disjoint splits into training, validation, and test sets. While this setting is widely adopted, it remains unclear whether it provides a comprehensive assessment of FM generalization and whether it aligns with practical deployment requirements.

LOSO Scenario. The LOSO scenario evaluates cross-subject generalization within the same task and headset configuration. Concretely, a model is fine-tuned using labeled data from a subset of subjects recorded with the same EEG device and paradigm, and is subsequently evaluated on held-out subjects. The primary advantage of LOSO is that the target subject is evaluated in a zero-calibration manner, as no labeled data from the test subject are used during fine-tuning. However, LOSO has two important limitations. First, it typically requires a substantial amount of labeled data from multiple subjects, which increases the fine-tuning cost. Second, it implicitly assumes the availability of a corpus collected with the same device configuration and task as the target deployment setting, which may not hold in practice, particularly for new devices or customized paradigms.

In our benchmark, LOSO fine-tuning followed the common practice of using all available trials from the fine-tuning subjects for most datasets. However, for the MI and P300 datasets (BNCI2014001, BNCI2014004, BNCI2015001, BNCI2014009, and BNCI2014008), we used only a single session from each subject for fine-tuning. For CHB-MIT, we used the ictal segments together with the 10-minute pre-ictal segments of each seizure for model adaptation and evaluation. For TUAB, we used the first 3 minutes of each recording as input segments. For Things-EEG2, we used three out of ten images per class. For Sleep-EDFx and TUAB, which contain a large number of subjects, we adopted a ten-fold subject split, where subjects were evenly partitioned into ten folds for cross-validation.

Within-Subject Few-Shot Scenario. To better reflect deployment settings where only limited calibration data are available for a new user, we designed a within-subject few-shot evaluation protocol. In this setting, the model was fine-tuned using a small labeled subset from the target subject and evaluated on the remaining data of the same subject. The within-subject scenario offers

two advantages. First, it substantially reduces the amount of fine-tuning data and lowers the adaptation cost. Second, it does not require an external training corpus that matches the target device and paradigm, thereby supporting rapid personalization for new devices, tasks, and users. The primary limitation is that it requires collecting labeled calibration data from the target user during deployment, which may be inconvenient in certain applications.

The fine-tuning ratio adopted for each dataset is determined by dataset-specific constraints, including the total number of trials available per subject, the trial length, and the class distribution of the paradigm. Detailed protocols are summarized in Table 12 of the supplementary information.

Specifically, for the MI datasets (BNCI2014001, BNCI2014004, and BNCI2015001), we fine-tuned using 30% of one session from the target subject, with fewer than 30 trials per class. For P300 datasets, we fine-tuned using 10% of one session for BNCI2014009 and 5% of one session for BNCI2014008. For CHB-MIT, we fine-tuned using the first seizure's ictal segments from the target subject together with its 10-minute pre-ictal segment, and evaluated on the remaining seizures' ictal segments with their corresponding pre-ictal segments. For Sleep-EDFx, we used 10% of the target subject's data for fine-tuning. For SEED, we used one video per class for fine-tuning. For Nakanishi2015 and EEGMat, we fine-tuned using 80% and 60% of the target subject's data, respectively. For Things-EEG2, we used three out of ten images per class for each subject. For SEED-VIG, we fine-tuned using 10% of the data from a single subject. We did not include a within-subject few-shot setting for TUAB, as each subject is associated with a single diagnostic label (normal or abnormal).

Fine-Tuning Strategies. For both LOSO and within-subject few-shot scenarios, we compared two fine-tuning strategies to assess the quality of pre-trained representations. The first strategy, *full-parameter fine-tuning*, updated all model parameters during fine-tuning, allowing the entire network to be optimized for the downstream task. The second strategy, *linear probing*, froze the pre-trained encoder and trained only the classification head, which directly evaluated the transferability of the learned representations without task-specific feature adaptation. By comparing these two strategies, we aimed to disentangle the contributions of pre-trained

representations from those of end-to-end fine-tuning.

Summary. The LOSO and within-subject few-shot protocols evaluate complementary aspects of generalization. LOSO measures cross-subject transfer under a fixed paradigm and device configuration without test-subject calibration, whereas within-subject few-shot evaluates rapid personalization with limited calibration data. Furthermore, the comparison between full-parameter fine-tuning and linear probing provides insights into the quality and transferability of pre-trained representations. By reporting results under both scenarios with both fine-tuning strategies, our benchmark provides a more comprehensive assessment of EEG FM generalization and better reflects practical deployment constraints.

3.3. Evaluated Approaches

In this benchmark, we compared traditional machine learning baselines, 7 deep learning models (including 4 CNN-based and 3 Transformer-based architectures), and 12 EEG FMs. The following provides a detailed description of each category.

Traditional Machine Learning Baselines. For each dataset, we selected a paradigm-specific traditional machine learning algorithm as the baseline, which remains competitive against deep learning methods in the respective domain. CSP+LDA (linear discriminant analysis) [101] was used for BNCI2014001, BNCI2014004, BNCI2015001, TUAB, SEED, and EEG-Mat. xDAWN+LDA [102] was employed for the P300 datasets BNCI2014009 and BNCI2014008. PSD+SVM (Power Spectral Density features and Support Vector Machine classifier) [103] was used for CHB-MIT. PSD+LDA [104] was applied to Sleep-EDFx. TRCA (task-related component analysis) [105] was used for Nakanishi2015. PSD+Ridge (ridge regression) [106] was employed for SEED-VIG.

Deep Learning Baselines. We evaluated 7 task-specific deep learning models trained from scratch. For CNN-based architectures, we included EEGNet [11], ShallowConvNet [10], LMDA-Net [107], and TSception [108]. For Transformer-based architectures, we included CNN-Transformer [109], Deformer [110], and Conformer [14].

These models are widely adopted in EEG

decoding and serve as strong baselines for comparison with FMs. We selected general-purpose models that are applicable across paradigms, rather than algorithms designed around the neurophysiological principles of a specific paradigm. Consequently, the chosen baselines are competitive but not necessarily the state-of-the-art for every individual paradigm. Our aim is to contrast EEG FMs against strong, representative task-specific models, rather than to quantify the full gap between FMs and the best-performing specialized method for each task.

EEG FMs. We evaluated all EEG FMs with publicly available code and pre-trained weights, including BENDR [28], BIOT [31], LaBraM [33], Neuro-GPT [35], EEGPT [42], CBraMod [45], TFM [50], BrainOmni [53], EEGMamba [60], MIRepNet [61], SingLEM [66], and LUNA [76]. Among these, MIRepNet is a paradigm-specific FM designed exclusively for MI tasks, while the remaining 11 models are general-purpose EEG FMs intended to support multiple BCI paradigms.

3.4. Main Results

We performed all benchmarking experiments on 24 NVIDIA A800 GPUs and 8 NVIDIA A100 GPUs. All results are averaged over 3 random seeds, and we reported the performance on the test set at the final epoch of fine-tuning. For fair comparison, the fine-tuning hyperparameters of each model were individually tuned rather than fixed to a single shared configuration.

For evaluation metrics, we reported balanced classification accuracy (BCA) as the primary metric for all classification tasks, 2-way accuracy for the Things-EEG2 dataset, and root mean squared error (RMSE) for the SEED-VIG regression task. Several settings specific to individual models and datasets are also noted here. BrainOmni was excluded from the CHB-MIT and Sleep-EDFx datasets, as its implementation does not support the bi-polar montage used in these datasets. In addition, BIOT includes three variants, BIOT-1D, BIOT-2D, and BIOT-6D, pre-trained on 1, 2, and 6 datasets, respectively. We reported results for BIOT-6D, the variant trained on the largest data scale, in the main tables, while results for BIOT-1D and BIOT-2D are provided in Section A4 of the supplementary information. The main results are summarized in Tables 2 and 3, and comprehensive per-subject results are provided in Section A4 of the supplementary in-

formation.

The detailed ranking results are provided in Tables 49 and 50 of the supplementary information. To assess the potential influence of pre-training–downstream data overlap on these rankings, we further reported a leakage-aware ranking in Table 51 of the supplementary information. Specifically, for each FM, we excluded downstream datasets utilized in its pre-training corpus (marked in Tables 2 and 3) before computing its average rank, yielding a more conservative estimate of generalization to unseen downstream distributions. The leakage-aware ranking preserves the relative ordering of the top-ranked models, with CBraMod and NeuroGPT retaining their positions among the top FMs. This suggests that the main findings reported in this section are not primarily driven by such overlap.

Can FMs learn generalized representations?

Tables 2 and 3 present the performance of 12 EEG FMs across 13 downstream datasets. A notable observation is that head-only fine-tuning (linear probing) consistently yielded inferior, and in many cases substantially lower, performance compared to full-parameter fine-tuning for most FMs. This finding suggested that adapting FMs to diverse downstream tasks cannot rely solely on features extracted by pre-trained encoders; task-specific fine-tuning of the encoder parameters remains essential. While several models such as EEGPT exhibited superior head-only performance relative to full fine-tuning, neither adaptation strategy achieved consistently strong results across the benchmark.

EEG FMs also exhibited considerable variability in their task-specific performance. For instance, CBraMod demonstrated competitive results across most tasks, achieving first and second place on the SEED dataset under LOSO and few-shot scenarios, respectively. However, it yielded the highest RMSE among all evaluated methods on SEED-VIG in the LOSO scenario. Similarly, LUNA attained state-of-the-art performance on TUAB but failed to generalize effectively to paradigms beyond clinical applications, a limitation likely attributable to its pre-training exclusively on TUEG and Siena datasets.

An encouraging finding emerged from the Nakanishi2015 dataset, an SSVEP paradigm with extremely limited fine-tuning data (12 trials per class). Despite this constraint, several FMs, including BENDR, EEGPT, and NeuroGPT, achieved strong performance. Since the SSVEP paradigm relies on decoding neural responses to

target stimuli flickering at distinct frequencies, the resulting signals exhibit pronounced periodicity and temporal structure. Consequently, the masked reconstruction objectives employed during pre-training may endow these models with enhanced capability to capture temporal dynamics of EEG signals.

To qualitatively examine the separability of the learned representations, we visualized the feature embeddings using *t*-SNE. We considered three representative datasets, SEED, TUAB, and BNCI2014008, corresponding to emotion recognition, clinical abnormality detection, and ERP/RSVP decoding, respectively. For compactness, Fig. 4 presents a representative subset, namely four top-performing specialist models (EEGNet, Conformer, ShallowConv, and Deformer) and three EEG FMs (CBraMod, NeuroGPT, and LaBraM) on SEED and TUAB, with both full-parameter fine-tuning and linear probing shown for each FM. The complete visualizations are provided in Section A5 of the supplementary information. Full-parameter fine-tuning generally yields more discriminative structures than linear probing, with clearer between-class separation and more compact within-class clusters. This qualitative evidence is consistent with our quantitative results, indicating that the frozen representations of current EEG FMs are not always sufficiently separable for downstream decoding.

We further conducted a scenario-level paired Wilcoxon signed-rank analysis to quantify whether full-parameter adaptation consistently improves over linear probing, with detailed results reported in Table 55 of the supplementary information. The results show that full fine-tuning provides a statistically significant advantage for the majority of EEG FMs. Specifically, 9 out of 12 FMs exhibited significant positive median Δ values, indicating that full fine-tuning outperformed linear probing across dataset–scenario pairs. Several models showed particularly strong directional consistency: BENDR and CBraMod favored full fine-tuning in 23 out of 25 scenarios, while TFM favored full fine-tuning in 24 scenarios with one tie. In contrast, EEGPT was the only model for which linear probing showed a significant advantage, whereas BrainOmni-Tiny and SingLEM did not exhibit significant differences between the two adaptation strategies. These findings reinforce our conclusion that current EEG FMs have not yet learned sufficiently universal representations for robust downstream decoding. Although pre-training provides useful initialization, task-

Table 2. Benchmark performance. The best metrics are marked in bold, and the second best by an underline. ‡ indicates that the corresponding dataset was used during pre-training of the model.

Approach	BNCI2014001	BNCI2014004	BNCI2015001	BNCI2014009	BNCI2014008	CHB-MIT	TUAB
Cross-subject (LOSO) — Full fine-tuning							
<i>Specialist models</i>							
Traditional ML	38.19	73.24	56.42	65.14	58.69	69.09 $\pm$ 0.04	66.03 $\pm$ 0.25
EEGNet	44.97 $\pm$ 0.57	76.38 $\pm$ 0.59	63.40 $\pm$ 1.32	78.39 $\pm$ 0.44	72.29 $\pm$ 0.17	77.36 $\pm$ 0.54	77.03 $\pm$ 0.67
ShallowConv	44.80 $\pm$ 0.50	74.23 $\pm$ 0.39	63.89 $\pm$ 0.80	78.05 $\pm$ 0.62	69.92 $\pm$ 0.05	80.18 $\pm$ 0.22	79.81 $\pm$ 0.12
LMDA	46.80 $\pm$ 0.31	74.88 $\pm$ 0.67	61.40 $\pm$ 0.94	78.45 $\pm$ 0.46	71.74 $\pm$ 0.12	77.47 $\pm$ 0.41	62.69 $\pm$ 1.52
CNN-T	39.15 $\pm$ 0.56	71.20 $\pm$ 0.42	59.64 $\pm$ 1.41	61.50 $\pm$ 1.71	51.93 $\pm$ 0.16	76.34 $\pm$ 0.61	73.20 $\pm$ 0.73
Deformer	41.53 $\pm$ 0.67	74.86 $\pm$ 0.82	63.06 $\pm$ 1.17	76.89 $\pm$ 0.14	57.75 $\pm$ 1.21	79.77 $\pm$ 0.14	81.48 $\pm$ 0.21
Conformer	41.64 $\pm$ 1.23	74.17 $\pm$ 0.49	59.10 $\pm$ 2.14	76.78 $\pm$ 0.33	67.87 $\pm$ 0.11	78.58 $\pm$ 0.56	77.78 $\pm$ 0.04
TSception	41.58 $\pm$ 0.63	69.51 $\pm$ 0.99	57.65 $\pm$ 1.51	51.77 $\pm$ 0.70	50.40 $\pm$ 0.09	74.37 $\pm$ 0.53	79.26 $\pm$ 0.49
<i>Foundation models</i>							
BENDR	<u>51.11</u> $\pm$ 0.25	73.35 $\pm$ 0.06	62.68 $\pm$ 0.49	73.46 $\pm$ 0.28	65.01 $\pm$ 0.18	75.50 $\pm$ 0.93	‡79.09 $\pm$ 0.16
BIOT-6D	34.27 $\pm$ 0.93	70.22 $\pm$ 1.32	63.94 $\pm$ 1.20	58.14 $\pm$ 0.33	54.77 $\pm$ 0.21	‡74.85 $\pm$ 0.33	‡77.90 $\pm$ 0.14
LaBraM	46.93 $\pm$ 1.43	<u>76.97</u> $\pm$ 1.08	64.14 $\pm$ 1.03	70.31 $\pm$ 0.24	63.07 $\pm$ 0.97	70.87 $\pm$ 0.59	76.23 $\pm$ 0.27
Neuro-GPT	46.97 $\pm$ 0.71	77.70 $\pm$ 0.70	60.62 $\pm$ 1.63	75.97 $\pm$ 0.53	68.59 $\pm$ 0.25	73.27 $\pm$ 0.27	‡79.50 $\pm$ 0.17
EEGPT	32.24 $\pm$ 1.45	71.37 $\pm$ 0.16	59.88 $\pm$ 1.39	62.77 $\pm$ 1.85	58.24 $\pm$ 0.40	66.91 $\pm$ 2.89	77.67 $\pm$ 0.17
CBraMod	53.03 $\pm$ 0.22	75.45 $\pm$ 0.35	63.47 $\pm$ 0.36	77.30 $\pm$ 0.28	69.91 $\pm$ 0.06	74.23 $\pm$ 0.19	‡79.98 $\pm$ 0.11
TFM	32.02 $\pm$ 0.66	60.12 $\pm$ 3.00	55.35 $\pm$ 1.46	53.10 $\pm$ 0.48	52.99 $\pm$ 0.29	‡63.46 $\pm$ 0.60	‡75.65 $\pm$ 0.07
BrainOmni-Tiny	41.58 $\pm$ 0.80	70.13 $\pm$ 0.89	61.88 $\pm$ 0.30	70.48 $\pm$ 0.23	61.40 $\pm$ 0.17	—	72.92 $\pm$ 0.25
BrainOmni-Base	40.93 $\pm$ 0.83	69.30 $\pm$ 0.89	60.64 $\pm$ 0.39	70.87 $\pm$ 0.29	59.31 $\pm$ 0.06	—	80.49 $\pm$ 0.29
EEGMamba	45.72 $\pm$ 0.54	73.30 $\pm$ 0.57	61.53 $\pm$ 0.50	76.01 $\pm$ 0.05	68.18 $\pm$ 0.05	75.92 $\pm$ 0.21	‡80.90 $\pm$ 0.10
SingLEM	30.57 $\pm$ 0.10	67.31 $\pm$ 0.18	54.47 $\pm$ 0.62	71.98 $\pm$ 0.56	63.42 $\pm$ 0.13	60.78 $\pm$ 1.82	50.80 $\pm$ 1.02
LUNA-Base	28.86 $\pm$ 0.50	56.17 $\pm$ 3.14	55.71 $\pm$ 1.37	51.67 $\pm$ 0.54	50.00 $\pm$ 0.04	78.12 $\pm$ 0.61	‡ 81.92 $\pm$ 0.07
Cross-subject (LOSO) — Linear probing							
<i>Foundation models</i>							
BENDR	32.18 $\pm$ 0.41	<u>60.46</u> $\pm$ 1.06	53.85 $\pm$ 1.11	61.24 $\pm$ 2.07	67.11 $\pm$ 0.76	53.22 $\pm$ 0.68	‡58.33 $\pm$ 0.66
BIOT-6D	30.88 $\pm$ 1.00	61.35 $\pm$ 2.11	63.43 $\pm$ 0.63	51.04 $\pm$ 0.12	53.19 $\pm$ 0.66	‡69.79 $\pm$ 0.62	‡73.46 $\pm$ 0.25
LaBraM	42.59 $\pm$ 0.27	65.05 $\pm$ 0.23	61.97 $\pm$ 0.17	67.75 $\pm$ 0.32	56.82 $\pm$ 0.60	68.84 $\pm$ 0.36	78.32 $\pm$ 0.11
Neuro-GPT	48.24 $\pm$ 1.04	75.57 $\pm$ 1.26	61.24 $\pm$ 0.50	58.70 $\pm$ 0.26	50.08 $\pm$ 0.04	70.45 $\pm$ 0.24	‡79.64 $\pm$ 0.15
EEGPT	37.37 $\pm$ 1.25	72.08 $\pm$ 2.07	63.00 $\pm$ 2.89	66.53 $\pm$ 0.05	57.26 $\pm$ 0.17	70.94 $\pm$ 0.69	77.51 $\pm$ 0.09
CBraMod	41.45 $\pm$ 0.50	69.27 $\pm$ 0.55	59.93 $\pm$ 0.29	58.63 $\pm$ 0.54	53.31 $\pm$ 0.14	75.21 $\pm$ 0.52	‡78.04 $\pm$ 0.04
TFM	28.34 $\pm$ 0.30	50.97 $\pm$ 1.13	53.43 $\pm$ 0.53	51.56 $\pm$ 0.51	51.93 $\pm$ 0.63	‡56.83 $\pm$ 0.75	‡68.25 $\pm$ 0.09
BrainOmni-Tiny	39.78 $\pm$ 0.36	66.05 $\pm$ 0.53	61.54 $\pm$ 0.54	61.34 $\pm$ 0.42	51.33 $\pm$ 0.14	—	77.03 $\pm$ 0.37
BrainOmni-Base	39.63 $\pm$ 0.58	67.48 $\pm$ 0.25	60.38 $\pm$ 0.19	69.50 $\pm$ 0.69	58.93 $\pm$ 0.43	—	79.32 $\pm$ 0.30
EEGMamba	34.32 $\pm$ 0.20	61.94 $\pm$ 0.31	56.38 $\pm$ 0.27	74.13 $\pm$ 0.30	63.65 $\pm$ 0.11	71.81 $\pm$ 0.26	‡78.24 $\pm$ 0.04
SingLEM	33.42 $\pm$ 0.22	63.69 $\pm$ 0.36	54.08 $\pm$ 0.20	72.52 $\pm$ 0.05	61.89 $\pm$ 0.03	50.50 $\pm$ 0.11	51.53 $\pm$ 0.13
LUNA-Base	29.21 $\pm$ 0.57	51.10 $\pm$ 0.65	56.72 $\pm$ 0.41	52.16 $\pm$ 0.50	49.99 $\pm$ 0.02	67.43 $\pm$ 0.45	‡75.88 $\pm$ 0.17
Within-subject (Few-shot) — Full fine-tuning							
<i>Specialist models</i>							
Traditional ML	60.62	79.20	<u>75.89</u>	54.45	56.65	77.71 $\pm$ 0.35	—
EEGNet	50.22 $\pm$ 1.14	75.53 $\pm$ 3.67	72.08 $\pm$ 0.39	68.99 $\pm$ 0.51	70.91 $\pm$ 1.17	88.45 $\pm$ 0.48	—
ShallowConv	52.87 $\pm$ 0.88	74.52 $\pm$ 0.71	73.79 $\pm$ 0.66	57.13 $\pm$ 0.93	55.97 $\pm$ 0.20	85.81 $\pm$ 0.44	—
LMDA	51.13 $\pm$ 0.76	75.11 $\pm$ 1.63	73.79 $\pm$ 1.50	60.28 $\pm$ 0.82	63.58 $\pm$ 0.30	86.80 $\pm$ 0.55	—
CNN-T	51.27 $\pm$ 1.10	75.77 $\pm$ 0.38	71.63 $\pm$ 1.36	50.32 $\pm$ 0.52	50.23 $\pm$ 0.29	87.66 $\pm$ 0.33	—
Deformer	42.21 $\pm$ 0.73	73.02 $\pm$ 2.13	70.32 $\pm$ 1.28	65.38 $\pm$ 0.32	<u>64.36</u> $\pm$ 0.93	86.88 $\pm$ 0.33	—
Conformer	57.19 $\pm$ 1.32	<u>80.17</u> $\pm$ 0.25	77.10 $\pm$ 1.49	65.43 $\pm$ 0.90	58.34 $\pm$ 0.23	91.58 $\pm$ 0.34	—
TSception	<u>58.68</u> $\pm$ 0.86	80.36 $\pm$ 0.65	74.35 $\pm$ 1.12	63.99 $\pm$ 0.43	51.73 $\pm$ 0.30	<u>90.92</u> $\pm$ 0.74	—
<i>Foundation models</i>							
BENDR	44.90 $\pm$ 1.31	71.70 $\pm$ 1.46	56.59 $\pm$ 0.25	59.27 $\pm$ 1.03	58.33 $\pm$ 1.00	54.14 $\pm$ 0.33	—
BIOT-6D	48.60 $\pm$ 0.72	68.43 $\pm$ 0.42	68.43 $\pm$ 1.48	52.19 $\pm$ 0.16	52.12 $\pm$ 0.27	‡79.60 $\pm$ 0.51	—
LaBraM	37.13 $\pm$ 0.92	69.76 $\pm$ 1.23	61.39 $\pm$ 0.65	60.06 $\pm$ 0.58	54.67 $\pm$ 0.67	71.31 $\pm$ 0.29	—
Neuro-GPT	46.39 $\pm$ 1.92	75.25 $\pm$ 1.58	61.90 $\pm$ 0.90	55.02 $\pm$ 0.97	61.61 $\pm$ 0.14	83.09 $\pm$ 0.59	—
EEGPT	34.99 $\pm$ 0.25	62.07 $\pm$ 0.74	59.17 $\pm$ 0.87	52.02 $\pm$ 1.14	51.93 $\pm$ 0.41	63.74 $\pm$ 0.28	—
CBraMod	50.34 $\pm$ 1.18	77.39 $\pm$ 0.35	70.30 $\pm$ 0.72	56.85 $\pm$ 1.03	58.00 $\pm$ 0.99	88.54 $\pm$ 0.16	—
TFM	33.30 $\pm$ 0.46	58.92 $\pm$ 2.41	55.34 $\pm$ 0.28	51.42 $\pm$ 0.87	51.21 $\pm$ 0.28	‡59.55 $\pm$ 0.57	—
BrainOmni-Tiny	39.00 $\pm$ 0.19	63.30 $\pm$ 0.98	61.59 $\pm$ 1.54	57.02 $\pm$ 0.36	56.34 $\pm$ 0.33	—	—
BrainOmni-Base	37.60 $\pm$ 0.27	61.84 $\pm$ 0.76	59.86 $\pm$ 0.83	59.75 $\pm$ 1.14	56.18 $\pm$ 0.14	—	—
EEGMamba	38.04 $\pm$ 0.45	65.07 $\pm$ 1.86	60.24 $\pm$ 0.48	65.50 $\pm$ 0.46	61.12 $\pm$ 0.35	79.50 $\pm$ 0.42	—
SingLEM	28.94 $\pm$ 0.73	56.75 $\pm$ 1.66	52.12 $\pm$ 0.63	57.70 $\pm$ 0.41	56.38 $\pm$ 0.28	56.70 $\pm$ 1.49	—
LUNA-Base	31.94 $\pm$ 1.24	56.50 $\pm$ 0.74	60.36 $\pm$ 1.69	51.66 $\pm$ 0.99	50.29 $\pm$ 0.12	72.30 $\pm$ 0.26	—
Within-subject (Few-shot) — Linear probing							
<i>Foundation models</i>							
BENDR	31.43 $\pm$ 0.85	55.27 $\pm$ 2.40	50.95 $\pm$ 0.90	52.03 $\pm$ 0.18	51.75 $\pm$ 0.70	50.07 $\pm$ 0.18	—
BIOT-6D	47.95 $\pm$ 0.53	67.15 $\pm$ 1.64	67.80 $\pm$ 1.21	53.03 $\pm$ 0.78	51.62 $\pm$ 0.47	‡73.41 $\pm$ 1.06	—
LaBraM	35.38 $\pm$ 0.45	59.83 $\pm$ 1.22	59.74 $\pm$ 0.86	55.15 $\pm$ 0.21	52.96 $\pm$ 0.24	66.93 $\pm$ 0.88	—
Neuro-GPT	49.82 $\pm$ 1.55	76.69 $\pm$ 1.35	65.60 $\pm$ 0.86	50.65 $\pm$ 0.31	50.88 $\pm$ 0.02	71.68 $\pm$ 1.57	—
EEGPT	35.86 $\pm$ 0.54	65.88 $\pm$ 2.18	60.87 $\pm$ 2.36	53.18 $\pm$ 0.97	51.34 $\pm$ 0.18	66.71 $\pm$ 1.20	—
CBraMod	40.74 $\pm$ 1.06	70.38 $\pm$ 0.90	60.81 $\pm$ 0.06	50.18 $\pm$ 0.15	51.43 $\pm$ 0.09	87.77 $\pm$ 0.73	—
TFM	27.78 $\pm$ 1.05	51.30 $\pm$ 1.08	53.73 $\pm$ 1.18	51.12 $\pm$ 0.83	50.93 $\pm$ 0.36	‡53.32 $\pm$ 0.37	—
BrainOmni-Tiny	40.61 $\pm$ 0.31	59.79 $\pm$ 0.60	63.21 $\pm$ 0.76	56.33 $\pm$ 0.16	52.96 $\pm$ 0.05	—	—
BrainOmni-Base	38.71 $\pm$ 0.36	59.10 $\pm$ 0.30	60.89 $\pm$ 1.07	58.37 $\pm$ 0.35	54.33 $\pm$ 0.30	—	—
EEGMamba	33.84 $\pm$ 0.40	59.02 $\pm$ 0.34	54.25 $\pm$ 0.65	65.79 $\pm$ 0.16	61.45 $\pm$ 0.09	82.18 $\pm$ 0.17	—
SingLEM	29.87 $\pm$ 0.18	56.51 $\pm$ 1.27	53.75 $\pm$ 0.32	63.66 $\pm$ 0.96	58.27 $\pm$ 0.40	51.07 $\pm$ 0.11	—
LUNA-Base	34.73 $\pm$ 0.05	55.61 $\pm$ 0.89	57.56 $\pm$ 0.18	51.42 $\pm$ 0.19	50.79 $\pm$ 0.15	71.11 $\pm$ 0.20	—

specific full-parameter adaptation remains important for most EEG FMs.

In summary, pre-trained EEG FMs demonstrated a capacity to extract transferable representations to a certain extent. However, this generalization capability remained insufficiently robust: the majority of models required full-

parameter fine-tuning on downstream tasks and could not directly leverage pre-trained encoders to obtain features for effective decoding. Even the best-performing FMs in our evaluation exhibited notable performance degradation on specific tasks, indicating that achieving truly universal EEG representations remains an open chal-

Table 3. Benchmark performance. The best metrics are marked in bold, and the second best by an underline. ‡ indicates that the corresponding dataset was used during pre-training of the model (continued).

Approach	Sleep-EDFx	SEED	Nakanishi2015	EEGMat	Things-EEG2	SEED-VIG
Cross-subject (LOSO) — Full fine-tuning						
<i>Specialist models</i>						
Traditional ML	51.78 $\pm$ 0.22	48.91	94.07	67.41	—	0.2489
EEGNet	73.75 $\pm$ 0.19	48.57 $\pm$ 1.34	95.88 $\pm$ 0.18	66.60 $\pm$ 0.63	74.42 $\pm$ 3.67	0.2561 $\pm$ 0.0092
ShallowConv	74.86 $\pm$ 0.42	53.41 $\pm$ 0.12	69.61 $\pm$ 0.86	72.22 $\pm$ 1.24	72.03 $\pm$ 0.38	0.2290 $\pm$ 0.0029
LMDA	74.58 $\pm$ 0.34	50.12 $\pm$ 0.43	85.12 $\pm$ 0.94	67.47 $\pm$ 1.21	78.72 $\pm$ 0.86	0.2389 $\pm$ 0.0029
CNN-T	75.74 $\pm$ 0.48	44.56 $\pm$ 1.40	46.34 $\pm$ 0.34	70.77 $\pm$ 2.49	59.05 $\pm$ 2.13	0.2556 $\pm$ 0.0140
Deformer	78.73 $\pm$ 0.09	51.05 $\pm$ 0.97	97.18 $\pm$ 0.13	71.73 $\pm$ 0.37	78.47 $\pm$ 0.65	0.2512 $\pm$ 0.0053
Conformer	68.40 $\pm$ 2.87	48.76 $\pm$ 1.23	91.34 $\pm$ 1.34	70.49 $\pm$ 0.93	64.00 $\pm$ 0.91	0.2405 $\pm$ 0.0044
TSception	76.41 $\pm$ 0.43	41.82 $\pm$ 0.79	60.51 $\pm$ 0.88	74.10 $\pm$ 0.04	63.33 $\pm$ 1.47	0.2302 $\pm$ 0.0019
<i>Foundation models</i>						
BENDR	71.45 $\pm$ 0.43	52.50 $\pm$ 0.85	92.94 $\pm$ 0.30	54.32 $\pm$ 0.71	71.47 $\pm$ 0.39	0.2412 $\pm$ 0.0025
BIOT-6D	66.35 $\pm$ 0.23	49.04 $\pm$ 0.90	72.35 $\pm$ 3.04	70.77 $\pm$ 1.49	50.57 $\pm$ 0.22	0.2374 $\pm$ 0.0033
LaBraM	63.56 $\pm$ 0.03	$\S$ 52.23 $\pm$ 0.92	79.18 $\pm$ 0.73	65.74 $\pm$ 1.61	75.15 $\pm$ 0.93	0.2281 $\pm$ 0.0035
Neuro-GPT	59.09 $\pm$ 0.25	49.67 $\pm$ 0.38	87.35 $\pm$ 0.40	72.62 $\pm$ 1.35	80.28 $\pm$ 1.08	0.2509 $\pm$ 0.0055
EEGPT	62.93 $\pm$ 1.06	$\S$ 48.74 $\pm$ 3.41	88.31 $\pm$ 3.30	58.02 $\pm$ 1.02	74.90 $\pm$ 1.35	0.2402 $\pm$ 0.0025
CBraMod	72.30 $\pm$ 0.18	53.61 $\pm$ 0.61	85.39 $\pm$ 0.60	68.43 $\pm$ 0.72	75.88 $\pm$ 0.89	0.2718 $\pm$ 0.0019
TFM	67.38 $\pm$ 0.25	36.66 $\pm$ 0.26	12.84 $\pm$ 0.68	63.02 $\pm$ 1.95	50.48 $\pm$ 0.45	0.2283 $\pm$ 0.0026
BrainOmni-Tiny	—	38.02 $\pm$ 0.03	78.33 $\pm$ 0.23	57.50 $\pm$ 0.80	66.83 $\pm$ 1.16	0.2434 $\pm$ 0.0003
BrainOmni-Base	—	44.72 $\pm$ 0.18	50.88 $\pm$ 1.99	51.51 $\pm$ 0.76	67.13 $\pm$ 0.39	0.2549 $\pm$ 0.0055
EEGMamba	66.68 $\pm$ 0.42	52.19 $\pm$ 0.04	70.08 $\pm$ 0.94	49.97 $\pm$ 0.04	74.48 $\pm$ 0.12	0.2297 $\pm$ 0.0010
SingLEM	67.23 $\pm$ 0.48	50.16 $\pm$ 1.48	33.54 $\pm$ 1.23	49.85 $\pm$ 0.16	61.97 $\pm$ 4.80	0.2349 $\pm$ 0.0004
LUNA-Base	71.03 $\pm$ 0.36	49.96 $\pm$ 0.35	8.52 $\pm$ 0.22	50.00 $\pm$ 0.00	63.68 $\pm$ 0.55	0.2342 $\pm$ 0.0043
Cross-subject (LOSO) — Linear probing						
<i>Foundation models</i>						
BENDR	54.78 $\pm$ 0.12	34.69 $\pm$ 0.10	57.61 $\pm$ 4.18	51.51 $\pm$ 1.57	54.87 $\pm$ 0.75	0.3068 $\pm$ 0.0017
BIOT-6D	60.59 $\pm$ 0.19	51.08 $\pm$ 0.77	68.31 $\pm$ 3.34	66.45 $\pm$ 1.24	48.35 $\pm$ 3.15	0.2349 $\pm$ 0.0011
LaBraM	62.73 $\pm$ 0.24	$\S$ 52.46 $\pm$ 0.05	53.22 $\pm$ 0.83	64.48 $\pm$ 0.56	59.65 $\pm$ 1.31	0.2334 $\pm$ 0.0010
Neuro-GPT	57.95 $\pm$ 0.58	49.67 $\pm$ 0.38	72.39 $\pm$ 1.75	68.58 $\pm$ 1.89	66.20 $\pm$ 0.99	0.2443 $\pm$ 0.0035
EEGPT	55.67 $\pm$ 0.87	$\S$ 50.04 $\pm$ 1.03	85.47 $\pm$ 4.26	59.38 $\pm$ 0.53	66.42 $\pm$ 0.49	0.2335 $\pm$ 0.0043
CBraMod	52.56 $\pm$ 0.30	51.83 $\pm$ 0.07	17.41 $\pm$ 0.28	60.34 $\pm$ 0.27	75.80 $\pm$ 1.19	0.2765 $\pm$ 0.0020
TFM	56.95 $\pm$ 0.46	35.28 $\pm$ 0.04	10.88 $\pm$ 0.58	62.44 $\pm$ 0.09	49.43 $\pm$ 0.95	0.2417 $\pm$ 0.0004
BrainOmni-Tiny	—	44.27 $\pm$ 0.11	73.77 $\pm$ 0.66	62.50 $\pm$ 0.50	60.75 $\pm$ 0.25	0.2447 $\pm$ 0.0012
BrainOmni-Base	—	44.06 $\pm$ 0.26	24.94 $\pm$ 0.15	50.40 $\pm$ 0.69	63.85 $\pm$ 0.95	0.2360 $\pm$ 0.0032
EEGMamba	51.96 $\pm$ 0.13	51.90 $\pm$ 0.15	17.82 $\pm$ 0.03	50.06 $\pm$ 0.09	73.40 $\pm$ 0.29	0.2343 $\pm$ 0.0002
SingLEM	34.77 $\pm$ 0.13	35.69 $\pm$ 0.04	17.90 $\pm$ 0.22	50.40 $\pm$ 0.43	63.12 $\pm$ 0.48	0.2632 $\pm$ 0.0006
LUNA-Base	58.14 $\pm$ 0.22	42.77 $\pm$ 0.22	9.34 $\pm$ 0.34	49.97 $\pm$ 0.39	49.20 $\pm$ 0.32	0.2367 $\pm$ 0.0013
Within-subject (Few-shot) — Full fine-tuning						
<i>Specialist models</i>						
Traditional ML	59.00	53.38	98.77	95.60	—	0.1764 $\pm$ 0.0013
EEGNet	48.29 $\pm$ 0.54	52.12 $\pm$ 0.47	66.67 $\pm$ 4.00	60.57 $\pm$ 1.29	89.25 $\pm$ 0.60	0.2082 $\pm$ 0.0045
ShallowConv	55.29 $\pm$ 0.49	51.97 $\pm$ 0.26	51.23 $\pm$ 0.02	69.37 $\pm$ 1.43	64.52 $\pm$ 0.56	0.3839 $\pm$ 0.0063
LMDA	46.75 $\pm$ 2.12	53.20 $\pm$ 1.39	49.49 $\pm$ 1.96	54.78 $\pm$ 0.22	84.53 $\pm$ 0.94	0.2027 $\pm$ 0.0110
CNN-T	64.77 $\pm$ 0.61	51.95 $\pm$ 1.06	61.63 $\pm$ 1.19	51.08 $\pm$ 0.95	57.50 $\pm$ 0.94	0.1538 $\pm$ 0.0100
Deformer	52.26 $\pm$ 0.38	52.19 $\pm$ 0.30	71.60 $\pm$ 1.90	52.01 $\pm$ 0.39	82.95 $\pm$ 0.72	0.2902 $\pm$ 0.0167
Conformer	63.31 $\pm$ 0.48	55.67 $\pm$ 1.63	41.36 $\pm$ 3.07	68.33 $\pm$ 1.71	59.90 $\pm$ 0.19	0.1421 $\pm$ 0.0034
TSception	60.31 $\pm$ 0.29	52.70 $\pm$ 1.21	72.74 $\pm$ 2.79	<u>89.12</u> $\pm$ 0.95	63.23 $\pm$ 0.86	0.1851 $\pm$ 0.0063
<i>Foundation models</i>						
BENDR	37.34 $\pm$ 0.24	41.03 $\pm$ 0.54	85.91 $\pm$ 0.29	52.16 $\pm$ 0.22	65.85 $\pm$ 0.49	0.2436 $\pm$ 0.0023
BIOT-6D	61.75 $\pm$ 0.48	48.41 $\pm$ 0.56	84.67 $\pm$ 1.48	86.11 $\pm$ 3.16	49.22 $\pm$ 2.46	0.2230 $\pm$ 0.0414
LaBraM	35.99 $\pm$ 0.23	$\S$ 47.00 $\pm$ 0.92	77.88 $\pm$ 2.14	65.74 $\pm$ 1.50	83.75 $\pm$ 0.74	0.1956 $\pm$ 0.0017
Neuro-GPT	54.50 $\pm$ 0.29	55.90 $\pm$ 0.09	76.44 $\pm$ 1.48	71.22 $\pm$ 3.69	81.02 $\pm$ 1.10	0.1880 $\pm$ 0.0035
EEGPT	56.38 $\pm$ 0.16	$\S$ 40.48 $\pm$ 0.44	75.72 $\pm$ 8.11	65.59 $\pm$ 2.51	64.35 $\pm$ 1.28	0.1990 $\pm$ 0.0007
CBraMod	57.72 $\pm$ 0.08	55.82 $\pm$ 0.39	62.35 $\pm$ 1.15	79.32 $\pm$ 1.15	84.83 $\pm$ 0.43	0.2051 $\pm$ 0.0036
TFM	58.82 $\pm$ 0.14	35.40 $\pm$ 0.15	11.73 $\pm$ 1.26	77.55 $\pm$ 0.95	50.40 $\pm$ 0.55	0.2208 $\pm$ 0.0058
BrainOmni-Tiny	—	46.74 $\pm$ 0.39	82.00 $\pm$ 0.52	58.72 $\pm$ 0.79	67.95 $\pm$ 0.25	0.2146 $\pm$ 0.0089
BrainOmni-Base	—	44.41 $\pm$ 0.12	52.88 $\pm$ 1.29	51.08 $\pm$ 1.86	67.70 $\pm$ 0.21	0.1923 $\pm$ 0.0062
EEGMamba	61.50 $\pm$ 0.62	51.33 $\pm$ 0.27	44.03 $\pm$ 0.15	50.08 $\pm$ 0.11	86.12 $\pm$ 0.65	0.1774 $\pm$ 0.0018
SingLEM	29.71 $\pm$ 1.75	47.01 $\pm$ 2.01	33.02 $\pm$ 2.63	50.46 $\pm$ 0.19	52.00 $\pm$ 1.27	0.2271 $\pm$ 0.0021
LUNA-Base	66.02 $\pm$ 0.06	46.06 $\pm$ 0.68	8.95 $\pm$ 0.50	50.62 $\pm$ 0.44	51.87 $\pm$ 0.80	0.1676 $\pm$ 0.0034
Within-subject (Few-shot) — Linear probing						
<i>Foundation models</i>						
BENDR	21.58 $\pm$ 0.06	34.00 $\pm$ 0.16	29.12 $\pm$ 2.45	49.46 $\pm$ 1.52	56.87 $\pm$ 1.17	0.2271 $\pm$ 0.0015
BIOT-6D	59.42 $\pm$ 0.10	48.10 $\pm$ 0.65	78.19 $\pm$ 5.09	77.08 $\pm$ 3.95	51.83 $\pm$ 0.53	0.2937 $\pm$ 0.0419
LaBraM	49.20 $\pm$ 0.10	$\S$ 49.46 $\pm$ 0.14	66.98 $\pm$ 0.50	70.22 $\pm$ 2.94	60.67 $\pm$ 0.47	0.1935 $\pm$ 0.0020
Neuro-GPT	49.69 $\pm$ 0.62	52.03 $\pm$ 0.10	52.67 $\pm$ 1.24	70.60 $\pm$ 1.43	65.37 $\pm$ 0.33	0.2015 $\pm$ 0.0051
EEGPT	61.99 $\pm$ 0.47	$\S$ 42.90 $\pm$ 0.42	77.78 $\pm$ 3.81	67.05 $\pm$ 1.22	66.30 $\pm$ 1.97	0.2004 $\pm$ 0.0062
CBraMod	41.33 $\pm$ 0.26	52.32 $\pm$ 0.39	24.28 $\pm$ 1.77	60.49 $\pm$ 2.25	81.67 $\pm$ 0.38	0.1895 $\pm$ 0.0075
TFM	49.56 $\pm$ 0.26	34.28 $\pm$ 0.44	10.08 $\pm$ 1.72	58.26 $\pm$ 6.15	49.77 $\pm$ 1.25	0.2208 $\pm$ 0.0058
BrainOmni-Tiny	—	43.54 $\pm$ 0.14	51.95 $\pm$ 1.39	59.41 $\pm$ 1.22	63.77 $\pm$ 0.50	0.2202 $\pm$ 0.0035
BrainOmni-Base	—	43.62 $\pm$ 0.15	23.56 $\pm$ 2.52	52.16 $\pm$ 1.47	63.27 $\pm$ 0.72	0.1861 $\pm$ 0.0031
EEGMamba	46.30 $\pm$ 0.18	49.36 $\pm$ 0.03	20.99 $\pm$ 0.50	50.15 $\pm$ 0.22	78.50 $\pm$ 0.33	0.1712 $\pm$ 0.0014
SingLEM	22.51 $\pm$ 0.24	34.62 $\pm$ 0.05	19.14 $\pm$ 0.91	49.69 $\pm$ 1.42	75.90 $\pm$ 0.53	0.2752 $\pm$ 0.0075
LUNA-Base	61.99 $\pm$ 0.07	42.27 $\pm$ 0.19	9.47 $\pm$ 1.72	50.08 $\pm$ 0.11	50.43 $\pm$ 0.56	0.1654 $\pm$ 0.0009

lenge.

Can FMs consistently outperform specialist models?

With the proliferation of EEG FMs, whether traditional deep learning methods remain competi-

tive is a question worth investigating. We compared existing FMs against eight specialist models, including one traditional machine learning method, four CNN-based methods, and three Transformer-based methods. These specialist models were trained from scratch using the same

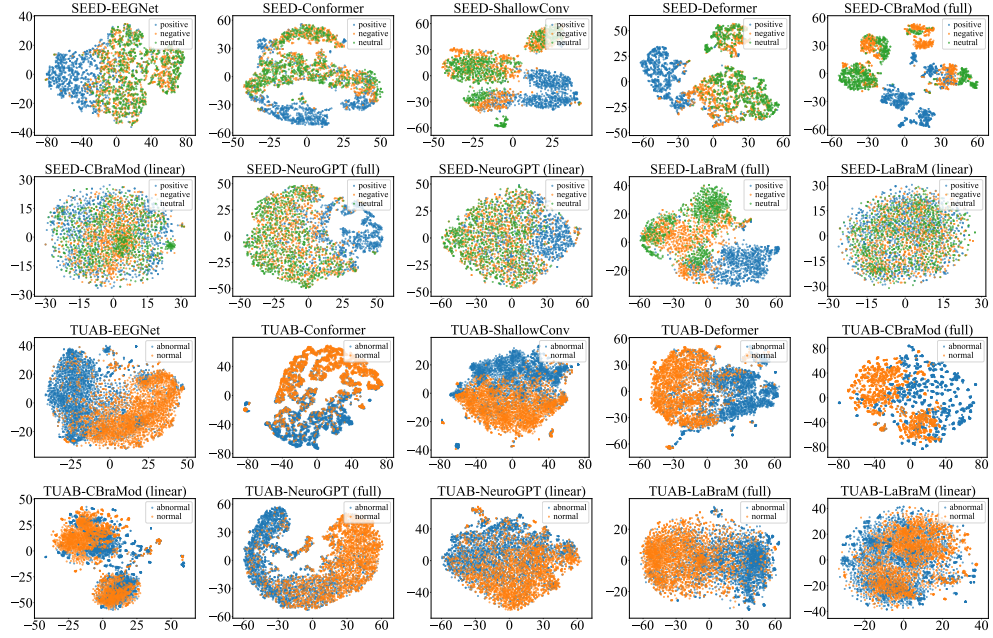


Figure 4. t -SNE visualization of the SEED and TUAB datasets.

fine-tuning data as the FMs and evaluated on identical test sets. Fig. 5a and b present the ranking of each model based on the number of top-1 and top-3 placements across all tasks and scenarios. Notably, EEGNet achieved the highest number of top-1 placements, demonstrating remarkable performance despite having only 2K parameters. ShallowConv obtained the highest number of top-3 placements. Among the top five models in both Fig. 5a and b, four were specialist models. Specifically, EEGNet, Conformer, Traditional ML, and Deformer ranked 1st, 2nd, 4th, and 5th in top-1 counts, respectively, while ShallowConv, EEGNet, Conformer, and Deformer ranked 1st to 4th in top-3 counts.

Furthermore, Fig. 5c and d compare the total number of top-1 and top-3 placements achieved by specialist models versus EEG FMs. To ensure a fair comparison given the larger number of FMs, we selected the seven FMs with the highest top-3 counts for this analysis. Specialist models achieved 16 first-place finishes and 47 top-3 placements, outperforming the selected FMs.

Overall, specialist models achieved higher average decoding accuracy than EEG FMs. A closer examination of the two evaluation scenarios reveals a more nuanced pattern. In the within-subject few-shot scenario, specialist models obtained the majority of top-1 placements, whereas in the LOSO scenario the two groups performed comparably. This contrast suggests that current EEG FMs rely on a relatively large amount of fine-tuning data to re-

alize their advantage, and that their benefit diminishes when only limited calibration data are available. We emphasize, however, that both scenarios correspond to realistic deployment settings: the LOSO scenario reflects applications with access to a labeled multi-subject corpus, while the few-shot scenario reflects rapid personalization for new users with minimal calibration.

Beyond accuracy, the two groups also differ markedly in computational cost. As measured in Section A4 of the supplementary information, the fine-tuning cost of EEG FMs is orders of magnitude higher than that of lightweight specialist models. For example, under the 4s, 10-channel setting, EEGPT requires approximately 361 GFLOPs per forward pass, more than three orders of magnitude larger than the 0.35 GFLOPs of EEGNet, and its fine-tuning throughput is about 46 times lower. This substantial overhead may limit the practicality of large EEG FMs in resource-constrained or real-time BCI applications. Taken together with the accuracy comparison, these results indicate that lightweight specialist architectures remain highly competitive in the era of FMs, particularly when computational efficiency is a critical deployment consideration.

To statistically verify whether EEG FMs consistently outperform specialist models, we performed a rank-based analysis using the Friedman test followed by the Nemenyi post-hoc test, with the critical difference (CD) diagram [111] reported in Figure 1 of the supplementary in-

formation. The Friedman test revealed significant overall differences among the model rankings, confirming that the observed ranking structure was not due to random variation. Although CBraMod attained the best average rank among all compared models, the Nemenyi post-hoc test showed that its difference from several top-ranked specialist models, including EEGNet, ShallowConv, Conformer, Deformer, and LMDA, was not statistically significant; these specialist architectures belonged to the same non-significant group as the best-performing FM. This provides statistical support for our observation that current EEG FMs do not consistently and significantly outperform competitive specialist models trained from scratch, and that specialist architectures remain highly competitive under the present data regimes and evaluation scenarios.

Do EEG FMs exhibit behavior consistent with scaling laws?

Fig. 6 presents the average rankings of EEG FMs across the 13 datasets under both the LOSO and few-shot scenarios. Detailed ranking results are provided in Table 49 of the supplementary information. CBraMod achieved the highest overall ranking with an average rank of 6.56 and 4.0M parameters. EEGNet, a widely adopted lightweight EEG decoding backbone, attained the second position with only 2K parameters. Among the EEG FMs, the top six overall, CBraMod (4.0M), Neuro-GPT (0.16M), LaBraM (5.8M), EEGMamba (3.3M), BENDR (4.0M), and BIOT-6D (3.2M), exhibit no clear correlation between model size and downstream performance.

These observations indicate that, under current data regimes and training practices, increasing the parameter count of an EEG FM does not yet guarantee better downstream generalization, and the scaling behavior commonly observed in vision and language foundation models has not been clearly established for EEG. This may be attributed to two factors. First, EEG data acquisition is costly in terms of time, labor, and resources, resulting in limited data availability and substantial measurement noise, with a notable shortage of large-scale, high-quality corpora [86] that may be insufficient to elicit emergent capabilities. Second, the pre-training objectives currently adopted by EEG FMs may not effectively translate increased model capacity into richer representations, so that the performance gains from enlarging the model are weaker and less consistent than the steady improvements predicted by scaling laws. Constructing large-

scale, high-quality EEG corpora and developing pre-training strategies that can better exploit larger model capacity therefore remain essential directions for future research, and whether scaling laws can emerge for EEG FMs is an open question worth further investigation.

3.5. Comparison of Different Fine-Tuning Ratios

In the within-subject few-shot scenario, a pre-trained model is expected to achieve satisfactory performance with minimal calibration data. However, many models exhibited suboptimal performance under this setting. To investigate whether this limitation is primarily attributable to insufficient fine-tuning data, we conducted an analysis across different fine-tuning data ratios, varying from 10% to 90% in increments of 20%. We selected three specialist models (EEGNet, ShallowConv, and LMDA) and the top three EEG FMs (CBraMod, Neuro-GPT, and LaBraM). The results are presented in Fig. 7, with detailed results for all models across various datasets provided in Section A4 of the supplementary information. Most models exhibited consistent performance improvements as the amount of fine-tuning data increased, which aligns with intuitive expectations. However, minimal calibration or even calibration-free adaptation remains a critical requirement for practical deployment. Developing models that can rapidly adapt to downstream tasks with limited data remains an open and pressing challenge.

4. DISCUSSION

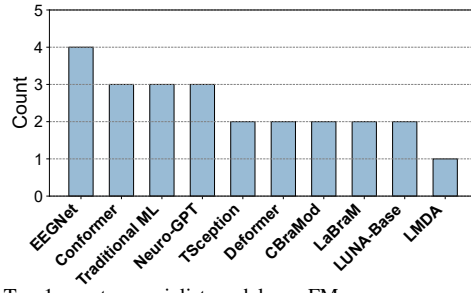
This section presents additional discussions.

4.1. Pre-training Objectives vs. Downstream Performance

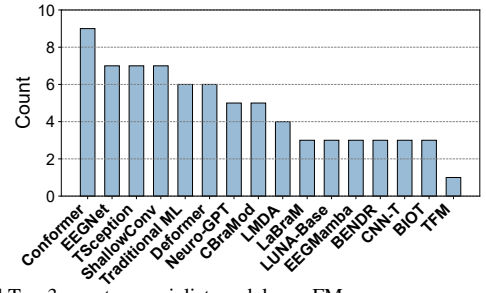
A natural question following the taxonomy in Section 'Overview OF EEG FMs' and the benchmark results in Section 'Benchmark of EEG FMs' is whether specific categories of pre-training objectives systematically yield superior downstream performance, and whether they exhibit paradigm-specific advantages. To address this question, we organize the 12 evaluated FMs by their pre-training objectives in Table 50 of the supplementary information, and analyze their performance across the 13 downstream datasets.

Several observations can be drawn from this analysis. First, no single category of pre-training objective dominates the overall ranking. Models

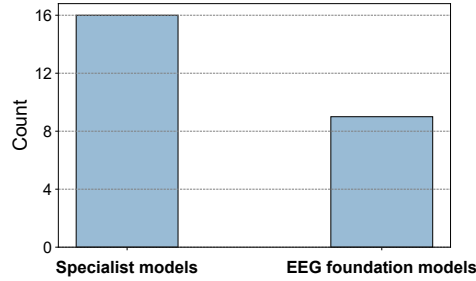
a Ranking by top-1 counts for each model.



b Ranking by top-3 counts for each model.



c Top-1 counts: specialist models vs. FMs.



d Top-3 counts: specialist models vs. FMs.

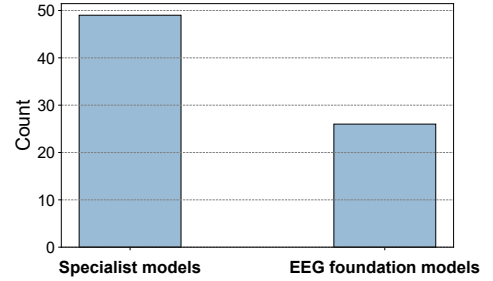


Figure 5. Comparison of ranking performance between specialist models and FMs. (a) and (b) show the number of top-1 and top-3 placements for individual models across all tasks and scenarios. (c) and (d) compare the aggregate top-1 and top-3 counts between specialist models and the seven best-performing FMs.

adopting raw-signal reconstruction (CBraMod, EEGMamba, EEGPT) span ranks from 1 to 15, while models adopting embedded-token reconstruction or prediction (Neuro-GPT, BENDR, BIOT-6D, LUNA-Base) span ranks from 7 to 18. Codebook-based objectives, including LaBraM, BrainOmni-Tiny, BrainOmni-Base, and TFM, span ranks from 9 to 20. This dispersion suggests that the choice of reconstruction or prediction target alone is not a decisive factor for downstream transferability; rather, the combined effect of pre-training corpus scale, data curation, masking strategy, and architectural inductive bias appears to be more influential. Second, the top-ranked FM in each major objective category, namely CBraMod (raw signals, 4.0M), Neuro-GPT (embedded tokens, 0.16M), and LaBraM (codebook index, 5.8M), achieves a competitive overall rank. This indicates that different objective families can yield useful transferable representations when paired with appropriate design choices. Third, within the codebook-based family, LaBraM, BrainOmni-Tiny, BrainOmni-Base, and TFM obtain average ranks of 9.36, 12.33, 12.76, and 16.20, respectively, which do not exhibit a monotonic trend with respect to model size. This is consistent with the broader observation in Section ‘Benchmark of EEG FMs’ that scaling alone does not currently guarantee improved generalization.

A further question of practical interest is whether specific pre-training objectives are systematically advantageous for specific BCI paradigms. The per-subject results in Tables 13–48 of the supplementary information do not reveal a consistent pairing between objective category and paradigm. For instance, although SSVEP signals are inherently frequency-coded, FMs employing explicit frequency-domain supervision are not among the top performers on Nakanishi2015. Instead, BENDR, EEGPT, and Neuro-GPT, which adopt time-domain or token-level objectives, achieve the strongest performance on this dataset. Similarly, on clinical paradigms such as TUAB and CHB-MIT, both raw-signal and embedded-token objectives can attain competitive performance, with no single family demonstrating a consistent advantage. These findings suggest that, under current data regimes and training practices, the alignment between pre-training objective and paradigm-specific signal structure is weaker than might be expected.

In summary, the reconstruction objective adopted by existing EEG FMs is not a decisive factor for downstream performance. Developing more reliable pre-training strategies that can extract universal and transferable EEG representations is therefore of greater importance than the choice of reconstruction target itself, and re-

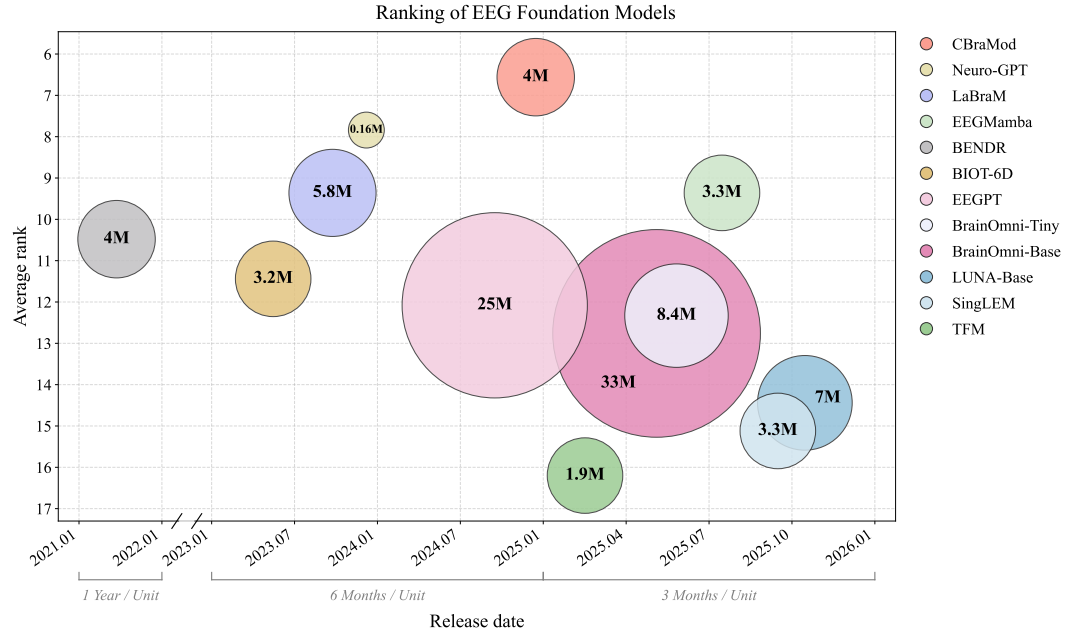


Figure 6. Overall ranking of EEG FMs with respect to release date and model size (bubble size indicates parameter count; lower rank is better).

mains an essential direction for future research.

4.2. Paradigm-Specific FMs

In real-world applications, the paradigm for a downstream task is generally determined prior to data collection. For example, stroke patients requiring exoskeleton-assisted rehabilitation are naturally suited to MI-based systems [112], while epilepsy monitoring demands epilepsy-specific approaches [113]. Therefore, when user information such as patient demographics and target applications is available, employing a paradigm-specific FM for direct adaptation represents a practical and effective strategy.

In recent years, several researchers have attempted to develop paradigm-specific FMs tailored to particular tasks, such as MEET [36] for emotion recognition, MIRepNet [61] for MI, PSGFM [62] for sleep staging, and EpilepsyFM [65] for epilepsy detection. Among these, we compared MIRepNet, which provides open source pre-trained weights, against existing general-purpose FMs on MI tasks. Section A4 of the supplementary information presents detailed results on three MI datasets: BNCI2014001, BNCI2014004, and BNCI2015001. MIRepNet achieved state-of-the-art performance in terms of both subject-averaged accuracy and Cohen's Kappa. This superior performance may be attributed to the fact that paradigm-specific FMs are pre-trained exclusively on datasets from the target task and incorporate neurophysiological

principles relevant to that paradigm in their pre-training strategies. Consequently, the pre-trained encoder is capable of extracting task-specific representations that facilitate rapid adaptation to downstream applications.

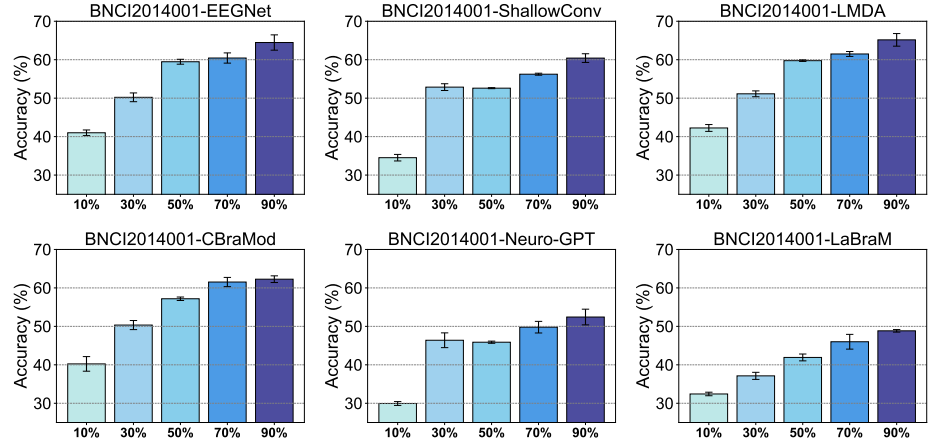
Given that the required paradigm is typically known before data acquisition in practical BCI deployment, developing paradigm-specific pre-trained FMs represents a viable and promising research direction. Furthermore, whether auxiliary data from other paradigms can enhance pre-training for a target paradigm remains an open question worthy of further investigation.

4.3. Effectiveness of EA

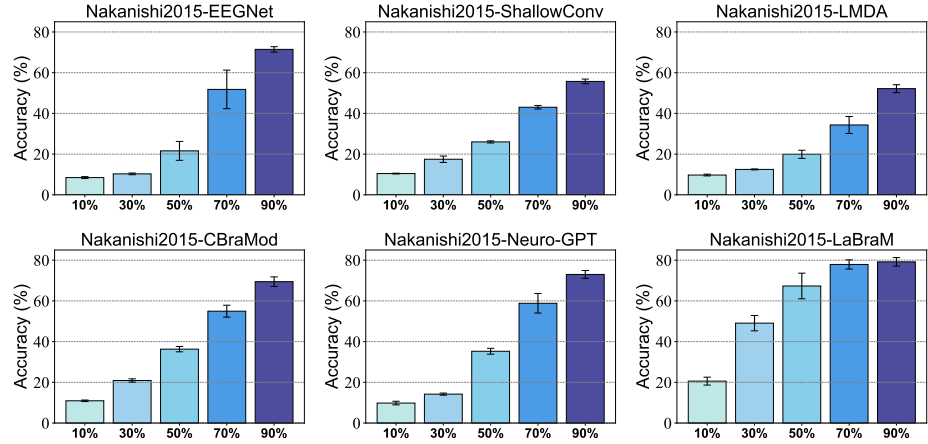
As mentioned in Section 'Data preprocessing', EA [87,88] aligns the marginal distributions across EEG trials. We compared model performance with and without EA on the BNCI2014001 dataset. As shown in Fig. 7, incorporating EA during training or fine-tuning improved generalization performance for the majority of models.

EA normalizes the average trial covariance of each subject to the identity matrix, thereby reducing inter-subject discrepancies in second-order statistics. From a Riemannian geometry perspective, the covariance matrices of different subjects occupy distinct regions on the symmetric positive-definite (SPD) manifold. EA re-centers each subject's covariance distribution to the identity matrix, which serves as the canoni-

a BNCI2014001 (MI paradigm).



b Nakanishi2015 (SSVEP paradigm).



c Impact of EA on BNCI2014001.

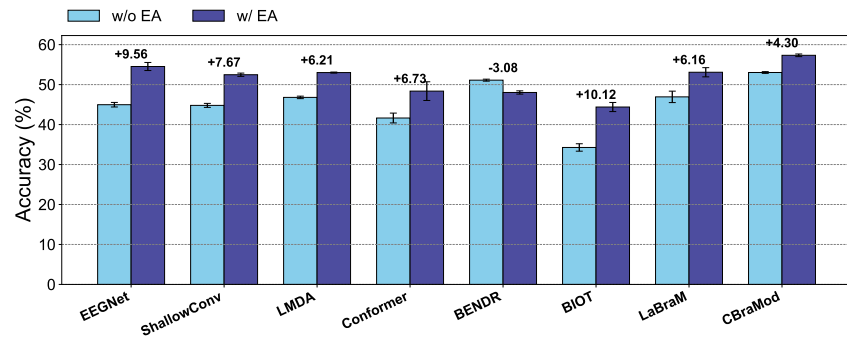


Figure 7. Performance comparison across different fine-tuning data ratios and the impact of EA. (a) and (b) Accuracy comparison on the BNCI2014001 dataset (MI paradigm) and the Nakanishi2015 dataset (SSVEP paradigm) with fine-tuning ratios varying from 10% to 90%. (c) Impact of EA on model performance on the BNCI2014001 dataset.

cal reference point on the SPD manifold, so that the aligned trials of all subjects share a common distribution after being mapped back to the Euclidean space.

To ensure a fair evaluation, EA was applied only to the FMs whose original implementations adopt it, namely EEGPT and MIREPNet; the remaining models were evaluated without EA. This is because applying EA during fine-tuning while it was absent during pre-training, or vice versa, would introduce a mismatch between the aligned and unaligned representation spaces, thereby confounding the comparison.

Despite its effectiveness, EA has potential limitations. As a whitening transformation, EA mixes the original channels into linear combinations, which may distort the spatial structure of the electrode layout. This could be detrimental to models that explicitly rely on spatial information, such as those employing electrode coordinate encoding or graph-based representations, and may also reduce the interpretability of the learned spatial patterns. The trade-off between the cross-subject alignment provided by EA and the preservation of spatial structure therefore warrants careful consideration when applying EA to spatially aware EEG FMs.

4.4. Future Research Directions

Large-scale high-quality data construction

Non-invasive EEG signals inherently suffer from low signal-to-noise ratios due to hardware limitations, environmental interference, and variations in subject attention during acquisition. Several existing approaches attempt to reconstruct raw signals during pre-training, which may inadvertently encourage models to fit noise patterns rather than learn generalizable representations beneficial for downstream tasks. Furthermore, current EEG FMs do not exhibit scaling law behavior, potentially due to the lack of large-scale, high-quality EEG datasets. Liu et al. [86] demonstrated the importance of data quality in the MI paradigm by performing channel selection based on neurophysiological principles and removing low-quality subjects. Models trained on this cleaner and smaller dataset achieved superior performance compared to those trained on uncleaned data. Therefore, constructing large-scale, high-quality EEG corpus through systematic data collection and rigorous cleaning procedures across diverse paradigms represents a critical direction for future research.

Paradigm-specific FMs

As discussed above, the target paradigm is typically known prior to downstream data acquisition, making paradigm-specific FMs a practical and well-motivated approach. Recent works have explored this direction, including MEET [36] for emotion recognition, MIREPNet [61] for MI, PSGFM [62] for sleep staging, and EpilepsyFM [65] for epilepsy detection. The results reported in Section A4 of the supplementary information for MIREPNet demonstrate its superior performance on MI tasks. This advantage may stem from pre-training exclusively on paradigm-specific data while incorporating neurophysiological principles into the pre-training pipeline, enabling the model to learn more effective representations for the target paradigm. These findings support the validity, feasibility, and practicality of paradigm-specific FMs as a promising research direction.

Efficient pre-training strategies

Most existing approaches adopt masked reconstruction as the primary pre-training objective, targeting raw signals, frequency-domain representations, or embedded tokens. However, no single model has demonstrated consistently strong performance across all tasks. Future research should address the following challenges: 1) developing more effective solutions for cross-device heterogeneity; 2) designing pre-training strategies that enable models to learn truly universal and transferable representations; and 3) exploring efficient fine-tuning strategies that facilitate rapid adaptation to new tasks, including methods to achieve competitive performance with less calibration data and techniques to accelerate distribution alignment between pre-trained models and downstream data. As a preliminary step in the latter direction, we conducted a parameter-efficient fine-tuning experiment, reported in Section A4 of the supplementary information.

5. CONCLUSION

This paper has presented EEG-FM-Compass, a comprehensive benchmark for EEG FMs in BCIs. We reviewed 55 studies and distilled their common pipeline components and pre-training objectives into a unified framework that enables structured comparison across heterogeneous devices and paradigms. Based on this analysis, we established a benchmark that evaluates 12 open source EEG FMs alongside competitive specialist baselines across 13 datasets spanning 9 representative BCI paradigms, under both cross-

subject LOSO and within-subject few-shot evaluation protocols.

The experimental results indicate that current EEG FMs have not yet achieved universally transferable representations. Specifically, full-parameter fine-tuning consistently provides substantial advantages over linear probing, suggesting that pre-trained encoders cannot be directly employed as fixed feature extractors across diverse downstream tasks. Furthermore, specialist models trained from scratch remain highly competitive, and increasing model size alone does not guarantee improved generalization. These findings highlight the need for future research on advancing pre-training strategies, as well as enhancing robustness to noise and cross-task heterogeneity.

6. DATA AVAILABILITY

The source code will be available on GitHub <https://github.com/Dingkun0817/EEG-FM-Benchmark>.

7. FUNDING

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8. AUTHOR CONTRIBUTIONS

D.L. conceived the study, conducted the analysis, and wrote the manuscript. Y.C. and Z.C. performed the experiments and contributed to the analysis. Z.C. and Y.W. contributed to the literature review and data collection. J.A. and J.L. revised the manuscript. D.W. supervised the study and revised the manuscript.

Conflict of interest statement. None declared.

A SUPPLEMENTARY INFORMATION

A1. SUMMARY OF EEG FMs

Table 4 summarizes the preprocessing details adopted by the EEG FMs, while Table 5 provides an overview of their pre-training strategies and backbone architectural configurations.

A2. PRE-TRAINING AND DOWNSTREAM DATASETS

Tables 6–13 summarize the pre-training and downstream datasets used by existing EEG FMs. For each model, these tables document both the datasets used during pre-training and those adopted for downstream evaluation, providing a systematic overview of the data resources underlying current EEG FMs.

A3. DATASET DESCRIPTIONS

Tables 14 and 15 summarize the characteristics of the 13 benchmark datasets and the corresponding within-subject few-shot fine-tuning protocols, respectively. The 13 datasets used in this benchmark are summarized below.

1. **BNCI2014001** contains EEG data from 9 subjects performing four MI tasks: left hand, right hand, both feet, and tongue. Each subject participated in two sessions, with each session consisting of 6 runs, yielding a total of 288 trials per session.
2. **BNCI2015001** contains EEG data from 12 subjects performing sustained MI of the right hand and both feet. The data were recorded at 512 Hz using 13 electrodes, with a bandpass filter between 0.5 and 100 Hz and a notch filter at 50 Hz.
3. **BNCI2014004** contains EEG data from 9 right-handed subjects performing two MI tasks: left hand and right hand. Each subject participated in five sessions, with the first two sessions for screening without feedback and the remaining three sessions with feedback. The data were recorded using three bipolar EEG channels (C3, Cz, C4) at 250 Hz, with 120 trials per subject for each MI class.
4. **BNCI2014009** contains P300 evoked potentials from 10 healthy subjects performing a 6×6 matrix speller task under overt attention conditions. EEG was recorded from 16 channels (Fz, FCz, Cz, CPz, Pz, Oz, F3, F4, C3, C4, CP3, CP4, P3, P4, PO7, PO8) at 256 Hz with 0.1–20 Hz bandpass filtering. Each subject completed four sessions with three runs per session.
5. **BNCI2014008** contains P300 evoked potentials from 8 subjects with amyotrophic lateral sclerosis (ALS) performing a 6×6 matrix speller task. EEG was recorded from 8 channels (Fz, Cz, Pz, Oz, P3, P4, PO7, PO8) at 256 Hz with 0.1–30 Hz bandpass filtering. Each subject completed seven runs of five-character spelling, yielding 35 trials in total.
6. **CHB-MIT** contains EEG recordings from 23 pediatric subjects with intractable seizures, recorded using a 16-channel bipolar montage at 256 Hz. The dataset is used for seizure detection, a binary classification task to identify the presence of epileptic seizures from EEG signals.
7. **TUAB** (Temple University Hospital Abnormal) is a large-scale clinical EEG dataset from the TUH EEG Corpus, containing recordings from 2,383 adult patients with over 1,000 hours of data in total. The dataset is used for abnormal EEG detection, a binary classification task to distinguish pathological brain activity from normal recordings.
8. **Sleep-EDFx** (Sleep-EDF Expanded) is a polysomnographic dataset containing 197 whole-night recordings from 78 healthy subjects. Each recording includes EEG signals from Fpz–Cz and Pz–Oz derivations, annotated into five sleep stages: Wake, N1, N2, N3, and REM. The dataset serves as a standard benchmark for automatic sleep stage classification.
9. **SEED** (SJTU Emotion EEG Dataset) is a benchmark dataset for EEG-based emotion recognition, containing recordings from 15 subjects who watched 15 film clips across three sessions spaced one week apart. The 62-channel EEG was recorded at 1,000 Hz using an ESI NeuroScan system, with each clip labeled as positive, neutral, or negative.
10. **Nakanishi2015** is an SSVEP benchmark dataset for multi-class target identification. It contains EEG recordings from 9 subjects responding to 12 visual stimuli with frequencies ranging from 9.25 to 14.75 Hz. Each subject completed 15 blocks of 12 trials, yielding 180 trials per subject. EEG was recorded at 256 Hz using 8 occipital channels.
11. **EEGMat** is a cognitive workload dataset collected from 36 subjects during mental arithmetic tasks. EEG was recorded using 19 channels at 500 Hz following the international 10–20 system. Subjects were categorized into good and poor performers based on task accuracy, enabling analysis of individual differences in workload-related brain activity.

Table 4. Preprocessing of EEG FMs.

Model	Resampling (Hz)	Filtering	Alignment	Channel Mapping	Patch	Stride	Overlap
BENDR	256	≤ 120 Hz (P300)	Norm $\rightarrow [-1,1]$	19	375ms	375ms	✗
BrainBERT	—	≥ 0.1 Hz + 60 Hz notch	STFT + z -score	—	200ms	25ms	✓
MBrain	—	—	Norm $\rightarrow \mathcal{N}(0, 1)$	19 / — (EEG / iEEG)	1s	1s	✗
BIOT	200 / 500 (EEG / ECG)	—	Norm $\rightarrow \frac{x}{P_{95}(x)}$	16 / 12 (EEG / ECG)	1s	0.5s	✓
Brant	250	—	—	—	6 s	—	—
LaBraM	200	0.1-75 Hz + 50 Hz notch	Norm $\rightarrow [-1,1]$	—	1s	1s	✗
Mentality	200	60&120 Hz notch	—	19	10s	10s	✗
Neuro-GPT	250	0.5-100 Hz + 60 Hz notch	z -score	22	2s	1.8s	✓
MEET	200	1-50 Hz	—	32×32 Map	—	—	✗
EEGFormer	250	—	Instance Norm	—	—	—	—
BrainWave	>1000 $\rightarrow$ 1000	$0.01 - \frac{f_c}{3}$ Hz + 50/60 Hz notch	—	—	1 s	1 s	✗
NeuroLM	200	0.1-75 Hz + 50 Hz notch	Norm $\rightarrow [-1,1]$	—	1s	1s	✗
Brant-X	—	≤ 45 Hz (EM)	z -score (only EM)	—	—	—	✗
FoME	250	0.5-100.5 Hz + 50 / 60 Hz notch	EMA	—	6s	6s	✗
EEGPT	256	≤ 38 Hz / ≤ 30 Hz / ≤ 120 Hz	EA / CAR / z -score	—	250ms	250ms	✗
BrainGPT	256	0.1-100 Hz	z -score	—	1s	125ms	✓
GEFM	256	—	—	19	—	—	—
CBraMod	200	0.3-75 Hz + 60 Hz notch	Norm $\rightarrow [-1,1]$	19	1s	1s	✗
CEReBrO	—	—	—	64	L = 64	S = 64	✗
LEAD	128 / 64 / 32	0.5-45 Hz	z -score	19	L = 128	S = 64	✓
FEMBA	250	—	Quantile Norm	22	128 ms	—	—
LCM	256	0-38 Hz (MI)	CAR	—	—	—	—
TFM	200	0.1-75 Hz + 50 Hz notch (TUH)	—	16	1s	0.5s	✓
Gram	200	0.1-75 Hz + 50 / 60 Hz notch	Norm $\rightarrow [-1,1]$	130	1s	1s	✗
ALFEE	256	50 / 60 Hz notch	z -score	90	1s	1s	✗
BrainOmni	256	0.1-96 Hz + 50 / 60 Hz notch	z -score	16	2s	1s	✓
E3GT	125	0.1-50 Hz	CAR	8	4s	1s	✓
CodeBrain	200	0.3-75 Hz + 60 Hz notch	Norm $\rightarrow \frac{x}{100}$	19	1s	1s	✗
DIVER-0	200	0.3-75 Hz + 60 Hz notch	—	19	1s	1s	✗
UniMind	200	0.1-75 Hz + 50 / 60 Hz notch	z -score	—	—	—	✗
CSBrain	200	0.3-75 Hz + 60 Hz notch	Norm $\rightarrow [-1,1]$	19	NA	NA	NA
DMAE-EEG	—	—	—	10 Regions	—	—	✗
EEGMamba	200	0.3-75 Hz + 50 / 60 Hz notch	Norm $\rightarrow [-1,1]$	—	1s	1s	✗
MIRepNet	250	8-32 Hz	Zero-mean + EA	45	NA	NA	NA
PSGFM	100	—	IQR Scaling	1	30s	30s	✗
EEGDM	200	0.1-75 Hz + 50 Hz notch	Norm $\rightarrow [-1,1]$	—	NA	NA	NA
CoMET	200	0.5-70 Hz	Norm $\rightarrow [-1,1]$	62	250ms	250ms	✗
EpilepsyFM	200	0.5-70 Hz + 50 Hz notch	z -score	8 Regions	1s	1s	✗
SingLEM	128	0.5-50 Hz + 50 Hz notch	Norm $\rightarrow (-1,1)$	1	1s	0.75s	✓
BrainPro	200	—	Norm $\rightarrow [-1,1]$	60	0.1s	0.1s	✗
Uni-NTFM	200	0.5-50 Hz + 50 Hz notch	—	5 Regions	NA	NA	NA
ELASTIQ	200	0.3-40 Hz (MI) / 0.3-70 Hz	—	65	0.5s	0.5s	✗
BioCodec	250 / 1000 (EEG / EMG)	0.5-100 Hz	z -score	1	NA	NA	NA
HEAR	200	1-75 Hz	CAR	—	—	—	✗
NeuroRVQ	200	—	—	—	1s	—	—
NeurIPT	64	0.1-30 Hz + 50 Hz notch	A-Law Companding	20	4s	4s	✗
REVE	200	0.5-99.5 Hz	z -score	—	1s	0.1s	✓
mdJPT	125	0.5-47 Hz	ICA + CAR	60	5s	2s	✓
LUNA	256	0.1-75 Hz + 50 / 60 Hz notch	z -score	—	L = 40	L = 40	✗
THD-BAR	200	0.1-75 Hz + 50 / 60 Hz notch	IQR Scaling	—	1s	1s	✗
EEG-X	128	—	—	—	1s	0.75s	✓
SAMBA	—	—	—	—	—	—	—
DeeperBrain	200	0.3-75 Hz + 50 / 60 Hz notch	Norm $\rightarrow [-1,1]$	—	1 s	1 s	✗
EEGMoE	—	1-45 Hz	Gaussian Filter / z -score	9×9 Map	—	—	✗
Brain-OF	200	1-80 Hz + 50 / 60 Hz notch	z -score + CAR	—	L = 64	L = 64	✗

Table 5. Pre-training strategy of EEG FMs.

Model	Masking strategy	Reconstruction objective	Loss function	Encoder depth	Attn-head	d_model	FFN
BENDR	Random Mask	Embedded Tokens	$\mathcal{L}_{cl}^{et}$	8	8	1536	3076
BrainBERT	Random Mask	Spectrogram	$\mathcal{L}_{mse}^{spec}$	6	12	768	—
MBrain	Random Mask	NA	NA	NA	NA	NA	NA
BIOT	Random Mask	Embedded Tokens	$\mathcal{L}_{cl}^{et}$	4	8	256	1024
Brant	Random Mask	Raw Signals	$\mathcal{L}_{mse}^{rs}$	17	16	2048	3072
LaBraM	Random Mask	EEG Codebook Index	$\mathcal{L}_{cls}^{ci}$	12 / 24 / 48	10 / 16 / 16	200 / 400 / 800	800 / 1600 / 3200
Mentality	—	Raw Signals	$\mathcal{L}_{mse}^{rs} + \mathcal{L}_{mse}^{spec}$	NA	NA	NA	NA
Neuro-GPT	Causal Mask	Embedded Tokens	$\mathcal{L}_{mse}^{et}$	6	—	1080	—
MEET	NA	NA	$\mathcal{L}_{cls}$	3 / 6 / 12	3 / 12 / 16	768 / 768 / 1024	3072 / 3072 / 4096
EEGFormer	—	Spectral Amplitude	$\mathcal{L}_{mse}^{sa} + \mathcal{L}_{mse}^{cbe}$	6 / 8 / 12	—	128	—
BrainWave	Random Mask	Spectrogram	—	10	16	768	2048
NeuroLM	Causal Mask	Codebook Index	$\mathcal{L}_{nll}^{ci}$	12 / 24 / 48	12 / 16 / 25	768 / 1024 / 1600	3072 / 4096 / 6400
Brant-X	NA	NA	$\mathcal{L}_{cl}^{coarse} + \mathcal{L}_{cl}^{fine}$	—	—	—	—
FoME	Random Mask	Raw Signals	$\mathcal{L}_{mse}^{rs}$	16	—	—	3072 / 7168
EEGPT	Random Mask	Raw Signals	$\mathcal{L}_{mse}^{rs} + \mathcal{L}_{mse}^{emb}$	—	—	—	—
BrainGPT	Causal Mask	Raw Signals	$\mathcal{L}_{mse}^{rs}$	3 / 9 / 12 / 20	4 / 8 / 14 / 28	128 / 256 / 896 / 1792	512 / 1024 / 3584 / 7168
GEFM	Random Mask	Embedded Tokens	$\mathcal{L}_{cl}^{et}$	8	8	1536	3076
CBraMod	Random Mask	Raw Signals	$\mathcal{L}_{mse}^{rs}$	12	8	200	800
CEReBrO	Random Mask	Embedded Tokens	$\mathcal{L}_{mse}^{et} + \mathcal{L}_{aux}$	8 / 10 / 12	12	192 / 576 / 768	768 / 2304 / 3072
LEAD	NA	NA	$\mathcal{L}_{cls}^{emb} + \mathcal{L}_{cls}$	12	8	128	256
FEMBA	Random Mask	Raw Signals	$\mathcal{L}_{s-l}^{rs}$	NA	NA	NA	NA
LCM	Random Mask	Embedded Tokens	$\mathcal{L}_{cl}^{emb} + \lambda \mathcal{L}_{mse}^{et}$	—	—	—	—
TFM	Random Mask	Codebook Index	$\mathcal{L}_{cls}^{ci}$	4	8	64	—
Gram	Random Mask	Spectral Amp.+ Codebook Index	$\mathcal{L}_{mse}^{sa} + \mathcal{L}_{cls}^{ci}$	12 / 24 / 32	10 / 16 / 16	200 / 400 / 800	800 / 1600 / 3200
ALFEE	Random & Causal Mask	Raw Signals + PSD	$\mathcal{L}_{mse}^{rs} + \mathcal{L}_{mse}^{rs \oplus PSD} + \mathcal{L}_{cls}^{dt}$	6 / 8 / 16 / 18 / 22	4 / 4 / 8 / 8 / 12	384 / 512 / 640 / 896 / 1152	256 / 512 / 512 / 768 / 768
BrainOmni	Random Mask	Codebook Index	$\mathcal{L}_{cls}^{ci}$	12	8 / 16	256 / 512	1024 / 2048
E3GT	Random Mask	SpecKMeansLabels	$\mathcal{L}_{cls}^{skl}$	12	12	768	3072
CodeBrain	Random Mask	Codebook Index	$\mathcal{L}_{cls}^{ci}$	NA	NA	NA	NA
DIVER-0	Random Mask	Raw Signals	—	12	10	200	800
UniMind	NA	NA	$\mathcal{L}_{cce}$	12	10	1152	—
CSBrain	Random Mask	Raw Signals	$\mathcal{L}_{mse}^{rs}$	12	8	200	800
DMAE-EEG	Random Mask	Raw Signals + Embedded Tokens	$\mathcal{L}_{mse}^{rs} + \mathcal{L}_{mse}^{et}$	—	—	—	—
EEGMamba	Random Mask	Raw Signals	$\mathcal{L}_{mse}^{rs}$	NA	NA	NA	NA
MIRepNet	Random Mask	Embedded Tokens	$\mathcal{L}_{mse} + \mathcal{L}_{cls}$	6	8	256	1024
PSGFM	Random Mask	SpecKMeansLabels	$\mathcal{L}_{cls}^{skl}$	12	12	768	3072
EEGDM	NA	Velocity	$\mathcal{L}_{mse}^v$	NA	NA	NA	NA
CoMET	Random Mask	Raw Signals	$\mathcal{L}_{mse}^{rs} + \mathcal{L}_{cl}^{glob}$	6 / 6 / 12	4 / 8 / 16	256 / 512 / 1024	1024 / 2048 / 4096
EpilepsyFM	Random Mask	EEG Codebook Index	$\mathcal{L}_{cls}^{ci}$	12	10	200	800
SingLEM	Random Mask	Raw Signals + Frequency	$\mathcal{L}_{huber}^{ms} + \mathcal{L}_{huber}^{ums} + \mathcal{L}_{l_2}^{freq}$	4 + 12	4 + 8	128	—
BrainPro	Random Mask	Raw Signals	$\mathcal{L}_{mse}^{rs} + \mathcal{L}_{mse}^{nw-mse} + \mathcal{L}_{dec}$	4	32	32	64
Uni-NTFM	Random Mask	Time + Band Power	$\mathcal{L}_{mse}^{emb} + \mathcal{L}_{mse}^{bp} + \mathcal{L}_{aux}$	12 / 12 / 26 / 24	—	256 / 512 / 512 / 768	—
ELASTIQ	Random & Causal Mask	Embedded Tokens	$\mathcal{L}_{mse}^{et}$	12	8	256	1024
BioCodec	NA	Time + Frequency	$\mathcal{L}_{huber}^{rs} + \mathcal{L}_{l_1}^{stf1} + \mathcal{L}_{l_2}^{stf2} + \mathcal{L}_{aux}$	2	8	128	—
HEAR	NA	Codebook + Frequency	$\mathcal{L}_{mse}^{cbe} + \mathcal{L}_{mse}^{freq}$	6 / 12	4 / 8	—	—
NeuroRVQ	Random Mask	EEG Codebook Index	$\mathcal{L}_{cls}^{ci}$	12	10	200	800
NeurIPT	Amplitude-Aware Mask	Raw Signals	$\mathcal{L}_{l_p}^{rs} + \mathcal{L}_{aux}$	6	8	768	512 / 768
REVE	Random Mask	Raw Signals	$\mathcal{L}_{mse}^{rs} + \mathcal{L}_{aux}$	4 / 22 / 22	8 / 8 / 19	512 / 512 / 1250	1365 / 1365 / 3333
mdJPT	NA	NA	$\mathcal{L}_{CDA} + \mathcal{L}_{ISA}$	2	8	128	—
LUNA	Random Mask	Embedded Tokens	$\mathcal{L}_{mse}^{et}$	8 / 10 / 24	8 / 12 / 16	256 / 576 / 1024	1024 / 2304 / 4096
THD-BAR	Causal Mask	Codebook Index	$\mathcal{L}_{cls}^{ci}$	12 / 24 / 48	12 / 16 / 25	768 / 1024 / 1600	3072 / 4096 / 6400
EEG-X	Random Mask	Noised-Removed Signals	$\mathcal{L}_{mse}^{nrs} + \mathcal{L}_{kd}^{lea-stu} + \mathcal{L}_{aux}$	4	8	16	64
SAMBA	Random Mask	Time + Frequency	$\mathcal{L}_{mse}^{no} + \mathcal{L}_{mse}^{freq}$	NA	NA	NA	NA
DeeperBrain	Random Mask	Raw Signals + Neuro Info	$\mathcal{L}_{huber}^{rs} + \mathcal{L}_{huber}^{ni}$	12	8	200	800
EEGMoE	Random Mask	Embedded Tokens	$\mathcal{L}_{l_1}^{et} + \mathcal{L}_{aux}$	3	4	512	—
Brain-OF	Random Mask	Raw Signals + Spectral Amp.	$\mathcal{L}_{s-l}^{rs} + \mathcal{L}_{s-l}^{sa}$	12 / 24 / 36	4 / 8 / 12	256 / 512 / 768	—

12. **Things-EEG2** is a large-scale dataset for visual object decoding, containing EEG recordings from 10 participants viewing natural object images. The dataset comprises 1,854 object classes from the THINGS image collection, supporting research on neural representations of visual semantics.
13. **SEED-VIG** is a dataset for EEG-based vigilance estimation collected during simulated driving. Vigilance levels are quantified using the PERCLOS (percentage of eye closure) metric derived from eye-tracking data. EEG was recorded at 200 Hz using 17 channels and segmented into 8-second epochs, supporting continuous vigilance prediction as a regression task.

A4. BENCHMARK RESULTS

Main Results

The detailed benchmark results for each subject are presented in Tables 16 - 51.

Table 6. Summary of the pre-trained and downstream datasets utilized in EEG FMs.

Models	Pre-training		Downstream	
	Datasets	Paradigms	Datasets	Paradigms
BENDR	TUEG	Clinic	PhysionetMI	MI / ME
			BCIC-IV-2A	MI / ME
			Margaux2012	ERN / ERP
			Citi2010	ERN / ERP
			Sleep-EDF	Sleep
BrainBERT	Brain TreeBank	Clinic	Brain TreeBank	Clinic
MBrain	TUSZ	Clinic	TUSZ	Clinic
	Private	Clinic	Private	Clinic
BIOT	CHB-MIT	Clinic	CHB-MIT	Clinic
	IIIC Seizure	Clinic	IIIC Seizure	Clinic
	TUAB	Clinic	TUAB	Clinic
	TUEV	Clinic	TUEV	Clinic
	SHHS	Sleep	HAR	MI / ME
	PREST	Resting	PTB-XL	(ECG)
	Cardiology	ECG		
Brant	Private	Clinic	MAYO	Clinic
			FNUSA	Clinic
			Private	Clinic
LaBraM	Siena	Clinic	TUAB	Clinic
	TUAR	Clinic	TUEV	Clinic
	TUEP	Clinic	MoBI	MI / ME
	TUSL	Clinic	SEED-V	Emotion
	TUSZ	Clinic		
	BCIC IV-1	MI / ME		
	Grasp and Lift	MI / ME		
	PhysionetMI	MI / ME		
	Emobrain	Emotion		
	SEED	Emotion		
	SEED-IV	Emotion		
	SEED-GER	Emotion		
	SEED-FRA	Emotion		
	RS-EEG	Resting		
	SPIS	Resting		
	InriaBCI	ERN / ERP		
	TVNT	ERN / ERP		
	RAW	—		
	Private	—		
Mentality	TUSZ	Clinic	TUSZ	Clinic
Neuro-GPT	TUEG	Clinic	BCIC-IV-2A	MI / ME
MEET	SEED	Emotion	SEED-IV	Emotion
EEGFormer	TUEG	Clinic	TUAB	Clinic
			TUAR	Clinic
			TUSL	Clinic
			TUSZ	Clinic
			Neonate	Clinic
BrainWave	Siena	Clinic	AD-65	Clinic
	TUEG	Clinic	CHB-MIT	Clinic
	Schizophrenia-81	Clinic	MDD-64	Clinic
	Stroke-50	Clinic	Depression-122	Clinic
	PD-31	Clinic	Schizophrenia-28	Clinic
	AD-184	Clinic	ADHD-Adult	Clinic
	CAP	Sleep	ADHD-Child	Clinic
	HMC	Sleep	SD-71	Sleep
	Sleep-EDF	Sleep		
	SRM	Resting		
	Private	—		
	IowaDataset	—		
	UNMDataset	—		

Table 7. Summary of the pre-trained and downstream datasets utilized in EEG FMs (continued).

Models	Pre-training		Downstream	
	Datasets	Paradigms	Datasets	Paradigms
NeuroLM	TUEG	Clinic	TUAB	Clinic
	Siena	Clinic	TUEV	Clinic
	BCIC IV-1	MI / ME	TUSL	Clinic
	Grasp and Lift	MI / ME	SEED	Emotion
	PhysionetMI	MI / ME	HMC	Sleep
	SEED-FRA	Emotion	EEGMat	Workload
	SEED-GER	Emotion		
	SEED-IV	Emotion		
	SEED-V	Emotion		
	Emobrain	Emotion		
	RS-EEG	Resting		
	SPIS	Resting		
	Inria BCI	ERN / ERP		
	TVNT	ERN / ERP		
	Private	—		
	RAW	—		
Brant-X	CAP	Sleep	FoG	Clinic
	ISRUC	Sleep	DREAMER	Emotion
	HMC	Sleep	Sleep-EDF-20	Sleep
			Sleep-EDF-78	Sleep
			Jaramillo2021	Clinic (EEG+EOG)
FoME	TUEG	Clinic	TUEV	Clinic
	CHB-MIT	Clinic	MAYO	Clinic
	MAYO	Clinic	FNUSA	Clinic
	FNUSA	Clinic	SEED	Emotion
	PhysionetMI	MI / ME	Sleep-EDFx	Sleep
	SEED	Emotion		
	SEED-IV	Emotion		
	Sleep-EDFx	Sleep		
EEGPT	PhysionetMI	MI / ME	TUAB	Clinic
	HGD	MI / ME	TUEV	Clinic
	SEED	Emotion	BCIC-IV-2A	MI / ME
	TSU	SSVEP	BCIC-IV-2B	MI / ME
	M3CV	—	Sleep-EDFx	Sleep
			KaggleERN	ERN / ERP
BrainGPT			PhysioP300	ERN / ERP
	FACED	Emotion	MIBCI	MI / ME
	SEED	Emotion	BCIC IV-1	MI / ME
	SEED-FRA	Emotion	DEAP	Emotion
	SEED-GER	Emotion	FACED	Emotion
	SEED-IV	Emotion	SEED-IV	Emotion
	SEED-V	Emotion	SEED-V	Emotion
	THINGS-EEG-10Hz	Visual	Sleep-EDF	Sleep
	THINGS-EEG-5Hz	Visual	HMC	Sleep
	IMG (Private)	Cross-modal	EEGMat	Workload
			STEW	Workload
			IMG (Private)	Cross-modal
GEFM			SPE	Cross-modal
	TUEG	Clinic	PhysionetMI	MI / ME
			PhysionetP300	ERN / ERP
			Perrin2012	ERN / ERP
CBraMod	TUEG	Clinic	CHB-MIT	Clinic
			TUEV	Clinic
			TUAB	Clinic
			PhysionetMI	MI / ME
			SHU-MI	MI / ME
			FACED	Emotion
			SEED-V	Emotion
			ISRUC	Sleep

Table 8. Summary of the pre-trained and downstream datasets utilized in EEG FMs (continued).

Models	Pre-training		Downstream	
	Datasets	Paradigms	Datasets	Paradigms
CEReBrO	TUEG	Clinic	TUAB	Clinic
			Neonate	Clinic
			MoBI	MI / ME
			SEED	Emotion
LEAD	BrainLat	Clinic	ADFTD	Clinic
	P-ADIC	Clinic	CNBPM	Clinic
	Depression	Clinic	Cognition	—
	FEPCR	Clinic	CAUEEG	—
	PD-RS	Clinic		
	TDBrain	Clinic		
	TUEP	Clinic		
	REEG-BACA	Resting		
	MCEF-RS	Resting		
	PEARL-Neuro	Resting		
	SRM-RS	Resting		
	AD-Auditory	ASSR		
FEMBA	TUEG	Clinic	TUAB	Clinic
			TUAR	Clinic
			TUSL	Clinic
LCM	PhysionetMI	MI / ME	BCIC-IV-2A	MI / ME
	SEED	Emotion	BCIC-IV-2B	MI / ME
	TSU	SSVEP		
TFM	TUAB	Clinic	TUAB	Clinic
	TUEV	Clinic	TUEV	Clinic
	CHB-MIT	Clinic	CHB-MIT	Clinic
	IIIC Seizure	Clinic	IIIC Seizure	Clinic
			EESM23	Sleep
Gram	TUAB	Clinic	TUEV	Clinic
	TUSZ	Clinic	SHU-MI	MI / ME
	Siena	Clinic	SEED-V	Emotion
	RUIJIN	Clinic	HMC	Sleep
	BCIC IV-1	MI / ME		
	PhysionetMI	MI / ME		
	Grasp and Lift	MI / ME		
	SEED	Emotion		
	SEED-IV	Emotion		
	SEED-GER	Emotion		
	SEED-FRA	Emotion		
	Emobrain	Emotion		
	RS-EEG	Resting		
	SPIS	Resting		
	InriaBCI	ERN / ERP		
	TVNT	ERN / ERP		
	RAW	—		
	Private	—		
ALFEE	TUEG	Clinic	TUAB	Clinic
	Siena	Clinic	TUEV	Clinic
	BCIC IV-1	MI / ME	TUSL	Clinic
	Grasp and Lift	MI / ME	SEED	Emotion
	PhysionetMI	MI / ME	HMC	Sleep
	SEED-IV	Emotion	EEGMat	Workload
	SEED-V	Emotion		
	SEED-GER	Emotion		
	SEED-FRA	Emotion		
	Emobrain	Emotion		
	RS-EEG	Resting		
	SPIS	Resting		
	InriaBCI	ERN / ERP		
	TVNT	ERN / ERP		
	RAW	—		

Table 9. Summary of the pre-trained and downstream datasets utilized in EEG FMs (continued).

Models	Pre-training		Downstream	
	Datasets	Paradigms	Datasets	Paradigms
BrainOmni	MusicEEG	Emotion	AD65	Clinic
	HFO	Sleep	MDD	Clinic
	Awakening	Sleep	PD31	Clinic
	Go-Nogo	Visual	TUAB	Clinic
	Features-EEG	Visual	TUEV	Clinic
	SRM	Resting	WBCIC.SHU	MI / ME
	PEARL-Neuro	—	PhysionetMI	MI / ME
	RestCog	—	FACED	Emotion
	HBN-EEG	—	SomatoMotor	MI / ME (EMEG)
	MEG-MASC	Listening (MEG)	MEG-MMI	Emotion (MEG)
	MEG-Narrative	Listening (MEG)	ASD74	ASD (MEG)
	SMN4Lang	Listening (MEG)		
	ASWR-MEG	Listening (MEG)		
	Kymata-SOTO	Listening (MEG)		
	MIND	Clinic (MEG)		
	THINGS-MEG	Visual (MEG)		
	ImageLine	Visual (MEG)		
	OMEGA	Resting (MEG)		
	CC700	(MEG)		
	AversiveMEG	(MEG)		
	ASWR-MEG	(MEG)		
	NeuroMorph	(MEG)		
E3GT	TUEG	Clinic	PhysionetMI	MI / ME
			PhysioP300	ERN / ERP
			Won2022	—
CodeBrain	TUEG	Clinic	CHB-MIT	Clinic
			TUEV	Clinic
			TUAB	Clinic
			SHU-MI	MI / ME
			FACED	Emotion
			SEED-V	Emotion
			ISRUC S1	Sleep
			ISRUC S3	Sleep
			BCIC2020-3	Imagined Speech
			MentalArithmetic	Mental Stress
DIVER-0	TUEG	Clinic	PhysionetMI	MI / ME
			FACED	Emotion
UniMind	NA	NA	TUAB	Clinic
			TUEV	Clinic
			TUSL	Clinic
			SHU-MI	MI / ME
			SEED	Emotion
			SEED-IV	Emotion
			HMC	Sleep
			Sleep-EDF	Sleep
			SHHS	Sleep
			EEGMat	Workload
CSBrain	TUEG	Clinic	CHB-MIT	Clinic
			Siena	Clinic
			TUEV	Clinic
			TUAB	Clinic
			TUSL	Clinic
			BCIC-IV-2A	MI / ME
			PhysionetMI	MI / ME
			SHU-MI	MI / ME
			FACED	Emotion
			SEED-V	Emotion
			ISRUC	Sleep
			HMC	Sleep
			BCIC2020-3	Imagined Speech
			SEED-VIG	Vigilance
			MentalArithmetic	Mental Stress
			Mumtaz2016	Mental Disorder
			PhysionetMI	MI / ME
			MultiMI1	MI / ME
DMAE-EEG	—	—	PhysionetMI	MI / ME
			MultiMI1	MI / ME

Table 10. Summary of the pre-trained and downstream datasets utilized in EEG FMs (continued).

Models	Pre-training		Downstream	
	Datasets	Paradigms	Datasets	Paradigms
EEGMamba	TUEG	Clinic	CHB-MIT	Clinic
	Siena	Clinic	PhysionetMI	MI / ME
	Physionet	Sleep	FACED	Emotion
	B-SNIP1	Resting	ISRUC	Sleep
	RAW	—	BCIC2020-3	Imagined Speech
			MODMA	MDD Diagnosis
MIRepNet	BNCI2014002	MI / ME	BCIC-IV-2A	MI / ME
	PhysionetMI	MI / ME	BNCI2015001	MI / ME
	Dreyer2023	MI / ME	BCIC-IV-2B	MI / ME
	Weibo2014	MI / ME	AlexMI	MI / ME
	Zhou2016	MI / ME		
	Lee2019	MI / ME		
	Cho2017	MI / ME		
PSGFM	SHHS	Sleep	Sleep-EDF	Sleep
	MESA	Sleep	Dreem	Sleep
	MrOS	Sleep	HomePAP	Sleep
	WSC	Sleep	APPLES	Sleep
	SOF	Sleep		
	CFS	Sleep		
	NCHSDB	Sleep		
EEGDM	TUEV	Clinic	TUEV	Clinic
			CHB-MIT	Clinic
CoMET	Stieger2021	MI / ME	TUAB	Clinic
	SEED	Emotion	TUEV	Clinic
	HBN	—	BCIC-IV-2A	MI / ME
	M3CV	—	BCIC-IV-2B	MI / ME
			Large-5F	MI / ME
			FACED	Emotion
			THUBenchmark	SSVEP
			PhysionetP300	ERN / ERP
			KaggleERN	ERN / ERP
			BCIC2020-3	Imagined Speech
EpilepsyFM	TUEP	Clinic	TUAB	Clinic
	TUSL	Clinic	TUEV	Clinic
	TUSZ	Clinic	CHB-MIT	Clinic
	Private-1	Clinic	Private-1	Clinic
			Private-2	Clinic
			Private-3	Clinic
SingLEM	Lin2025	Clinic	Dreyer2023	MI / ME
	Lopez2015	Clinic	WBCIC-MI-2C	MI / ME
	Veloso2017	Clinic	WBCIC-MI-3C	MI / ME
	Cho2017	MI / ME	N-back-2C	Cognitive
	Kaya2017	MI / ME	DSR-2C	DSR
	Schalk2009	MI / ME	WG-2C	Word Generation
	Xiang2024	Sleep		
	Babayan2021	Sleep		
	Gu2024	SSVEP		
	Mou2024	Cognitive		
	Xue2025	RSVP		
	...	...		
BrainPro	TUEP	Clinic	BCIC-IV-2A	MI / ME
	TUSZ	Clinic	SHU-MI	MI / ME
	TUSL	Clinic	FACED	Emotion
	Grasp and Lift	MI / ME	SEED-V	Emotion
	PhysionetMI	MI / ME	SEED-VII	Emotion
	Lee2019	MI / ME		
	HGD	MI / ME		
	Emobrain	Emotion		
	SEED	Emotion		
	SEED-IV	Emotion		
	SEED-GER	Emotion		
	SEED-FRA	Emotion		
	RS-EEG	Resting		
	SPIS	Resting		
	RAW	—		
	Private	—		

Table 11. Summary of the pre-trained and downstream datasets utilized in EEG FMs (continued).

Models	Pre-training		Downstream	
	Datasets	Paradigms	Datasets	Paradigms
Uni-NTFM	CAUEEG	Clinic	TUAB	Clinic
	TUEG	Clinic	TUEV	Clinic
	Siena	Clinic	TUSL	Clinic
	BCIC-IV-1	MI / ME	BCIC-IV-2A	MI / ME
	Emobrain	Emotion	SEED	Emotion
	SEED-IV	Emotion	HMC	Sleep
	SEED-V	Emotion	EEGMat	Workload
	SEED-GER	Emotion	ADFTD	NDD
	SEED-FRA	Emotion	TDBrain	Mental Disorder
	REEG-BACA	Resting		
ELASTIQ	RS-EEG	Resting		
	RAW	—		
	Stieger2021	MI / ME	OpenBMI	MI / ME
	SEED-FRA	Emotion	BCIC-IV-2A	MI / ME
	SEED-GER	Emotion	BCIC-Upperlimb	MI / ME
	SEED-SD	Sleep & Emotion	SHU-MI	MI / ME
	Chisco	Imagined Speech	HighGamma	MI / ME
	ThinkOutLoud	—	Cho2017	MI / ME
			Shin2017A	MI / ME
			PhysionetMI	MI / ME
BioCodec			FACED	Emotion
			SEED	Emotion
			SEED-IV	Emotion
			SEED-V	Emotion
			SEED-VII	Emotion
			OpenBMI	SSVEP
			eldBETA	SSVEP
			Wang2016	SSVEP
			BETA	SSVEP
			EEGMat	Workload
HEAR			BCIC2020-3	Imagined Speech
			ADHD-AliMotie	ADHD
	TUEG	Clinic	TUAB	Clinic
	emg2qwerty	EMG	TUEV	Clinic
			PhysionetMI	MI / ME
			BCIC-IV-2A	MI / ME
			Sleep-EDF	Sleep
			KaggleERN	ERN / ERP
			N400	Speech
NeuroRVQ				
	TUEP	Clinic	BCI-IV-1	MI / ME
	TUEV	Clinic	BCI-IV-2B	MI / ME
	TUAB	Clinic	EEGMMIDB	MI / ME
	CHB-MIT	Clinic	LargeMI	MI / ME
	TUSL	Clinic	SHUDB	MI / ME
	OpenBMI	MI / ME	BCI-IV-2A	MI / ME
	HMC	Sleep	HGD	MI / ME
	Sleep-EDFx	Sleep		
	CAP	Sleep		
EEG-X	PhysionetP300	ERN / ERP		
	KaggleERN	ERN / ERP		
	EEGMAT	Workload		
	Migrainedb	—		
	TUAR	Clinic	HighGamma	MI / ME
	TUEP	Clinic	Sleep-EDF	Sleep
	TUSZ	Clinic	Pavlov2022	Resting
	Siena	Clinic	Schalk2004	—
	BCIC-IV-1	MI / ME		
	Grasp and Lift	MI / ME		
SAMB	PhysionetMI	MI / ME		
	SPIS	Resting		
	Trujillo2017	Resting		
	Inria BCI	ERN / ERP		
	bi2015a	ERN / ERP		
	Trujillo2020	—		
	Private	MI / ME		

Table 12. Summary of the pre-trained and downstream datasets utilized in EEG FMs (continued).

Models	Pre-training		Downstream	
	Datasets	Paradigms	Datasets	Paradigms
NeurIPT	TUAR	Clinic	TUAB	Clinic
	TUEP	Clinic	TUEV	Clinic
	TUSZ	Clinic	BCIC-IV-2A	MI / ME
	TUSL	Clinic	SEED-V	Emotion
	Siena	Clinic	Sleep-EDFx	Sleep
	PhysionetMI	MI / ME	PhysionetP300	ERN/ERP
	Grasp and Lift	MI / ME	MentalArithmetic	Mental Stress
	SEED	Emotion	Mumtaz2016	Mental Disorder
	SEED-IV	Emotion		
	SEED-GER	Emotion		
REVE	SEED-FRA	Emotion		
	Emobrain	Emotion		
	RS-EEG	Resting		
	SPIS	Resting		
	InriaBCI	ERN/ERP		
	TVNT	ERN/ERP		
	RAW	—		
mdJPT	TUEG	Clinic	TUEV	Clinic
	Physionet	Clinic	TUAB	Clinic
	OpenNeuro	Clinic	PhysionetMI	MI / ME
	MOABB	MI / ME	BCIC-IV-2A	MI / ME
	MOABB	ERN / ERP	FACED	Emotion
			HMC	Sleep
			ISRUC	Sleep
			BCIC2020-3	Imagined Speech
			MAT	Mental Stress
			Mumtaz2016	Mental Disorder
LUNA	SEED	Emotion	SEED	Emotion
	SEED-IV	Emotion	SEED-IV	Emotion
	SEED-V	Emotion	SEED-V	Emotion
	SEED-VII	Emotion	SEED-VII	Emotion
	FACED	Emotion	FACED	Emotion
	DEAP	Emotion	DEAP	Emotion
THD-BAR	TUAR	Clinic	TUAB	Clinic
	TUEP	Clinic	TUEV	Clinic
	TUSZ	Clinic	BCIC-IV-1	MI / ME
	TUSL	Clinic	SEED	Emotion
	Siena	Clinic	DEAP	Emotion
	PhysionetMI	MI / ME	Sleep-EDF	Sleep
	Grasp and Lift	MI / ME	HMC	Sleep
	SEED-IV	Emotion	EEGMat	Workload
	SEED-V	Emotion	STEW	Workload
	SEED-GER	Emotion		
EEG-X	SEED-FRA	Emotion		
	Emobrain	Emotion		
	RS-EEG	Resting		
	SPIS	Resting		
	InriaBCI	ERN / ERP		
	TVNT	ERN / ERP		
	RAW	—		
SAMB	TUAR	Clinic	TUAB	Clinic
	DREAMER	Emotion	PhysionetMI	MI / ME
	EEGMat	Workload	GrosseWentrup	MI / ME
	STEW	Workload	BCIC-IV-2A	MI / ME
	Attention	Attention	BCIC-III-II	ERN / ERP
	Crowdsourced	—	BCIC-II-Iib	ERN / ERP
	DriverDistraction	—	STEW	Workload
			Crowdsourced	—
			DriverDistraction	—

Table 13. Summary of the pre-trained and downstream datasets utilized in EEG FMs (continued).

Models	Pre-training		Downstream	
	Datasets	Paradigms	Datasets	Paradigms
DeeperBrain	TUEG	Clinic	CHB-MIT	Clinic
	Siena	Clinic	PhysionetMI	MI / ME
	PhysioNet2018	Sleep	BCIC-IV-2A	MI / ME
	ds006171	Visual	SHU-MI	MI / ME
	ds006547	Visual	FACED	Emotion
	ds006480	Sleep	SEED-V	Emotion
	ds006525	Sleep	SEED-VII	Emotion
	ds006317	Imagined Speech	ISRUC	Sleep
	RAW	—	SEED-VIG	Vigilance
	ds006367	—	BCIC2020-3	Imagined Speech
	ds006370	—	MODMA	MDD Diagnosis
	ds006437	—	MentalArithmetic	Mental Stress
	ds006446	—		
	ds006466	—		
EEGMoE	BCIC-IV-1	MI / ME	BCIC-IV-2A	MI / ME
	EEGMMIDB	MI / ME	DEAP	Emotion
	DREAMER	Emotion	STEW	Workload
	SEED	Emotion		
	MAHNOB-HCI	Emotion		
	EEGMat	Workload		
Brain-OF	TUEG	Clinic	TUAB	Clinic
	PhysionetMI	MI / ME	CHB-MIT	Clinic
	WBCIC-MI	MI / ME	SEED-V	Emotion
	SEED	Emotion	LEMON-EEG	Aging
	SEED-DV	Emotion		
	SEED-SD	Emotion		
	SEED-VII	Emotion		
	FACED	Emotion		
	REEG-BACA	Resting		
	TDBRAIN	Resting		
	PREDICT	Resting		
	SEED-VIG	Vigilance		
	SEED-VLA/VRW	Vigilance		
	...	(MEG)		
	...	(fMRI)		

Table 14. Summary of the EEG datasets in benchmarking.

BCI Paradigm	Dataset	Number of Subjects	Number of Channels	Sampling Rate (Hz)	Trial Length (seconds)	Number of Trials	Tasks	Labels
MI	BNCI2014001 [89]	9	22	250	4	2,592	Classification	left / right hand, feet, tongue
	BNCI2014004 [90]	9	3	250	4.5	1,400	Classification	left hand, right hand
	BNCI2015001 [91]	12	13	512	5	2,400	Classification	right hand, both feet
P300	BNCI2014009 [92]	10	16	256	0.8	5,760	Classification	target, non-target
	BNCI2014008 [92]	8	8	256	1	33,600	Classification	target, non-target
Clinic	CHB-MIT [93]	23	18	256	4	29,840	Classification	interictal, ictal
	TUAB [94]	2,383	21	250	10	53,604	Classification	normal, abnormal
Sleep	Sleep-EDFx [95]	78	2	100	30	414,961	Classification	W, N1, N2, N3, REM
Emotion	SEED [96]	15	62	200	1	50,910	Classification	positive, neutral, negative
SSVEP	Nakanishi2015 [97]	9	8	256	4	1,620	Classification	9.25–14.75 Hz (0.5 Hz interval)
Workload	EEGMat [98]	36	19	500	4	1,080	Classification	low, high
Visual Decoding	Things-EEG2 [99]	10	63	1000	1	18,540	Retrieve	200 images matching
Fatigue	SEED-VIG [100]	21	17	200	8	18,585	Regression	PERCLOS

Table 15. Summary of the within-subject few-shot fine-tuning protocol across the evaluated datasets.

BCI Paradigm	Dataset	Few-shot Protocol	Total Fine-tuning Trials	Fine-tuning Trials per Class	Trial Length (seconds)
MI	BNCI2014001	30% of one session per subject	84	21	4
	BNCI2014004	30% of one session per subject	36–48	18–24	4
	BNCI2015001	30% of one session per subject	60	30	4
P300	BNCI2014008	5% of one session per subject	210	35 / 175	1
	BNCI2014009	10% of one session per subject	57	9 / 48	0.8 [†]
Clinic	CHB-MIT	First seizure with 10-min pre-ictal segment [‡]	152–192	150 / 2–42	4
Sleep	Sleep-EDFx	10% of the target subject	223–574	59–413 / 3–88 / 49–128 / 0–63 / 6–61 [§]	30
Emotion	SEED	One video per class	674	206 / 233 / 235	1
SSVEP	Nakanishi2015	80% of the target subject	144	12	4
Workload	EEGMat	60% of the target subject	18	9	4
Visual Decoding	Things-EEG2	3 of 10 images per class	4,962	3	1
Fatigue	SEED-VIG	10% of the target subject	88	—	8

[†] The original epoch length is 0.8s; signals are extended to 1s by prepending the 0.2s segment preceding each epoch, in order to satisfy the minimum input-length requirement of the evaluated EEG FMs.

[‡] For each subject, the model is fine-tuned on the ictal segments of the first seizure together with its 10-min pre-ictal segment, and evaluated on the remaining seizures with their corresponding pre-ictal segments.

[§] Trials per class correspond to the five sleep stages (W / N1 / N2 / N3 / REM).

SEED-VIG is a regression task (continuous PERCLOS values), and therefore the trials-per-class column is not applicable.

Table 16. Accuracies (%) on BNCI2014001. The best accuracies are marked in bold, and the second best by an underline.

Scenario	Tuning	Model Type	Approach	S0	S1	S2	S3	S4	S5	S6	S7	S8	Avg.
Cross-subject (LOSO)	Specialist Models	CSP+LDA	45.83	27.43	52.78	30.21	30.21	23.61	36.81	57.29	39.58		38.19
		EEGNet	61.23	26.85	69.56	35.07	25.12	28.47	32.18	60.88	65.39		44.97 $\pm$ 0.57
		ShallowConv	68.17	31.13	52.89	37.62	27.43	27.78	31.25	65.74	61.23		44.80 $\pm$ 0.50
		LMDA	66.55	32.64	66.09	35.65	27.78	31.02	29.51	67.36	64.58		46.80 $\pm$ 0.31
		CNN-T	56.83	31.48	50.93	32.29	26.04	28.24	27.20	51.04	48.26		39.15 $\pm$ 0.56
		Deformer	56.48	24.54	60.53	34.26	26.62	29.28	31.02	55.90	55.09		41.53 $\pm$ 0.67
		Conformer	60.88	26.27	57.29	31.48	26.85	30.56	23.84	63.77	53.82		41.64 $\pm$ 1.23
		MSCFormer	52.66	27.43	62.04	36.46	26.39	28.82	25.58	50.46	47.69		39.72 $\pm$ 2.54
		FBCNet	56.94	28.70	60.42	39.12	26.04	24.54	31.13	63.54	54.63		42.79 $\pm$ 0.77
		TSception	63.19	27.66	48.50	34.03	25.46	27.78	33.33	64.93	49.31		41.58 $\pm$ 0.63
	Foundation Models	BENDR	47.92	<u>43.98</u>	56.60	40.74	59.26	46.18	64.00	51.50	49.77		51.11 $\pm$ 0.25
		BIOT-1D	42.82	28.12	32.75	30.90	28.94	30.56	31.02	34.03	30.44		32.18 $\pm$ 0.54
		BIOT-2D	35.88	32.18	41.32	31.25	28.24	29.05	31.13	38.66	32.18		33.32 $\pm$ 2.04
		BIOT-6D	37.15	31.60	39.35	33.45	33.10	29.17	29.63	43.06	31.94		34.27 $\pm$ 0.93
		LaBraM	51.97	37.04	59.03	36.57	43.17	40.28	50.81	50.58	52.89		46.93 $\pm$ 1.43
		Neuro-GPT	59.72	32.18	62.62	34.49	31.71	37.27	39.81	65.62	59.26		46.97 $\pm$ 0.71
		EEGPT	39.81	26.50	34.14	28.47	27.66	30.90	30.79	34.49	37.38		32.24 $\pm$ 1.45
		CBraMod	54.51	46.99	63.19	47.92	<u>43.29</u>	<u>44.91</u>	54.75	60.19	61.57		<u>53.03</u> $\pm$ 0.22
		TFM	32.87	30.79	33.91	32.41	28.01	27.08	29.86	37.50	35.76		32.02 $\pm$ 0.66
		BrainOmni-Tiny	43.17	35.3	49.88	32.99	38.08	37.38	41.78	48.15	47.45		41.58 $\pm$ 0.80
		BrainOmni-Base	42.82	32.52	49.42	32.18	37.85	38.19	42.82	47.11	45.49		40.93 $\pm$ 0.83
		EEGMamba	54.75	38.19	64.12	37.15	29.86	38.54	42.59	50.35	55.90		45.72 $\pm$ 0.54
		MIRepNet	<u>72.11</u>	39.58	78.36	46.88	37.73	40.74	51.39	70.72	50.35		54.21 $\pm$ 0.24
		SingLEM	34.26	26.39	30.21	30.90	33.56	30.21	30.79	29.75	29.05		30.57 $\pm$ 0.10
		LUNA-Base	31.94	26.16	36.69	28.01	25.00	28.59	25.00	28.36	29.98		28.86 $\pm$ 0.50
	Foundation Models	BENDR	31.37	26.16	35.76	31.48	33.22	27.20	37.96	33.22	33.22		32.18 $\pm$ 0.41
		BIOT-1D	38.31	25.35	29.51	31.83	24.88	29.63	25.00	36.11	26.74		29.71 $\pm$ 0.42
		BIOT-2D	36.11	26.74	32.52	30.79	24.88	24.88	27.08	34.95	36.57		30.50 $\pm$ 0.46
		BIOT-6D	34.95	32.41	30.67	29.40	25.00	31.48	23.50	39.81	30.67		30.88 $\pm$ 1.00
		LaBraM	47.92	33.22	49.42	37.62	40.86	39.35	41.67	45.25	48.03		42.59 $\pm$ 0.27
		Neuro-GPT	54.63	38.43	62.85	<u>47.45</u>	32.87	33.33	39.81	64.12	60.65		48.24 $\pm$ 1.04
		EEGPT	46.30	30.79	38.77	34.38	31.83	34.49	36.11	42.48	41.20		37.37 $\pm$ 1.25
		CBraMod	49.31	31.37	55.67	34.61	29.28	29.17	32.75	56.60	54.28		41.45 $\pm$ 0.50
		TFM	32.87	27.66	33.22	28.70	26.85	24.65	25.93	26.97	28.24		28.34 $\pm$ 0.30
		BrainOmni-Tiny	45.37	32.29	48.84	31.25	34.61	36.11	39.00	45.25	45.25		39.78 $\pm$ 0.36
		BrainOmni-Base	43.17	32.29	46.64	32.52	36.00	36.11	38.66	45.60	45.72		39.63 $\pm$ 0.58
		EEGMamba	39.35	30.56	40.97	30.32	28.12	35.42	36.57	30.32	37.27		34.32 $\pm$ 0.20
MIRepNet	73.26	25.93	<u>75.12</u>	39.70	36.34	33.10	<u>59.49</u>	64.70	46.64		50.48 $\pm$ 0.28		
SingLEM	37.04	33.45	34.72	34.72	35.76	33.10	35.30	31.25	25.46		33.42 $\pm$ 0.22		
LUNA-Base	38.31	25.46	39.00	27.66	25.00	25.12	27.43	28.36	26.50		29.21 $\pm$ 0.57		
Within-subject (Few-shot)	Specialist Models	CSP+LDA	78.43	54.90	76.96	<u>46.57</u>	34.80	42.16	<u>77.45</u>	75.49	58.82		<u>60.62</u>
		EEGNet	62.42	33.66	66.34	37.42	29.90	31.05	56.21	68.46	66.50		50.22 $\pm$ 1.14
		ShallowConv	61.11	47.22	65.69	45.92	31.05	37.75	52.45	68.30	<u>66.34</u>		52.87 $\pm$ 0.88
		LMDA	62.25	43.63	65.52	41.01	28.59	34.48	49.67	71.08	63.89		51.13 $\pm$ 0.76
		CNN-T	60.78	46.24	68.46	34.64	31.37	37.42	62.25	69.28	50.98		51.27 $\pm$ 1.10
		Deformer	51.47	36.93	57.84	32.35	23.04	26.96	34.80	60.95	55.56		42.21 $\pm$ 0.73
		Conformer	63.24	50.98	<u>77.12</u>	44.61	32.52	44.93	65.52	77.45	58.33		57.19 $\pm$ 1.32
		MSCFormer	63.24	47.06	<u>74.35</u>	39.71	37.91	40.52	64.38	70.10	59.48		55.19 $\pm$ 1.20
		FBCNet	63.07	49.35	69.12	46.41	34.15	38.07	60.46	75.82	63.40		55.54 $\pm$ 1.70
		TSception	71.73	45.42	71.57	44.61	42.65	41.83	69.44	<u>78.92</u>	61.93		58.68 $\pm$ 0.86
	Foundation Models	BENDR	36.76	38.89	48.69	38.40	55.88	42.65	46.57	45.75	50.49		44.90 $\pm$ 1.31
		BIOT-1D	57.52	41.18	51.96	31.54	39.05	30.72	47.88	57.52	52.78		45.57 $\pm$ 0.85
		BIOT-2D	54.58	37.91	44.28	31.54	32.52	34.80	46.73	54.08	55.56		43.55 $\pm$ 0.75
		BIOT-6D	61.93	43.79	56.54	30.07	36.76	34.31	59.97	59.31	54.74		48.60 $\pm$ 0.72
		LaBraM	43.46	32.03	37.75	30.07	34.15	30.39	35.13	43.95	47.22		37.13 $\pm$ 0.92
		Neuro-GPT	44.77	40.20	62.42	37.91	35.29	39.54	38.40	64.54	54.41		46.39 $\pm$ 1.92
		EEGPT	46.08	28.27	33.66	31.05	26.96	32.84	31.37	41.01	43.63		34.99 $\pm$ 0.25
		CBraMod	56.37	46.57	69.61	38.40	39.38	30.07	56.54	61.93	54.25		50.34 $\pm$ 1.18
		TFM	38.73	29.74	35.13	28.92	21.73	29.25	35.13	39.54	41.50		33.30 $\pm$ 0.46
		BrainOmni-Tiny	45.10	36.76	43.46	39.05	31.54	29.08	37.42	43.14	45.42		39.00 $\pm$ 0.19
		BrainOmni-Base	47.22	33.99	39.87	35.95	30.07	28.59	37.75	41.67	43.30		37.60 $\pm$ 0.27
		EEGMamba	45.42	33.82	44.12	37.09	31.86	32.35	29.74	45.92	41.99		38.04 $\pm$ 0.45
		MIRepNet	<u>72.28</u>	<u>53.96</u>	82.67	50.00	<u>47.19</u>	<u>43.23</u>	80.20	79.70	60.23		63.27 $\pm$ 0.47
		SingLEM	33.82	26.14	28.43	27.29	26.14	30.07	32.03	27.12	29.41		28.94 $\pm$ 0.73
		LUNA-Base	46.90	25.33	39.38	24.18	26.96	25.00	31.54	33.17	34.97		31.94 $\pm$ 1.24
	Foundation Models	BENDR	31.86	30.72	33.66	33.33	30.07	26.31	34.48	27.61	34.80		31.43 $\pm$ 0.85
		BIOT-1D	56.05	41.34	49.02	33.99	45.92	33.82	51.63	54.25	55.39		46.82 $\pm$ 0.38
		BIOT-2D	53.76	39.54	41.99	32.68	35.46	30.23	43.14	56.86	52.78		42.94 $\pm$ 0.33
		BIOT-6D	56.21	41.34	53.43	37.25	41.01	38.56	52.94	60.46	50.33		47.95 $\pm$ 0.53
		LaBraM	39.54	29.25	39.38	28.76	36.44	29.58	35.46	37.75	42.32		35.38 $\pm$ 0.45
		Neuro-GPT	55.23	41.01	60.29	42.81	36.11	41.18	46.08	71.24	54.41		49.82 $\pm$ 1.55
		EEGPT	46.08	26.80	35.29	32.68	28.27	29.25	33.99	44.93	45.42		35.86 $\pm$ 0.54
		CBraMod	34.97	37.42	43.95	35.78	27.94	32.52	34.15	60.62	59.31		40.74 $\pm$ 1.06
		TFM	26.47	26.80	31.70	27.94	22.88	26.14	24.18	29.90	33.99		27.78 $\pm$ 1.05
		BrainOmni-Tiny	47.06	33.66	44.44	34.48	34.31	30.72	43.30	48.69	48.86		40.61 $\pm$ 0.31
		BrainOmni-Base	44.28	35.13	40.85	35.62	28.92	29.74	40.20	47.88	45.75		38.71 $\pm$ 0.36
		EEGMamba	35.95	29.08	42.32	31.05	28.92	30.56	29.25	37.58	39.87		33.84 $\pm$ 0.40
Linear Probing	MIRepNet	69.97	47.03	71.78	34.49	37.13	35.31	58.91	59.74	53.14		51.94 $\pm$ 1.84	
	SingLEM	32.35	30.23	30.07	25.00	31.21	27.29	32.19	28.76	31.70		29.87 $\pm$ 0.18	
	LUNA-Base	48.37	26.63	45.10	27.61	23.86	26.31	37.91	38.07	38.73		34.73 $\pm$ 0.05	

Table 17. Cohen's Kappa (%) on BNCI2014001. The best metrics are marked in bold, and the second best by an underline.

Scenario	Tuning	Model Type	Approach	S0	S1	S2	S3	S4	S5	S6	S7	S8	Avg.
Cross-subject (LOSO)		Specialist Models	CSP+LDA	27.78	3.24	37.04	6.94	6.94	-1.85	15.74	43.06	19.44	17.59
			EEGNet	48.30	2.47	59.41	13.42	0.92	4.63	9.57	47.84	53.86	26.63 $\pm$ 0.75
			ShallowConv	57.56	8.18	37.19	16.82	3.24	3.70	8.33	54.32	48.30	26.40 $\pm$ 0.67
			LMDA	55.40	10.19	54.78	14.20	3.70	8.02	6.02	56.48	52.78	29.06 $\pm$ 0.40
			CNN-T	42.44	8.64	34.57	9.72	1.39	4.32	2.93	34.72	31.02	18.86 $\pm$ 0.74
			Deformer	41.98	0.15	47.38	12.34	2.16	5.71	8.02	41.20	40.12	22.03 $\pm$ 0.89
			Conformer	47.84	1.85	43.06	8.64	2.47	7.41	0.46	51.70	38.43	22.19 $\pm$ 1.64
			MSCFormer	36.88	3.24	49.38	15.28	1.85	5.09	0.77	33.95	30.25	19.63 $\pm$ 3.39
			FBCNet	42.59	4.94	47.22	18.83	1.39	-0.62	8.18	51.39	39.51	23.71 $\pm$ 1.03
			TSception	50.93	3.55	31.33	12.04	0.62	3.70	11.11	53.24	32.41	22.10 $\pm$ 0.84
	Full Fine-tuning	Foundation Models	BENDR	30.56	<u>25.31</u>	42.13	20.99	45.68	28.24	52.00	35.34	33.02	34.81 $\pm$ 0.34
			BIOT-1D	23.77	4.17	10.34	7.87	5.25	7.41	8.02	12.04	7.25	9.57 $\pm$ 0.72
			BIOT-2D	14.51	9.57	21.76	8.33	4.32	5.40	8.18	18.21	9.57	11.09 $\pm$ 2.72
			BIOT-6D	16.21	8.80	19.14	11.26	10.80	5.56	6.17	24.08	9.26	12.36 $\pm$ 1.23
			LaBraM	35.96	16.05	45.37	15.43	24.23	20.37	34.42	34.11	37.19	29.24 $\pm$ 1.91
			Neuro-GPT	46.30	9.57	50.15	12.66	8.95	16.36	19.75	54.17	45.68	29.29 $\pm$ 0.94
			EEGPT	19.75	2.47	12.19	4.63	3.55	7.87	7.71	12.65	16.51	9.65 $\pm$ 1.93
			CBraMod	39.35	29.32	50.93	30.55	<u>24.38</u>	<u>26.54</u>	39.66	46.91	48.77	<u>37.38</u> $\pm$ 0.30
			TFM	10.49	7.72	11.88	9.88	4.01	2.78	6.48	16.67	14.35	9.36 $\pm$ 0.87
			BrainOmni-Tiny	24.23	13.74	33.18	10.65	17.44	16.51	22.38	30.86	29.94	22.10 $\pm$ 1.07
			BrainOmni-Base	23.77	10.03	32.56	9.57	17.13	17.59	23.77	29.48	27.31	21.24 $\pm$ 1.10
			EEGMamba	39.66	17.59	52.16	16.20	6.48	18.06	23.46	33.80	41.20	27.62 $\pm$ 0.72
			MIRepNet	<u>62.81</u>	20.37	71.30	28.40	17.75	21.76	30.25	59.72	33.95	38.48 $\pm$ 0.45
			SingLEM	12.35	1.85	6.94	7.87	11.42	6.94	7.72	6.33	5.40	7.42 $\pm$ 0.13
			LUNA-Base	9.26	1.54	15.59	4.01	0.00	4.78	0.00	4.48	6.64	5.14 $\pm$ 0.66
	Linear Probing	Foundation Models	BENDR	8.49	1.85	14.35	8.64	10.96	3.70	17.28	10.96	10.96	9.57 $\pm$ 0.54
			BIOT-1D	17.75	0.46	6.02	9.10	-0.15	6.17	0.00	14.81	2.31	6.28 $\pm$ 0.56
			BIOT-2D	14.81	2.31	10.03	7.72	-0.15	-0.15	2.78	13.27	15.43	7.34 $\pm$ 0.61
			BIOT-6D	13.27	9.88	7.56	5.86	0.00	8.64	-2.01	19.75	7.56	7.84 $\pm$ 1.34
			LaBraM	30.56	10.96	32.56	16.82	21.14	19.14	22.22	27.01	30.71	23.46 $\pm$ 0.37
			Neuro-GPT	39.51	17.90	50.46	<u>29.94</u>	10.49	11.11	19.75	52.16	47.53	30.98 $\pm$ 1.39
			EEGPT	28.40	7.72	18.36	12.50	9.10	12.65	14.81	23.30	21.60	16.50 $\pm$ 1.66
			CBraMod	32.41	8.49	40.90	12.81	5.71	5.56	10.34	42.13	39.04	21.93 $\pm$ 0.66
			TFM	10.49	3.55	10.96	4.94	2.47	-0.46	1.23	2.62	4.32	4.46 $\pm$ 0.40
			BrainOmni-Tiny	27.16	9.72	31.79	8.33	12.81	14.81	18.67	27.01	27.01	19.70 $\pm$ 0.48
			BrainOmni-Base	24.23	9.72	28.86	10.03	14.66	14.81	18.21	27.47	27.62	19.51 $\pm$ 0.77
			EEGMamba	19.14	7.41	21.30	7.10	4.17	13.89	15.43	7.10	16.36	12.43 $\pm$ 0.27
			MIRepNet	64.35	1.23	<u>66.82</u>	19.60	15.12	10.80	<u>45.99</u>	52.93	28.86	33.97 $\pm$ 0.37
			SingLEM	16.05	11.27	12.96	12.96	14.35	10.80	13.73	8.33	0.62	11.23 $\pm$ 0.30
			LUNA-Base	17.75	0.62	18.67	3.55	0.00	0.15	3.24	4.48	2.01	5.61 $\pm$ 0.76
Within-subject (Few-shot)		Specialist Models	CSP+LDA	71.24	39.87	69.28	28.76	13.07	22.88	69.93	67.32	45.10	47.49
			EEGNet	49.89	11.55	55.12	16.56	6.54	8.06	41.61	57.95	55.34	33.62 $\pm$ 1.51
			ShallowConv	48.15	29.63	54.25	27.89	8.06	16.99	36.60	57.73	55.12	37.16 $\pm$ 1.17
			LMDA	49.67	24.84	54.03	21.35	4.79	12.63	32.90	61.44	51.85	34.83 $\pm$ 1.01
			CNN-T	47.71	28.32	57.95	12.85	8.50	16.56	49.67	59.04	34.64	35.03 $\pm$ 1.47
			Deformer	35.29	15.90	43.79	9.80	0.00	2.62	13.07	47.93	40.74	22.95 $\pm$ 0.97
			Conformer	50.98	34.64	<u>69.50</u>	26.14	10.02	26.58	54.03	69.93	44.45	42.92 $\pm$ 1.75
			MSCFormer	50.98	29.41	65.80	19.61	17.21	20.70	52.51	60.13	45.97	40.26 $\pm$ 0.60
			FBCNet	50.76	32.46	58.82	28.54	12.20	17.43	47.28	67.76	51.20	40.72 $\pm$ 2.26
			TSception	62.31	27.23	62.09	26.14	23.53	22.44	59.26	<u>71.90</u>	49.24	44.90 $\pm$ 1.14
	Full Fine-tuning	Foundation Models	BENDR	15.68	18.52	31.59	17.86	41.17	23.53	28.76	27.67	33.99	26.53 $\pm$ 1.76
			BIOT-1D	43.36	21.57	35.95	8.71	18.74	7.63	30.50	43.36	37.04	27.43 $\pm$ 1.13
			BIOT-2D	39.43	17.21	25.71	8.71	10.02	13.07	28.98	38.78	40.74	24.74 $\pm$ 1.00
			BIOT-6D	49.24	25.06	42.05	6.75	15.69	12.42	46.62	45.75	39.65	31.47 $\pm$ 0.95
			LaBraM	24.62	9.37	16.99	6.75	12.20	7.19	13.51	25.27	29.63	16.17 $\pm$ 1.22
			Neuro-GPT	26.36	20.26	49.89	17.21	13.73	19.39	17.86	52.72	39.22	28.52 $\pm$ 2.55
			EEGPT	28.10	4.36	11.55	8.06	2.83	10.46	8.49	21.35	24.84	13.31 $\pm$ 0.34
			CBraMod	41.83	28.76	59.48	17.87	19.17	6.75	42.05	49.24	39.00	33.79 $\pm$ 1.58
			TFM	18.30	6.32	13.51	5.23	-4.36	5.66	13.51	19.39	22.00	11.06 $\pm$ 0.61
			BrainOmni-Tiny	26.80	15.69	24.62	18.74	8.71	5.45	16.56	24.18	27.23	18.67 $\pm$ 0.26
			BrainOmni-Base	29.63	11.98	19.83	14.60	6.75	4.79	16.99	22.22	24.40	16.80 $\pm$ 0.36
			EEGMamba	27.23	11.76	25.49	16.12	9.15	9.80	6.32	27.89	22.66	17.38 $\pm$ 0.59
			MIRepNet	<u>63.02</u>	<u>38.62</u>	76.89	33.61	<u>29.79</u>	<u>24.40</u>	73.61	72.94	47.15	51.11 $\pm$ 0.55
			SingLEM	11.76	1.53	4.58	3.05	1.53	6.75	9.37	2.83	5.88	5.25 $\pm$ 0.98
			LUNA-Base	29.19	0.44	19.17	-1.09	2.61	0.00	8.71	10.89	13.29	9.25 $\pm$ 1.66
	Linear Probing	Foundation Models	BENDR	9.15	7.63	11.55	11.11	6.75	1.74	12.64	3.49	13.07	8.57 $\pm$ 1.13
			BIOT-1D	41.39	21.79	32.03	11.98	27.89	11.76	35.51	39.00	40.52	29.10 $\pm$ 0.50
			BIOT-2D	38.34	19.39	22.66	10.24	13.94	6.97	24.18	42.48	37.04	23.92 $\pm$ 0.45
			BIOT-6D	41.61	21.79	37.91	16.34	21.35	18.08	37.25	47.28	33.77	30.60 $\pm$ 0.71
			LaBraM	19.39	5.66	19.17	5.01	15.25	6.10	13.94	16.99	23.09	13.85 $\pm$ 0.60
			Neuro-GPT	40.31	21.35	47.06	23.75	14.81	21.57	28.10	61.66	39.22	33.09 $\pm$ 2.07
			EEGPT	28.10	2.40	13.73	10.24	4.36	5.66	11.98	26.58	27.23	14.48 $\pm$ 0.72
			CBraMod	13.29	16.56	25.27	14.38	3.92	10.02	12.20	47.49	45.75	20.99 $\pm$ 1.42
			TFM	1.96	2.40	8.93	3.92	-2.83	1.53	-1.09	6.54	11.98	3.70 $\pm$ 1.39
			BrainOmni-Tiny	29.41	11.55	25.93	12.64	12.42	7.63	24.40	31.59	31.81	20.82 $\pm$ 0.42
			BrainOmni-Base	25.71	13.51	21.13	14.16	5.23	6.32	20.26	30.50	27.67	18.28 $\pm$ 0.48
			EEGMamba	14.60	5.45	23.09	8.06	5.23	7.41	5.66	16.78	19.83	11.79 $\pm$ 0.54
			MIRepNet	60.11	29.52	62.45	14.06	17.48	14.22	45.71	46.64	37.55	36.42 $\pm$ 2.25
			SingLEM	9.80	6.97	6.75	0.00	8.28	3.05	9.59	5.01	8.93	6.49 $\pm$ 0.24
			LUNA-Base	31.15	2.18	26.80	3.49	-1.53	1.74	17.21	17.43	18.30	12.98 $\pm$ 0.07

Table 18. Accuracies (%) on BNCI2014004. The best accuracies are marked in bold, and the second best by an underline.

Scenario	Tuning	Model Type	Approach	S0	S1	S2	S3	S4	S5	S6	S7	S8	Avg.
Cross-subject (LOSO)		Specialist Models	CSP+LDA	77.50	54.17	53.75	91.25	85.62	<u>83.12</u>	63.12	68.75	81.88	73.24
			EEGNet	73.33	59.72	55.42	94.58	82.92	83.96	73.54	76.67	87.29	76.38 ± 0.59
			ShallowConv	72.71	60.56	55.83	91.46	77.50	70.83	74.38	77.92	86.88	74.23 ± 0.39
			LMDA	75.42	54.17	52.71	96.46	81.67	82.08	73.12	73.96	84.38	74.88 ± 0.67
			CNN-T	66.04	53.06	51.04	<u>97.08</u>	78.12	76.88	68.12	70.83	79.58	71.20 ± 0.42
			Deformer	68.12	56.67	57.50	97.50	80.62	82.71	70.42	75.42	84.79	74.86 ± 0.82
			Conformer	73.54	51.11	54.17	97.50	87.92	76.25	72.29	74.17	80.62	74.17 ± 0.49
			MSCFormer	63.12	58.06	58.13	93.54	80.00	78.96	72.71	76.46	85.21	74.02 ± 0.70
			FBCNet	79.17	51.39	56.04	96.25	81.46	63.33	65.83	70.00	<u>88.75</u>	72.47 ± 0.74
			TSception	70.83	54.72	50.42	95.21	75.83	74.17	58.12	66.88	79.38	69.51 ± 0.99
	Full Fine-tuning	Foundation Models	BENDR	68.75	<u>68.06</u>	71.04	68.75	96.67	68.96	87.71	74.79	55.42	73.35 ± 0.06
			BIOT-1D	66.88	53.89	52.71	90.83	80.21	70.42	67.08	75.42	75.83	70.36 ± 1.82
			BIOT-2D	70.42	55.00	53.96	92.71	75.42	72.92	66.46	75.83	73.54	70.69 ± 0.98
			BIOT-6D	68.75	57.78	51.46	91.25	73.75	64.38	72.71	72.50	79.38	70.22 ± 1.32
			LaBraM	80.21	64.17	58.54	91.88	88.54	77.08	78.12	77.29	76.88	76.97 ± 1.08
			Neuro-GPT	73.12	66.39	59.38	87.92	95.42	68.75	<u>85.00</u>	80.42	82.92	77.70 ± 0.70
			EEGPT	62.92	66.11	61.88	72.71	90.00	64.17	83.12	77.50	63.96	71.37 ± 0.16
			CBraMod	73.33	64.44	54.17	92.71	91.25	81.25	76.25	72.08	73.54	75.45 ± 0.35
			TFM	54.17	51.11	60.83	63.33	72.71	58.12	63.12	62.71	55.00	60.12 ± 3.00
			BrainOmni-Tiny	68.75	58.89	62.50	63.75	92.29	63.12	81.25	76.04	64.58	70.13 ± 0.89
			BrainOmni-Base	63.75	61.39	58.54	65.62	90.21	62.08	79.58	76.88	65.62	69.30 ± 0.89
			EEGMamba	66.46	70.56	<u>70.21</u>	64.58	95.00	68.12	83.54	80.21	61.04	73.30 ± 0.57
			MIRepNet	77.50	66.94	60.21	96.04	85.62	79.79	69.79	<u>85.00</u>	84.79	<u>78.41</u> ± 0.52
			SingLEM	64.79	60.56	56.46	69.79	76.88	73.33	72.29	70.00	61.67	67.31 ± 0.18
			LUNA-Base	58.54	53.61	52.50	66.25	52.08	50.21	55.21	50.42	66.67	56.17 ± 3.14
	Linear Probing	Foundation Models	BENDR	55.62	55.00	58.96	56.25	71.25	57.71	69.38	67.92	52.08	60.46 ± 1.06
			BIOT-1D	70.21	51.11	52.71	94.58	64.17	66.88	58.12	74.17	78.54	67.83 ± 2.78
			BIOT-2D	63.96	50.56	49.58	90.62	57.71	62.50	56.67	73.33	57.92	62.54 ± 1.29
			BIOT-6D	63.96	51.94	49.38	80.00	63.12	65.62	52.71	61.88	63.54	61.35 ± 2.11
			LaBraM	66.25	47.78	52.71	83.54	71.04	60.21	62.71	70.00	71.25	65.05 ± 0.23
			Neuro-GPT	68.12	65.56	58.75	79.79	<u>96.46</u>	65.21	81.67	85.21	79.38	75.57 ± 1.26
			EEGPT	65.21	65.00	65.62	74.58	89.17	63.12	79.17	79.58	67.29	72.08 ± 2.07
			CBraMod	62.91	53.89	49.38	96.66	66.87	81.05	64.58	69.79	78.33	69.27 ± 0.55
			TFM	53.75	46.67	56.46	49.79	51.04	51.46	52.92	46.88	49.79	50.97 ± 1.13
			BrainOmni-Tiny	66.46	55.28	57.50	58.12	89.38	56.67	80.00	69.17	61.88	66.05 ± 0.53
			BrainOmni-Base	65.21	60.00	59.58	59.38	90.42	59.17	80.42	68.54	64.58	67.48 ± 0.25
			EEGMamba	59.17	61.39	55.62	55.42	76.25	63.54	67.71	63.54	54.79	61.94 ± 0.31
			MIRepNet	<u>78.75</u>	62.50	64.17	95.62	84.17	83.96	72.08	83.96	89.38	79.40 ± 0.06
			SingLEM	58.75	53.61	59.17	64.38	72.50	59.38	74.79	77.29	53.33	63.69 ± 0.36
			LUNA-Base	47.08	51.94	49.38	48.33	53.96	53.12	50.00	58.54	47.50	51.10 ± 0.65
Within-subject (Few-shot)		Specialist Models	CSP+LDA	83.93	58.33	51.79	100.00	87.50	88.39	67.86	87.50	87.50	79.20
			EEGNet	73.81	53.57	52.08	98.21	84.23	77.68	70.83	84.23	85.12	75.53 ± 3.67
			ShallowConv	74.11	56.35	44.35	98.51	78.57	78.87	64.58	82.74	92.56	74.52 ± 0.71
			LMDA	81.25	52.78	45.54	98.51	78.57	75.60	67.56	84.52	91.67	75.11 ± 1.63
			CNN-T	71.13	57.54	52.68	97.92	87.20	76.49	65.77	84.82	88.39	75.77 ± 0.38
			Deformer	63.39	48.81	47.92	97.92	84.52	71.73	70.83	86.01	86.01	73.02 ± 2.13
			Conformer	88.69	55.16	54.76	97.62	89.29	81.85	75.89	88.69	89.58	80.17 ± 0.25
			MSCFormer	88.69	48.02	63.10	97.62	86.61	80.65	77.08	82.14	90.48	79.38 ± 0.83
			FBCNet	75.60	48.41	55.36	97.92	83.33	76.49	62.80	82.44	<u>92.26</u>	74.96 ± 0.94
			TSception	<u>84.52</u>	57.14	57.14	99.11	92.56	<u>84.52</u>	69.64	87.80	90.77	<u>80.36</u> ± 0.65
	Full Fine-tuning	Foundation Models	BENDR	65.18	56.35	79.76	53.87	100.0	83.63	91.67	63.69	51.19	71.70 ± 1.46
			BIOT-1D	58.33	48.81	47.32	70.24	64.88	54.17	60.42	71.13	64.58	59.99 ± 0.87
			BIOT-2D	59.23	53.17	52.68	75.00	62.50	55.65	51.79	69.05	69.64	60.97 ± 0.72
			BIOT-6D	69.35	49.21	51.49	90.48	88.10	63.69	59.82	73.81	69.94	68.43 ± 0.42
			LaBraM	61.61	48.41	58.04	96.43	97.02	50.30	74.11	83.93	58.04	69.76 ± 1.23
			Neuro-GPT	66.07	57.94	61.61	78.57	<u>98.81</u>	76.49	79.46	86.01	72.32	75.25 ± 1.58
			EEGPT	58.63	55.95	55.36	57.14	83.04	61.01	63.69	62.20	61.61	62.07 ± 0.74
			CBraMod	66.37	50.40	54.17	98.21	<u>98.81</u>	82.14	78.27	82.74	85.42	77.39 ± 0.35
			TFM	54.17	51.98	55.36	53.87	92.56	45.54	55.36	62.80	58.63	58.92 ± 2.41
			BrainOmni-Tiny	55.36	56.35	49.40	65.48	92.86	63.10	65.18	66.96	55.06	63.30 ± 0.98
			BrainOmni-Base	50.30	54.76	47.02	60.71	92.86	63.39	77.38	59.52	50.60	61.84 ± 0.76
			EEGMamba	56.55	56.75	59.82	62.50	92.26	65.77	74.70	69.64	47.62	65.07 ± 1.86
			MIRepNet	<u>84.52</u>	57.54	60.42	98.81	88.99	82.44	81.25	86.90	88.99	81.10 ± 1.37
			SingLEM	57.44	52.38	52.38	61.31	53.87	55.06	61.90	63.10	53.27	56.75 ± 1.66
			LUNA-Base	51.79	51.98	51.79	50.30	75.00	49.70	52.38	68.75	56.85	56.50 ± 0.74
	Linear Probing	Foundation Models	BENDR	52.68	51.59	51.79	50.00	68.75	53.27	61.90	58.04	49.40	55.27 ± 2.40
			BIOT-1D	67.26	50.00	45.24	80.36	78.87	57.44	57.14	71.43	64.58	63.59 ± 0.80
			BIOT-2D	52.98	53.97	49.70	81.85	68.45	55.06	53.27	72.02	66.96	61.59 ± 2.40
			BIOT-6D	68.15	50.79	52.08	83.63	82.74	61.61	52.08	77.68	75.60	67.15 ± 1.64
			LaBraM	56.85	49.21	56.25	77.08	65.48	47.92	59.82	73.51	52.38	59.83 ± 1.22
			Neuro-GPT	70.24	65.48	<u>65.18</u>	75.00	97.32	84.23	<u>81.55</u>	81.25	69.94	76.69 ± 1.35
			EEGPT	59.23	<u>61.11</u>	58.93	68.45	88.39	65.18	61.61	67.56	62.50	65.88 ± 2.18
			CBraMod	56.55	52.78	53.27	89.88	95.54	65.18	56.85	83.93	79.46	70.38 ± 0.90
			TFM	52.38	55.16	53.57	49.70	62.20	42.86	51.49	51.19	43.15	51.30 ± 1.08
			BrainOmni-Tiny	56.85	58.33	54.46	52.08	80.36	59.23	61.31	65.77	49.70	59.79 ± 0.60
			BrainOmni-Base	52.98	55.16	46.43	61.01	79.46	61.01	66.96	59.52	49.40	59.10 ± 0.30
			EEGMamba	50.60	57.94	51.79	57.74	83.33	60.12	58.63	66.96	44.05	59.02 ± 0.34
			MIRepNet	73.21	58.33	63.99	96.43	85.42	70.83	67.26	<u>88.39</u>	87.50	76.82 ± 1.41
			SingLEM	58.33	46.43	56.55	54.46	57.74	54.76	65.77	59.52	55.06	56.51 ± 1.27
			LUNA-Base	50.60	50.79	51.79	49.70	73.21	44.94	51.49	70.24	57.74	55.61 ± 0.89

Table 19. Cohen's Kappa (%) on BNCI2014004. The best metrics are marked in bold, and the second best by an underline.

Scenario	Tuning	Model Type	Approach	S0	S1	S2	S3	S4	S5	S6	S7	S8	Avg.
Cross-subject (LOSO)	Specialist Models	Specialist Models	CSP+LDA	55.00	8.33	7.50	82.50	71.25	<u>66.25</u>	26.25	37.50	63.75	46.48
			EEGNet	46.67	19.44	10.83	89.17	65.83	67.92	47.08	53.33	74.58	52.76 $\pm$ 1.17
			ShallowConv	45.42	21.11	11.67	82.92	55.00	41.67	48.75	55.83	73.75	48.46 $\pm$ 0.78
			LMDA	50.83	8.33	5.42	92.92	63.33	64.17	46.25	47.92	68.75	49.77 $\pm$ 1.34
			CNN-T	32.08	6.11	2.08	<u>94.17</u>	56.25	53.75	36.25	41.67	59.17	42.39 $\pm$ 0.84
			Deformer	36.25	13.33	15.00	95.00	61.25	65.42	40.83	50.83	69.58	49.72 $\pm$ 1.64
			Conformer	47.08	2.22	8.33	95.00	75.83	52.50	44.58	48.33	61.25	48.35 $\pm$ 0.97
			MSCFormer	26.25	16.11	16.25	87.08	60.00	57.92	45.42	52.92	70.42	48.04 $\pm$ 1.39
			FBCNet	58.33	2.78	12.08	92.50	62.92	26.67	31.67	40.00	<u>77.50</u>	44.94 $\pm$ 1.48
			TSception	41.67	9.44	0.83	90.42	51.67	48.33	16.25	33.75	58.75	39.01 $\pm$ 1.99
	Full Fine-tuning	Foundation Models	BENDR	37.50	<u>36.11</u>	42.08	37.50	93.33	37.92	75.42	49.58	10.83	46.70 $\pm$ 0.12
			BIOT-1D	33.75	7.78	5.42	81.67	60.42	40.83	34.17	50.83	51.67	40.73 $\pm$ 0.64
			BIOT-2D	40.83	10.00	7.92	85.42	50.83	45.83	32.92	51.67	47.08	41.39 $\pm$ 1.96
			BIOT-6D	37.50	15.56	3.33	82.50	47.50	28.75	45.42	45.00	58.75	40.43 $\pm$ 2.63
			LaBraM	60.42	28.33	17.08	83.75	77.08	54.17	56.25	54.58	53.75	53.94 $\pm$ 2.16
			Neuro-GPT	46.25	32.78	18.75	75.83	90.83	37.50	<u>70.00</u>	60.83	65.83	55.40 $\pm$ 1.40
			EEGPT	25.83	32.22	23.75	45.42	80.00	28.33	66.25	55.00	27.92	42.75 $\pm$ 0.32
			CBraMod	46.67	28.89	8.33	85.42	82.50	62.50	52.50	44.17	47.08	50.89 $\pm$ 0.70
			TFM	8.33	2.22	21.67	26.67	45.42	16.25	26.25	25.42	10.00	20.25 $\pm$ 6.00
			BrainOmni-Tiny	37.50	17.78	25.00	27.50	84.58	26.25	62.50	52.08	29.17	40.26 $\pm$ 1.78
			BrainOmni-Base	27.50	22.78	17.08	31.25	80.42	24.17	59.17	53.75	31.25	38.60 $\pm$ 1.79
			EEGMamba	32.92	41.11	<u>40.42</u>	29.17	90.00	36.25	67.08	60.42	22.08	46.60 $\pm$ 1.13
			MIRepNet	55.00	33.89	20.42	92.08	71.25	59.58	39.58	<u>70.00</u>	69.58	<u>56.82</u> $\pm$ 1.04
			SingLEM	29.58	21.11	12.92	39.58	53.75	46.67	44.58	40.00	23.33	34.61 $\pm$ 0.36
			LUNA-Base	17.08	7.22	5.00	32.50	4.17	0.42	10.42	0.83	33.33	12.33 $\pm$ 6.28
	Linear Probing	Foundation Models	BENDR	11.25	10.00	17.92	12.50	42.50	15.42	38.75	35.83	4.17	20.93 $\pm$ 2.12
			BIOT-1D	40.42	2.22	5.42	89.17	28.33	33.75	16.25	48.33	57.08	35.66 $\pm$ 5.57
			BIOT-2D	27.92	1.11	-0.83	81.25	15.42	25.00	13.33	46.67	15.83	25.08 $\pm$ 2.58
			BIOT-6D	27.92	3.89	-1.25	60.00	26.25	31.25	5.42	23.75	27.08	22.70 $\pm$ 4.23
			LaBraM	32.50	-4.44	5.42	67.08	42.08	20.42	25.42	40.00	42.50	30.11 $\pm$ 0.46
			Neuro-GPT	36.25	31.11	17.50	59.58	<u>92.92</u>	30.42	63.33	70.42	58.75	51.14 $\pm$ 2.52
			EEGPT	30.42	30.00	31.25	49.17	78.33	26.25	58.33	59.17	34.58	44.17 $\pm$ 4.14
			CBraMod	25.83	7.78	-1.25	93.33	33.75	62.08	29.17	39.58	56.67	38.55 $\pm$ 1.11
			TFM	7.50	-6.67	12.92	-0.42	2.08	2.92	5.83	-6.25	-0.42	1.94 $\pm$ 2.27
			BrainOmni-Tiny	32.92	10.56	15.00	16.25	78.75	13.33	60.00	38.33	23.75	32.10 $\pm$ 1.06
			BrainOmni-Base	30.42	20.00	19.17	18.75	80.83	18.33	60.83	37.08	29.17	34.95 $\pm$ 0.51
			EEGMamba	18.33	22.78	11.25	10.83	52.50	27.08	35.42	27.08	9.58	23.87 $\pm$ 0.62
			MIRepNet	<u>57.50</u>	25.00	28.33	91.25	68.33	67.92	44.17	67.92	78.75	58.80 $\pm$ 0.12
			SingLEM	17.50	7.22	18.33	28.75	45.00	18.75	49.58	54.58	6.67	27.38 $\pm$ 0.72
			LUNA-Base	-5.83	3.89	-1.25	-3.33	7.92	6.25	0.00	17.08	-5.00	2.19 $\pm$ 1.30
Within-subject (Few-shot)	Specialist Models	Specialist Models	CSP+LDA	67.86	16.67	3.57	100.00	75.00	76.79	35.71	75.00	75.00	58.40
			EEGNet	47.62	7.14	4.17	96.43	68.45	55.36	41.67	68.45	70.24	51.06 $\pm$ 7.35
			ShallowConv	48.21	12.70	0.00	97.02	57.14	57.74	29.16	65.48	85.12	49.03 $\pm$ 1.42
			LMDA	62.50	7.14	0.60	97.02	57.14	51.19	35.12	69.05	83.33	50.22 $\pm$ 3.27
			CNN-T	42.26	15.08	5.36	95.83	74.41	52.97	31.55	69.65	76.78	51.55 $\pm$ 0.76
			Deformer	26.79	0.00	0.00	95.83	69.05	43.45	41.66	72.02	72.02	46.03 $\pm$ 4.26
			Conformer	77.38	10.31	9.53	95.24	78.57	63.69	51.78	77.38	79.17	60.34 $\pm$ 0.50
			MSCFormer	77.38	-3.97	26.19	95.24	73.21	61.31	54.17	64.29	80.95	58.75 $\pm$ 1.65
			FBCNet	51.19	-3.17	10.71	95.83	66.67	52.98	25.60	64.88	<u>84.52</u>	49.91 $\pm$ 1.89
			TSception	69.05	14.29	14.29	<u>98.21</u>	85.12	<u>69.05</u>	39.29	75.60	81.55	60.71 $\pm$ 1.30
	Full Fine-tuning	Foundation Models	BENDR	30.36	12.70	59.52	8.93	100.00	67.26	83.33	27.38	2.98	43.41 $\pm$ 2.92
			BIOT-1D	16.67	-2.38	-5.36	40.48	29.76	8.33	20.83	42.26	29.17	19.97 $\pm$ 1.75
			BIOT-2D	18.45	6.35	5.36	50.00	25.00	11.31	3.57	38.10	39.29	21.94 $\pm$ 1.45
			BIOT-6D	38.69	0.00	4.76	80.95	76.19	27.38	19.64	47.62	39.88	36.86 $\pm$ 0.84
			LaBraM	23.21	2.38	16.07	92.86	94.05	3.57	48.21	67.86	16.07	39.53 $\pm$ 2.45
			Neuro-GPT	32.14	15.87	23.22	57.14	<u>97.62</u>	52.97	58.93	72.02	44.64	50.51 $\pm$ 3.15
			EEGPT	17.26	11.90	10.71	14.29	66.07	22.03	27.38	24.41	23.21	24.14 $\pm$ 1.49
			CBraMod	32.74	3.97	8.34	96.43	<u>97.62</u>	64.28	56.55	65.48	70.83	54.78 $\pm$ 0.70
			TFM	8.33	3.97	10.71	7.74	85.12	-8.93	10.71	25.60	17.26	17.84 $\pm$ 4.81
			BrainOmni-Tiny	10.71	12.70	1.19	30.95	85.71	26.19	30.36	33.93	11.31	26.61 $\pm$ 1.97
			BrainOmni-Base	0.60	9.52	-5.95	21.43	85.71	26.79	54.76	19.05	1.19	23.68 $\pm$ 1.52
			EEGMamba	13.10	13.49	19.64	25.00	84.52	31.55	49.40	39.29	-4.76	30.14 $\pm$ 3.73
			MIRepNet	<u>69.23</u>	17.84	20.47	97.62	77.96	65.07	62.66	73.89	78.01	62.53 $\pm$ 2.66
			SingLEM	14.88	4.76	4.76	22.62	7.74	10.12	23.81	26.19	6.55	13.49 $\pm$ 3.33
			LUNA-Base	3.57	3.97	3.57	0.60	50.00	-0.60	4.76	37.50	13.69	13.01 $\pm$ 1.47
	Linear Probing	Foundation Models	BENDR	5.36	3.17	3.57	0.00	37.50	6.55	23.81	16.07	-1.19	10.54 $\pm$ 4.79
			BIOT-1D	34.52	0.00	-9.52	60.71	57.74	14.88	14.29	42.86	29.17	27.18 $\pm$ 1.59
			BIOT-2D	5.95	7.94	-0.60	63.69	36.90	10.12	6.55	44.05	33.93	23.17 $\pm$ 4.80
			BIOT-6D	36.31	1.59	4.17	67.26	65.48	23.21	4.17	55.36	51.19	34.30 $\pm$ 3.28
			LaBraM	13.69	-1.59	12.50	54.17	30.95	-4.17	19.64	47.02	4.76	19.66 $\pm$ 2.43
			Neuro-GPT	40.48	30.95	<u>30.36</u>	50.00	94.64	68.45	<u>63.10</u>	62.50	39.88	53.37 $\pm$ 2.71
			EEGPT	18.45	<u>22.22</u>	17.86	36.90	76.79	30.36	23.21	35.12	25.00	31.77 $\pm$ 4.35
			CBraMod	13.10	5.56	6.55	79.76	91.07	30.36	13.69	67.86	58.93	40.76 $\pm$ 1.80
			TFM	4.76	10.32	7.14	-0.60	24.40	-14.29	2.98	2.38	-13.69	2.60 $\pm$ 2.16
			BrainOmni-Tiny	13.69	16.67	8.93	4.17	60.71	18.45	22.62	31.55	-0.60	19.58 $\pm$ 1.20
			BrainOmni-Base	5.95	10.32	-7.14	22.02	58.93	22.02	33.93	19.05	-1.19	18.21 $\pm$ 0.59
			EEGMamba	1.19	15.87	3.57	15.48	66.67	20.24	17.26	33.93	-11.90	18.03 $\pm$ 0.69
			MIRepNet	47.17	18.15	27.99	92.85	71.02	43.23	35.41	<u>76.75</u>	75.06	54.18 $\pm$ 2.61
			SingLEM	16.67	-7.14	13.10	8.93	15.48	9.52	31.55	19.05	10.12	13.03 $\pm$ 2.53
			LUNA-Base	1.19	1.59	3.57	-0.60	46.43	-10.12	2.98	40.48	15.48	11.22 $\pm$ 1.77

Table 20. Accuracies (%) on BNCI2015001. The best accuracies are marked in bold, and the second best by an underline.

Scenario	Tuning	Model Type	Approach	S0	S1	S2	S3	S4	S5	S6	S7	S8	S9	S10	S11	Avg.				
Cross-subject (LOSO)		Specialist Models	CSP+LDA	58.00	55.50	51.50	82.00	64.00	58.00	49.00	50.00	57.00	50.00	50.00	52.00	56.42				
			EEGNet	91.50	96.50	72.00	72.00	60.17	54.67	61.50	51.33	49.67	49.67	50.50	51.33	63.40 _{±1.32}				
			ShallowConv	94.33	82.00	62.67	70.17	61.00	63.00	68.83	55.83	52.00	55.50	51.50	49.83	63.89 _{±0.80}				
			LMDA	65.67	89.50	56.00	73.67	55.33	63.17	66.50	57.83	51.00	57.67	50.83	49.67	61.40 _{±0.94}				
			CNN-T	53.00	87.33	62.67	69.67	69.00	58.50	54.83	54.00	53.67	51.00	51.17	50.83	59.64 _{±1.41}				
			Deformer	82.67	78.17	68.17	67.50	57.33	63.67	<u>72.33</u>	55.83	54.33	56.17	50.00	50.50	63.06 _{±1.17}				
			Conformer	52.83	64.83	52.00	67.50	73.17	57.50	65.00	53.33	56.83	55.50	59.00	51.67	59.10 _{±2.14}				
			MSCFormer	53.33	63.17	53.67	65.17	72.50	58.17	56.67	53.33	51.83	54.33	50.17	49.83	56.85 _{±0.63}				
			FBCNet	62.67	61.33	66.67	59.67	<u>75.67</u>	58.33	56.83	51.00	51.67	50.33	57.50	49.33	58.42 _{±3.43}				
			TSception	57.83	82.33	72.83	62.83	52.83	53.33	50.67	51.00	54.50	50.67	51.50	51.50	57.65 _{±1.51}				
		Full Fine-tuning	Foundation Models	BENDR	63.83	63.50	68.33	64.50	58.83	<u>68.67</u>	63.50	60.50	<u>63.67</u>	66.17	55.33	55.33	62.68 _{±0.49}			
				BIOT-1D	74.33	93.50	65.17	78.33	66.50	59.17	61.17	53.00	55.67	57.83	50.17	51.83	63.89 _{±1.03}			
				BIOT-2D	86.17	74.50	70.33	69.50	70.83	55.50	57.67	53.50	57.67	57.00	50.00	53.83	63.04 _{±2.30}			
				BIOT-6D	83.33	85.50	<u>76.50</u>	78.67	58.00	57.50	61.00	52.50	54.83	56.17	49.33	54.00	63.94 _{±1.20}			
				LaBraM	66.83	75.00	64.17	70.33	74.67	68.17	58.00	57.83	58.33	66.33	53.33	<u>56.67</u>	64.14 _{±1.03}			
				Neuro-GPT	74.00	71.83	56.33	74.33	59.00	61.17	65.50	53.83	54.67	55.33	51.17	50.33	60.62 _{±1.63}			
				EEGPT	67.67	72.67	63.83	69.33	57.33	57.00	59.33	55.67	52.17	56.5	52.00	55.00	59.88 _{±1.39}			
				CBraMod	54.83	78.17	62.67	69.50	75.17	65.17	60.83	<u>61.83</u>	59.00	66.83	51.50	56.17	63.47 _{±0.36}			
				TFM	63.83	64.33	60.33	59.67	51.83	53.33	55.00	49.50	52.00	54.33	50.50	49.50	55.35 _{±1.46}			
				BrainOmni-Tiny	63.00	68.17	65.50	66.50	67.00	67.50	58.00	55.50	57.00	61.83	55.83	<u>56.67</u>	61.88 _{±0.30}			
				BrainOmni-Base	67.17	61.17	65.33	66.17	64.33	58.83	56.17	56.83	55.33	61.50	<u>57.83</u>	57.00	60.64 _{±0.39}			
				EEGMamba	58.17	81.50	59.50	75.67	71.50	56.33	62.83	55.83	51.50	60.33	50.00	55.17	61.53 _{±0.50}			
				MIRepNet	70.50	<u>95.33</u>	61.17	83.33	84.50	69.83	73.33	58.17	64.17	<u>67.67</u>	55.00	51.83	<u>69.57</u> _{±0.54}			
				SingLEM	51.17	52.33	56.67	51.17	53.83	63.67	54.50	51.33	59.17	52.83	55.50	51.50	54.47 _{±0.62}			
				LUNA-Base	57.17	65.33	58.50	59.50	50.00	55.50	63.33	50.00	49.67	58.83	50.33	50.33	55.71 _{±1.37}			
							BENDR	53.83	53.67	57.33	55.00	51.83	55.00	57.17	47.33	53.50	53.17	56.00	52.33	53.85 _{±1.11}
							BIOT-1D	88.67	83.50	55.00	85.17	51.83	60.00	59.33	52.00	52.50	61.67	50.00	53.67	62.78 _{±0.44}
							BIOT-2D	<u>96.50</u>	84.83	73.00	79.33	51.00	59.50	62.67	51.50	54.33	56.67	50.17	55.67	64.60 _{±0.48}
							BIOT-6D	88.17	86.83	66.83	81.33	53.67	59.33	55.67	51.17	59.83	53.00	51.00	54.33	63.43 _{±0.63}
							LaBraM	68.67	66.33	64.17	66.83	64.33	64.17	62.67	60.83	58.83	59.00	50.83	57.00	61.97 _{±0.17}
	Neuro-GPT	81.00	74.67				66.67	64.83	63.50	56.00	59.00	<u>61.83</u>	53.67	54.33	50.33	49.00	61.24 _{±0.50}			
	EEGPT	80.00	77.50				68.83	73.67	61.33	59.17	60.67	55.83	57.17	58.33	53.67	49.83	63.00 _{±2.89}			
	CBraMod	58.00	84.67				65.67	63.17	50.17	64.67	61.67	49.00	51.67	64.00	51.83	54.67	59.93 _{±0.29}			
	TFM	55.50	66.67				57.17	60.33	50.33	50.50	53.50	49.50	49.83	50.67	45.50	51.67	53.43 _{±0.53}			
	BrainOmni-Tiny	61.67	66.83				62.00	70.50	66.00	65.50	60.50	57.83	57.33	61.00	54.67	54.67	61.54 _{±0.54}			
				BrainOmni-Base	61.17	64.33	63.17	66.00	68.83	62.67	59.17	52.83	56.17	61.67	52.50	56.00	60.38 _{±0.19}			
EEGMamba				51.50	60.67	61.00	59.33	51.00	60.50	59.67	57.50	58.50	54.67	51.33	50.83	56.38 _{±0.27}				
MIRepNet				98.83	<u>95.33</u>	77.00	<u>83.83</u>	<u>75.67</u>	66.67	66.00	66.67	59.67	68.50	57.17	52.33	<u>72.31</u> _{±0.64}				
SingLEM				46.83	49.83	55.50	53.50	53.83	61.83	54.83	55.00	56.17	55.17	53.17	53.33	54.08 _{±0.20}				
LUNA-Base				50.00	79.67	52.83	72.67	50.00	58.83	51.67	49.83	52.50	61.67	49.67	51.33	56.72 _{±0.41}				
						BENDR	53.83	53.67	57.33	55.00	51.83	55.00	57.17	47.33	53.50	53.17	56.00	52.33	53.85 _{±1.11}	
						BIOT-1D	88.67	83.50	55.00	85.17	51.83	60.00	59.33	52.00	52.50	61.67	50.00	53.67	62.78 _{±0.44}	
						BIOT-2D	<u>96.50</u>	84.83	73.00	79.33	51.00	59.50	62.67	51.50	54.33	56.67	50.17	55.67	64.60 _{±0.48}	
						BIOT-6D	88.17	86.83	66.83	81.33	53.67	59.33	55.67	51.17	59.83	53.00	51.00	54.33	63.43 _{±0.63}	
						LaBraM	68.67	66.33	64.17	66.83	64.33	64.17	62.67	60.83	58.83	59.00	50.83	57.00	61.97 _{±0.17}	
	Neuro-GPT	81.00	74.67			66.67	64.83	63.50	56.00	59.00	<u>61.83</u>	53.67	54.33	50.33	49.00	61.24 _{±0.50}				
	EEGPT	80.00	77.50			68.83	73.67	61.33	59.17	60.67	55.83	57.17	58.33	53.67	49.83	63.00 _{±2.89}				
	CBraMod	58.00	84.67			65.67	63.17	50.17	64.67	61.67	49.00	51.67	64.00	51.83	54.67	59.93 _{±0.29}				
	TFM	55.50	66.67			57.17	60.33	50.33	50.50	53.50	49.50	49.83	50.67	45.50	51.67	53.43 _{±0.53}				
	BrainOmni-Tiny	61.67	66.83			62.00	70.50	66.00	65.50	60.50	57.83	57.33	61.00	54.67	54.67	61.54 _{±0.54}				
			BrainOmni-Base	61.17	64.33	63.17	66.00	68.83	62.67	59.17	52.83	56.17	61.67	52.50	56.00	60.38 _{±0.19}				
			EEGMamba	51.50	60.67	61.00	59.33	51.00	60.50	59.67	57.50	58.50	54.67	51.33	50.83	56.38 _{±0.27}				
			MIRepNet	98.83	<u>95.33</u>	77.00	<u>83.83</u>	<u>75.67</u>	66.67	66.00	66.67	59.67	68.50	57.17	52.33	<u>72.31</u> _{±0.64}				
			SingLEM	46.83	49.83	55.50	53.50	53.83	61.83	54.83	55.00	56.17	55.17	53.17	53.33	54.08 _{±0.20}				
			LUNA-Base	50.00	79.67	52.83	72.67	50.00	58.83	51.67	49.83	52.50	61.67	49.67	51.33	56.72 _{±0.41}				
						CSP+LDA	97.14	94.29	89.29	<u>86.43</u>	80.00	59.29	84.29	54.29	59.29	65.00	73.57	67.86	75.89	
						EEGNet	50.00	95.71	91.67	87.14	83.81	72.86	85.00	47.86	55.24	59.52	85.95	50.24	72.08 _{±0.39}	
						ShallowConv	70.71	94.05	83.81	79.52	80.95	<u>72.62</u>	76.19	64.76	50.48	64.52	91.43	56.43	73.79 _{±0.66}	
						LMDA	50.00	<u>97.38</u>	87.38	82.38	88.10	67.38	81.90	<u>64.05</u>	55.24	65.95	91.90	53.81	73.79 _{±1.50}	
						CNN-T	62.62	93.10	87.62	75.00	82.86	69.29	72.38	57.14	<u>64.05</u>	64.29	76.67	54.52	71.63 _{±1.36}	
Deformer	50.00	94.05				91.67	79.76	80.00	65.00	81.90	49.52	58.33	<u>68.33</u>	70.95	54.29	70.32 _{±1.28}				
Conformer	75.48	<u>97.38</u>				92.62	83.33	<u>89.76</u>	69.05	84.76	55.95	58.57	64.52	<u>92.38</u>	61.43	<u>77.10</u> _{±1.49}				
MSCFormer	79.05	95.00				90.95	85.95	86.43	68.10	<u>85.95</u>	59.76	57.14	60.00	93.57	61.19	76.92 _{±2.03}				
FBCNet	95.24	95.00				86.19	82.14	85.00	66.90	81.19	54.05	51.67	63.81	76.90	<u>63.81</u>	75.16 _{±0.85}				
TSception	71.19	97.86				90.00	85.24	87.38	70.95	77.86	55.95	59.29	63.10	75.00	58.33	74.35 _{±1.12}				
Within-subject (Few-shot)		Specialist Models	CSP+LDA	97.14	94.29	89.29	<u>86.43</u>	80.00	59.29	84.29	54.29	59.29	65.00	73.57	67.86	75.89				
			EEGNet	50.00	95.71	91.67	87.14	83.81	72.86	85.00	47.86	55.24	59.52	85.95	50.24	72.08 _{±0.39}				
			ShallowConv	70.71	94.05	83.81	79.52	80.95	<u>72.62</u>	76.19	64.76	50.48	64.52	91.43	56.43	73.79 _{±0.66}				
			LMDA	50.00	<u>97.38</u>	87.38	82.38	88.10	67.38	81.90	<u>64.05</u>	55.24	65.95	91.90	53.81	73.79 _{±1.50}				
			CNN-T	62.62	93.10	87.62	75.00	82.86	69.29	72.38	57.14	<u>64.05</u>	64.29	76.67	54.52	71.63 _{±1.36}				
			Deformer	50.00	94.05	91.67	79.76	80.00	65.00	81.90	49.52	58.33	<u>68.33</u>	70.95	54.29	70.32 _{±1.28}				
			Conformer	75.48	<u>97.38</u>	92.62	83.33	<u>89.76</u>	69.05	84.76	55.95	58.57	64.52	<u>92.38</u>	61.43	<u>77.10</u> _{±1.49}				
			MSCFormer	79.05	95.00	90.95	85.95	86.43	68.10	<u>85.95</u>	59.76	57.14	60.00	93.57	61.19	76.92 _{±2.03}				
			FBCNet	95.24	95.00	86.19	82.14	85.00	66.90	81.19	54.05	51.67	63.81	76.90	<u>63.81</u>	75.16 _{±0.85}				
			TSception	71.19	97.86	90.00	85.24	87.38	70.95	77.86	55.95	59.29	63.10	75.00	58.33					

Table 21. Cohen's Kappa (%) on BNCI2015001. The best metrics are marked in bold, and the second best by an underline.

Scenario	Tuning	Model Type	Approach	S0	S1	S2	S3	S4	S5	S6	S7	S8	S9	S10	S11	Avg.
Cross-subject (LOSO)	Specialist Models	Specialist Models	CSP+LDA	16.00	11.00	3.00	64.00	28.00	16.00	-2.00	0.00	14.00	0.00	0.00	4.00	12.83
			EEGNet	83.00	93.00	44.00	44.00	20.33	9.33	23.00	2.67	0.00	0.33	1.00	2.67	26.80 _{±2.63}
			ShallowConv	88.67	64.00	25.33	40.33	22.00	26.00	37.67	11.67	4.00	11.00	3.00	0.67	27.78 _{±1.59}
			LMDA	31.33	79.00	12.00	47.33	10.67	26.33	33.00	15.67	2.00	15.67	1.67	1.00	22.81 _{±1.88}
			CNN-T	6.00	74.67	25.33	39.33	38.00	17.00	9.67	8.00	7.33	2.33	2.33	2.00	19.28 _{±2.82}
			Deformer	65.33	56.33	36.33	35.00	14.67	27.33	<u>44.67</u>	11.67	8.67	12.67	0.00	3.00	26.11 _{±2.35}
			Conformer	5.67	29.67	4.00	35.00	46.33	15.00	30.00	6.67	13.67	11.33	18.00	3.33	18.19 _{±4.28}
			MSCFormer	6.67	26.33	7.33	30.33	45.00	16.33	13.33	6.67	3.67	8.67	0.33	-0.33	13.69 _{±1.26}
			FBCNet	25.33	22.67	33.33	19.33	<u>51.33</u>	16.67	13.67	2.00	3.33	0.67	15.00	-1.33	16.83 _{±6.85}
			TSception	15.67	64.67	45.67	25.67	5.67	6.67	1.33	2.00	9.00	1.33	3.00	3.00	15.31 _{±3.02}
	Full Fine-tuning	Foundation Models	BENDR	27.67	27.00	36.67	29.00	17.67	<u>37.33</u>	27.00	21.00	<u>27.33</u>	32.33	10.67	10.67	25.36 _{±0.98}
			BIOT-1D	48.67	87.00	30.33	56.67	33.00	18.33	22.33	6.00	11.33	15.67	0.33	3.67	27.78 _{±2.07}
			BIOT-2D	72.33	49.00	40.67	39.00	41.67	11.00	15.33	7.00	15.33	14.00	0.00	7.67	26.08 _{±4.60}
			BIOT-6D	66.67	71.00	<u>53.00</u>	57.33	16.00	16.33	22.00	5.33	9.67	12.33	0.00	8.00	27.89 _{±2.40}
			LaBraM	33.67	50.00	28.33	40.67	49.33	36.33	16.00	15.67	16.67	32.67	6.67	13.33	28.28 _{±2.05}
			Neuro-GPT	55.00	52.00	17.67	49.00	23.67	24.33	28.00	11.00	9.00	6.00	2.33	5.67	23.64 _{±2.90}
			EEGPT	35.33	45.33	27.67	38.67	14.67	14.00	18.67	11.33	4.33	13.00	5.00	10.00	19.75 _{±2.77}
			CBraMod	9.67	56.33	25.33	39.00	50.33	30.33	21.67	<u>23.67</u>	18.00	33.67	3.00	12.33	26.95 _{±0.73}
			TFM	27.67	28.67	20.67	19.33	3.67	6.67	10.00	-1.00	4.00	8.67	1.00	-1.00	10.69 _{±2.93}
			BrainOmni-Tiny	26.00	36.33	31.00	33.00	34.00	35.00	16.00	11.00	14.00	23.67	11.67	<u>13.33</u>	23.75 _{±0.60}
			BrainOmni-Base	34.33	22.33	30.67	32.33	28.67	17.67	12.33	13.67	10.67	23.00	15.67	14.00	21.28 _{±0.79}
			EEGMamba	16.33	63.00	19.00	51.33	43.00	12.67	25.67	11.67	3.00	20.67	0.00	10.33	23.06 _{±0.99}
			MIRepNet	41.00	<u>90.67</u>	22.33	66.67	69.00	39.67	46.67	16.33	28.33	<u>35.33</u>	10.00	3.67	<u>39.14</u> _{±1.08}
			SingLEM	2.33	4.67	13.33	2.33	7.67	27.33	9.00	2.67	18.33	5.67	11.00	3.00	8.94 _{±1.24}
			LUNA-Base	14.33	30.67	17.00	19.00	0.00	11.00	26.67	0.00	-0.67	17.67	0.67	0.67	11.42 _{±2.74}
	Linear Probing	Foundation Models	BENDR	7.67	7.33	14.67	10.00	3.67	10.00	14.33	-5.33	7.00	6.33	12.00	4.67	7.69 _{±2.23}
			BIOT-1D	77.33	67.00	10.00	70.33	3.67	20.00	18.67	4.00	5.00	23.33	0.00	7.33	25.56 _{±0.89}
			BIOT-2D	<u>93.00</u>	69.67	46.00	58.67	2.00	19.00	25.33	3.00	8.67	13.33	0.33	11.33	29.19 _{±0.97}
			BIOT-6D	76.33	73.67	33.67	62.67	7.33	18.67	11.33	2.33	19.67	6.00	2.00	8.67	26.86 _{±1.27}
			LaBraM	37.33	32.67	28.33	33.67	28.67	28.33	25.33	21.67	17.67	18.00	1.67	14.00	23.94 _{±0.34}
			Neuro-GPT	62.00	49.33	33.33	29.67	27.00	12.00	18.00	<u>23.67</u>	7.33	8.67	0.67	-2.00	22.47 _{±1.01}
			EEGPT	60.00	55.00	37.67	47.33	22.67	18.33	21.33	11.67	14.33	16.67	7.33	-0.33	26.00 _{±5.79}
			CBraMod	16.00	69.33	31.33	26.33	0.33	29.33	23.33	-2.00	3.33	28.00	3.67	9.33	19.86 _{±0.57}
			TFM	11.00	33.33	14.33	20.67	0.67	1.00	7.00	-1.00	-0.33	1.33	-9.00	3.33	6.86 _{±1.07}
			BrainOmni-Tiny	23.33	33.67	24.00	41.00	32.00	31.00	21.00	15.67	14.67	22.00	9.33	9.33	23.08 _{±1.09}
			BrainOmni-Base	22.33	28.67	26.33	32.00	37.67	25.33	18.33	5.67	12.33	23.33	5.00	12.00	20.75 _{±0.38}
			EEGMamba	3.00	21.33	22.00	18.67	2.00	21.00	19.33	15.00	17.00	9.33	2.67	1.67	12.75 _{±0.54}
			MIRepNet	97.67	<u>90.67</u>	54.00	<u>67.67</u>	<u>51.33</u>	33.33	32.00	33.33	19.33	37.00	14.33	4.67	44.61 _{±1.28}
			SingLEM	-6.33	-0.33	11.00	7.00	7.67	23.67	9.67	10.00	12.33	10.33	6.33	6.67	8.17 _{±0.41}
			LUNA-Base	0.00	59.33	5.67	45.33	0.00	17.67	3.33	-0.33	5.00	23.33	-0.67	2.67	13.44 _{±0.82}
Within-subject (Few-shot)	Specialist Models	Specialist Models	CSP+LDA	94.29	88.57	78.57	<u>72.86</u>	60.00	18.57	68.57	8.57	18.57	30.00	47.14	35.71	51.79
			EEGNet	0.00	91.43	83.33	74.29	67.62	45.71	70.00	0.00	10.95	19.05	71.91	0.95	44.17 _{±0.77}
			ShallowConv	41.43	88.09	67.62	59.05	61.91	<u>45.24</u>	52.38	29.53	3.33	29.05	82.86	12.86	47.58 _{±1.33}
			LMDA	0.00	<u>94.76</u>	74.76	64.76	76.19	34.76	63.81	<u>28.10</u>	10.48	31.90	83.81	7.62	47.58 _{±0.31}
			CNN-T	25.24	86.19	75.24	50.00	65.71	38.57	44.76	14.29	<u>28.09</u>	28.57	53.33	9.05	43.25 _{±2.72}
			Deformer	0.00	88.09	83.33	59.52	60.00	30.00	63.81	6.19	16.67	<u>36.67</u>	41.91	8.57	40.63 _{±2.56}
			Conformer	50.95	<u>94.76</u>	85.24	66.67	<u>79.52</u>	38.10	69.53	11.91	17.14	29.05	<u>84.76</u>	22.86	<u>54.21</u> _{±2.98}
			MSCFormer	58.10	90.00	81.90	71.90	72.86	36.19	<u>71.90</u>	19.52	14.29	20.00	87.14	22.38	53.85 _{±4.07}
			FBCNet	90.48	90.00	72.38	64.29	70.00	33.81	62.38	8.10	3.33	27.62	53.81	<u>27.62</u>	50.32 _{±1.71}
			TSception	42.38	95.71	80.00	70.48	74.76	41.90	55.71	11.90	18.57	26.19	50.00	16.67	48.69 _{±2.24}
	Full Fine-tuning	Foundation Models	BENDR	23.33	23.34	8.57	7.62	7.62	10.96	17.62	13.81	7.14	22.86	10.00	8.10	13.17 _{±0.50}
			BIOT-1D	47.14	91.43	50.00	55.71	25.24	16.67	44.29	3.33	2.38	16.19	27.62	-2.86	31.43 _{±2.61}
			BIOT-2D	73.33	89.05	48.57	53.33	17.62	12.38	36.19	0.48	3.81	17.14	7.62	-1.43	29.84 _{±2.90}
			BIOT-6D	61.43	87.14	78.57	60.00	23.33	24.29	41.90	4.76	8.57	25.71	28.09	2.86	36.94 _{±2.63}
			LaBraM	68.57	62.38	26.19	23.81	6.19	21.43	29.05	10.48	10.95	0.48	12.86	7.14	22.78 _{±1.30}
			Neuro-GPT	35.24	77.15	34.76	36.67	14.76	2.86	31.43	10.95	3.33	22.38	9.52	13.33	23.81 _{±1.80}
			EEGPT	44.76	57.62	16.19	40.95	2.38	3.81	18.57	9.05	9.05	10.95	10.95	3.33	18.33 _{±1.75}
			CBraMod	91.90	81.43	50.47	50.48	58.09	24.29	50.95	9.52	10.95	26.19	22.38	10.95	40.60 _{±1.43}
			TFM	43.33	51.43	13.33	6.19	0.00	10.00	5.24	4.76	-1.43	-1.90	1.43	-4.29	10.67 _{±0.56}
			BrainOmni-Tiny	50.00	51.91	30.00	19.05	11.91	15.71	27.62	18.57	17.14	17.62	13.33	6.19	23.18 _{±3.08}
			BrainOmni-Base	36.19	58.57	28.57	22.86	4.29	13.33	22.86	11.43	14.29	3.81	13.33	7.14	19.72 _{±1.65}
			EEGMamba	67.14	51.43	21.43	21.90	2.86	8.10	18.10	8.10	14.29	19.05	9.05	4.29	20.48 _{±0.96}
			MIRepNet	90.96	90.49	83.41	70.13	84.78	<u>45.24</u>	76.23	25.55	35.11	44.92	80.01	15.20	61.84 _{±0.88}
			SingLEM	8.57	7.62	0.00	0.48	2.38	7.62	9.05	-0.48	3.33	-4.29	18.57	-1.90	4.25 _{±1.25}
			LUNA-Base	54.29	80.00	21.90	54.76	2.86	2.86	21.43	1.90	-2.86	11.90	0.00	-0.48	20.71 _{±3.37}
	Linear Probing	Foundation Models	BENDR	4.76	0.00	9.52	-4.29	8.10	-10.00	-1.43	1.43	-4.76	4.76	9.05	5.71	1.90 _{±1.80}
			BIOT-1D	65.24	89.05	77.62	64.76	50.48	26.19	68.10	0.00	5.24	20.95	47.14	1.43	43.02 _{±2.01}
			BIOT-2D	60.95	89.52	88.57	57.62	27.62	19.52	61.43	-1.90	0.00	3.33	23.81	3.33	36.15 _{±1.13}
BIOT-6D			48.09	90.00	77.62	60.95	27.14	25.24	56.19	1.91	16.19	6.19	18.57	2.38	35.59 _{±2.42}	
LaBraM			55.24	43.81	27.14	25.71	0.95	23.81	18.57	14.76	9.52	0.00	7.62	6.67	19.48 _{±1.72}	
Neuro-GPT			0.48	83.33	79.05	48.10	27.62	-3.81	59.05	14.29	7.14	32.38	21.90	4.76	31.19 _{±1.73}	
EEGPT			57.62	59.05	38.57	35.71	7.14	15.71	27.14	-4.76	7.62	13.81	6.19	-2.86	21.75 _{±4.72}	
CBraMod			83.33	15.71	15.71	20.48	4.76	44.29	20.95	7.14	6.19	-0.48	30.48	10.95	21.63 _{±0.11}	
TFM			39.05	18.10	4.29	12.38	2.86	-4.76	5.71	9.05	0.00	1.90	-1.90	2.86	7.46 _{±2.37}	
BrainOmni-Tiny			50.48	45.71	35.24	37.14	28.10	5.24	34.76	25.24	20.00	16.67	19.05	-0.48	26.43 _{±1.53}	
BrainOmni-Base			38.57	54.29	23.81	28.57	8.10	10.48	26.19	15.24	11.90	17.62	20.48	6.19	21.79 _{±2.14}	
EEGMamba			33.33	20.00	16.67	8.57	-0.48	6.19	6.19	0.95	2.38	-5.71	13.33	0.48	8.49 _{±2.29}	
MIRepNet			<u>92.38</u>	79.69	<u>86.21</u>	63.84	70.64	35.29	40.62	9.47	21.67	34.42	32.33	-1.15	47.12 _{±1.34}	

Table 22. Classification AUCs (%) on BNCI2014009. The best metrics are marked in bold, and the second best by an underline.

Scenario	Tuning	Model Type	Approach	S0	S1	S2	S3	S4	S5	S6	S7	S8	S9	Avg.
Cross-subject (LOSO)	Specialist Models	Specialist Models	xDAWN+LDA	72.41	78.53	68.27	72.34	84.86	71.73	69.54	78.35	88.25	90.35	77.46
			EEGNet	81.81	88.18	73.04	86.03	92.46	85.42	84.73	85.83	<u>96.47</u>	90.22	86.42 _{±0.22}
			ShallowConv	85.92	86.56	74.76	83.29	93.78	82.81	<u>85.49</u>	85.70	95.04	92.26	86.56 _{±0.15}
			LMDA	80.62	89.16	<u>73.42</u>	<u>88.89</u>	92.10	84.28	82.66	86.18	96.37	93.13	86.68 _{±0.34}
			CNN-T	58.56	71.51	60.30	64.32	76.97	65.45	61.77	68.54	67.77	62.83	65.80 _{±2.39}
			Deformer	<u>82.99</u>	88.38	73.35	85.65	92.20	<u>84.76</u>	85.98	<u>86.37</u>	97.36	<u>93.55</u>	87.06 _{±0.14}
			Conformer	79.48	85.20	73.07	83.14	<u>92.62</u>	78.09	77.34	81.27	95.82	92.63	83.87 _{±0.31}
			TSception	49.32	54.37	51.78	53.92	53.92	51.73	56.37	51.38	56.85	56.46	53.61 _{±0.66}
	Full Fine-tuning	Foundation Models	BENDR	78.31	83.86	73.26	85.00	91.37	83.54	78.39	84.11	94.71	93.95	84.65 _{±0.36}
			BIOT-1D	61.96	62.41	61.44	58.11	56.42	58.36	63.63	60.78	79.99	71.42	63.45 _{±0.19}
			BIOT-2D	53.75	51.77	54.16	56.83	56.32	53.84	54.07	51.53	70.12	59.73	56.21 _{±0.25}
			BIOT-6D	58.00	66.47	63.50	58.13	58.09	59.38	65.13	58.95	81.05	66.74	63.55 _{±0.13}
			LaBraM	73.85	84.09	72.61	86.72	87.02	79.56	77.32	79.80	94.53	89.39	82.49 _{±0.35}
			Neuro-GPT	79.72	87.30	72.79	80.07	91.13	76.83	80.56	82.15	94.97	90.50	83.60 _{±0.26}
			EEGPT	67.26	77.29	67.04	73.50	75.28	66.88	72.27	69.97	86.35	84.37	74.02 _{±3.43}
			CBraMod	80.73	<u>89.29</u>	<u>72.23</u>	89.93	92.35	82.36	84.61	86.71	96.44	92.66	<u>86.73</u> _{±0.10}
			TFM	49.69	51.74	52.98	55.67	54.71	57.17	51.10	52.59	58.46	52.63	53.67 _{±0.78}
			BrainOmni-Tiny	72.28	82.17	69.53	81.15	82.47	70.85	73.67	74.37	91.60	77.75	77.58 _{±0.23}
			BrainOmni-Base	73.87	81.03	68.66	82.95	84.86	69.26	73.45	74.63	91.75	79.86	78.03 _{±0.15}
			EEGMamba	73.81	89.35	72.51	87.50	91.77	77.34	80.38	82.52	95.76	84.63	83.56 _{±0.02}
			SingLEM	77.88	76.73	70.61	88.82	83.36	74.81	71.88	78.38	87.36	82.58	79.24 _{±0.90}
			LUNA-Base	54.24	48.72	56.50	54.74	54.66	59.86	56.15	51.96	65.72	57.55	56.01 _{±0.52}
	Linear Probing	Foundation Models	BENDR	74.20	75.67	69.76	78.38	85.47	77.36	72.86	76.41	91.71	84.57	78.64 _{±1.45}
			BIOT-1D	58.22	55.25	63.42	60.24	61.14	57.35	61.75	57.53	74.10	61.56	61.05 _{±0.06}
			BIOT-2D	52.73	51.67	56.28	58.60	61.80	54.00	53.22	53.32	72.21	67.58	58.14 _{±0.20}
			BIOT-6D	55.88	57.05	63.37	61.23	59.04	57.23	59.71	57.36	73.74	62.47	60.71 _{±0.30}
			LaBraM	74.94	76.95	64.55	78.26	83.32	70.28	71.23	69.23	86.20	79.90	75.49 _{±0.15}
			Neuro-GPT	75.31	86.01	71.95	79.15	90.47	77.15	80.02	79.46	93.85	89.64	82.30 _{±0.29}
			EEGPT	74.75	85.42	72.22	83.44	91.49	79.25	80.64	81.86	93.72	92.22	83.50 _{±0.50}
			CBraMod	69.70	72.28	63.64	71.76	82.54	74.06	72.56	64.58	80.25	77.94	72.93 _{±0.20}
			TFM	50.03	49.56	51.46	55.07	53.09	56.83	51.10	48.81	56.73	53.13	52.58 _{±0.30}
			BrainOmni-Tiny	75.20	77.97	66.15	78.96	80.52	69.37	71.52	71.29	89.12	79.51	75.96 _{±0.58}
			BrainOmni-Base	73.62	80.05	68.11	80.28	83.82	68.00	73.01	72.47	90.25	76.35	76.60 _{±0.10}
			EEGMamba	76.84	83.58	72.91	83.06	87.13	73.25	75.06	77.07	94.11	84.91	80.79 _{±0.06}
			SingLEM	78.88	79.35	69.98	87.55	83.42	74.32	71.85	79.16	89.25	87.09	80.09 _{±0.04}
			LUNA-Base	52.89	48.66	51.25	56.03	54.36	58.31	55.55	52.62	65.43	55.48	55.06 _{±0.33}
Within-subject (Few-shot)	Specialist Models	Specialist Models	xDAWN+LDA	55.19	59.02	53.24	73.40	63.96	61.38	57.65	55.08	68.84	76.21	62.40
			EEGNet	71.22	76.32	68.12	79.35	85.28	74.85	62.88	77.21	88.36	<u>90.02</u>	77.36 _{±0.73}
			ShallowConv	59.54	62.57	58.28	63.52	73.78	54.73	54.91	54.78	70.45	75.23	62.78 _{±1.47}
			LMDA	65.62	63.31	66.49	70.39	77.57	60.54	56.76	72.92	82.93	79.94	69.65 _{±2.03}
			CNN-T	47.29	49.01	50.65	46.04	50.61	50.52	45.63	49.38	55.93	52.39	49.74 _{±0.31}
			Deformer	68.34	75.70	<u>67.71</u>	<u>73.86</u>	86.82	71.00	60.93	<u>75.58</u>	85.37	92.15	<u>75.75</u> _{±1.50}
			Conformer	66.87	72.73	64.80	<u>76.40</u>	78.07	69.46	65.48	61.79	84.93	81.58	72.21 _{±0.52}
			TSception	68.43	71.91	57.05	67.98	82.87	66.50	57.24	62.68	82.20	78.23	69.51 _{±1.11}
	Full Fine-tuning	Foundation Models	BENDR	62.93	58.96	61.53	67.61	74.63	57.10	55.81	61.91	77.51	71.95	64.99 _{±1.11}
			BIOT-1D	56.97	53.55	55.57	52.82	54.11	53.05	47.41	48.40	68.94	62.76	55.36 _{±1.65}
			BIOT-2D	52.08	55.25	54.17	55.41	59.50	53.04	51.01	47.26	71.11	60.00	55.88 _{±0.47}
			BIOT-6D	49.31	53.39	51.76	50.41	59.51	56.36	45.72	51.87	67.56	57.72	54.36 _{±1.13}
			LaBraM	<u>68.94</u>	72.99	60.47	71.84	78.41	63.67	61.57	64.12	82.12	81.43	70.56 _{±0.70}
			Neuro-GPT	63.17	78.70	64.51	72.27	79.53	70.86	66.18	69.63	79.64	80.49	72.50 _{±1.90}
			EEGPT	56.05	64.50	51.78	59.12	70.92	55.86	50.45	57.35	68.83	65.32	60.02 _{±2.03}
			CBraMod	58.27	55.52	58.25	63.13	70.60	60.99	57.38	59.66	71.97	69.82	62.56 _{±1.80}
			TFM	51.30	53.42	51.04	54.98	56.78	52.78	48.27	44.78	56.60	54.20	52.42 _{±0.48}
			BrainOmni-Tiny	61.56	81.78	59.59	74.27	82.33	69.25	58.25	65.71	<u>88.30</u>	79.21	72.02 _{±3.18}
			BrainOmni-Base	62.39	80.21	61.06	73.73	76.53	67.14	56.35	64.90	84.76	73.76	70.09 _{±0.36}
			EEGMamba	56.84	80.08	57.89	73.40	84.78	71.79	63.04	59.72	81.74	84.24	71.35 _{±0.40}
			SingLEM	61.35	67.66	61.88	67.01	68.94	63.03	58.83	64.67	68.84	80.25	66.25 _{±0.32}
			LUNA-Base	45.90	52.36	59.66	47.29	58.67	50.55	51.34	51.13	63.63	60.21	54.08 _{±0.13}
	Linear Probing	Foundation Models	BENDR	47.64	64.17	63.67	58.95	56.83	55.84	48.53	64.81	69.68	73.84	60.39 _{±0.47}
			BIOT-1D	52.10	52.69	54.23	52.97	56.00	51.28	50.64	48.56	62.09	60.87	54.14 _{±0.51}
			BIOT-2D	49.72	53.36	54.48	56.21	57.82	51.32	52.29	50.38	68.97	63.33	55.79 _{±0.68}
			BIOT-6D	54.73	55.57	50.34	51.40	60.48	50.60	48.57	45.91	65.09	57.85	54.05 _{±0.34}
			LaBraM	60.11	67.83	50.38	67.57	74.32	56.70	56.21	53.23	72.56	68.55	62.74 _{±0.35}
			Neuro-GPT	62.93	71.00	60.88	64.46	69.47	64.67	60.13	64.22	75.83	79.71	67.33 _{±2.05}
			EEGPT	56.46	66.61	55.62	60.09	72.44	62.87	55.45	61.25	67.55	70.13	62.85 _{±4.35}
			CBraMod	52.81	51.58	54.14	54.06	60.21	50.07	56.48	52.98	65.38	59.94	55.77 _{±0.37}
			TFM	50.80	50.04	52.62	52.35	50.48	50.77	53.45	48.62	50.67	52.99	51.28 _{±1.92}
			BrainOmni-Tiny	59.53	76.17	58.10	70.16	74.43	66.24	55.79	62.22	82.03	71.57	67.62 _{±0.35}
			BrainOmni-Base	58.29	76.41	59.31	71.18	72.51	64.51	54.36	58.25	77.76	71.78	66.44 _{±0.66}
			EEGMamba	56.71	<u>81.48</u>	58.31	71.55	<u>85.81</u>	<u>73.59</u>	<u>64.42</u>	60.49	84.51	84.31	72.12 _{±0.16}
			SingLEM	65.80	67.14	62.28	72.75	71.41	60.12	60.60	69.07	74.56	84.70	68.84 _{±1.09}
			LUNA-Base	46.18	51.89	49.47	46.49	59.73	45.02	52.91	52.90	65.85	58.88	52.93 _{±0.19}

Table 23. Balanced classification accuracies (%) on BNCI2014009. The best BCAs are marked in bold, and the second best by an underline.

Scenario	Tuning	Model Type	Approach	S0	S1	S2	S3	S4	S5	S6	S7	S8	S9	Avg.
Cross-subject (LOSO)	Specialist Models	Specialist Models	xDAWN+LDA	59.27	69.79	62.71	59.17	74.27	54.48	55.31	67.29	76.04	73.02	65.14
			EEGNet	73.96	79.17	68.09	77.60	83.89	77.12	74.90	75.69	90.76	82.74	<u>78.39</u> _{±0.44}
			ShallowConv	73.51	<u>79.83</u>	68.02	76.32	84.24	<u>76.84</u>	71.91	75.83	90.56	83.44	<u>78.05</u> _{±0.62}
			LMDA	70.14	79.13	69.72	<u>81.56</u>	84.72	75.62	70.24	78.58	88.72	<u>86.11</u>	78.45 _{±0.46}
			CNN-T	57.33	66.67	56.77	60.49	67.12	59.62	58.75	63.58	63.78	60.90	<u>61.50</u> _{±1.71}
			Deformer	71.22	77.47	67.36	73.19	84.55	72.43	68.92	75.42	91.81	86.56	<u>76.89</u> _{±0.14}
			Conformer	56.25	56.56	60.21	61.77	72.22	65.45	58.65	58.89	70.14	62.05	<u>62.22</u> _{±0.97}
			TSception	<u>73.54</u>	76.84	67.81	76.25	<u>85.73</u>	72.05	71.63	71.88	86.04	86.01	<u>76.78</u> _{±0.33}
	Full Fine-tuning	Foundation Models	BENDR	64.79	73.54	68.68	72.33	80.17	72.05	64.44	70.21	85.66	82.74	<u>73.46</u> _{±0.28}
			BIOT-1D	57.88	57.43	58.78	55.35	52.50	53.68	58.99	58.09	72.26	65.03	<u>59.00</u> _{±0.29}
			BIOT-2D	52.22	51.63	52.71	56.84	53.82	54.62	52.78	51.25	64.31	55.45	<u>54.56</u> _{±0.27}
			BIOT-6D	53.61	57.08	57.22	56.15	54.86	56.08	55.24	56.39	73.09	61.67	<u>58.14</u> _{±0.33}
			LaBraM	61.56	69.20	65.28	73.96	75.76	64.13	60.38	69.51	84.86	78.40	<u>70.31</u> _{±0.24}
			Neuro-GPT	72.05	80.52	66.42	72.33	83.89	69.62	72.40	73.85	86.74	81.91	<u>75.97</u> _{±0.53}
			EEGPT	57.50	62.67	59.27	62.29	64.24	57.12	57.43	59.86	73.89	73.40	<u>62.77</u> _{±1.85}
			CBraMod	66.42	74.76	67.81	81.98	85.76	69.83	72.22	<u>77.88</u>	<u>91.67</u>	84.65	<u>77.30</u> _{±0.28}
			TFM	50.49	52.08	52.40	55.07	54.51	54.13	50.38	52.78	57.26	51.94	<u>53.10</u> _{±0.48}
			BrainOmni-Tiny	65.52	71.74	64.65	73.44	75.45	65.14	68.30	64.93	84.20	71.42	<u>70.48</u> _{±0.23}
			BrainOmni-Base	67.47	73.26	62.47	75.35	76.35	63.54	66.98	67.60	82.99	72.71	<u>70.87</u> _{±0.29}
			EEGMamba	69.76	78.99	67.64	79.76	84.51	67.53	<u>73.65</u>	73.51	87.85	76.91	<u>76.01</u> _{±0.05}
			SingLEM	68.99	69.58	64.41	80.87	76.22	67.78	65.56	71.01	79.51	75.83	<u>71.98</u> _{±0.56}
			LUNA-Base	50.76	49.17	52.95	52.50	53.96	51.35	52.43	51.42	50.00	52.15	<u>51.67</u> _{±0.54}
	Linear Probing	Foundation Models	BENDR	57.71	57.99	60.87	60.80	63.44	59.24	55.56	62.19	70.28	64.38	<u>61.24</u> _{±2.07}
			BIOT-1D	53.16	55.28	58.58	55.38	58.02	56.39	54.41	54.79	66.84	58.26	<u>57.11</u> _{±0.35}
			BIOT-2D	51.01	49.03	54.90	55.97	60.52	52.95	52.67	51.74	64.13	57.40	<u>55.03</u> _{±0.43}
			BIOT-6D	50.00	49.93	50.62	50.80	50.45	50.49	50.66	50.21	54.06	53.19	<u>51.04</u> _{±0.12}
			LaBraM	67.50	66.91	60.80	66.98	74.06	61.94	64.51	63.30	76.88	74.62	<u>67.75</u> _{±0.32}
			Neuro-GPT	55.45	54.48	56.42	59.17	67.05	54.06	51.91	54.41	65.83	68.23	<u>58.70</u> _{±0.26}
			EEGPT	57.15	62.99	63.12	65.83	77.53	59.83	58.37	62.43	78.96	79.06	<u>66.53</u> _{±0.05}
			CBraMod	57.15	55.52	58.09	56.98	65.87	55.21	54.93	53.40	67.53	61.63	<u>58.63</u> _{±0.54}
			TFM	49.90	48.96	50.83	52.95	52.36	54.79	51.53	48.75	54.51	51.04	<u>51.56</u> _{±0.51}
			BrainOmni-Tiny	58.85	59.13	60.90	64.17	63.54	56.56	55.62	57.43	70.80	66.35	<u>61.34</u> _{±0.42}
			BrainOmni-Base	65.49	71.56	61.46	73.85	76.53	65.24	64.34	65.52	81.70	69.27	<u>69.50</u> _{±0.69}
			EEGMamba	70.31	74.17	<u>68.78</u>	75.00	79.06	69.03	69.03	70.49	87.26	78.12	<u>74.13</u> _{±0.30}
			SingLEM	71.35	70.45	65.87	79.83	73.75	69.31	64.51	71.77	81.28	77.12	<u>72.52</u> _{±0.05}
			LUNA-Base	53.30	48.78	48.37	53.44	52.78	51.98	51.49	50.56	58.40	52.53	<u>52.16</u> _{±0.50}
Within-subject (Few-shot)	Specialist Models	Specialist Models	xDAWN+LDA	50.43	54.57	51.34	61.02	56.30	53.77	50.77	50.32	57.23	58.72	54.45
			EEGNet	<u>62.85</u>	65.24	61.19	71.32	<u>76.36</u>	<u>67.33</u>	57.62	68.86	78.37	80.73	68.99 _{±0.51}
			ShallowConv	53.29	55.03	54.18	57.65	65.09	52.43	54.58	52.72	63.84	62.50	<u>57.13</u> _{±0.93}
			LMDA	59.16	53.58	57.09	59.86	65.92	54.85	51.06	60.77	70.08	70.40	<u>60.28</u> _{±0.82}
			CNN-T	48.50	51.38	49.66	47.02	50.15	50.82	47.94	52.71	53.82	51.22	<u>50.32</u> _{±0.52}
			Deformer	58.02	66.61	<u>59.21</u>	62.45	74.99	61.87	54.14	<u>64.29</u>	75.74	76.44	<u>65.38</u> _{±0.32}
			Conformer	59.49	66.63	59.13	<u>68.60</u>	71.18	64.52	60.30	55.58	75.82	73.01	<u>65.43</u> _{±0.90}
			TSception	63.06	63.44	55.19	62.09	74.97	60.54	55.44	58.56	74.44	72.14	<u>63.99</u> _{±0.43}
	Full Fine-tuning	Foundation Models	BENDR	59.56	54.20	54.40	61.02	63.49	54.13	53.49	57.34	70.00	65.06	<u>59.27</u> _{±1.03}
			BIOT-1D	51.82	51.20	54.37	54.13	53.61	51.96	48.52	49.92	61.57	57.80	<u>53.49</u> _{±0.73}
			BIOT-2D	51.59	53.56	51.97	54.41	56.20	51.74	49.91	48.64	64.30	56.03	<u>53.83</u> _{±0.35}
			BIOT-6D	49.87	49.27	51.07	50.83	53.14	52.65	49.56	49.87	62.18	53.50	<u>52.19</u> _{±0.16}
			LaBraM	57.82	63.17	51.74	63.08	64.48	56.50	52.43	56.50	70.20	64.69	<u>60.06</u> _{±0.58}
			Neuro-GPT	52.39	55.36	50.00	54.17	62.09	60.17	51.19	51.76	52.26	60.83	<u>55.02</u> _{±0.97}
			EEGPT	51.87	51.64	51.22	52.90	53.48	52.53	49.27	50.95	54.90	51.45	<u>52.02</u> _{±1.14}
			CBraMod	57.18	53.56	53.63	55.73	60.83	54.67	53.08	55.10	64.30	60.47	<u>56.85</u> _{±1.03}
			TFM	50.77	52.42	49.99	53.23	54.41	51.37	48.58	45.41	54.59	53.43	<u>51.42</u> _{±0.87}
			BrainOmni-Tiny	50.87	61.21	51.33	56.93	65.92	56.73	52.23	52.60	59.19	63.16	<u>57.02</u> _{±0.36}
			BrainOmni-Base	54.61	64.46	55.86	62.42	63.99	57.21	53.12	54.05	67.49	64.34	<u>59.75</u> _{±1.14}
			EEGMamba	55.23	72.14	55.76	64.76	76.41	66.25	57.79	54.57	74.20	77.86	<u>65.50</u> _{±0.46}
			SingLEM	56.73	56.82	56.86	58.08	58.43	54.13	53.03	55.31	59.80	67.79	<u>57.70</u> _{±0.41}
			LUNA-Base	47.46	51.80	54.36	50.08	50.78	49.92	50.03	49.95	55.52	56.75	<u>51.66</u> _{±0.99}
	Linear Probing	Foundation Models	BENDR	49.95	50.42	49.77	50.50	53.21	52.45	50.53	51.56	53.44	58.50	<u>52.03</u> _{±0.18}
			BIOT-1D	52.09	52.59	50.59	50.60	53.01	51.64	49.87	49.76	59.48	56.93	<u>52.66</u> _{±0.12}
			BIOT-2D	49.54	51.34	53.05	54.53	56.70	51.11	51.73	49.08	62.34	56.86	<u>53.63</u> _{±0.30}
			BIOT-6D	52.97	53.45	51.38	50.48	56.05	50.62	49.50	50.97	59.41	55.49	<u>53.03</u> _{±0.78}
			LaBraM	52.90	61.96	50.24	53.93	61.84	53.71	52.23	50.85	55.47	58.40	<u>55.15</u> _{±0.21}
			Neuro-GPT	50.00	50.00	49.96	50.00	50.34	50.46	49.96	50.38	51.34	54.02	<u>50.65</u> _{±0.31}
			EEGPT	50.64	56.81	50.95	53.98	56.85	51.67	49.68	50.75	57.68	52.75	<u>53.18</u> _{±0.97}
			CBraMod	50.00	50.22	49.96	50.00	51.10	49.92	49.65	49.81	50.15	50.95	<u>50.18</u> _{±0.15}
			TFM	51.59	49.46	53.40	51.21	51.54	51.93	52.00	48.15	50.67	51.21	<u>51.12</u> _{±0.83}
			BrainOmni-Tiny	52.59	61.71	52.06	56.65	61.29	57.22	52.27	52.10	58.27	59.18	<u>56.33</u> _{±0.16}
			BrainOmni-Base	53.92	65.95	50.86	59.42	64.25	59.25	51.97	50.83	62.13	65.12	<u>58.37</u> _{±0.35}
			EEGMamba	52.98	<u>71.77</u>	56.72	63.48	75.39	68.06	<u>59.53</u>	55.70	<u>77.93</u>	76.33	<u>65.79</u> _{±0.16}
			SingLEM	60.53	62.98	56.82	67.14	65.37	56.58	58.62	61.38	69.01	<u>78.15</u>	<u>63.66</u> _{±0.96}
			LUNA-Base	46.49	51.77	49.55	47.62	53.42	46.18	52.86	50.71	60.88	54.72	<u>51.42</u> _{±0.19}

Table 24. Classification AUCs (%) on BNCI2014008. The best metrics are marked in bold, and the second best by an underline.

Scenario	Tuning	Model Type	Approach	S0	S1	S2	S3	S4	S5	S6	S7	Avg.
Cross-subject (LOSO)		Specialist Models	xDAWN+LDA	74.40	73.89	82.98	73.59	73.49	75.63	76.78	83.04	76.72
			EEGNet	76.93	76.03	84.89	<u>75.86</u>	<u>79.61</u>	79.52	<u>81.46</u>	84.91	79.90 _{±0.11}
			ShallowConv	73.45	73.71	81.35	73.90	76.49	78.36	78.21	84.30	77.47 _{±0.02}
			LMDA	<u>76.73</u>	74.53	<u>83.30</u>	75.88	79.79	<u>78.38</u>	81.69	<u>84.50</u>	<u>79.35</u> _{±0.10}
			CNN-T	53.91	57.47	53.13	51.42	60.06	51.51	57.31	53.63	54.80 _{±0.35}
			Deformer	75.63	71.69	80.11	71.61	73.60	72.84	78.54	81.41	75.68 _{±0.48}
			Conformer	71.50	72.36	80.10	70.38	76.80	73.74	73.76	80.32	74.87 _{±0.15}
			TSception	51.44	51.25	49.39	51.22	49.70	50.45	51.05	50.11	50.57 _{±0.09}
	Full Fine-tuning	Foundation Models	BENDR	69.76	72.57	77.16	71.93	77.50	69.69	74.84	80.44	74.24 _{±0.10}
			BIOT-1D	52.80	62.35	54.41	52.76	63.00	53.80	64.55	53.31	57.12 _{±0.19}
			BIOT-2D	51.11	56.91	51.31	48.82	58.83	52.62	58.74	52.44	53.85 _{±0.27}
			BIOT-6D	55.13	60.40	55.25	53.79	62.31	54.02	64.09	55.59	57.57 _{±0.07}
			LaBraM	67.66	69.99	76.62	67.90	70.97	69.31	73.28	70.34	70.76 _{±0.04}
			Neuro-GPT	73.03	72.86	80.00	72.65	76.13	74.66	73.44	81.46	75.53 _{±0.08}
			EEGPT	65.53	66.03	70.92	62.90	66.93	62.77	68.20	69.59	66.61 _{±0.69}
			CBraMod	75.20	<u>74.54</u>	79.23	74.19	79.31	76.98	81.00	80.77	77.65 _{±0.09}
			TFM	52.43	57.74	51.39	50.33	57.63	52.84	58.92	52.72	54.25 _{±0.29}
			BrainOmni-Tiny	61.85	67.89	71.98	63.91	62.48	59.92	70.40	71.91	66.29 _{±0.16}
			BrainOmni-Base	63.64	59.21	66.65	62.53	57.31	60.15	62.83	70.58	62.86 _{±0.12}
			EEGMamba	76.25	71.80	83.19	71.68	69.22	70.17	75.15	77.70	74.40 _{±0.04}
			SingLEM	63.73	65.23	70.46	69.31	65.62	71.48	71.66	76.71	69.28 _{±0.25}
			LUNA-Base	50.38	52.96	50.87	49.99	58.14	52.36	56.34	54.30	53.17 _{±0.10}
	Linear Probing	Foundation Models	BENDR	69.81	69.32	75.91	71.31	75.25	74.20	73.40	77.15	73.29 _{±1.01}
			BIOT-1D	52.92	59.90	51.49	51.80	63.02	50.45	61.47	54.96	55.75 _{±0.15}
			BIOT-2D	51.20	57.21	52.14	49.71	59.63	52.58	57.81	53.34	54.20 _{±0.31}
			BIOT-6D	51.05	59.85	52.65	51.88	62.32	52.36	57.78	52.14	55.00 _{±0.81}
			LaBraM	58.78	62.25	63.97	60.45	58.25	57.67	66.59	64.03	61.50 _{±0.24}
			Neuro-GPT	65.75	66.77	72.18	65.16	54.56	68.31	71.08	79.97	67.97 _{±0.50}
			EEGPT	76.53	73.50	82.98	71.55	76.15	70.34	75.59	79.15	75.72 _{±0.32}
			CBraMod	52.54	59.48	53.90	53.03	61.97	53.24	57.98	56.46	56.07 _{±0.10}
			TFM	52.58	53.73	49.84	50.19	55.35	51.70	57.11	52.42	52.87 _{±0.94}
			BrainOmni-Tiny	58.63	62.34	67.43	59.75	55.87	56.22	67.23	68.05	61.94 _{±0.27}
			BrainOmni-Base	59.65	63.74	68.53	60.32	57.93	57.99	68.12	70.87	63.39 _{±0.33}
			EEGMamba	70.53	66.92	76.71	66.68	61.34	64.16	71.47	75.36	69.14 _{±0.05}
			SingLEM	57.87	64.73	66.46	67.22	59.39	71.82	72.02	76.35	66.98 _{±0.04}
			LUNA-Base	49.31	55.48	50.80	48.82	58.71	51.64	54.56	58.13	53.43 _{±0.08}
Within-subject (Few-shot)		Specialist Models	xDAWN+LDA	54.34	56.74	64.61	48.32	62.07	67.53	63.52	71.84	61.12
			EEGNet	69.36	75.53	81.07	71.74	76.27	81.01	80.66	90.82	78.31 _{±1.49}
			ShallowConv	51.21	62.73	54.72	53.05	64.58	60.74	62.24	74.24	60.44 _{±0.23}
			LMDA	57.96	66.22	68.01	67.30	64.38	72.99	70.10	87.72	69.34 _{±0.54}
			CNN-T	49.60	53.85	50.80	49.75	52.90	49.39	52.63	53.06	51.50 _{±0.64}
			Deformer	59.07	<u>68.02</u>	<u>78.98</u>	<u>69.27</u>	65.80	<u>75.80</u>	<u>73.40</u>	<u>89.34</u>	<u>72.46</u> _{±1.09}
			Conformer	57.69	55.62	66.52	49.97	60.09	71.46	67.46	70.87	62.46 _{±0.46}
			TSception	49.97	53.20	50.87	48.05	54.61	52.34	53.45	57.99	52.56 _{±0.25}
	Full Fine-tuning	Foundation Models	BENDR	55.33	60.50	67.41	60.25	61.02	61.34	64.66	78.47	63.62 _{±1.49}
			BIOT-1D	51.41	57.40	50.83	50.30	60.72	52.60	57.37	55.44	54.51 _{±0.34}
			BIOT-2D	51.46	55.23	50.61	48.51	55.37	51.02	49.05	52.15	51.67 _{±0.08}
			BIOT-6D	51.29	57.25	50.39	49.53	61.01	52.20	58.05	55.60	54.42 _{±0.29}
			LaBraM	51.41	61.08	58.25	56.45	57.63	63.12	59.22	79.52	60.83 _{±0.52}
			Neuro-GPT	64.53	59.94	72.83	51.91	<u>66.77</u>	71.69	69.54	80.14	67.17 _{±0.42}
			EEGPT	52.92	52.85	57.11	49.41	59.04	54.15	61.42	59.06	55.75 _{±1.32}
			CBraMod	53.08	61.36	57.86	57.06	60.81	62.99	62.99	85.15	62.66 _{±0.96}
			TFM	49.76	54.41	51.70	50.80	52.11	51.90	54.73	50.36	51.97 _{±0.35}
			BrainOmni-Tiny	56.33	55.36	60.86	54.91	57.39	68.76	67.12	74.79	61.94 _{±0.44}
			BrainOmni-Base	56.10	56.42	59.19	53.10	58.56	67.88	65.27	73.90	61.30 _{±0.47}
			EEGMamba	65.63	59.70	69.71	49.43	64.32	70.32	65.26	81.23	65.70 _{±0.47}
			SingLEM	56.31	54.49	52.20	50.96	58.81	64.05	65.36	79.35	60.19 _{±0.91}
			LUNA-Base	48.68	54.51	49.73	49.23	57.45	48.67	53.02	53.32	51.83 _{±0.13}
	Linear Probing	Foundation Models	BENDR	50.42	52.76	52.32	50.11	57.56	57.65	59.41	60.24	55.06 _{±1.08}
			BIOT-1D	51.42	57.51	51.74	50.71	57.61	50.93	52.12	53.70	53.22 _{±0.26}
			BIOT-2D	51.09	55.85	50.47	49.67	54.93	50.51	50.23	51.53	51.79 _{±0.18}
			BIOT-6D	49.70	56.72	51.06	49.68	56.15	51.22	51.36	52.95	52.36 _{±0.64}
			LaBraM	52.15	54.67	56.10	52.32	55.50	54.56	61.65	60.06	55.88 _{±0.23}
			Neuro-GPT	63.01	54.45	64.93	55.73	62.24	69.05	65.60	78.36	64.17 _{±1.36}
			EEGPT	51.47	52.84	54.44	48.58	60.60	53.71	59.70	59.68	55.13 _{±0.40}
			CBraMod	51.17	57.01	50.38	50.53	59.07	48.51	52.74	53.45	52.86 _{±0.34}
			TFM	50.98	52.79	50.53	48.91	52.89	50.60	53.10	50.83	51.33 _{±0.50}
			BrainOmni-Tiny	51.51	54.63	55.09	51.08	56.79	61.07	63.49	64.41	57.26 _{±0.22}
			BrainOmni-Base	51.91	55.07	55.71	50.92	57.40	60.98	62.71	66.05	57.59 _{±0.23}
			EEGMamba	<u>67.98</u>	63.07	71.60	48.57	64.95	68.83	68.46	80.03	66.69 _{±0.32}
			SingLEM	55.98	54.40	58.84	51.83	60.19	69.30	65.37	76.93	61.61 _{±0.45}
			LUNA-Base	48.46	54.51	49.62	49.05	57.05	48.26	52.57	54.69	51.78 _{±0.05}

Table 25. Balanced classification accuracies (%) on BNCI2014008. The best BCAs are marked in bold, and the second best by an underline.

Scenario	Tuning	Model Type	Approach	S0	S1	S2	S3	S4	S5	S6	S7	Avg.	
Cross-subject (LOSO)		Specialist Models	xDAWN+LDA	55.06	58.16	62.73	63.89	53.66	55.24	63.33	57.50	58.69	
			EEGNet	69.66	69.00	76.86	<u>69.85</u>	71.71	71.87	<u>73.18</u>	76.22	72.29 _{±0.17}	
			ShallowConv	66.29	<u>68.30</u>	74.02	67.11	67.45	<u>70.63</u>	70.27	75.25	69.92 _{±0.05}	
			LMDA	68.94	67.62	<u>75.95</u>	70.29	<u>71.17</u>	69.65	74.13	<u>76.21</u>	<u>71.74</u> _{±0.12}	
			CNN-T	50.16	53.36	51.38	50.58	54.02	50.65	54.78	50.54	51.93 _{±0.16}	
			Deformer	57.00	58.14	61.30	59.57	54.70	50.77	63.29	57.27	57.75 _{±1.21}	
			Conformer	62.48	66.20	72.50	65.30	69.10	67.80	67.15	72.43	67.87 _{±0.11}	
			TSception	50.77	50.63	49.53	50.89	49.43	50.92	50.85	50.18	50.40 _{±0.09}	
		Full Fine-tuning	Foundation Models	BENDR	59.21	64.15	66.81	64.35	70.22	58.96	66.45	69.95	65.01 _{±0.18}
				BIOT-1D	51.93	59.30	53.56	51.77	60.15	51.89	59.35	51.14	54.88 _{±0.12}
				BIOT-2D	51.11	55.27	50.87	49.38	55.54	51.68	56.27	50.95	52.63 _{±0.45}
				BIOT-6D	52.90	57.26	53.87	52.56	58.98	52.24	58.96	51.41	54.77 _{±0.21}
				LaBraM	59.58	61.95	68.58	61.94	63.16	60.98	66.89	61.51	63.07 _{±0.97}
				Neuro-GPT	66.03	67.03	72.65	66.24	69.41	67.57	66.98	72.84	68.59 _{±0.25}
				EEGPT	56.11	58.62	62.15	56.48	57.66	55.26	60.02	59.65	58.24 _{±0.40}
				CBraMod	66.92	67.88	71.50	68.20	70.32	69.99	72.82	71.65	69.91 _{±0.06}
				TFM	51.49	55.53	51.09	50.32	55.43	52.25	56.01	51.80	52.99 _{±0.29}
				BrainOmni-Tiny	57.80	61.86	65.92	60.26	59.61	56.32	64.17	65.29	61.40 _{±0.17}
				BrainOmni-Base	59.39	56.77	62.07	59.81	55.00	56.82	59.36	65.26	59.31 _{±0.06}
				EEGMamba	<u>69.25</u>	65.84	74.47	66.76	64.54	64.53	69.34	70.68	68.18 _{±0.05}
				SingLEM	56.49	60.53	65.22	63.80	60.73	65.08	65.87	69.61	63.42 _{±0.13}
				LUNA-Base	50.00	50.00	50.00	50.02	50.00	50.00	50.00	50.00	50.00 _{±0.04}
		Linear Probing	Foundation Models	BENDR	64.33	63.35	69.30	65.75	68.62	67.86	67.14	70.50	67.11 _{±0.76}
				BIOT-1D	51.56	57.86	50.43	50.66	60.12	49.50	56.89	51.43	53.56 _{±0.58}
				BIOT-2D	49.95	53.52	51.47	50.07	54.52	51.83	53.09	50.90	51.92 _{±0.14}
				BIOT-6D	50.73	56.44	51.64	50.55	60.25	51.45	54.00	50.48	53.19 _{±0.66}
				LaBraM	54.62	56.61	58.50	56.89	55.49	54.98	59.66	57.83	56.82 _{±0.60}
				Neuro-GPT	50.00	50.24	50.30	50.10	49.98	50.00	50.02	50.00	50.08 _{±0.04}
				EEGPT	55.44	56.53	64.61	57.69	56.46	52.20	57.57	57.56	57.26 _{±0.17}
				CBraMod	50.07	57.29	51.28	52.24	58.46	51.78	55.31	50.05	53.31 _{±0.14}
				TFM	51.27	52.89	49.68	49.72	53.99	51.22	54.94	51.72	51.93 _{±0.63}
				BrainOmni-Tiny	50.26	51.07	52.79	51.28	51.27	50.51	51.63	51.80	51.33 _{±0.14}
				BrainOmni-Base	56.63	59.12	63.25	57.02	55.28	55.03	62.17	62.94	58.93 _{±0.43}
EEGMamba	65.02	62.49	68.57	62.04	57.49	59.56	65.53	68.50	63.65 _{±0.11}				
SingLEM	54.60	59.72	62.24	62.32	56.31	65.07	66.01	68.84	61.89 _{±0.03}				
LUNA-Base	49.99	49.94	49.98	49.95	50.02	50.00	49.99	50.03	49.99 _{±0.02}				
Within-subject (Few-shot)		Specialist Models	xDAWN+LDA	52.71	52.42	59.10	48.81	56.75	61.05	58.51	63.88	56.65	
			EEGNet	63.04	68.10	73.39	65.52	68.72	72.74	73.07	82.72	70.91 _{±1.17}	
			ShallowConv	49.95	58.63	52.22	50.88	59.61	54.61	58.67	63.20	55.97 _{±0.20}	
			LMDA	55.96	60.99	61.85	<u>62.08</u>	61.40	64.93	63.50	<u>77.89</u>	63.58 _{±0.30}	
			CNN-T	50.12	50.64	50.17	49.88	50.23	49.48	50.70	50.65	50.23 _{±0.29}	
			Deformer	54.62	<u>62.33</u>	<u>70.14</u>	61.34	59.04	<u>65.89</u>	<u>64.68</u>	76.84	<u>64.36</u> _{±0.93}	
			Conformer	54.54	<u>54.53</u>	60.96	49.93	56.58	64.93	61.58	63.65	58.34 _{±0.23}	
			TSception	50.17	52.47	50.24	49.98	52.41	51.23	52.48	54.91	51.73 _{±0.30}	
		Full Fine-tuning	Foundation Models	BENDR	52.64	56.84	61.07	56.18	55.45	56.28	58.83	69.32	58.33 _{±1.00}
				BIOT-1D	50.39	55.56	50.65	50.53	51.96	50.98	54.93	51.89	52.11 _{±0.37}
				BIOT-2D	51.59	53.43	50.56	49.23	53.07	50.34	50.23	50.95	51.17 _{±0.10}
				BIOT-6D	50.20	55.10	49.66	50.13	52.91	50.91	54.39	53.66	52.12 _{±0.27}
				LaBraM	49.86	55.73	52.27	51.50	52.61	55.32	54.77	65.33	54.67 _{±0.67}
				Neuro-GPT	58.67	56.80	65.52	51.33	<u>61.79</u>	64.22	61.80	72.72	61.61 _{±0.14}
				EEGPT	50.60	50.60	51.75	49.83	53.40	50.88	55.34	53.07	51.93 _{±0.41}
				CBraMod	51.03	57.18	53.97	54.02	56.93	57.87	58.13	74.89	58.00 _{±0.99}
				TFM	49.45	52.69	51.14	50.18	50.86	51.76	53.41	50.16	51.21 _{±0.28}
				BrainOmni-Tiny	53.73	53.38	55.43	52.97	54.04	59.83	58.32	63.01	56.34 _{±0.33}
				BrainOmni-Base	53.31	54.06	54.04	52.63	53.99	59.03	58.32	64.07	56.18 _{±0.14}
				EEGMamba	60.07	57.24	64.04	49.64	61.00	64.54	58.47	73.97	61.12 _{±0.35}
				SingLEM	53.14	52.69	50.47	51.33	56.75	57.25	59.74	69.67	56.38 _{±0.28}
				LUNA-Base	49.97	51.72	50.07	49.97	50.00	50.04	50.33	50.20	50.29 _{±0.12}
		Linear Probing	Foundation Models	BENDR	50.25	50.85	50.50	49.71	52.93	52.50	53.81	53.42	51.75 _{±0.70}
				BIOT-1D	50.95	55.89	51.17	50.27	55.66	50.22	52.14	52.61	52.36 _{±0.22}
				BIOT-2D	50.90	53.60	50.43	49.89	54.18	50.32	50.67	51.24	51.40 _{±0.30}
				BIOT-6D	49.51	54.23	50.87	49.62	54.17	50.94	51.57	52.07	51.62 _{±0.47}
				LaBraM	49.79	52.81	53.39	51.28	52.85	52.33	55.10	56.15	52.96 _{±0.24}
				Neuro-GPT	50.02	50.31	50.00	50.00	50.18	50.51	55.80	50.26	50.88 _{±0.02}
				EEGPT	49.86	51.31	50.72	49.87	52.07	50.06	55.06	51.76	51.34 _{±0.18}
				CBraMod	50.87	55.36	50.77	50.37	52.10	49.45	50.95	51.56	51.43 _{±0.09}
				TFM	50.78	52.23	50.50	49.23	51.73	50.20	52.67	50.11	50.93 _{±0.36}
				BrainOmni-Tiny	51.22	51.93	51.29	50.60	53.53	54.18	56.41	54.55	52.96 _{±0.05}
				BrainOmni-Base	51.70	53.75	53.18	49.96	53.79	56.21	57.93	58.11	54.33 _{±0.30}
EEGMamba	<u>62.67</u>	59.14	64.82	48.77	60.81	63.33	59.65	72.44	61.45 _{±0.09}				
SingLEM	53.67	53.54	55.99	51.26	57.60	63.63	60.76	69.69	58.27 _{±0.40}				
LUNA-Base	49.43	52.76	50.34	49.40	52.74	50.00	51.11	50.53	50.79 _{±0.15}				

Table 26. Classification AUCs (%) on CHB-MIT. The best AUCs are marked in bold, and the second best by an underline.

Scenario	Tuning	Model Type	Approach	S0	S1	S2	S3	S4	S5	S6	S7	S8	S9	S10	S11
Cross-subject (LOSO)		Specialist Models	PSD+SVM	87.68	87.33	90.45	56.32	96.28	64.13	91.22	95.00	98.39	96.49	97.08	63.53
			EEGNet	99.50	98.04	95.99	95.61	91.61	78.02	91.87	75.30	98.44	98.44	99.55	58.59
			ShallowConv	99.90	97.39	<u>99.03</u>	95.29	94.16	86.96	<u>99.07</u>	94.44	<u>98.42</u>	98.26	99.28	58.85
			LMDA	99.88	96.44	93.43	90.55	95.63	80.83	94.44	86.41	97.54	97.76	97.48	64.77
			CNN-T	98.86	90.07	97.24	95.99	95.70	85.79	95.67	85.60	97.95	96.05	98.83	57.31
			Deformer	99.93	98.25	96.21	97.13	93.73	87.57	95.76	90.85	96.99	98.02	99.02	61.13
			Conformer	<u>99.91</u>	<u>98.15</u>	98.53	95.30	86.09	85.77	97.42	96.56	91.95	97.13	99.00	55.21
			TSception	99.27	81.69	97.13	<u>97.46</u>	92.70	66.27	95.13	93.08	96.71	96.41	97.14	56.86
	Full Fine-tuning	Foundation Models	BENDR	99.26	96.67	96.22	96.93	73.91	84.40	98.02	87.97	93.33	96.41	<u>99.58</u>	59.86
			BIOT-1D	99.09	96.05	94.58	94.71	94.31	75.45	96.13	79.90	85.72	74.47	95.73	62.56
			BIOT-2D	99.51	96.69	96.44	95.25	92.59	75.24	95.32	86.34	84.47	68.81	96.08	68.62
			BIOT-6D	99.17	98.07	98.59	97.49	89.86	65.48	97.15	89.82	88.69	75.08	96.92	66.43
			LaBraM	98.19	97.61	97.36	92.58	71.01	84.55	93.17	78.85	90.15	91.28	99.39	63.84
			Neuro-GPT	97.21	90.37	94.46	91.04	70.62	73.47	94.77	78.83	90.29	80.11	98.52	50.57
			EEGPT	88.74	85.16	65.54	78.52	50.77	74.26	83.16	<u>77.23</u>	83.38	70.23	95.59	59.38
			CBraMod	98.96	95.82	96.63	95.90	94.48	73.59	97.04	98.42	96.71	95.97	97.64	66.18
			TFM	75.76	82.89	69.66	81.30	81.27	68.19	87.39	68.74	91.08	57.70	79.92	52.44
			EEGMamba	99.61	91.50	98.58	94.08	90.37	82.35	91.60	95.39	97.88	92.01	98.57	64.19
			SingLEM	79.46	87.84	63.77	70.88	60.02	59.43	74.08	81.41	82.35	50.70	79.07	49.69
			LUNA-Base	98.25	96.77	99.22	90.03	85.50	<u>89.08</u>	96.92	<u>97.66</u>	85.37	88.83	96.14	61.76
			LUNA-Large	98.65	97.42	98.41	94.03	84.89	90.11	95.63	97.00	89.46	86.30	96.05	58.64
			LUNA-Huge	98.64	97.94	98.13	90.34	88.10	84.74	97.33	97.42	89.56	91.52	97.06	52.97
	Linear Probing	Foundation Models	BENDR	53.46	55.72	52.28	57.24	55.77	50.80	59.08	54.17	56.71	51.49	59.02	50.23
			BIOT-1D	93.79	88.40	69.15	88.88	<u>97.69</u>	64.92	91.24	73.39	59.54	91.20	<u>77.60</u>	
			BIOT-2D	91.88	90.79	69.47	91.03	98.44	71.46	97.87	72.08	97.56	70.19	90.58	79.39
			BIOT-6D	93.74	91.33	62.81	90.72	97.45	61.84	99.17	80.13	97.37	62.72	92.10	75.87
			LaBraM	92.29	88.15	74.41	80.66	57.29	64.97	76.06	52.17	88.28	88.12	98.01	60.15
			Neuro-GPT	95.83	79.16	93.39	79.76	71.20	70.61	94.51	72.06	88.13	77.61	99.59	55.63
			EEGPT	94.81	80.71	87.60	78.83	55.21	79.92	87.74	86.32	86.77	85.13	95.89	57.72
			CBraMod	97.95	90.81	89.81	90.56	96.02	84.76	95.73	93.18	89.75	99.27	96.60	66.89
			TFM	47.32	76.34	64.40	63.69	81.11	58.40	80.31	47.82	55.14	75.19	70.56	57.49
			EEGMamba	93.51	89.46	86.45	87.72	88.37	62.88	92.60	87.38	94.52	89.55	96.26	73.41
			SingLEM	52.74	50.77	51.07	50.80	42.11	45.02	62.60	55.24	48.86	49.90	49.25	49.66
			LUNA-Base	75.69	93.80	80.35	96.30	92.13	62.60	91.93	80.79	80.94	78.13	95.60	64.80
			LUNA-Large	76.19	95.50	93.44	91.86	93.76	66.75	90.99	77.86	77.15	74.12	91.86	63.35
			LUNA-Huge	91.59	<u>98.15</u>	97.13	95.18	92.54	59.66	92.06	94.27	68.69	76.78	97.02	68.87
Within-subject (Few-shot)		Specialist Models	PSD+SVM	98.58	94.59	98.79	82.77	99.57	93.77	<u>97.76</u>	98.34	98.83	99.78	96.81	59.56
			EEGNet	<u>99.81</u>	96.52	99.85	<u>97.02</u>	99.91	53.09	96.43	98.94	99.01	99.43	<u>99.72</u>	67.67
			ShallowConv	98.34	95.74	99.85	87.44	99.71	70.89	95.90	98.35	97.65	98.12	99.45	62.96
			LMDA	99.47	95.94	99.30	84.72	99.82	77.09	99.29	99.27	98.13	97.58	98.74	67.15
			CNN-T	99.35	96.64	97.82	90.82	99.61	97.18	93.42	99.43	98.34	99.81	99.57	78.58
			Deformer	99.26	96.78	<u>99.61</u>	95.38	<u>99.90</u>	51.78	96.72	99.40	98.80	<u>99.92</u>	99.87	57.90
			Conformer	99.37	99.58	99.55	98.13	98.48	62.42	93.66	98.56	97.92	99.82	98.30	<u>80.79</u>
			TSception	99.25	<u>97.79</u>	92.37	83.42	99.91	99.26	94.05	99.86	99.35	99.93	99.67	68.43
	Full Fine-tuning	Foundation Models	BENDR	53.76	68.04	61.26	57.05	64.08	51.19	81.93	95.17	48.64	53.93	53.94	49.82
			BIOT-1D	84.41	79.83	81.42	48.74	92.54	60.33	85.58	93.77	95.19	89.86	75.33	64.87
			BIOT-2D	82.72	89.11	83.49	57.08	94.78	57.83	87.70	94.62	94.60	88.63	82.10	64.03
			BIOT-6D	92.02	50.00	85.71	40.45	94.53	<u>98.94</u>	91.07	97.56	94.03	93.47	96.42	70.11
			LaBraM	98.34	93.17	65.99	56.11	91.27	84.01	93.56	99.66	81.83	99.48	97.39	62.90
			Neuro-GPT	99.48	95.26	94.37	55.86	96.54	91.80	90.82	<u>99.83</u>	94.39	98.98	99.11	66.33
			EEGPT	86.93	84.66	61.10	61.43	92.47	70.21	72.78	97.38	60.24	93.93	67.69	51.36
			CBraMod	99.67	93.69	88.07	45.04	99.57	98.39	96.89	99.65	98.76	99.49	99.54	81.17
			TFM	55.04	59.21	76.23	60.82	73.79	64.11	74.19	69.25	57.86	70.40	59.36	49.02
			EEGMamba	98.92	95.98	86.95	53.01	98.27	94.27	89.08	98.74	83.53	96.83	72.48	62.98
			SingLEM	53.19	82.59	53.09	53.18	53.13	53.98	65.12	98.40	52.43	55.54	51.80	51.52
			LUNA-Base	94.89	88.20	89.20	64.80	85.31	47.45	92.95	98.36	69.39	78.08	79.28	63.65
			LUNA-Large	94.75	95.88	94.73	63.40	86.30	47.78	89.05	98.48	63.36	83.66	94.50	64.34
			LUNA-Huge	98.06	95.88	97.83	74.12	81.84	49.98	90.26	99.07	65.69	83.03	83.98	63.70
	Linear Probing	Foundation Models	BENDR	53.28	49.62	50.83	48.64	49.56	48.41	48.93	50.91	49.45	49.08	48.03	49.88
			BIOT-1D	72.43	66.81	73.87	51.12	76.25	82.85	74.89	95.34	80.82	89.01	71.27	63.15
			BIOT-2D	90.86	65.95	72.71	73.66	76.86	87.93	67.18	95.52	83.04	83.92	91.90	63.39
			BIOT-6D	98.00	50.00	85.89	44.89	85.59	95.38	78.63	97.77	94.92	88.80	81.32	70.13
			LaBraM	96.60	95.70	67.80	54.56	94.17	76.81	90.01	98.61	81.62	97.60	97.21	62.54
			Neuro-GPT	88.61	88.53	79.21	40.76	88.51	78.41	85.02	99.68	84.00	92.01	72.17	56.71
			EEGPT	87.74	88.75	60.77	62.50	92.90	78.64	74.25	97.52	60.32	95.79	79.10	48.54
			CBraMod	99.89	93.35	91.55	33.65	99.05	97.50	96.67	99.32	<u>99.07</u>	98.93	99.07	79.98
			TFM	45.89	51.30	70.16	50.16	61.08	58.58	70.24	55.01	49.21	61.55	55.29	49.03
			EEGMamba	99.33	92.52	85.89	50.78	98.59	95.78	89.06	99.78	92.38	97.51	89.90	72.53
			SingLEM	52.46	62.35	47.89	51.50	50.44	55.79	54.21	52.94	50.65	50.56	53.99	50.41
			LUNA-Base	77.55	94.93	74.12	48.65	80.25	59.35	84.25	98.37	60.27	80.43	73.09	69.08
			LUNA-Large	98.33	95.96	95.28	69.36	87.05	42.18	86.47	97.78	65.36	79.38	90.45	69.34
			LUNA-Huge	94.98	96.37	84.74	54.79	77.99	55.12	85.78	97.90	61.12	82.75	71.38	70.15

Table 27. Classification AUCs (%) on CHB-MIT. The best AUCs are marked in bold, and the second best by an underline (continued).

Scenario	Tuning	Model Type	Approach	S12	S13	S14	S15	S16	S17	S18	S19	S20	S21	S22	Avg.
Cross-subject (LOSO)		Specialist Models	PSD+SVM	68.26	73.36	39.34	81.41	62.20	15.81	92.62	57.13	56.00	97.75	91.97	76.51
			EEGNet	59.13	7.95	83.28	84.11	97.47	<u>93.76</u>	92.25	92.38	83.81	91.65	98.38	85.44 _{±0.97}
			ShallowConv	63.10	8.07	79.66	85.00	99.49	95.08	100.00	95.95	95.70	99.74	99.33	88.79 _{±0.12}
			LMDA	75.46	18.44	69.35	78.94	97.12	85.97	99.87	82.07	84.44	97.61	99.50	86.26 _{±0.01}
			CNN-T	79.56	56.84	47.28	82.91	90.73	87.12	99.47	84.34	95.06	96.58	<u>99.40</u>	87.58 _{±0.21}
			Deformer	64.15	25.06	88.04	84.77	<u>99.14</u>	82.23	<u>99.96</u>	92.32	92.61	99.58	96.10	<u>88.63</u> _{±0.53}
			Conformer	67.74	8.40	73.25	76.97	98.65	87.60	98.93	93.77	83.62	99.63	98.24	86.43 _{±0.39}
			TSception	70.35	14.96	83.26	91.39	98.98	91.77	99.66	77.70	96.51	<u>99.78</u>	<u>99.40</u>	86.68 _{±0.22}
	Full Fine-tuning	Foundation Models	BENDR	63.68	32.12	85.02	77.61	98.98	93.09	99.37	88.40	94.60	97.52	97.90	87.43 _{±0.07}
			BIOT-1D	71.83	15.65	75.93	80.56	97.47	72.23	97.61	66.91	86.82	96.98	95.76	82.89 _{±0.38}
			BIOT-2D	70.89	14.79	76.99	79.73	98.32	73.03	98.67	64.43	95.32	98.98	95.96	83.59 _{±0.29}
			BIOT-6D	73.96	13.90	67.90	85.80	97.53	76.55	97.39	68.56	92.92	99.00	97.66	84.08 _{±0.12}
			LaBraM	75.72	35.40	82.94	84.35	95.49	71.33	95.06	74.66	83.18	99.27	95.41	84.82 _{±0.27}
			Neuro-GPT	66.13	31.27	85.30	71.79	92.15	91.42	95.93	72.97	92.43	99.18	96.17	82.83 _{±0.16}
			EEGPT	71.10	34.81	76.17	60.93	90.35	88.34	88.96	71.69	74.35	88.14	80.28	75.53 _{±1.72}
			CBraMod	74.84	19.72	67.06	88.07	97.02	84.66	97.76	80.27	<u>97.02</u>	99.84	97.37	87.43 _{±0.51}
			TFM	65.60	41.65	65.14	50.38	80.20	65.13	75.32	51.56	57.36	87.37	74.74	69.16 _{±0.82}
			EEGMamba	61.15	31.59	85.91	85.54	94.49	92.53	99.63	84.35	86.69	96.74	98.72	87.54 _{±0.16}
			SingLEM	55.28	50.81	72.68	53.21	73.99	56.72	57.71	48.51	65.38	88.56	82.37	67.13 _{±1.99}
			LUNA-Base	68.04	39.11	69.01	<u>88.97</u>	98.95	92.07	99.80	69.00	94.73	97.07	98.60	86.99 _{±0.62}
			LUNA-Large	62.97	41.05	83.61	82.67	98.94	89.75	99.29	68.96	97.56	95.73	98.70	87.21 _{±0.44}
			LUNA-Huge	57.43	35.44	67.08	84.21	96.96	92.78	98.57	63.34	95.99	94.48	99.19	85.62 _{±0.69}
	Linear Probing	Foundation Models	BENDR	53.41	51.43	52.23	53.63	57.24	52.30	60.38	47.84	53.59	55.26	55.99	54.32 _{±0.66}
			BIOT-1D	84.62	8.22	72.19	74.48	85.32	73.19	99.31	75.78	70.17	93.72	90.13	79.13 _{±0.29}
			BIOT-2D	<u>83.30</u>	8.54	68.92	72.19	83.49	80.38	99.41	79.25	49.52	95.87	93.69	79.80 _{±1.20}
			BIOT-6D	80.09	9.96	62.18	78.07	85.66	79.55	99.75	81.58	79.34	94.93	83.41	79.99 _{±1.11}
			LaBraM	70.92	<u>58.04</u>	76.62	78.26	90.36	57.60	92.27	85.35	78.85	97.37	92.73	78.21 _{±0.13}
			Neuro-GPT	70.81	25.74	82.76	61.37	93.88	91.34	99.28	79.47	96.54	98.48	95.69	81.43 _{±0.57}
			EEGPT	77.51	41.70	80.12	72.23	93.71	88.10	95.90	80.21	79.63	93.93	81.32	80.91 _{±1.09}
			CBraMod	68.58	13.82	81.13	82.68	96.59	77.46	95.80	74.47	89.88	95.08	93.98	85.25 _{±0.26}
			TFM	43.87	33.47	66.93	51.13	71.34	53.24	71.83	38.47	53.35	65.81	50.06	59.88 _{±0.55}
			EEGMamba	53.85	47.82	86.12	68.98	97.42	76.34	99.60	<u>95.36</u>	84.50	93.23	94.45	84.34 _{±0.03}
			SingLEM	49.99	43.31	48.21	47.28	61.54	50.78	40.26	49.01	53.52	60.32	48.98	50.49 _{±0.20}
			LUNA-Base	65.82	22.63	82.27	72.67	88.55	85.59	97.58	58.69	83.77	90.36	90.67	79.64 _{±0.11}
			LUNA-Large	57.92	19.95	<u>86.45</u>	74.95	77.47	85.99	80.45	70.31	81.33	91.26	86.80	78.51 _{±0.03}
			LUNA-Huge	67.72	28.91	79.75	82.25	96.64	90.89	99.80	70.85	90.63	96.21	95.07	83.94 _{±0.05}
Within-subject (Few-shot)		Specialist Models	PSD+SVM	96.33	69.40	91.55	75.86	95.94	85.20	99.67	93.67	98.83	99.17	97.96	92.29 _{±0.02}
			EEGNet	96.93	83.43	97.17	75.83	<u>99.51</u>	94.10	100.00	98.53	99.90	99.93	99.79	93.59 _{±0.69}
			ShallowConv	97.63	81.17	91.85	69.77	99.33	93.63	99.74	93.87	99.88	99.14	99.52	92.61 _{±0.25}
			LMDA	95.30	84.65	95.69	76.04	98.77	93.21	100.00	99.48	99.76	<u>99.94</u>	99.66	93.87 _{±0.31}
			CNN-T	93.80	73.78	91.16	83.92	97.18	88.56	100.00	99.05	99.88	94.91	99.68	<u>94.46</u> _{±0.43}
			Deformer	96.64	89.42	95.46	55.01	99.33	92.67	<u>99.99</u>	97.70	99.79	99.96	99.50	92.21 _{±0.41}
			Conformer	95.00	87.37	96.09	78.66	99.43	93.22	100.00	92.58	99.59	99.91	98.77	94.21 _{±0.79}
			TSception	<u>96.98</u>	88.76	93.77	83.86	98.65	94.50	100.00	98.08	99.92	99.92	<u>99.74</u>	95.11 _{±0.13}
	Full Fine-tuning	Foundation Models	BENDR	76.86	55.84	52.49	54.21	77.27	54.16	74.81	51.31	43.70	81.53	73.46	62.37 _{±0.37}
			BIOT-1D	54.45	91.30	85.69	83.72	87.19	77.66	99.56	97.15	91.29	90.90	90.31	82.66 _{±0.15}
			BIOT-2D	53.43	92.11	81.47	77.97	91.98	73.28	99.78	95.85	91.34	94.74	92.33	83.52 _{±1.04}
			BIOT-6D	85.87	93.03	50.00	89.78	96.56	81.85	97.00	50.00	98.09	98.89	97.68	84.48 _{±0.30}
			LaBraM	90.70	<u>97.67</u>	94.16	77.93	90.44	88.25	100.00	98.62	88.15	97.46	98.29	88.93 _{±0.39}
			Neuro-GPT	88.55	98.17	<u>96.39</u>	<u>88.90</u>	99.46	96.33	99.98	99.62	99.56	99.20	99.66	93.42 _{±0.52}
			EEGPT	82.33	94.76	76.12	86.34	80.22	93.63	94.67	78.40	71.69	97.28	95.42	80.48 _{±0.28}
			CBraMod	92.69	96.61	94.40	85.43	99.60	90.00	100.00	<u>99.69</u>	<u>99.91</u>	99.79	99.29	93.80 _{±0.38}
			TFM	58.01	62.15	61.50	57.19	74.44	60.04	81.00	66.93	65.13	75.84	59.96	64.85 _{±1.19}
			EEGMamba	84.82	96.39	81.14	86.08	90.32	86.04	100.00	99.64	96.64	90.19	95.41	88.60 _{±0.52}
			SingLEM	74.59	62.04	60.38	57.25	55.75	47.44	42.17	49.21	59.10	87.34	81.66	60.91 _{±2.32}
			LUNA-Base	63.97	78.91	71.15	81.41	90.96	88.23	99.94	96.92	92.46	82.64	94.11	82.27 _{±0.58}
			LUNA-Large	67.68	90.63	85.48	84.62	89.57	84.62	99.78	96.04	92.29	84.98	98.54	84.80 _{±0.18}
			LUNA-Huge	65.51	88.30	87.24	84.63	94.44	89.29	99.86	96.68	92.61	89.89	98.98	85.69 _{±0.59}
	Linear Probing	Foundation Models	BENDR	50.23	50.70	50.30	59.10	53.22	51.39	47.61	51.77	57.71	50.76	51.00	50.89 _{±1.07}
			BIOT-1D	52.84	92.24	74.65	82.29	87.78	66.84	61.33	92.25	88.98	88.06	90.63	77.20 _{±0.38}
			BIOT-2D	53.74	96.03	67.21	87.15	88.40	62.90	81.00	93.87	88.46	93.13	90.91	80.68 _{±0.30}
			BIOT-6D	83.09	93.82	50.00	83.90	96.70	68.42	98.78	50.00	94.59	97.24	93.89	81.82 _{±0.88}
			LaBraM	90.38	95.95	88.68	75.47	89.19	82.41	99.81	96.59	83.43	96.64	96.82	87.33 _{±0.23}
			Neuro-GPT	78.44	95.31	92.23	71.54	87.41	87.16	98.73	93.40	90.63	89.02	93.54	83.96 _{±1.21}
			EEGPT	84.79	96.39	76.34	85.03	82.38	<u>94.73</u>	96.83	83.24	79.61	97.98	97.04	82.66 _{±0.79}
			CBraMod	92.00	97.24	94.87	86.58	99.49	89.31	100.00	99.73	99.74	99.72	99.06	93.29 _{±0.27}
			TFM	52.49	48.71	53.03	50.75	68.51	44.51	51.32	61.30	58.84	59.34	55.39	55.73 _{±0.29}
			EEGMamba	90.79	93.02	95.29	88.34	90.05	86.23	100.00	99.57	98.60	93.04	96.83	91.12 _{±0.17}
			SingLEM	54.09	46.60	51.30	58.78	54.67	40.81	39.92	47.19	49.12	54.19	58.32	51.66 _{±0.52}
			LUNA-Base	66.57	71.07	82.08	77.34	91.05	85.86	98.77	94.58	92.73	75.94	74.93	78.75 _{±0.43}
			LUNA-Large	57.34	88.85	87.79	72.83	90.76	80.51	97.90	98.08	91.10	84.48	92.89	83.45 _{±0.20}
			LUNA-Huge	67.43	80.55	82.59	76.63	87.04	90.38	100.00	89.56	92.84	80.20	78.07	80.80 _{±0.19}

Table 28. Balanced classification accuracies (%) on CHB-MIT. The best BCAs are marked in bold, and the second best by an underline.

Scenario	Tuning	Model Type	Approach	S0	S1	S2	S3	S4	S5	S6	S7	S8	S9	S10	S11
Cross-subject (LOSO)	Specialist Models	Specialist Models	PSD+SVM	83.80	79.44	74.27	64.11	91.05	49.18	87.97	84.67	96.18	89.68	93.17	51.31
			EEGNet	96.15	92.11	93.94	<u>91.62</u>	61.96	69.03	88.62	72.41	75.50	95.05	94.92	52.97
			ShallowConv	97.66	93.91	95.79	89.89	61.29	68.82	93.27	82.61	81.86	<u>93.33</u>	93.44	52.26
			LMDA	98.31	93.15	91.70	87.03	67.41	72.49	84.68	76.37	65.53	89.04	88.50	56.30
			CNN-T	92.69	77.10	84.21	87.60	81.60	<u>78.76</u>	89.43	73.36	74.23	88.74	92.49	55.71
			Deformer	<u>98.18</u>	93.62	82.27	92.27	66.37	69.06	91.90	78.01	65.28	92.26	90.50	62.41
			Conformer	95.97	92.37	94.42	90.89	64.39	65.20	90.40	86.45	65.36	88.16	84.17	58.91
			TSception	93.49	68.67	92.68	90.35	69.01	54.74	89.73	72.69	73.25	91.38	90.73	51.00
	Full Fine-tuning	Foundation Models	BENDR	96.07	89.98	78.67	89.45	58.40	68.98	92.05	66.06	73.98	88.54	<u>95.97</u>	52.85
			BIOT-1D	95.13	92.66	84.23	82.29	<u>83.14</u>	69.74	86.32	70.58	50.02	58.75	89.93	62.42
			BIOT-2D	94.81	92.97	85.87	82.09	72.35	69.52	87.63	71.52	49.55	53.61	91.10	63.95
			BIOT-6D	93.93	<u>94.48</u>	88.06	83.23	77.96	65.00	<u>93.16</u>	74.04	51.74	63.94	91.76	63.23
			LaBraM	82.99	87.90	75.55	79.82	66.49	56.23	78.67	58.77	81.98	81.58	96.37	54.74
			Neuro-GPT	92.16	80.79	82.30	84.46	58.62	52.46	88.80	59.54	<u>82.21</u>	72.32	85.59	51.73
			EEGPT	80.62	75.67	61.96	68.62	48.06	62.70	75.41	68.25	73.45	62.44	84.84	55.70
			CBraMod	87.52	84.25	78.73	87.87	82.14	66.90	90.43	73.33	68.42	87.55	89.74	62.01
			TFM	68.67	76.22	60.27	74.65	69.98	62.86	78.78	62.46	62.90	50.73	72.20	52.66
			EEGMamba	92.03	85.59	90.79	84.62	75.15	71.44	81.23	86.87	63.50	61.82	88.02	54.04
			SingLEM	72.05	76.01	58.94	64.11	54.97	55.70	70.74	67.29	63.90	48.13	69.17	51.86
			LUNA-Base	90.29	89.19	<u>95.65</u>	81.01	73.98	78.78	90.50	86.84	56.53	74.93	85.73	51.18
			LUNA-Large	92.80	91.48	92.18	83.40	76.08	75.76	89.69	85.24	55.75	61.71	87.51	52.93
			LUNA-Huge	89.77	93.34	91.51	80.89	75.04	72.13	90.61	85.76	56.48	75.19	88.89	54.03
	Linear Probing	Foundation Models	BENDR	50.97	55.34	51.59	55.19	54.31	49.61	55.77	52.89	54.93	52.34	55.94	50.62
			BIOT-1D	87.10	84.14	72.24	83.05	65.92	57.80	75.00	65.18	50.67	47.85	79.45	67.90
			BIOT-2D	81.54	84.17	73.25	85.10	72.03	62.77	75.34	65.55	50.17	47.61	82.67	<u>68.53</u>
			BIOT-6D	85.78	87.80	68.50	84.14	80.38	60.43	74.07	70.26	49.86	47.56	79.93	68.38
			LaBraM	76.69	81.85	73.29	73.65	54.99	60.14	67.12	52.01	77.56	73.23	79.17	52.97
			Neuro-GPT	88.11	72.28	76.64	72.84	65.41	52.87	83.54	57.58	79.13	69.78	92.85	51.33
			EEGPT	87.92	74.02	73.92	65.67	54.19	71.95	80.38	71.99	78.37	75.48	88.65	52.99
			CBraMod	96.26	86.69	86.91	85.63	74.08	73.41	86.23	80.12	58.13	84.60	76.93	61.03
			TFM	46.55	72.25	54.36	59.03	75.62	55.27	72.93	49.54	52.17	69.43	61.61	58.55
			EEGMamba	86.62	82.99	68.37	80.77	76.49	59.60	84.86	77.01	61.75	69.53	76.88	61.92
			SingLEM	52.27	50.25	50.76	50.85	45.58	45.80	57.44	55.36	49.32	49.81	51.08	50.24
			LUNA-Base	70.11	84.40	59.97	70.97	71.00	58.60	89.25	67.99	55.20	54.78	88.85	66.85
			LUNA-Large	72.25	84.52	72.84	74.07	65.79	55.45	87.29	70.42	52.95	50.75	83.63	64.65
			LUNA-Huge	85.62	94.78	63.81	82.43	69.33	55.28	89.18	86.51	49.53	48.86	92.52	70.39
Within-subject (Few-shot)	Specialist Models	Specialist Models	PSD+SVM	93.97	82.81	82.67	77.48	87.12	52.94	87.42	91.42	<u>95.21</u>	96.48	51.74	55.22
			EEGNet	90.59	96.22	<u>98.79</u>	84.66	97.01	49.54	93.05	95.08	95.02	<u>98.16</u>	<u>97.23</u>	68.02
			ShallowConv	81.89	92.12	98.96	77.22	95.89	52.63	90.68	91.91	93.06	90.21	93.07	58.79
			LMDA	88.59	94.17	96.92	76.72	96.38	49.99	<u>93.79</u>	96.46	92.95	95.30	94.24	63.08
			CNN-T	<u>95.66</u>	<u>96.63</u>	91.14	<u>85.64</u>	86.36	68.00	90.79	90.42	93.42	94.09	96.76	66.75
			Deformer	90.89	93.07	97.02	74.60	94.21	49.78	88.98	94.66	95.17	97.65	96.09	54.26
			Conformer	96.04	96.24	98.19	92.46	95.73	69.49	93.85	94.73	96.28	97.94	97.34	<u>71.35</u>
			TSception	90.98	97.39	88.81	78.77	98.47	95.48	92.05	95.21	92.13	98.63	94.49	72.82
	Full Fine-tuning	Foundation Models	BENDR	50.18	54.87	52.72	51.01	53.11	49.99	54.75	82.77	49.64	50.07	50.09	49.99
			BIOT-1D	83.85	74.81	80.69	62.32	91.72	55.43	85.75	92.26	94.80	88.22	72.30	62.69
			BIOT-2D	82.06	85.68	77.63	67.40	93.94	50.51	87.39	92.97	94.43	87.82	78.02	62.07
			BIOT-6D	86.35	50.00	80.90	48.12	94.14	<u>90.22</u>	90.30	91.46	92.94	86.33	92.63	67.36
			LaBraM	89.06	69.99	59.09	61.78	80.11	50.00	63.73	93.76	70.19	86.27	51.28	51.00
			Neuro-GPT	90.61	85.40	81.19	56.44	89.08	67.68	69.10	97.83	83.02	95.58	63.75	68.08
			EEGPT	77.72	57.30	50.49	62.97	84.20	49.95	58.49	<u>94.73</u>	58.51	76.71	50.00	50.39
			CBraMod	91.26	91.40	83.63	61.49	<u>98.41</u>	89.16	93.12	96.46	94.95	95.64	96.71	67.18
			TFM	51.67	56.76	70.01	57.10	68.68	52.43	69.25	63.71	55.24	64.10	53.91	49.83
			EEGMamba	89.97	89.77	77.92	55.27	94.09	62.38	83.44	96.28	72.58	88.84	68.37	57.43
			SingLEM	51.22	69.87	49.74	53.13	53.25	50.51	55.46	96.91	51.62	50.60	50.15	50.16
			LUNA-Base	81.92	77.14	78.15	56.84	79.01	50.00	88.80	95.85	67.22	72.54	50.09	54.14
			LUNA-Large	77.65	81.81	88.58	57.85	79.87	50.00	86.69	95.61	62.16	73.85	76.66	55.32
			LUNA-Huge	90.76	89.68	94.85	73.54	80.22	50.00	88.79	95.34	60.06	66.89	59.57	50.48
	Linear Probing	Foundation Models	BENDR	50.15	50.62	49.63	49.74	49.64	50.00	51.17	49.69	50.42	50.17	50.00	49.78
			BIOT-1D	68.54	66.10	72.99	56.03	76.01	69.68	79.29	92.44	78.08	85.54	66.44	57.67
			BIOT-2D	87.19	65.23	70.40	72.86	77.19	81.39	69.70	93.14	75.96	80.70	87.46	57.32
			BIOT-6D	91.53	50.00	77.66	61.26	78.34	86.58	79.48	92.96	84.46	82.97	80.58	58.80
			LaBraM	67.03	76.20	57.39	59.57	85.01	50.00	61.98	95.53	71.52	77.04	53.15	51.82
			Neuro-GPT	75.12	72.57	71.03	46.60	78.58	50.49	70.24	94.82	74.76	80.37	50.83	51.68
			EEGPT	78.97	60.86	50.76	64.15	83.27	50.46	60.99	94.98	60.02	81.63	50.00	50.30
			CBraMod	91.35	87.73	87.02	52.68	98.32	83.53	93.19	95.78	93.35	95.04	95.56	70.16
			TFM	47.83	48.42	63.07	49.74	57.68	50.52	64.20	52.96	50.73	58.08	50.50	49.17
			EEGMamba	95.21	81.44	78.26	55.34	93.23	81.08	79.96	96.79	77.01	90.65	71.41	63.98
			SingLEM	50.93	53.25	49.89	50.06	52.58	50.38	52.52	52.09	50.03	50.52	50.24	50.19
			LUNA-Base	72.77	86.10	66.68	42.88	72.23	49.83	81.11	96.62	62.58	74.73	57.18	63.19
			LUNA-Large	73.22	80.44	89.27	61.22	79.54	49.89	88.10	<u>97.04</u>	63.10	73.65	77.28	52.91
			LUNA-Huge	79.34	84.29	76.66	47.20	74.92	54.53	84.44	95.70	60.97	76.82	55.12	64.17

Table 29. Balanced classification accuracies (%) on CHB-MIT. The best BCAs are marked in bold, and the second best by an underline (continued).

Scenario	Tuning	Model Type	Approach	S12	S13	S14	S15	S16	S17	S18	S19	S20	S21	S22	Avg.
Cross-subject (LOSO)		Specialist Models	PSD+SVM	56.12	51.04	49.86	49.27	50.00	50.00	84.62	49.96	50.00	87.22	66.25	69.09 _{±0.04}
			EEGNet	54.37	30.62	54.02	75.54	94.87	86.18	82.05	75.56	57.77	90.78	93.24	77.36 _{±0.54}
			ShallowConv	54.55	38.57	69.35	75.29	91.68	86.05	95.30	76.33	64.94	95.19	92.68	80.18 _{±0.22}
			LMDA	61.56	39.29	58.09	74.02	88.82	79.03	96.15	69.55	56.69	92.44	95.52	77.47 _{±0.41}
			CNN-T	70.90	55.64	51.79	57.07	64.55	73.09	93.48	59.63	68.34	89.22	96.10	76.34 _{±0.61}
			Deformer	55.98	41.81	74.85	74.82	96.25	80.38	96.15	81.35	71.57	96.19	83.17	79.77 _{±0.14}
			Conformer	56.59	21.68	68.84	72.63	95.31	80.66	98.29	83.68	64.63	95.56	92.78	78.58 _{±0.56}
			TSception	57.01	45.34	52.22	67.60	89.22	78.90	85.83	62.07	53.81	93.70	87.00	74.37 _{±0.53}
	Full Fine-tuning	Foundation Models	BENDR	56.70	48.57	62.03	58.28	90.60	83.32	83.22	66.16	65.96	86.56	84.20	75.50 _{±0.93}
			BIOT-1D	61.58	35.64	63.36	69.40	87.59	70.51	90.94	71.80	64.26	89.74	90.98	74.83 _{±0.30}
			BIOT-2D	58.10	37.64	60.93	73.94	80.98	71.98	94.36	69.76	58.00	94.15	87.88	74.03 _{±0.47}
			BIOT-6D	62.23	37.74	55.77	72.12	76.85	71.03	91.91	71.12	57.94	95.30	88.98	74.85 _{±0.33}
			LaBraM	59.47	50.92	72.57	60.03	71.66	58.89	79.03	52.87	55.44	91.48	76.45	70.87 _{±0.59}
			Neuro-GPT	58.81	45.64	70.32	64.96	81.19	83.05	75.90	72.43	66.88	95.37	79.72	73.27 _{±0.27}
			EEGPT	63.01	42.65	67.15	54.61	76.75	74.61	68.75	63.55	57.37	79.89	72.82	66.91 _{±2.89}
			CBraMod	58.18	44.28	54.76	71.82	60.88	63.14	90.60	59.83	66.58	93.11	85.19	74.23 _{±0.19}
			TFM	61.89	44.32	61.89	47.14	68.85	59.67	71.44	53.00	52.12	78.89	67.94	63.46 _{±0.60}
			EEGMamba	52.26	43.64	76.33	78.58	83.41	70.16	91.66	75.38	54.09	92.33	93.32	75.92 _{±0.21}
			SingLEM	52.45	42.78	60.07	54.61	62.24	55.30	58.55	52.69	54.98	81.78	69.67	60.78 _{±1.82}
			LUNA-Base	61.62	52.89	63.44	79.78	94.30	79.55	91.03	63.06	70.98	91.41	94.20	78.12 _{±0.61}
			LUNA-Large	55.83	54.06	61.23	75.13	94.60	80.31	92.74	67.19	63.18	90.59	94.81	77.14 _{±0.14}
			LUNA-Huge	54.96	42.71	53.50	76.69	83.27	78.94	90.12	62.94	67.41	86.67	93.39	75.84 _{±0.81}
	Linear Probing	Foundation Models	BENDR	51.95	51.62	51.68	55.87	53.94	51.14	59.84	49.31	51.49	52.11	55.64	53.22 _{±0.68}
			BIOT-1D	67.33	19.24	61.33	62.06	77.11	65.91	94.92	73.52	58.20	86.74	81.69	68.88 _{±0.22}
			BIOT-2D	66.79	19.22	60.23	62.21	77.47	75.22	93.15	80.15	47.68	81.15	84.81	69.43 _{±0.52}
			BIOT-6D	56.87	24.85	52.78	66.40	78.24	73.37	93.68	80.90	55.73	87.37	77.86	69.79 _{±0.62}
			LaBraM	65.36	56.90	69.54	62.44	81.37	55.99	73.90	54.55	65.02	91.00	84.51	68.84 _{±0.36}
			Neuro-GPT	61.24	39.35	73.59	49.10	84.57	84.47	74.73	62.60	58.39	93.67	76.37	70.45 _{±0.24}
			EEGPT	67.50	48.01	72.61	59.64	84.60	81.87	65.05	65.41	56.95	82.67	71.72	70.94 _{±0.69}
			CBraMod	60.81	33.79	54.34	77.23	78.56	65.85	93.16	69.86	71.09	88.04	87.10	75.21 _{±0.52}
			TFM	46.44	41.11	62.13	47.62	58.64	52.52	66.76	45.70	52.52	59.15	47.10	56.83 _{±0.75}
			EEGMamba	50.99	48.51	65.25	63.26	86.79	65.32	88.24	67.19	61.27	86.41	81.55	71.81 _{±0.26}
			SingLEM	50.10	45.47	48.77	48.19	56.62	51.80	43.40	49.47	53.37	56.63	49.01	50.50 _{±0.11}
			LUNA-Base	63.78	31.50	50.69	53.68	70.40	71.90	90.42	65.27	60.91	83.07	71.19	67.43 _{±0.45}
			LUNA-Large	55.78	29.40	54.01	57.88	67.62	68.28	70.80	71.98	56.27	84.56	73.95	66.31 _{±0.27}
			LUNA-Huge	59.00	32.93	50.80	70.66	67.34	72.91	90.17	77.87	59.91	90.85	74.79	71.11 _{±0.88}
Within-subject (Few-shot)		Specialist Models	PSD+SVM	72.84	53.15	77.52	50.00	86.44	68.64	87.50	84.23	74.17	93.54	84.93	77.71 _{±0.35}
			EEGNet	87.90	70.06	91.11	52.91	96.50	86.13	95.83	95.03	99.44	97.52	98.59	88.45 _{±0.48}
			ShallowConv	90.18	78.94	81.99	53.10	96.94	86.10	97.56	85.88	97.98	92.17	96.28	85.81 _{±0.44}
			LMDA	85.23	65.05	89.46	50.00	95.22	86.23	100.00	94.58	94.90	99.31	97.83	86.80 _{±0.55}
			CNN-T	89.72	68.65	86.32	74.69	91.67	83.71	96.67	89.06	97.36	91.50	91.30	87.66 _{±0.33}
			Deformer	89.18	79.20	89.72	49.93	96.17	87.18	97.50	91.24	98.49	97.63	95.57	86.88 _{±0.33}
			Conformer	88.84	82.31	93.00	74.27	96.67	87.45	100.00	91.91	99.33	97.93	95.09	91.58 _{±0.34}
			TSception	89.93	81.02	81.83	65.88	95.50	88.44	99.78	96.88	99.63	98.52	98.54	90.92 _{±0.74}
	Full Fine-tuning	Foundation Models	BENDR	56.44	50.00	50.77	51.54	60.61	50.44	57.06	50.27	49.36	60.72	58.78	54.14 _{±0.33}
			BIOT-1D	54.03	89.25	77.61	75.91	82.33	75.80	99.56	95.82	88.08	88.72	88.76	80.90 _{±0.37}
			BIOT-2D	53.22	90.30	73.66	75.16	86.33	72.36	99.67	94.78	88.81	93.61	89.24	81.61 _{±0.95}
			BIOT-6D	76.14	68.66	50.00	78.04	91.44	77.56	87.44	50.00	92.63	96.87	91.24	79.60 _{±0.51}
			LaBraM	79.61	53.60	81.53	50.00	72.67	77.81	97.50	77.32	50.88	90.52	82.35	71.31 _{±0.29}
			Neuro-GPT	77.87	80.63	86.13	56.67	90.72	89.30	99.56	91.95	98.53	93.83	98.09	83.09 _{±0.59}
			EEGPT	73.68	54.50	59.71	50.00	63.83	60.27	76.67	50.49	49.96	76.46	78.91	63.74 _{±0.28}
			CBraMod	86.16	76.56	88.94	66.94	96.50	84.86	99.17	92.99	92.75	96.37	95.89	88.54 _{±0.16}
			TFM	55.78	55.30	57.39	49.93	67.72	55.30	73.78	58.52	59.83	69.15	54.30	59.55 _{±0.57}
			EEGMamba	58.35	79.47	74.28	70.99	79.89	77.94	99.06	92.23	90.11	81.87	88.07	79.50 _{±0.42}
			SingLEM	63.97	49.97	56.27	49.96	51.61	49.18	46.50	50.05	55.30	78.35	70.35	56.70 _{±1.49}
			LUNA-Base	52.65	51.80	65.62	53.96	85.00	83.68	82.89	88.49	83.20	75.37	88.48	72.30 _{±0.26}
			LUNA-Large	55.44	61.44	75.29	57.81	82.33	72.40	64.78	85.15	83.61	73.48	88.11	73.30 _{±1.66}
			LUNA-Huge	56.41	57.84	79.71	63.88	88.33	84.20	80.78	69.02	83.48	81.76	91.46	75.52 _{±0.92}
	Linear Probing	Foundation Models	BENDR	49.98	50.00	50.50	50.00	51.06	50.42	49.06	49.92	49.81	49.83	49.96	50.07 _{±0.18}
			BIOT-1D	52.60	80.41	66.32	77.41	76.72	69.65	55.44	89.17	86.72	84.44	85.70	73.63 _{±0.33}
			BIOT-2D	53.11	84.83	58.49	78.94	74.33	63.82	63.78	91.49	85.56	89.44	87.63	76.08 _{±0.48}
			BIOT-6D	58.04	74.08	50.00	73.09	83.83	65.05	53.22	50.00	87.78	82.26	86.56	73.41 _{±1.06}
			LaBraM	77.29	50.00	71.94	50.00	69.89	55.71	88.33	55.65	52.78	86.67	74.87	66.93 _{±0.88}
			Neuro-GPT	71.10	81.42	79.58	51.09	75.44	77.08	92.11	75.87	72.92	79.20	75.76	71.68 _{±1.57}
			EEGPT	71.44	66.12	53.47	52.20	73.44	69.48	85.00	54.47	50.40	85.44	86.44	66.71 _{±1.20}
			CBraMod	75.72	80.55	89.33	72.25	96.33	79.69	100.00	95.90	93.55	96.54	95.17	87.77 _{±0.73}
			TFM	52.53	50.95	51.50	50.31	61.61	44.13	52.33	55.47	53.97	56.20	54.41	53.32 _{±0.37}
			EEGMamba	69.41	77.32	86.65	80.53	81.94	76.84	100.00	88.06	94.66	79.78	90.57	82.18 _{±0.17}
			SingLEM	54.63	50.07	51.88	49.77	52.78	50.24	45.33	50.06	50.70	51.50	54.89	51.07 _{±0.11}
			LUNA-Base	53.68	62.17	64.11	66.69	80.94	77.06	97.39	80.67	86.55	71.28	69.15	71.11 _{±0.20}
			LUNA-Large	53.99	70.07	77.90	58.85	82.78	74.70	54.11	92.74	85.01	74.78	84.43	73.70 _{±0.41}
			LUNA-Huge	53.27	70.92	64.94	66.86	77.39	80.23	95.78	78.41	85.88	73.37	72.65	72.78 _{±0.30}

Table 30. Classification accuracies (%) on TUAB. The best accuracies are marked in bold, and the second best by an underline.

Scenario	Tuning	Model Type	Approach	S0	S1	S2	S3	S4	S5	S6	S7	S8	S9	Avg.
Cross-subject (LOSO)		Specialist Models	CSP+LDA	67.01	65.16	67.06	65.41	65.19	63.65	68.40	65.69	66.18	66.58	66.03 $\pm$ 0.25
			EEGNet	75.88	78.02	75.68	77.91	77.16	79.49	76.19	77.17	77.54	75.23	77.03 $\pm$ 0.67
			ShallowConv	80.01	79.37	79.00	79.34	80.04	80.58	79.94	79.67	79.91	80.19	79.81 $\pm$ 0.12
			LMDA	55.08	63.90	65.58	55.30	70.41	60.98	67.43	53.15	61.63	73.46	62.69 $\pm$ 1.52
			CNN-T	75.73	77.69	71.97	68.19	73.36	75.70	73.68	68.20	75.16	72.29	73.20 $\pm$ 0.73
			Deformer	<u>82.61</u>	81.94	80.53	79.54	81.01	<u>82.45</u>	<u>82.11</u>	<u>82.14</u>	<u>81.39</u>	81.10	<u>81.48</u> $\pm$ 0.21
			Conformer	78.26	78.34	77.25	77.56	77.95	77.85	77.48	77.71	77.11	78.29	77.78 $\pm$ 0.04
			TSception	80.28	<u>81.75</u>	77.74	78.15	79.41	79.38	80.02	76.32	79.57	79.92	79.26 $\pm$ 0.49
	Full Fine-tuning	Foundation Models	BENDR	78.78	78.45	78.43	79.67	78.81	79.28	79.42	79.44	79.17	79.47	79.09 $\pm$ 0.16
			BIOT-1D	78.79	80.17	78.99	79.51	77.58	78.85	78.98	79.14	78.87	78.35	78.92 $\pm$ 0.16
			BIOT-2D	78.47	78.87	77.94	77.90	76.68	78.08	77.51	78.76	78.01	77.43	77.97 $\pm$ 0.10
			BIOT-6D	78.33	79.39	77.43	76.97	76.98	78.47	77.59	78.87	77.58	77.38	77.90 $\pm$ 0.14
			LaBraM	75.85	76.55	76.26	75.85	76.64	76.57	76.12	76.34	76.05	76.09	76.23 $\pm$ 0.27
			Neuro-GPT	79.28	79.78	79.64	78.66	79.69	80.40	79.46	80.61	78.95	78.52	79.50 $\pm$ 0.17
			EEGPT	76.73	77.90	77.95	78.49	77.39	79.23	76.97	78.21	77.58	76.26	77.67 $\pm$ 0.17
			CBraMod	80.14	80.70	79.93	78.99	79.87	80.33	79.68	80.28	80.29	79.59	79.98 $\pm$ 0.11
			TFM	74.76	76.97	74.90	75.45	75.44	76.29	75.92	75.60	75.76	75.36	75.65 $\pm$ 0.07
			BrainOmni-Tiny	70.69	74.42	69.65	73.90	73.17	75.88	72.61	74.50	71.60	72.79	72.92 $\pm$ 0.25
			BrainOmni-Base	81.17	81.31	78.52	79.33	<u>81.24</u>	82.21	81.58	77.69	80.51	<u>81.30</u>	80.49 $\pm$ 0.29
			EEGMamba	80.44	80.49	<u>80.81</u>	81.19	81.18	81.96	81.23	81.23	80.61	79.81	80.90 $\pm$ 0.10
			SingLEM	49.49	49.21	50.81	51.85	51.06	52.01	48.64	55.79	49.69	49.49	50.80 $\pm$ 1.02
			LUNA-Base	83.15	81.29	81.68	<u>80.42</u>	82.34	82.69	82.34	82.25	81.47	81.55	81.92 $\pm$ 0.07
			LUNA-Large	80.12	81.17	79.70	78.66	80.00	81.91	80.07	80.40	80.38	79.82	80.22 $\pm$ 0.17
			LUNA-Huge	81.53	80.45	80.13	80.00	80.98	81.79	80.90	81.54	80.44	80.97	80.87 $\pm$ 0.24
	Linear Probing	Foundation Models	BENDR	57.77	59.62	58.21	57.96	58.64	58.32	58.31	57.84	58.43	58.26	58.33 $\pm$ 0.66
			BIOT-1D	70.15	71.61	69.34	72.68	70.05	73.03	70.97	69.09	72.66	70.55	71.01 $\pm$ 1.52
			BIOT-2D	71.32	72.42	68.89	73.05	70.70	71.57	71.59	72.71	71.93	71.26	71.54 $\pm$ 0.78
			BIOT-6D	72.42	75.65	73.35	73.52	71.70	74.64	73.42	74.48	74.29	71.09	73.46 $\pm$ 0.25
			LaBraM	77.90	78.34	77.82	78.93	78.75	78.78	78.80	78.55	77.85	77.48	78.32 $\pm$ 0.11
			Neuro-GPT	79.61	79.90	79.43	79.05	79.73	80.22	79.25	80.91	78.91	79.40	79.64 $\pm$ 0.15
			EEGPT	76.41	77.90	78.00	77.88	77.81	78.83	77.31	77.26	77.31	76.43	77.51 $\pm$ 0.09
			CBraMod	78.62	79.14	76.99	77.92	77.31	78.26	78.36	78.38	77.51	77.94	78.04 $\pm$ 0.04
			TFM	66.08	68.84	69.24	67.93	68.99	69.25	68.68	67.91	67.99	67.58	68.25 $\pm$ 0.09
			BrainOmni-Tiny	76.74	77.79	76.61	76.24	77.33	78.15	76.74	77.53	76.57	76.56	77.03 $\pm$ 0.37
			BrainOmni-Base	79.48	77.44	79.51	79.69	80.01	80.80	79.45	78.04	79.39	79.36	79.32 $\pm$ 0.30
			EEGMamba	78.21	78.86	77.43	77.75	78.48	79.00	78.95	78.48	77.50	77.71	78.24 $\pm$ 0.04
			SingLEM	51.57	51.64	51.82	51.63	51.43	51.84	51.36	51.76	51.13	51.11	51.53 $\pm$ 0.13
			LUNA-Base	75.12	75.76	76.09	75.81	76.54	77.85	74.75	76.33	75.75	74.77	75.88 $\pm$ 0.17
			LUNA-Large	74.98	76.50	75.10	75.08	76.00	77.77	73.89	77.31	75.55	75.03	75.72 $\pm$ 0.07
			LUNA-Huge	75.53	75.76	75.55	75.04	76.71	77.70	74.52	77.36	75.96	75.06	75.92 $\pm$ 0.07

Table 31. Cohen's Kappa (%) on TUAB. The best metrics are marked in bold, and the second best by an underline.

Scenario	Tuning	Model Type	Approach	S0	S1	S2	S3	S4	S5	S6	S7	S8	S9	Avg.
Cross-subject (LOSO)		Specialist Models	CSP+LDA	33.95	29.97	34.34	31.04	30.14	27.16	36.63	30.71	32.16	32.80	31.89 $\pm$ 0.52
			EEGNet	51.73	55.97	51.53	55.97	54.28	58.67	52.50	54.22	55.06	50.76	54.07 $\pm$ 1.29
			ShallowConv	59.95	58.57	58.09	58.75	59.92	61.08	59.51	59.19	59.64	60.14	59.48 $\pm$ 0.24
			LMDA	9.59	26.14	31.61	11.37	39.51	20.50	33.66	0.74	22.81	47.21	24.31 $\pm$ 3.77
			CNN-T	51.32	54.80	44.72	37.22	46.31	51.00	46.44	34.39	50.03	44.05	46.03 $\pm$ 1.34
			Deformer	<u>65.17</u>	63.61	61.18	59.41	61.77	<u>64.83</u>	<u>63.94</u>	<u>63.86</u>	<u>62.70</u>	61.91	<u>62.84</u> $\pm$ 0.41
			Conformer	56.44	56.50	54.59	55.22	55.66	55.53	54.63	55.02	54.02	56.18	55.38 $\pm$ 0.09
			TSception	60.53	<u>63.40</u>	55.56	56.45	58.75	58.67	59.72	52.95	59.04	59.66	58.47 $\pm$ 0.96
	Full Fine-tuning	Foundation Models	BENDR	57.50	56.83	56.94	59.37	57.51	58.48	58.63	58.66	58.12	58.63	58.07 $\pm$ 0.33
			BIOT-1D	57.51	60.23	58.02	59.08	55.02	57.59	57.64	58.05	57.65	56.51	57.73 $\pm$ 0.32
			BIOT-2D	56.86	57.61	55.95	55.75	53.20	56.07	54.64	57.21	55.93	54.67	55.79 $\pm$ 0.19
			BIOT-6D	56.58	58.64	54.92	53.98	53.82	56.77	55.05	57.52	55.12	54.50	55.69 $\pm$ 0.25
			LaBraM	51.67	52.98	52.54	51.79	53.24	53.09	51.88	52.46	52.01	51.90	52.36 $\pm$ 0.52
			Neuro-GPT	58.46	59.23	59.39	57.53	59.13	60.62	58.46	60.80	57.69	56.52	58.78 $\pm$ 0.35
			EEGPT	53.37	55.54	55.96	57.08	54.59	58.40	53.59	56.14	55.04	52.40	55.21 $\pm$ 0.33
			CBraMod	60.22	61.23	59.94	58.11	59.54	60.47	58.95	60.21	60.47	58.86	59.80 $\pm$ 0.22
			TFM	49.44	53.72	49.91	50.92	50.63	52.47	51.54	50.75	51.33	50.37	51.11 $\pm$ 0.10
			BrainOmni-Tiny	41.18	49.15	39.59	47.67	46.33	51.48	44.87	48.45	43.18	44.35	45.62 $\pm$ 0.50
			BrainOmni-Base	62.37	62.30	57.23	58.43	<u>62.29</u>	64.28	62.66	55.7	60.66	<u>62.26</u>	60.82 $\pm$ 0.65
			EEGMamba	60.80	60.85	<u>61.69</u>	62.44	62.22	63.79	62.15	62.22	61.09	59.32	61.66 $\pm$ 0.21
			SingLEM	0.00	0.00	0.00	0.00	0.00	0.00	0.00	13.48	0.00	0.00	1.35 $\pm$ 1.86
			LUNA-Base	66.23	62.47	63.45	<u>60.86</u>	64.50	65.19	64.37	64.28	62.85	62.85	63.71 $\pm$ 0.16
			LUNA-Large	60.11	62.23	59.44	57.35	59.68	63.61	60.01	60.66	60.68	59.09	60.29 $\pm$ 0.37
			LUNA-Huge	63.06	60.83	60.33	60.04	61.88	63.49	61.49	62.91	60.77	61.49	61.63 $\pm$ 0.43
	Linear Probing	Foundation Models	BENDR	15.62	19.27	16.49	15.96	17.44	16.44	16.83	15.63	16.73	16.62	16.70 $\pm$ 1.32
			BIOT-1D	39.95	42.95	38.49	45.34	39.92	45.98	42.28	37.22	44.94	40.70	41.78 $\pm$ 3.30
			BIOT-2D	42.67	44.43	38.24	46.06	41.07	43.31	42.36	45.01	43.82	42.40	42.94 $\pm$ 1.31
			BIOT-6D	44.79	50.98	46.56	47.37	43.08	49.19	45.86	48.69	48.32	42.35	46.72 $\pm$ 0.60
			LaBraM	55.75	56.54	55.72	57.90	57.36	57.42	57.27	56.88	55.55	54.66	56.50 $\pm$ 0.22
			Neuro-GPT	59.14	59.61	58.95	58.18	59.24	60.31	58.13	61.47	57.59	58.34	59.10 $\pm$ 0.29
			EEGPT	52.67	55.66	56.08	55.83	55.50	57.76	54.56	54.56	54.45	52.87	54.99 $\pm$ 0.21
			CBraMod	57.19	58.12	54.05	55.94	54.45	56.41	56.36	56.51	54.85	55.49	55.94 $\pm$ 0.07
			TFM	31.98	37.25	38.61	36.19	37.43	38.31	36.89	35.03	35.65	34.43	36.18 $\pm$ 0.16
			BrainOmni-Tiny	53.39	55.43	53.26	52.77	54.57	56.24	53.22	54.72	52.90	52.79	53.93 $\pm$ 0.67
			BrainOmni-Base	58.98	55.11	59.04	59.31	59.89	61.51	58.29	56.15	58.53	58.58	58.54 $\pm$ 0.64
			EEGMamba	56.34	57.53	54.96	55.62	56.77	57.88	57.51	56.63	54.78	55.06	56.31 $\pm$ 0.08
			SingLEM	3.12	3.22	3.67	3.35	2.83	3.61	2.78	3.36	2.15	2.16	3.03 $\pm$ 0.25
			LUNA-Base	50.19	51.59	52.16	51.61	52.90	55.62	49.10	52.58	51.32	49.21	51.63 $\pm$ 0.29
			LUNA-Large	49.85	52.83	50.32	50.35	51.77	55.37	47.30	54.14	50.91	49.68	51.25 $\pm$ 0.16
			LUNA-Huge	50.91	51.49	51.11	50.29	53.07	55.21	48.59	54.43	51.88	50.04	51.70 $\pm$ 0.11

Table 32. Cohen's Kappa (%) on Sleep-EDFx. The best metrics are marked in bold, and the second best by an underline.

Scenario	Tuning	Model Type	Approach	S0	S1	S2	S3	S4	S5	S6	S7	S8	S9	Avg.
Cross-subject (LOSO)		Specialist Models	PSD+LDA	56.10	59.27	63.88	56.79	57.29	57.88	61.43	61.09	54.35	60.02	58.81 _{±0.18}
			EEGNet	63.53	64.88	67.99	64.09	57.62	59.65	66.20	66.81	63.21	69.66	64.36 _{±0.46}
			ShallowConv	68.04	73.86	73.79	70.82	70.45	74.03	69.73	73.05	69.78	74.67	71.82 _{±0.32}
			LMDA	68.32	71.47	73.13	67.84	67.83	71.82	69.69	72.17	65.08	71.87	69.92 _{±0.12}
			CNN-T	67.52	72.69	73.21	69.17	70.71	70.47	69.42	73.14	66.21	72.74	70.53 _{±0.29}
			Deformer	<u>72.88</u>	<u>76.44</u>	<u>78.41</u>	<u>73.18</u>	<u>72.55</u>	<u>76.14</u>	<u>74.63</u>	<u>75.42</u>	71.62	<u>77.20</u>	<u>74.85</u> _{±0.23}
			Conformer	59.78	68.40	66.12	59.30	58.51	63.49	60.56	66.83	54.80	67.53	62.53 _{±0.54}
			TSception	71.24	75.14	76.00	71.70	71.56	73.40	72.20	75.52	<u>71.68</u>	76.46	73.49 _{±0.27}
	Full Fine-tuning	Foundation Models	BENDR	76.15	79.23	80.45	77.44	77.56	79.05	78.10	78.87	77.16	80.23	78.42 _{±0.19}
			BIOT-1D	64.92	68.89	69.40	66.40	64.71	68.02	66.71	70.65	63.80	68.97	67.25 _{±0.11}
			BIOT-2D	64.60	68.11	68.82	66.13	65.64	67.11	66.95	69.57	63.78	69.48	67.02 _{±0.03}
			BIOT-6D	65.95	69.48	69.78	67.20	66.54	68.20	66.66	71.01	65.31	70.37	68.05 _{±0.14}
			LaBraM	63.60	67.09	66.24	63.43	62.52	65.47	64.54	68.31	63.22	62.94	64.74 _{±0.60}
			Neuro-GPT	54.22	53.42	57.42	53.97	53.09	56.30	50.80	60.33	52.19	58.40	55.02 _{±0.12}
			EEGPT	60.45	61.84	63.40	59.13	62.45	63.43	59.18	63.50	61.10	65.52	62.00 _{±1.29}
			CBraMod	66.75	71.36	71.34	67.44	66.69	69.89	68.24	71.78	65.67	71.95	69.11 _{±0.28}
			TFM	62.76	63.90	65.59	60.60	60.84	60.71	61.70	65.07	61.96	64.56	62.77 _{±0.24}
			EEGMamba	60.39	62.47	63.43	58.05	60.18	61.11	58.83	68.97	59.06	66.28	61.88 _{±0.34}
			SingLEM	61.85	63.99	65.96	61.68	60.32	64.20	62.02	68.07	61.75	67.16	63.70 _{±0.74}
			LUNA-Base	65.24	69.71	70.84	64.22	63.47	67.19	69.42	68.65	64.93	65.35	66.90 _{±0.18}
	Linear Probing	Foundation Models	BENDR	40.53	48.72	49.01	47.65	48.01	44.00	46.38	47.20	45.67	51.10	46.83 _{±0.11}
			BIOT-1D	35.85	44.89	45.24	41.27	41.23	43.07	43.19	46.75	37.47	47.04	42.60 _{±0.20}
			BIOT-2D	34.70	31.39	40.56	33.36	32.69	29.09	36.04	39.17	32.35	42.08	35.14 _{±0.27}
			BIOT-6D	47.70	48.52	51.40	47.02	46.62	46.09	47.44	53.73	46.74	54.46	48.97 _{±0.16}
			LaBraM	52.84	55.93	57.35	50.40	51.52	52.28	50.99	59.21	53.28	57.05	54.08 _{±0.18}
			Neuro-GPT	53.15	55.33	56.25	50.77	51.70	53.39	50.29	59.57	50.26	55.37	53.61 _{±0.67}
			EEGPT	52.92	60.78	50.73	49.04	58.73	41.03	47.84	40.59	51.59	51.98	50.52 _{±4.11}
			CBraMod	38.60	38.57	42.84	34.84	35.31	31.92	36.85	45.32	38.80	41.96	38.50 _{±0.12}
			TFM	49.83	51.59	53.69	50.69	49.36	49.95	50.26	56.49	49.58	55.97	51.74 _{±0.55}
			EEGMamba	33.69	29.94	35.53	31.04	30.86	26.89	30.37	33.56	32.59	32.30	31.68 _{±0.04}
			SingLEM	12.25	10.15	11.92	10.00	9.41	10.27	9.91	10.65	10.65	10.82	10.60 _{±0.10}
			LUNA-Base	47.75	52.82	56.15	47.18	50.25	48.95	49.64	56.21	50.40	55.43	51.48 _{±0.08}

Table 33. Balanced classification accuracies (%) on Sleep-EDFx. The best BCAs are marked in bold, and the second best by an underline.

Scenario	Tuning	Model Type	Approach	S0	S1	S2	S3	S4	S5	S6	S7	S8	S9	Avg.
Cross-subject		Specialist Models	PSD+LDA	50.10	51.22	54.46	52.93	50.65	50.79	53.69	51.11	49.63	53.23	51.78 _{±0.22}
			EEGNet	72.61	73.33	76.61	73.54	71.52	72.23	73.94	73.97	73.22	76.48	73.75 _{±0.19}
			ShallowConv	71.21	75.28	76.94	74.56	75.21	76.35	73.75	73.93	73.90	77.52	74.86 _{±0.42}
			LMDA	71.89	74.39	77.64	73.91	73.72	75.17	<u>74.84</u>	75.13	72.84	76.27	74.58 _{±0.34}
			CNN-T	73.65	75.88	76.88	74.88	76.47	76.70	73.84	<u>77.56</u>	74.49	77.08	75.74 _{±0.48}
			Deformer	76.20	79.32	81.04	78.35	78.19	79.99	77.53	78.57	76.94	81.19	78.73 _{±0.09}
			Conformer	65.74	69.70	71.25	69.89	67.08	70.07	68.00	68.04	63.71	70.53	68.40 _{±2.87}
			TSception	<u>73.68</u>	<u>76.70</u>	<u>78.41</u>	<u>75.61</u>	<u>76.66</u>	<u>77.23</u>	74.61	76.28	<u>76.01</u>	<u>78.87</u>	<u>76.41</u> _{±0.43}
	Full Fine-tuning	Foundation Models	BENDR	68.37	72.87	72.50	72.16	72.26	72.04	70.03	70.56	70.09	73.60	71.45 _{±0.43}
			BIOT-1D	67.34	70.00	71.01	67.88	68.28	70.47	68.02	70.00	66.93	70.96	69.09 _{±0.24}
			BIOT-2D	66.89	68.40	70.71	67.75	68.64	69.00	67.74	69.42	65.78	70.42	68.48 _{±0.18}
			BIOT-6D	64.23	67.27	67.63	66.25	66.37	67.16	64.43	67.88	63.95	68.33	66.35 _{±0.23}
			LaBraM	62.67	64.24	63.63	62.91	63.32	64.06	63.71	63.71	63.46	63.84	63.56 _{±0.03}
			Neuro-GPT	59.00	58.45	61.66	57.98	58.91	61.04	55.33	61.36	56.60	60.55	59.09 _{±0.25}
			EEGPT	57.46	62.29	64.94	62.73	64.02	65.59	60.59	64.60	62.29	64.78	62.93 _{±1.06}
			CBraMod	70.05	72.99	74.26	71.49	72.00	73.37	70.94	73.63	70.18	74.09	72.30 _{±0.18}
			TFM	65.93	67.27	70.51	65.77	67.62	66.81	66.18	67.78	66.63	69.29	67.38 _{±0.25}
			EEGMamba	65.22	66.62	69.51	65.03	65.41	66.35	64.16	68.44	66.15	69.90	66.68 _{±0.42}
			SingLEM	65.25	67.45	69.48	65.39	67.20	68.24	65.81	68.28	65.63	69.59	67.23 _{±0.48}
			LUNA-Base	68.29	69.91	74.62	69.50	69.31	72.39	72.09	71.24	69.85	73.10	71.03 _{±0.36}
	Linear Probing	Foundation Models	BENDR	48.81	55.96	56.68	55.64	56.52	53.80	55.01	53.53	53.71	58.18	54.78 _{±0.12}
			BIOT-1D	48.79	59.09	57.54	56.91	57.51	57.58	56.79	56.63	54.78	57.34	56.30 _{±0.33}
			BIOT-2D	54.60	51.95	59.96	54.62	53.28	50.77	55.17	56.48	54.05	58.76	54.97 _{±0.28}
			BIOT-6D	58.35	60.80	63.02	60.09	60.69	58.98	58.61	61.45	60.17	63.72	60.59 _{±0.19}
			LaBraM	60.17	63.07	65.30	61.61	62.19	62.75	61.71	63.88	61.49	65.09	62.73 _{±0.24}
			Neuro-GPT	56.23	58.46	60.02	57.11	58.46	58.54	56.62	58.15	56.22	59.70	57.95 _{±0.58}
			EEGPT	51.08	59.04	53.49	57.05	56.04	56.79	57.66	52.46	58.66	54.44	55.67 _{±0.87}
			CBraMod	53.13	50.79	55.43	51.87	50.52	48.93	51.30	55.53	53.30	54.76	52.56 _{±0.30}
			TFM	55.02	57.19	57.55	56.13	57.09	55.97	57.26	58.13	55.73	59.40	56.95 _{±0.46}
			EEGMamba	52.38	50.55	55.26	52.24	52.57	49.67	50.40	51.98	51.93	52.56	51.96 _{±0.13}
			SingLEM	35.47	34.32	36.56	34.81	34.83	34.43	33.72	34.35	34.26	34.98	34.77 _{±0.13}
			LUNA-Base	54.87	57.74	62.11	57.14	57.17	57.13	58.12	59.53	57.64	59.96	58.14 _{±0.22}

Table 34. Cohen's Kappa (%) on Sleep-EDFx. The best metrics are marked in bold, and the second best by an underline.

Scenario	Tuning	Model Type	Approach	S0	S1	S2	S3	S4	S5	S6	S7	S8	S9	S10	S11	S12	S13	S14	S15	S16	S17	S18	S19
Within-subject (Few-shot)		Specialist Models	PSD+LDA	76.05	76.34	71.08	72.37	70.51	71.57	78.11	75.49	69.06	73.75	74.44	54.00	64.18	62.49	81.67	70.96	60.14	60.24	75.96	64.20
			EEGNet	68.07	66.92	65.91	65.31	64.62	58.55	70.20	66.64	60.08	64.74	67.14	49.53	54.38	56.74	67.67	66.85	53.43	55.28	70.70	51.05
			ShallowConv	79.86	75.38	74.20	78.72	73.63	65.23	79.29	75.15	70.52	72.05	78.05	63.33	66.42	75.73	82.40	72.67	51.99	62.81	76.39	65.52
			LMDA	68.53	60.54	64.10	71.80	56.80	60.37	64.88	64.24	65.13	57.15	56.75	51.41	55.35	32.78	60.03	57.98	33.87	41.19	72.18	47.23
			CNN-T	78.99	79.04	83.43	82.28	76.73	64.38	77.43	81.90	76.95	75.15	79.48	65.79	57.00	60.84	72.67	74.04	75.59	64.08	79.52	63.98
			Deformer	74.48	76.89	76.21	74.37	71.64	75.37	77.22	76.09	69.98	69.73	75.19	48.38	65.95	63.33	70.70	74.74	65.84	61.88	75.98	55.29
			Conformer	80.60	68.84	80.33	81.97	73.85	69.94	73.32	82.02	65.89	68.86	75.48	68.09	65.76	67.50	86.49	74.70	68.47	49.82	73.78	61.65
			TSception	76.40	65.88	72.04	76.49	67.24	57.22	73.28	75.80	69.76	62.50	59.67	52.77	56.07	39.93	68.30	63.36	52.77	40.42	74.11	60.27
	Full Fine-tuning	Foundation Models	BENDR	55.59	68.93	66.95	63.25	61.82	31.11	63.89	57.70	50.14	64.17	69.36	43.97	49.18	15.14	59.99	59.58	45.07	30.47	64.01	54.91
			BIOT-1D	73.35	75.80	75.62	77.38	73.13	66.44	70.00	68.48	71.20	72.32	73.25	54.27	59.53	70.66	66.40	74.25	59.05	43.59	78.00	61.67
			BIOT-2D	71.21	75.33	76.67	81.37	73.44	62.90	60.46	73.77	72.74	61.01	71.45	53.80	64.13	69.39	72.22	63.88	61.24	43.07	79.39	51.49
			BIOT-6D	77.86	78.96	79.65	82.10	73.79	64.71	72.43	76.51	71.83	71.32	77.09	65.99	69.11	70.88	69.57	75.15	70.14	59.66	78.63	53.91
			LaBraM	52.97	57.26	42.52	47.91	38.57	41.87	34.22	56.70	47.83	57.29	48.81	50.48	23.72	14.45	59.28	51.19	23.61	32.71	60.63	26.48
			Neuro-GPT	70.34	64.86	72.32	75.45	67.42	58.35	65.32	67.14	63.94	66.34	71.14	47.27	57.54	71.05	70.95	67.26	63.64	48.46	71.57	45.27
			EEGPT	61.83	70.02	65.65	73.44	70.04	64.73	71.51	71.36	64.45	68.95	54.70	49.13	67.99	68.83	75.28	68.66	65.77	59.06	61.58	60.82
			CBraMod	73.38	71.99	75.45	76.20	67.65	60.18	59.49	71.28	70.19	67.81	72.29	59.26	57.82	56.25	69.93	66.55	64.52	55.59	73.31	52.59
			TFM	75.79	71.85	75.82	79.93	67.52	68.54	73.64	69.02	71.57	71.01	68.08	59.42	64.78	70.69	79.76	73.93	55.12	54.44	75.04	57.66
			EEGMamba	77.63	69.81	67.89	79.15	69.94	68.04	70.17	78.79	72.42	68.73	66.37	62.53	59.57	70.69	80.40	72.67	59.72	54.44	76.45	57.82
			SingLEM	4.50	16.19	9.17	22.44	26.75	3.48	20.81	21.67	4.84	38.48	2.99	14.48	1.94	4.99	16.88	43.65	4.43	1.69	32.92	8.30
			LUNA-Base	83.07	82.58	77.88	81.08	70.27	77.43	67.35	70.57	79.43	66.25	74.71	42.81	69.10	71.91	82.17	67.57	57.96	60.72	81.58	55.50
	Linear Probing	Foundation Models	BENDR	3.44	4.52	6.69	2.02	8.14	5.18	7.24	2.55	2.93	6.01	4.34	2.59	2.05	1.70	2.52	3.69	3.15	1.45	2.59	4.36
			BIOT-1D	68.32	67.43	52.61	76.35	61.01	52.26	60.84	62.82	57.57	56.96	36.35	38.75	53.33	77.08	67.95	64.22	53.04	41.81	74.93	44.89
			BIOT-2D	63.80	45.05	27.00	62.08	26.49	45.51	58.90	45.27	60.35	34.16	20.23	24.38	47.15	49.02	64.72	30.43	36.37	30.64	70.47	34.32
			BIOT-6D	73.21	70.24	70.06	73.59	70.03	58.03	67.18	59.88	64.22	69.90	60.71	41.38	61.11	74.19	67.18	65.66	68.44	52.24	73.55	44.73
			LaBraM	68.54	68.82	67.58	72.16	65.28	58.02	66.82	70.84	63.82	68.90	60.59	55.30	56.66	66.44	70.82	67.68	56.44	49.30	71.69	55.62
			Neuro-GPT	65.95	57.41	67.73	65.52	61.25	52.08	60.04	54.69	62.19	60.12	57.49	40.73	54.52	65.22	66.28	59.97	57.16	39.34	67.75	39.80
			EEGPT	81.36	80.86	81.19	84.84	75.15	81.15	80.40	86.59	84.55	80.78	76.89	34.70	79.14	80.91	90.95	71.38	74.36	67.78	70.50	65.17
			CBraMod	56.36	60.74	42.16	66.38	57.75	46.21	51.43	49.31	40.26	56.35	42.52	49.19	22.85	45.42	50.51	37.19	35.17	29.12	59.26	30.11
			TFM	63.21	55.67	60.55	65.88	55.74	49.10	63.18	54.81	58.06	53.77	44.22	53.40	50.61	44.00	63.11	62.62	43.07	37.79	67.39	51.84
			EEGMamba	45.54	39.27	41.52	52.38	48.35	31.13	41.62	42.18	41.06	41.53	30.41	48.26	23.87	34.60	39.61	34.67	21.88	18.55	46.05	21.97
			SingLEM	2.90	1.67	3.43	2.49	4.37	-0.16	2.13	2.13	3.59	0.90	2.87	1.13	1.08	1.73	1.83	1.03	1.80	0.13	1.34	1.33
			LUNA-Base	82.24	62.75	77.03	70.49	62.64	68.62	64.03	66.64	77.24	61.20	55.92	40.12	67.01	66.21	75.85	61.20	28.38	48.35	79.92	54.13
Scenario	Tuning	Model Type	Approach	S20	S21	S22	S23	S24	S25	S26	S27	S28	S29	S30	S31	S32	S33	S34	S35	S36	S37	S38	S39
Within-subject (Few-shot)		Specialist Models	PSD+LDA	56.91	70.77	43.60	64.76	71.03	65.69	61.75	65.46	70.38	56.23	63.89	60.88	64.81	64.82	69.85	60.83	68.23	49.04	69.01	54.77
			EEGNet	44.28	66.99	24.39	38.33	70.58	60.23	53.63	52.13	62.99	51.16	60.97	54.90	52.54	63.78	58.87	51.54	72.69	56.03	66.98	52.62
			ShallowConv	45.81	73.15	27.15	47.40	74.47	70.56	64.35	57.22	68.92	50.98	69.95	66.30	68.01	67.94	74.11	59.68	73.04	57.94	72.25	53.23
			LMDA	35.99	64.28	26.06	23.22	62.28	57.07	50.66	50.96	56.42	36.74	51.78	53.09	43.59	64.96	70.29	33.72	53.26	53.31	54.86	54.86
			CNN-T	41.63	75.49	46.46	56.48	77.43	71.11	69.96	51.61	77.69	60.33	68.56	64.71	65.69	68.83	78.87	54.12	44.41	59.72	61.77	56.33
			Deformer	47.56	73.72	54.94	55.71	76.61	72.33	65.82	52.38	72.68	50.21	70.40	59.09	62.71	71.55	74.21	63.50	72.04	62.42	70.62	65.35
			Conformer	54.91	63.04	29.08	67.98	74.26	73.97	67.92	64.09	75.67	55.48	64.22	61.94	58.42	70.13	75.56	49.96	63.64	56.34	71.84	56.33
			TSception	26.20	63.04	24.72	44.79	60.50	63.80	56.57	43.70	55.99	46.48	67.65	52.64	65.14	68.83	64.75	43.67	25.59	57.52	37.71	50.71
	Full Fine-tuning	Foundation Models	BENDR	55.80	57.68	39.03	26.46	73.48	67.98	56.02	41.51	61.39	47.62	46.17	54.12	44.69	66.84	72.96	23.72	26.99	50.10	72.78	48.20
			BIOT-1D	44.70	68.02	60.48	53.89	64.52	65.55	65.25	53.11	74.27	42.03	64.38	61.75	65.20	67.98	73.02	58.81	65.89	56.69	50.40	52.14
			BIOT-2D	43.45	65.18	48.87	61.34	61.73	72.26	63.28	47.05	74.99	49.89	61.49	60.83	58.70	68.30	66.88	49.72	57.86	53.77	52.21	52.36
			BIOT-6D	40.49	67.64	41.78	51.36	68.25	71.94	71.24	61.64	76.44	52.92	68.21	63.57	66.82	67.97	69.84	57.30	66.06	58.26	63.74	53.99
			LaBraM	16.85	44.12	40.11	38.68	55.82	37.66	45.85	26.30	57.02	35.74	26.04	52.51	26.58	43.68	50.47	38.76	51.96	40.98	48.12	45.10
			Neuro-GPT	37.23	61.90	31.29	59.55	68.92	71.17	58.31	45.43	66.43	51.18	62.86	56.44	58.62	66.04	68.86	53.01	69.49	58.88	61.84	53.06
			EEGPT	47.56	57.43	59.14	45.56	59.67	51.35	71.44	63.59	60.88	65.62	46.97	58.68	62.81	56.86	60.44	53.14	45.22	40.34	46.24	47.60
			CBraMod	43.87	65.12	21.66	53.89	65.89	66.22	56.14	61.64	69.89	53.75	59.87	58.69	66.41	70.45	62.37	56.99	59.26	59.35	58.00	54.06
			TFM	60.46	68.95	40.50	65.36	68.84	72.52	64.26	63.31	72.42	56.66	64.16	57.99	68.64	68.53	72.78	60.86	75.71	67.45	68.44	54.98
			EEGMamba	46.92	57.72	31.69	58.61	69.05	66.89	64.01	61.45	75.92	56.43	69.31	58.39	43.21	66.43	68.12	59.50	70.11	62.55	67.38	56.75
			SingLEM	15.99	8.68	18.09	22.70	23.05	7.10	40.27	2.62	35.41	9.93	2.58	40.08	19.41	53.17	37.36	4.58	1.31	8.05	18.10	16.50
			LUNA-Base	55.38	70.07	28.33	55.10	76.92	79.01	69.66	63.93	70.95	57.81	67.68	74.97	69.24	71.36	77.41	50.15	60.91	54.41	69.12	58.69
	Linear Probing	Foundation Models	BENDR	4.24	3.21	5.30	1.47	7.79	4.36	5.00	1.90	3.51	1.18	0.76	2.65	1.14	2.96	8.90	4.70.				

Table 35. Cohen's Kappa (%) on Sleep-EDFx. The best metrics are marked in bold, and the second best by an underline (continued).

Scenario	Tuning	Model Type	Approach	S40	S41	S42	S43	S44	S45	S46	S47	S48	S49	S50	S51	S52	S53	S54	S55	S56	S57	S58	S59
Within-subject (Few-shot)	Specialist Models	Specialist Models	PSD+LDA	60.81	<u>79.48</u>	57.11	<u>62.76</u>	68.15	72.18	34.91	70.09	65.06	35.52	<u>65.74</u>	68.05	67.14	61.19	60.39	63.43	38.45	<u>64.77</u>	55.56	56.93
			EEGNet	50.38	64.13	58.96	55.70	63.84	64.56	29.35	58.49	52.60	27.76	53.26	47.94	63.90	56.26	63.56	47.90	33.88	54.83	28.16	56.34
			ShallowConv	62.44	76.83	67.35	61.78	72.67	68.42	36.94	61.12	63.81	25.67	53.44	56.91	63.39	62.93	63.05	59.87	50.78	62.06	35.74	61.59
			LMDA	47.38	69.64	37.59	51.89	57.61	59.41	21.42	48.23	45.17	21.05	33.30	28.06	57.72	43.58	59.97	42.24	23.54	39.14	28.18	43.34
			CNN-T	58.41	74.38	66.03	56.76	79.10	57.96	26.74	54.55	<u>68.41</u>	24.70	59.89	63.16	61.25	61.30	53.36	60.27	49.76	45.95	35.16	65.55
			Deformer	63.82	75.53	67.95	59.28	67.85	65.33	37.83	67.75	61.26	30.30	30.92	<u>69.33</u>	<u>69.16</u>	62.23	60.77	66.81	52.39	59.73	31.10	61.72
			Conformer	56.19	69.18	67.74	48.95	62.50	64.98	40.67	77.18	56.99	28.16	60.73	56.52	59.94	61.42	52.65	61.10	45.71	50.14	47.21	62.99
			TSception	49.51	69.05	37.23	47.13	63.12	64.62	18.17	63.68	50.70	22.73	43.89	37.15	60.89	63.67	53.09	58.19	28.35	47.12	33.74	55.25
	Full Fine-tuning	Foundation Models	BENDR	24.16	55.50	65.60	55.61	57.28	67.43	10.71	57.21	55.80	39.68	26.37	23.95	49.50	43.92	65.75	49.79	35.91	59.58	21.22	51.09
			BIOT-1D	33.82	73.52	68.96	52.64	51.02	56.48	17.79	72.36	60.83	28.74	58.44	63.05	56.81	58.68	53.32	49.74	43.06	30.39	56.49	
			BIOT-2D	58.52	72.84	65.87	52.61	58.96	62.17	18.23	71.57	58.36	33.90	56.94	62.23	59.55	50.56	56.98	55.93	48.82	41.91	29.93	51.86
			BIOT-6D	64.49	75.89	<u>72.74</u>	55.64	66.89	67.98	30.87	<u>72.92</u>	63.19	32.39	62.53	63.80	65.50	59.80	58.37	58.55	49.49	51.21	34.91	51.62
			LaBraM	43.33	47.94	34.71	40.40	40.45	63.54	24.73	32.99	63.97	13.30	32.04	16.24	22.64	34.22	54.66	23.16	25.54	39.72	22.16	34.67
			Neuro-GPT	59.09	70.05	62.35	48.72	58.29	66.76	29.96	64.35	60.49	29.04	59.47	66.56	59.87	61.37	57.56	59.24	53.67	47.32	30.26	50.75
			EEGPT	62.65	67.78	57.64	50.32	64.54	55.30	49.60	68.83	50.79	<u>42.00</u>	47.60	44.08	56.67	61.17	39.16	48.04	37.62	42.35	57.78	56.72
			CBraMod	58.36	72.36	38.89	52.23	62.08	68.77	26.60	64.76	64.86	29.63	56.45	56.98	62.01	<u>65.59</u>	51.66	54.16	57.79	44.91	40.24	58.91
			TFM	64.10	74.65	62.82	54.17	65.94	<u>70.01</u>	36.03	67.63	65.10	35.05	64.78	63.78	65.80	62.84	58.26	58.55	45.61	52.29	42.90	60.20
			EEGMamba	51.02	77.73	50.95	54.77	71.38	62.59	24.99	71.74	61.30	30.19	53.90	63.07	55.75	54.75	60.02	40.34	50.58	40.79	53.67	
			SingLEM	26.36	24.24	7.97	12.68	29.89	30.22	3.59	44.01	32.08	1.48	25.42	3.51	16.37	7.59	20.93	2.49	5.55	13.42	8.42	8.78
			LUNA-Base	66.60	78.98	50.91	63.31	<u>73.80</u>	64.01	63.40	64.70	56.64	33.83	59.72	49.49	65.37	63.84	62.10	<u>64.86</u>	<u>60.17</u>	62.53	38.72	<u>63.81</u>
	Linear Probing	Foundation Models	BENDR	3.02	2.94	2.81	5.00	3.02	2.50	-0.83	3.11	1.46	2.00	3.63	0.60	5.59	2.29	2.85	6.16	11.61	4.20	3.33	1.97
			BIOT-1D	35.72	64.88	52.59	38.73	34.53	54.36	15.08	57.14	53.34	22.83	51.27	54.57	52.48	43.34	48.43	48.92	27.74	34.55	21.48	37.04
			BIOT-2D	52.46	58.59	43.48	37.66	32.78	51.41	11.12	50.56	44.25	23.65	26.17	29.90	48.67	33.35	26.12	33.79	39.46	26.05	15.84	35.59
			BIOT-6D	55.86	69.45	55.32	49.49	62.64	62.09	14.54	68.60	59.21	33.50	62.73	58.83	57.32	58.20	51.66	55.27	44.48	42.50	32.65	49.05
			LaBraM	56.34	69.03	53.79	54.73	61.84	66.17	31.72	62.37	64.14	32.64	59.52	67.60	62.44	56.09	53.11	48.02	34.22	50.16	38.46	50.83
			Neuro-GPT	55.26	64.23	56.97	47.14	55.05	59.89	29.11	59.13	55.39	27.25	50.53	63.04	56.03	53.78	49.95	49.49	36.83	43.48	28.66	41.97
			EEGPT	72.83	81.15	80.54	58.80	68.18	67.70	57.41	71.71	78.10	50.99	68.03	74.12	70.37	72.32	61.54	58.86	63.47	69.49	<u>56.20</u>	62.73
			CBraMod	36.44	62.94	40.24	47.19	31.63	53.57	18.68	52.33	54.39	14.64	43.71	22.77	39.71	44.01	49.55	11.80	18.73	24.64	31.24	36.08
			TFM	56.17	53.98	30.79	48.19	54.54	68.64	19.15	53.59	58.37	24.71	53.81	51.73	56.22	50.51	53.44	48.79	19.12	46.50	27.99	35.12
			EEGMamba	33.83	53.96	30.99	24.38	36.71	47.89	23.45	45.09	40.29	19.34	26.74	26.40	44.99	23.92	34.73	25.22	18.79	28.15	20.44	20.70
			SingLEM	2.16	2.20	0.97	2.47	2.89	3.04	2.16	2.55	0.69	0.33	2.74	1.66	1.38	2.40	2.21	2.86	2.95	1.68	1.50	1.39
			LUNA-Base	<u>67.89</u>	77.18	58.74	60.19	70.68	62.73	<u>57.25</u>	71.15	57.12	28.30	60.78	45.64	62.87	49.03	61.75	57.72	30.69	56.35	29.52	55.13
Within-subject (Few-shot)	Specialist Models	Specialist Models	PSD+LDA	62.80	53.51	61.26	56.24	61.89	59.01	46.51	64.63	54.97	49.33	56.25	53.87	62.94	61.34	48.01	58.08	62.80	69.64	62.90	
			EEGNet	59.52	47.53	67.36	59.21	53.58	55.27	42.55	57.49	55.15	48.69	51.51	55.28	46.34	54.77	46.91	59.37	50.64	53.98	55.85 _{a0.12}	
			ShallowConv	65.26	48.71	<u>74.65</u>	59.74	60.64	47.28	51.80	68.31	58.58	54.40	59.07	59.81	54.39	61.11	50.27	<u>68.64</u>	62.01	70.94	<u>63.13</u> _{a0.52}	
			LMDA	63.41	48.28	62.81	47.92	47.07	48.66	22.77	58.26	49.36	47.45	42.36	53.18	30.52	53.14	40.93	57.96	48.85	54.79	49.69 _{a2.06}	
			CNN-T	<u>67.76</u>	46.96	73.48	58.94	<u>61.83</u>	57.13	56.89	59.52	49.81	58.19	60.15	48.63	3.15	<u>63.43</u>	<u>55.09</u>	64.69	65.61	75.86	62.44 _{a0.15}	
			Deformer	64.71	51.12	73.00	58.13	60.58	52.78	48.79	65.23	54.80	58.59	54.11	59.20	57.37	56.87	48.00	67.09	60.25	71.50	63.03 _{a0.19}	
			Conformer	60.25	31.18	73.38	73.56	52.24	49.08	55.70	51.85	43.90	58.64	56.16	58.74	44.97	55.27	49.46	70.40	63.14	64.00	61.99 _{a0.67}	
			TSception	64.00	39.84	52.60	47.41	47.30	48.98	32.32	64.75	49.34	63.86	56.20	55.69	45.39	53.17	41.57	62.09	49.69	66.81	53.75 _{a0.49}	
	Full Fine-tuning	Foundation Models	BENDR	57.19	<u>60.27</u>	69.17	54.79	61.49	45.67	18.32	53.11	61.95	43.57	49.05	60.62	13.28	54.83	46.31	58.82	54.01	55.76	50.55 _{a0.52}	
			BIOT-1D	60.85	45.88	72.90	48.46	46.43	42.10	50.64	60.16	47.30	56.16	52.51	54.36	61.49	48.45	45.10	60.08	54.71	61.64	58.92 _{a0.10}	
			BIOT-2D	62.17	39.09	70.76	47.56	47.36	39.86	49.02	55.20	46.82	53.35	51.27	54.14	61.43	49.68	45.17	59.36	53.40	60.55	57.99 _{a0.12}	
			BIOT-6D	64.70	50.29	73.79	54.58	47.55	42.69	49.54	<u>67.05</u>	51.33	54.91	56.75	58.21	69.12	54.63	47.07	59.73	59.74	70.04	62.43 _{a0.68}	
			LaBraM	55.07	54.87	46.16	35.94	55.38	45.00	6.03	23.64	37.45	25.01	28.92	38.42	3.97	27.85	33.78	33.19	44.81	23.88	38.73 _{a0.33}	
			Neuro-GPT	57.03	44.41	65.17	56.46	47.86	49.83	51.14	61.63	52.84	60.28	53.13	56.91	57.19	57.43	42.13	63.95	55.36	60.22	58.13 _{a0.10}	
			EEGPT	54.08	41.03	45.14	57.85	53.36	48.53	46.64	43.99	44.20	51.11	56.62	58.75	41.57	53.75	52.11	47.17	65.83	60.97	56.38 _{a0.16}	
			CBraMod	58.44	46.68	61.27	53.43	48.14	48.21	45.38	64.94	51.57	<u>63.17</u>	60.63	52.82	57.19	59.74	42.43	65.58	55.96	61.88	58.59 _{a0.03}	
			TFM	63.04	44.72	66.50	54.04	56.62	51.51	56.53	62.55	52.12	57.76	50.21	59.29	56.72	57.70	43.89	64.21	58.73	67.15	62.31 _{a0.09}	
EEGMamba	61.95	46.89	56.56	56.95	51.12	49.35	<u>60.19</u>	60.88	52.47	56.40	54.58	57.89	55.44	62.92	48.62	65.27	59.51	63.92	60.13 _{a0.88}				
SingLEM	39.84	41.00	12.50	10.74	36.29	33.83	1.74	34.60	41.09	19.24	4.87	40.49	0.12	26.68	23.41	9.43	35.96	9.75	18.36 _{a4.96}				
LUNA-Base	62.14	60.25	71.10	<u>69.76</u>	48.76	<u>61.14</u>	47.53	58.33	63.47	58.99	68.31	<u>62.62</u>	67.36	59.83	51.61	62.24	67.51	<u>73.02</u>	64.38 _{a0.22}				
Linear Probing	Foundation Models	BENDR	9.17	4.60	2.67	0.75	4.75	2.15	1.30	1.31	2.10	3.18	4.21	8.89	0.04								

Table 36. Balanced classification accuracies (%) on Sleep-EDFx. The best BCAs are marked in bold, and the second best by an underline.

Scenario	Tuning	Model Type	Approach	S0	S1	S2	S3	S4	S5	S6	S7	S8	S9	S10	S11	S12	S13	S14	S15	S16	S17	S18	S19	
Within-subject (Few-shot)	Specialist Models	Specialist Models	PSD+LDA	71.43	73.06	67.93	64.96	68.87	62.69	68.73	64.23	66.30	66.07	54.66	51.15	67.60	62.80	73.51	70.79	54.38	59.56	63.48	63.97	
			EEGNet	58.14	54.34	53.86	50.42	54.41	54.70	56.53	58.14	55.69	53.49	46.63	42.77	49.06	43.13	55.60	61.21	48.94	46.11	58.51	47.10	
			ShallowConv	69.50	67.90	57.19	65.42	64.58	61.73	65.33	65.27	63.85	63.55	53.22	50.21	64.40	62.98	69.71	67.44	55.37	55.59	61.95	62.19	
			LMDA	59.61	53.53	52.33	58.97	46.58	52.21	56.92	56.80	62.22	52.15	46.31	44.57	55.39	35.66	54.66	59.04	43.11	42.31	60.06	47.75	
			CNN-T	74.44	78.51	73.99	73.40	76.30	67.55	72.20	73.76	76.20	69.53	68.73	64.50	65.99	67.89	69.73	77.11	70.39	62.68	72.12	65.82	
			Deformer	63.51	58.19	61.45	55.23	58.15	66.52	63.11	67.33	61.94	54.27	53.46	40.62	58.59	47.15	58.51	65.45	55.79	50.26	59.66	46.04	
			Conformer	72.04	70.56	70.92	75.80	75.44	66.47	65.62	76.07	69.83	58.89	60.29	57.31	67.35	72.64	80.12	72.55	67.28	56.72	64.59	72.69	
			TSception	71.71	74.17	72.23	72.58	72.21	67.22	69.96	72.80	75.22	63.53	70.53	52.84	66.52	57.53	61.73	66.06	58.94	52.79	67.15	62.38	
	Full Fine-tuning	Foundation Models	BENDR	41.33	41.63	44.26	39.63	39.79	28.08	39.81	42.19	39.41	41.39	38.87	30.60	36.16	23.54	42.18	42.19	37.61	28.67	44.39	37.69	
			BIOT-1D	68.45	73.95	70.41	70.44	70.61	61.97	62.51	65.54	67.45	66.86	68.50	49.97	62.23	74.13	64.07	70.77	61.08	58.30	65.56	67.30	
			BIOT-2D	67.83	76.97	72.02	74.36	72.59	65.77	57.18	69.76	70.98	62.40	73.56	54.24	62.57	76.62	71.87	65.59	57.60	55.89	68.80	61.61	
			BIOT-6D	70.66	74.93	71.62	71.67	72.42	64.21	63.56	71.11	68.01	66.84	68.67	52.75	67.29	73.97	63.09	70.72	66.35	56.95	67.44	60.54	
			LaBraM	47.00	40.82	41.57	37.84	36.90	33.22	39.13	49.57	41.99	45.54	33.61	38.68	31.36	28.25	46.40	44.90	32.42	32.59	48.69	31.90	
			Neuro-GPT	63.93	63.59	64.77	67.73	66.01	59.67	60.60	65.60	61.99	61.23	61.14	42.66	60.19	71.23	60.78	65.68	58.06	45.15	59.84	53.00	
			EEGPT	61.83	70.02	65.65	73.44	70.04	64.73	71.51	71.36	64.45	68.95	54.70	49.13	67.99	68.83	75.28	68.66	65.77	59.06	61.58	60.82	
			CBraMod	66.40	72.50	71.67	68.04	71.63	60.43	58.03	64.41	68.92	66.54	61.12	50.87	62.08	69.80	65.20	64.70	59.97	52.07	61.80	60.76	
			TFM	68.03	67.59	68.32	69.76	62.31	62.22	65.84	62.83	67.48	63.78	60.06	53.63	66.11	69.78	69.85	69.19	55.66	56.53	61.79	60.60	
			EEGMamba	71.71	73.13	63.93	73.99	71.07	70.88	76.88	73.53	71.36	67.14	58.61	57.41	68.08	74.31	72.99	60.86	58.27	67.17	64.43		
			SingLEM	22.74	27.26	28.85	30.49	33.02	22.78	30.11	32.80	23.93	36.50	23.92	27.84	21.11	23.64	29.83	39.08	24.26	21.47	35.73	24.80	
			LUNA-Base	73.02	79.75	64.45	74.31	76.29	73.38	67.14	66.86	74.96	69.57	75.10	47.60	70.83	74.02	74.56	71.74	62.72	60.59	75.08	67.71	
	Linear Probing	Foundation Models	BENDR	21.30	21.08	22.16	20.42	22.88	21.30	22.33	20.83	20.98	22.13	21.13	20.72	20.66	20.40	20.77	21.40	21.51	20.55	21.06	21.66	
			BIOT-1D	67.45	76.46	65.28	69.99	69.62	65.47	62.98	64.80	65.45	63.78	62.52	42.30	62.25	78.83	67.22	69.82	59.90	56.19	68.22	58.77	
			BIOT-2D	62.67	71.07	54.76	67.16	56.34	61.37	58.98	57.93	64.10	54.85	57.35	37.76	57.93	67.71	66.50	60.03	54.35	50.50	66.94	53.28	
			BIOT-6D	68.16	73.50	67.74	67.55	69.08	63.49	63.53	62.00	67.98	65.02	64.42	46.22	65.26	73.10	61.21	68.82	67.66	58.12	62.44	54.59	
			LaBraM	58.55	55.00	53.29	56.38	56.84	51.34	55.67	60.75	53.91	58.46	45.03	43.09	52.53	53.73	57.40	58.08	52.17	46.31	56.51	51.70	
			Neuro-GPT	58.88	54.71	55.81	54.71	57.25	50.89	55.14	54.65	61.11	52.29	51.32	37.51	54.50	61.05	56.74	56.98	51.63	41.27	56.42	44.11	
			EEGPT	68.87	72.70	70.07	74.38	72.60	72.40	75.90	77.16	72.93	74.06	63.97	39.91	70.58	76.17	77.90	70.82	66.06	59.09	64.15	73.34	
			CBraMod	48.55	43.39	37.37	49.31	51.53	39.71	47.20	48.90	40.98	46.10	34.78	36.77	29.06	38.33	43.10	40.65	43.08	36.59	50.31	36.76	
			TFM	60.47	56.56	58.80	60.58	57.52	54.86	58.36	53.35	60.11	55.83	51.34	50.30	57.04	40.08	61.82	61.60	47.90	45.29	55.17	52.11	
			EEGMamba	52.94	52.86	56.32	63.90	57.19	51.58	65.14	56.68	51.88	52.30	40.14	52.42	46.21	54.14	52.90	55.56	45.08	38.39	56.14	44.44	
			SingLEM	22.69	22.69	26.21	22.15	24.16	19.86	24.23	22.78	23.31	21.28	24.61	22.60	21.37	20.78	21.79	22.20	20.72	20.58	20.86	20.48	
			LUNA-Base	70.66	67.82	71.30	66.05	68.96	65.57	57.78	65.64	70.81	65.69	67.31	46.73	63.66	73.14	67.23	66.90	47.44	57.57	69.13	63.93	
Within-subject (Few-shot)	Specialist Models	Specialist Models	PSD+LDA	54.69	68.60	48.39	54.26	66.12	67.81	57.84	63.36	65.19	64.18	55.56	67.16	61.15	52.31	53.35	61.75	55.62	41.53	60.22	56.44	
			EEGNet	42.17	43.88	38.46	35.80	48.31	44.94	47.06	56.28	58.02	54.08	45.02	51.52	56.92	55.01	61.04	40.22	40.88	39.61	48.42	45.61	
			ShallowConv	39.95	53.58	45.30	41.39	52.00	63.52	63.78	57.52	66.14	59.44	54.03	62.14	64.27	58.05	66.18	53.28	41.40	42.92	50.36	49.34	
			LMDA	36.63	43.47	43.62	27.69	45.48	44.92	45.67	52.97	64.59	51.51	40.44	48.96	50.97	58.71	64.28	36.42	35.22	39.84	41.77	45.81	
			CNN-T	56.95	72.96	58.08	54.21	74.90	74.43	77.32	55.17	77.85	68.72	71.63	73.33	68.26	68.27	76.89	63.81	39.11	55.93	60.75	67.08	
			Deformer	40.67	45.60	47.49	42.25	49.11	53.21	55.35	51.78	65.85	55.54	49.50	55.72	60.47	61.16	66.60	45.46	39.74	41.91	44.62	47.75	
			Conformer	56.84	55.33	48.22	65.52	71.82	70.38	72.68	63.47	76.80	66.07	58.03	65.65	59.42	60.30	71.80	62.25	62.18	52.98	66.69	62.99	
			TSception	53.96	68.38	49.13	46.92	64.17	66.62	64.56	55.17	71.86	61.23	71.48	62.25	63.74	63.49	68.87	57.74	45.18	44.55	50.35	57.31	
	Full Fine-tuning	Foundation Models	BENDR	36.81	34.18	37.00	27.37	41.18	39.48	39.61	37.34	41.54	36.62	32.48	39.28	46.84	47.84	55.24	25.96	25.36	34.55	39.97	34.30	
			BIOT-1D	59.55	60.70	52.96	57.45	67.12	70.32	68.11	50.67	70.66	49.46	54.97	66.52	62.41	61.13	70.85	63.64	62.34	52.51	50.68	60.34	
			BIOT-2D	58.42	57.16	46.88	55.65	67.22	74.81	65.45	49.87	69.98	49.15	57.57	64.60	59.04	61.13	67.65	61.22	61.87	58.78	53.66	60.52	
			BIOT-6D	54.57	57.45	47.34	54.73	66.70	75.05	72.23	57.57	71.04	57.07	56.74	67.68	63.05	61.64	66.94	63.28	65.15	48.31	58.56	59.69	
			LaBraM	25.74	32.06	37.04	32.67	35.82	30.58	37.98	34.06	51.15	35.93	26.13	42.59	36.99	42.29	47.91	34.13	32.39	32.24	34.31	37.34	
			Neuro-GPT	36.57	54.77	47.04	46.82	56.53	64.71	62.92	45.51	70.81	54.01	50.94	60.86	56.94	57.03	64.72	56.80	48.49	43.54	52.47	52.49	
			EEGPT	47.56	57.43	59.14	45.56	59.67	51.35	71.44	63.59	60.88	65.62	46.97	58.68	62.81	56.86	60.44	53.14	45.22	40.34	46.24	47.60	
			CBraMod	49.51	61.89	43.23	51.93	64.33	72.27	60.73	60.97	73.21	58.11	52.69	63.50	60.42	58.42	65.95	58.76	60.79	45.16	57.77	59.92	
			TFM	52.63	59.03	46.29	51.26	62.02	70.00	64.08	60.66	68.80	61.68	58.16	61.00	65.55	59.84	68.04	60.11	55.25	48.43	61.72	58.54	
			EEGMamba	53.15	57.55	53.61	59.85	63.77	70.57	70.70	71.13	62.42	78.91	65.95	62.79	64.80	58.32	61.12	61.53	66.48	56.22	49.47	60.52	63.82
			SingLEM	26.41	24.57	30.85	28.43	31.31	23.99	39.97	21.48	38.85	25.10	22.40	38.59	35.61	47.94	45.24	23.28	21.08	24.90	29.34	26.83	
			LUNA-Base	68.35	68.45	48.80	58.97	72.00	74.14	71.46	59.23	74.22	68.28	69.98	67.94	63.13	59.78	73.15	64.61	62.44	66.74	67.03	59.36	
	Linear Probing	Foundation Models	BENDR	21.02	20.62	22.02	20.23	21.92	21.12	21.55	20.74	21.03	20.28	20.26	21.14	25.80	25.86	28.07	21.28	20.18	21.35	21.29	20.67	
			BIOT-1D	46.69	61.53	60.26	43.73	65.97	75.85	62.61	38.29	66.72	45.17	56										

Table 37. Balanced classification accuracies (%) on Sleep-EDFx. The best BCAs are marked in bold, and the second best by an underline (continued).

Scenario	Tuning	Model Type	Approach	S40	S41	S42	S43	S44	S45	S46	S47	S48	S49	S50	S51	S52	S53	S54	S55	S56	S57	S58	S59
Within-subject (Few-shot)	Specialist Models	Specialist Models	PSD+LDA	59.46	74.13	57.22	62.63	70.36	58.04	53.13	67.96	70.65	40.97	55.67	57.74	59.62	58.24	58.23	64.30	34.91	45.94	<u>59.78</u>	55.52
			EEGNet	47.10	53.53	44.73	46.44	61.93	44.32	42.13	54.34	45.10	35.09	38.44	35.91	49.83	44.05	44.62	43.49	34.12	43.43	39.93	51.22
			ShallowConv	60.75	70.65	56.67	58.53	70.56	46.68	51.98	61.58	55.58	33.39	38.76	38.40	47.89	52.59	53.30	52.45	39.61	47.89	43.81	57.49
			LMDA	49.21	62.67	42.43	50.75	60.69	43.64	42.73	48.26	40.86	28.87	30.79	29.19	44.80	39.68	47.44	39.42	29.73	36.64	39.23	48.87
			CNN-T	71.05	<u>76.11</u>	69.04	73.02	<u>79.44</u>	44.92	51.49	65.71	77.53	45.01	51.08	63.56	57.63	75.15	<u>65.70</u>	56.95	45.85	47.59	45.67	68.56
			Deformer	56.30	69.29	52.96	53.22	62.77	43.87	54.71	58.60	48.95	34.51	29.67	39.49	51.37	46.50	50.29	52.21	44.49	44.13	39.48	53.39
			Conformer	60.02	68.23	65.21	63.30	67.76	60.65	56.11	75.93	71.74	<u>47.85</u>	54.51	65.00	56.96	70.55	60.03	<u>67.22</u>	<u>58.71</u>	60.61	45.84	<u>64.46</u>
			TSception	61.92	74.68	57.72	61.71	73.48	55.52	46.84	70.69	64.63	46.95	45.96	41.56	63.40	64.36	55.45	73.16	48.92	46.83	49.98	57.82
	Full Fine-tuning	Foundation Models	BENDR	26.20	39.25	39.77	36.18	47.90	41.01	23.15	39.77	36.98	32.15	26.47	24.86	34.72	31.67	40.37	37.62	29.44	36.75	28.21	41.12
			BIOT-1D	53.08	70.98	62.05	55.92	63.53	51.94	44.57	71.08	<u>73.63</u>	47.66	62.83	<u>64.19</u>	55.73	71.04	62.73	48.05	46.32	50.58	40.45	59.43
			BIOT-2D	59.32	70.73	61.59	53.82	73.93	62.07	45.77	72.36	71.55	45.36	63.65	58.33	52.51	66.84	59.74	53.70	53.91	52.14	41.21	57.46
			BIOT-6D	64.10	71.91	61.93	56.34	70.69	58.65	47.73	73.28	70.75	38.66	56.97	62.12	57.03	73.67	69.96	51.49	49.11	50.22	47.67	56.31
			LaBraM	36.38	40.49	32.12	33.46	44.52	40.93	30.74	36.17	41.71	26.03	29.09	24.01	29.10	30.56	36.92	29.54	28.10	30.82	30.65	39.92
			Neuro-GPT	55.56	67.11	56.87	53.40	69.37	48.44	48.93	63.28	54.63	32.43	42.70	43.39	48.74	52.69	49.67	54.43	42.47	40.80	42.93	50.54
			EEGPT	62.65	67.78	57.64	50.32	64.54	55.30	49.60	68.83	50.79	42.00	47.60	44.08	56.67	61.17	39.16	48.04	37.62	42.35	57.78	56.72
			CBraMod	61.46	72.27	54.58	54.17	70.95	52.88	50.42	66.21	57.74	34.05	41.52	51.79	53.38	66.14	48.44	49.54	46.54	45.29	41.07	57.19
			TFM	64.81	70.08	58.67	60.04	70.39	53.51	50.30	68.24	61.13	39.62	51.76	46.94	56.34	66.35	54.15	55.68	41.92	45.40	45.73	60.17
			EEGMamba	61.74	76.66	61.31	64.53	79.58	52.31	51.51	<u>74.24</u>	60.82	34.92	42.73	<u>62.72</u>	65.01	56.61	63.51	55.68	41.92	45.40	45.73	60.17
			SingLEM	31.33	35.50	25.69	24.50	41.75	32.50	23.66	41.61	33.48	20.56	30.69	24.06	29.17	24.17	32.09	22.06	22.07	25.35	24.90	24.94
			LUNA-Base	65.57	72.51	<u>65.67</u>	<u>68.69</u>	72.90	66.71	58.23	65.83	73.54	49.38	56.58	58.92	68.09	<u>74.64</u>	61.07	66.75	60.55	69.79	54.23	60.34
	Linear Probing	Foundation Models	BENDR	20.70	20.77	20.65	21.55	22.05	20.88	19.73	20.69	20.68	20.50	21.04	20.08	21.63	20.59	20.73	22.24	23.36	21.15	21.20	20.77
			BIOT-1D	51.73	68.70	61.88	53.32	62.46	53.35	38.91	62.76	70.74	44.81	50.12	51.81	51.27	63.69	49.80	56.91	38.97	39.08	33.09	52.32
			BIOT-2D	55.44	64.27	56.13	48.02	59.65	47.74	33.76	60.92	66.20	30.60	46.19	49.91	46.59	62.62	45.82	49.64	41.48	36.21	27.21	47.41
			BIOT-6D	58.07	67.88	51.47	57.13	68.19	61.21	40.78	71.93	66.80	46.98	59.97	55.47	59.50	67.13	59.85	59.54	50.44	40.04	48.45	52.40
			LaBraM	48.17	59.70	47.90	45.96	61.16	48.60	45.77	59.29	50.60	33.19	42.12	38.96	47.54	45.35	47.42	43.32	34.81	39.36	39.60	50.20
			Neuro-GPT	51.32	59.31	54.66	46.89	59.17	46.54	45.84	57.62	51.41	31.74	39.87	43.92	45.47	47.06	48.21	48.19	36.70	40.47	36.68	44.46
			EEGPT	68.84	68.38	65.67	49.45	76.24	62.24	56.92	71.25	61.03	46.30	51.27	56.83	59.79	69.72	46.04	54.67	46.45	47.97	66.12	62.32
			CBraMod	40.74	50.15	41.32	45.81	49.45	43.43	39.31	48.71	48.51	25.43	33.07	29.44	34.55	40.52	44.47	31.31	29.04	30.19	31.40	40.71
			TFM	55.13	53.86	46.22	56.51	57.93	46.82	41.72	51.65	50.35	33.61	43.58	36.76	46.35	49.07	48.69	49.01	32.15	41.03	36.60	47.42
			EEGMamba	52.35	63.95	51.13	44.36	59.98	44.95	45.90	59.62	43.05	29.70	31.92	32.40	52.34	43.45	43.61	44.38	28.36	33.65	35.56	40.57
			SingLEM	22.16	23.53	21.40	21.52	24.44	23.46	22.09	22.49	20.23	19.44	22.93	20.97	23.65	21.91	22.61	23.15	21.60	23.00	21.64	21.92
			LUNA-Base	66.68	72.52	65.06	60.78	73.41	<u>65.69</u>	<u>57.20</u>	70.48	72.25	45.17	51.50	51.51	62.95	66.16	45.88	62.44	49.83	<u>64.07</u>	47.66	56.47
Within-subject (Few-shot)	Specialist Models	Specialist Models	BENDR	57.94	51.89	51.16	53.80	52.38	60.31	49.13	57.55	46.74	46.51	50.56	45.97	43.09	54.74	57.13	49.21	68.67	69.13		59.00
			EEGNet	51.35	39.86	48.26	61.88	46.68	49.67	40.97	43.36	46.51	50.60	53.91	54.57	36.35	48.85	47.81	43.78	51.36	49.22		48.29 _{a0.54}
			ShallowConv	58.81	43.80	52.18	64.33	54.97	42.65	43.25	48.52	48.78	54.99	55.63	58.16	38.84	50.74	50.27	50.94	61.77	63.64		55.29 _{a0.49}
			LMDA	56.78	40.88	47.57	57.94	46.92	44.92	29.74	42.96	44.27	48.39	49.38	53.81	27.94	48.76	45.02	41.87	50.12	54.38		46.75 _{a2.12}
			CNN-T	<u>63.12</u>	50.56	62.13	67.20	64.24	<u>64.05</u>	<u>60.96</u>	60.90	<u>50.66</u>	58.49	66.13	56.38	23.03	54.75	<u>68.19</u>	67.60	74.67	<u>71.48</u>		<u>64.77</u> _{a0.61}
			Deformer	58.95	43.00	52.05	62.33	50.79	48.26	41.87	47.62	48.04	54.09	53.76	58.46	40.62	49.66	46.30	48.60	57.85	62.58		52.26 _{a0.38}
			Conformer	59.16	40.41	71.33	72.61	57.02	48.56	62.95	57.95	41.74	61.06	57.13	59.60	58.43	49.32	63.54	61.48	71.00	61.97		63.31 _{a0.48}
			TSception	61.55	44.47	62.44	63.19	56.33	59.98	43.75	60.70	44.76	<u>59.68</u>	58.58	59.41	53.76	50.11	64.80	55.21	60.77	68.30		60.31 _{a0.29}
	Full Fine-tuning	Foundation Models	BENDR	44.21	40.99	39.61	44.69	41.17	35.85	25.23	34.58	39.62	42.07	44.96	51.62	24.38	46.44	36.55	36.45	39.00	38.93		37.34 _{a0.24}
			BIOT-1D	57.65	49.05	60.32	59.94	53.63	48.83	54.26	63.90	49.69	56.10	63.09	59.85	<u>59.42</u>	60.39	63.78	57.03	63.76	64.21		60.64 _{a0.29}
			BIOT-2D	58.48	49.23	61.55	59.76	54.91	45.77	53.37	65.23	42.98	54.50	62.73	56.05	52.53	58.88	66.68	57.72	62.04	64.62		60.79 _{a0.19}
			BIOT-6D	59.13	51.99	58.02	63.50	53.85	45.36	52.79	<u>65.30</u>	48.62	55.09	64.63	63.80	62.67	58.37	64.13	58.05	66.93	68.26		61.75 _{a0.48}
			LaBraM	46.88	40.31	32.62	39.58	43.94	38.30	23.19	29.18	33.20	34.99	37.62	42.14	21.44	36.42	34.27	29.40	39.29	31.37		35.99 _{a0.23}
			Neuro-GPT	53.29	41.63	54.89	59.50	54.06	49.78	45.11	48.27	44.51	57.27	55.65	57.89	41.77	51.41	55.32	48.97	57.69	59.06		54.50 _{a0.29}
			EEGPT	54.08	41.03	45.14	57.85	53.36	48.53	46.64	43.99	44.20	51.11	56.62	58.75	41.57	53.75	52.11	47.17	65.83	60.97		56.38 _{a0.16}
			CBraMod	56.79	41.98	54.93	63.40	56.10	49.37	41.30	53.05	44.40	58.76	63.30	54.45	49.71	52.63	55.13	48.59	63.11	63.28		57.72 _{a0.08}
			TFM	60.07	44.34	56.48	62.46	57.36	51.08	46.30	56.96	47.42	57.68	57.96	63.02	50.97	50.57	55.21	53.22	61.35	63.97		58.82 _{a0.14}
EEGMamba	59.00	47.39	63.63	59.58	57.88	57.21	52.25	56.88	49.43	54.89	62.45	59.18	46.38	53.62	67.21	48.17	69.07	65.74		61.50 _{a0.62}			
SingLEM	38.24	36.23	28.12	32.48	37.72	35.85	20.82	33.84	36.18	34.95	28.24	46.64	20.25	32.85	31.97	24.40	37.37	25.17		29.71 _{a1.75}			
LUNA-Base	60.22	58.03	74.22	<u>69.41</u>	54.17	65.36	49.04	59.74	<u>50.66</u>	59.64	71.28	<u>64.98</u>	53.04	69.84	70.01	<u>66.23</u>	<u>74.34</u>	71.841					

Table 38. Classification accuracies (%) on SEED. The best accuracies are marked in bold, and the second best by an underline.

Scenario	Tuning	Model Type	Approach	S0	S1	S2	S3	S4	S5	S6	S7	S8	S9	S10	S11	S12	S13	S14	Avg.
Cross-subject (LOSO)	Specialist Models	Specialist Models	CSP+LDA	33.68	46.91	45.02	58.93	37.07	58.66	38.66	59.63	41.07	57.13	62.67	48.00	59.13	52.68	34.47	48.91
			EEGNet	32.40	39.06	47.20	54.10	43.64	53.91	41.42	65.06	59.62	39.16	55.98	48.72	53.87	62.87	31.57	48.57 _{±1.34}
			ShallowConv	38.14	48.57	36.32	55.76	52.64	57.65	50.71	69.46	54.11	48.23	66.54	59.75	60.77	68.07	34.47	53.41 _{±0.12}
			LMDA	33.02	41.07	50.85	59.70	41.44	60.80	38.19	67.78	50.91	53.78	51.10	55.17	50.36	58.90	38.79	50.12 _{±0.43}
			CNN-T	31.58	33.36	44.49	56.87	37.94	32.16	38.35	60.03	61.22	51.66	52.70	44.89	43.86	45.98	33.34	44.56 _{±1.40}
			Deformer	33.70	52.11	38.85	58.35	47.96	56.47	44.99	67.22	52.46	42.97	66.27	49.66	59.12	66.39	29.31	51.05 _{±0.97}
			Conformer	34.32	38.19	38.48	44.62	45.55	55.77	43.40	67.18	53.04	36.27	65.67	46.66	61.01	60.07	41.12	48.76 _{±1.23}
			TSception	33.33	33.85	47.43	51.13	29.62	54.47	33.86	59.18	48.68	50.63	39.69	49.39	33.96	33.10	28.91	41.82 _{±0.79}
	Full Fine-tuning	Foundation Models	BENDR	35.59	51.03	48.96	63.97	44.02	47.92	50.96	70.85	52.18	53.67	61.15	55.65	52.97	62.99	35.64	52.50 _{±0.85}
			BIOT-1D	33.62	49.04	55.88	53.61	41.77	34.80	53.81	53.66	49.00	64.39	65.01	55.76	46.93	55.28	31.44	49.60 _{±1.51}
			BIOT-2D	33.16	46.60	56.94	58.69	45.52	36.30	52.45	66.07	47.42	60.94	57.27	46.56	53.35	58.14	32.67	50.14 _{±1.28}
			BIOT-6D	30.39	38.67	61.27	52.75	44.52	34.04	47.53	68.84	53.10	54.69	53.43	48.07	51.21	52.89	44.16	49.04 _{±0.90}
			LaBraM	50.65	49.88	49.68	58.21	46.86	51.24	46.39	68.73	51.46	49.35	62.18	44.49	48.67	49.86	55.87	52.23 _{±0.92}
			Neuro-GPT	59.11	43.80	41.78	55.96	41.30	51.03	40.25	64.99	49.29	38.73	61.13	37.47	57.62	52.38	50.18	49.67 _{±0.38}
			EEGPT	59.89	43.72	41.81	54.49	46.86	50.76	36.11	58.77	43.02	40.25	50.77	47.45	53.91	58.05	45.28	48.74 _{±3.41}
			CBraMod	52.38	48.88	57.81	68.99	54.67	60.00	39.78	58.92	52.37	54.65	60.68	49.42	54.47	54.59	36.50	53.61 _{±0.61}
			TFM	34.08	35.39	34.90	39.20	37.20	36.23	34.43	39.02	40.24	36.20	41.69	37.77	34.80	34.33	34.41	36.66 _{±0.26}
			BrainOmni-Tiny	40.33	36.59	36.06	36.52	36.01	40.60	37.03	38.53	36.91	37.96	38.99	36.64	37.99	40.71	39.43	38.02 _{±0.03}
			BrainOmni-Base	36.19	38.74	39.50	38.59	41.14	56.50	42.20	49.95	45.83	49.53	51.14	49.02	47.64	47.15	37.73	44.72 _{±0.18}
			EEGMamba	33.94	44.08	46.17	59.75	53.46	53.77	47.08	70.52	46.87	60.42	66.07	54.40	35.09	64.67	46.50	52.19 _{±0.04}
			SingLEM	63.73	44.47	43.62	51.07	44.40	50.55	40.42	51.30	39.72	54.86	60.26	54.29	42.03	53.52	58.10	50.16 _{±1.48}
			LUNA-Base	37.20	49.37	47.97	59.87	46.03	55.48	49.63	50.23	52.95	61.74	54.89	46.90	59.89	43.56	33.66	49.96 _{±0.35}
	Linear Probing	Foundation Models	BENDR	34.09	33.72	34.55	34.52	33.68	34.62	34.76	35.33	35.85	35.43	34.40	34.16	34.37	34.31	36.58	34.69 _{±0.10}
			BIOT-1D	33.17	40.13	48.60	63.33	40.12	56.68	51.37	60.44	50.95	49.04	61.98	49.31	57.08	64.15	33.48	50.66 _{±0.12}
			BIOT-2D	31.56	38.79	53.83	63.73	41.87	63.32	44.77	60.67	51.90	47.66	62.94	49.70	57.32	64.13	33.26	51.03 _{±0.76}
			BIOT-6D	33.32	42.47	47.91	58.52	43.13	62.02	47.60	63.37	50.75	50.40	58.79	50.96	56.36	66.41	34.16	51.08 _{±0.77}
			LaBraM	55.78	47.94	47.27	57.83	45.88	56.05	41.25	66.19	41.30	53.22	64.05	51.86	37.85	61.70	58.80	52.46 _{±0.05}
			Neuro-GPT	32.79	46.09	38.12	56.55	42.59	53.31	60.99	54.45	45.50	55.14	54.29	51.64	58.79	62.85	31.87	49.66 _{±0.79}
			EEGPT	36.59	41.88	44.39	56.24	43.17	43.02	47.87	68.63	50.60	52.82	66.23	56.32	50.62	57.24	34.93	50.04 _{±1.03}
			CBraMod	33.49	37.94	48.63	65.30	41.19	63.29	42.94	70.85	51.16	64.24	60.06	55.23	43.14	58.95	40.98	51.83 _{±0.07}
			TFM	33.07	35.35	34.18	36.41	35.44	36.15	34.14	36.20	37.35	34.86	37.30	34.86	34.63	34.32	34.96	35.28 _{±0.04}
			BrainOmni-Tiny	37.47	38.24	40.39	37.36	40.99	52.55	42.25	49.24	48.08	46.09	51.37	47.14	48.18	48.33	36.36	44.27 _{±0.11}
			BrainOmni-Base	36.63	38.84	40.58	37.95	41.00	52.06	42.94	47.14	46.52	47.51	51.06	46.40	46.30	49.60	36.42	44.06 _{±0.26}
			EEGMamba	30.25	39.07	49.33	58.90	52.50	56.51	48.48	71.22	45.05	58.24	62.82	53.20	40.51	61.75	50.71	51.90 _{±0.15}
			SingLEM	33.83	33.89	34.72	35.17	34.86	38.33	35.43	37.20	34.82	36.10	36.91	38.60	35.22	33.92	36.29	35.69 _{±0.04}
			LUNA-Base	33.07	33.22	43.24	33.40	41.40	63.09	37.33	47.05	45.59	49.29	47.15	45.78	35.52	50.41	35.97	42.77 _{±0.22}
Within-subject (Few-shot)	Specialist Models	Specialist Models	CSP+LDA	63.57	53.79	40.81	58.27	49.01	60.11	54.37	63.01	68.27	61.29	67.76	40.55	32.02	54.23	33.57	53.38
			EEGNet	34.47	45.10	37.60	61.48	43.63	33.21	54.85	68.52	75.38	56.71	63.06	53.58	60.94	53.38	39.84	52.12 _{±0.47}
			ShallowConv	33.54	62.63	35.87	71.63	45.31	38.73	48.97	67.01	67.45	61.42	64.67	54.79	32.16	54.91	40.49	51.97 _{±0.26}
			LMDA	36.76	57.05	45.86	74.88	43.53	37.88	46.24	66.04	60.88	58.44	64.91	50.81	42.43	55.59	56.69	53.20 _{±1.39}
			CNN-T	37.30	58.52	45.27	78.09	46.25	35.27	37.02	64.03	67.03	57.86	63.73	54.69	33.01	48.76	52.37	51.95 _{±1.06}
			Deformer	33.55	61.32	40.61	76.14	48.33	40.38	38.90	70.33	65.99	63.53	64.14	53.77	33.84	52.94	39.01	52.19 _{±0.30}
			Conformer	55.22	49.95	40.89	54.50	43.92	66.46	56.62	65.81	61.94	58.32	68.36	51.14	33.09	56.32	72.54	55.67 _{±1.63}
			TSception	33.54	53.93	45.22	77.94	45.34	35.07	36.90	67.72	70.33	47.78	62.24	52.83	61.65	44.80	55.18	52.70 _{±1.21}
	Full Fine-tuning	Foundation Models	BENDR	33.27	37.21	38.66	41.62	35.72	34.82	34.69	54.40	69.39	41.97	41.20	44.82	30.86	37.39	39.42	41.03 _{±0.54}
			BIOT-1D	31.46	53.54	44.49	58.87	34.04	39.31	36.27	63.16	69.72	53.71	61.43	53.95	39.49	43.41	39.85	48.18 _{±1.20}
			BIOT-2D	31.81	51.63	47.48	55.43	38.96	43.88	36.08	65.92	68.55	53.21	62.44	53.06	37.13	42.54	36.27	48.29 _{±0.67}
			BIOT-6D	32.62	49.79	41.24	64.11	37.25	40.31	34.14	58.95	69.30	55.93	68.63	51.96	32.45	41.72	47.75	48.41 _{±0.56}
			LaBraM	32.05	42.37	41.58	47.87	45.61	38.97	43.75	63.33	55.16	58.53	61.32	51.86	32.82	53.24	36.61	47.00 _{±0.92}
			Neuro-GPT	57.98	55.36	39.50	42.30	43.95	59.82	60.21	65.28	62.23	57.98	66.90	51.76	37.39	62.71	75.10	55.90 _{±0.09}
			EEGPT	35.48	33.31	36.79	34.40	36.21	40.40	37.59	55.39	53.88	37.78	51.41	38.81	38.87	40.32	36.52	40.48 _{±0.44}
			CBraMod	32.12	69.34	43.87	74.02	49.77	47.39	48.35	65.40	64.33	55.00	65.39	54.20	30.25	51.69	86.20	55.82 _{±0.39}
			TFM	38.70	34.72	35.01	35.34	36.40	33.32	34.38	37.51	38.17	33.38	36.52	35.65	32.66	33.81	35.39	35.40 _{±0.15}
			BrainOmni-Tiny	47.29	36.73	39.01	38.76	42.30	54.69	44.19	56.52	53.32	42.83	50.86	48.65	39.29	48.16	58.47	46.74 _{±0.39}
			BrainOmni-Base	32.44	39.40	40.54	38.49	43.54	35.66	41.36	56.03	54.58	44.99	53.32	49.95	44.22	47.72	43.84	44.41 _{±0.12}
			EEGMamba	27.25	58.84	43.28	69.66	50.78	51.68	53.24	63.81	61.58	54.33	62.01	52.90	29.26	50.29	41.09	51.33 _{±0.27}
			SingLEM	41.24	34.80	40.44	41.03	39.41	47.08	45.47	41.73	57.05	50.50	57.86	48.91	43.01	43.64	72.95	47.01 _{±0.21}
			LUNA-Base	33.69	43.44	40.87	50.12	41.95	35.25	40.69	57.61	55.75	52.02	63.38	45.39	31.03	56.02	43.65	46.06 _{±0.68}
	Linear Probing	Foundation Models	BENDR	32.50	32.33	34.42	34.58	35.58	33.04	33.70	34.23	34.58	33.58	35.85	33.22	34.13	33.95	34.39	34.00 _{±0.16}
			BIOT-1D	34.02	48.69	47.90	54.52	33.98	40.81	45.78	61.15	67.49	51.65	62.41	53.57	28.77	47.87	45.31	48.26 _{±0.53}
			BIOT-2D	30.80	40.74	48.28	58.96	33.54	50.34	33.92	59.68	67.65	55.25	61.14	51.21	31.67	41.09	47.22	47.43 _{±0.43}
			BIOT-6D	32.33	52.16	43.49	61.86	32.60	41.47	34.36	66.46	66.89	54.51	63.10	52.77	27.17	41.96	50.33	48.10 _{±0.65}
			LaBraM	39.01	53.55	45.36	53.08	41.03	42.30	35.67	63.30	62.40	47.94	64.67	55.07	33.80	44.45	60.23	49.46 _{±0.14}
			Neuro-GPT	34.77	55.84	35.97	59.23	50.97	51.82	49.25	68.88	42.36	53.10</						

Table 39. Cohen's Kappa (%) on SEED. The best metrics are marked in bold, and the second best by an underline.

Scenario	Tuning	Model Type	Approach	S0	S1	S2	S3	S4	S5	S6	S7	S8	S9	S10	S11	S12	S13	S14	Avg.
Cross-subject (LOSO)	Full Fine-tuning	Specialist Models	CSP+LDA	-1.19	20.93	17.19	38.53	4.22	37.92	8.86	39.11	10.36	35.85	44.10	21.48	38.18	29.26	0.00	22.99
			EEGNet	-0.79	9.48	20.03	31.48	15.43	31.17	12.77	47.38	39.23	9.62	34.28	23.18	30.47	44.16	-4.00	22.93 _{±1.95}
			ShallowConv	6.58	23.44	2.94	33.66	28.73	36.16	26.52	54.04	30.76	22.86	49.85	39.62	40.90	51.99	0.00	29.87 _{±0.15}
			LMDA	0.03	12.50	25.79	39.56	12.22	40.93	8.23	51.66	25.92	30.99	27.16	33.05	25.64	38.40	8.76	25.39 _{±0.62}
			CNN-T	-4.14	1.21	16.22	35.20	7.31	-2.98	8.46	39.94	41.69	27.67	29.11	17.63	16.44	18.86	-1.03	16.77 _{±2.09}
			Deformer	-0.06	28.65	6.86	37.54	21.60	34.60	18.05	50.65	28.22	15.10	49.46	24.57	38.29	49.53	-6.78	26.42 _{±1.49}
			Conformer	1.95	8.23	6.28	17.28	18.25	33.65	15.69	50.61	29.10	4.97	48.50	20.39	41.40	39.97	11.86	23.21 _{±1.81}
			TSception	0.00	0.87	21.37	27.48	-5.51	32.37	0.92	40.08	23.35	26.79	10.17	24.52	0.99	-0.23	-6.57	13.11 _{±1.22}
	Foundation Models	BENDR	3.31	27.01	22.60	46.05	16.30	21.77	26.73	56.23	27.63	30.68	41.57	33.32	29.51	44.16	3.63	28.70 _{±1.23}	
		BIOT-1D	0.86	24.07	33.40	30.72	13.06	3.26	31.01	30.59	22.81	46.66	47.49	33.71	20.62	32.79	-3.90	24.48 _{±2.25}	
		BIOT-2D	-0.27	20.41	35.12	38.20	18.71	5.39	28.94	48.97	20.98	41.58	35.65	20.06	30.04	37.27	-1.03	25.33 _{±1.89}	
		BIOT-6D	-3.51	8.58	41.83	29.27	17.28	1.56	21.75	53.17	29.02	32.03	29.99	21.74	27.13	29.35	16.43	23.71 _{±1.38}	
		LaBraM	25.95	25.15	23.81	37.55	19.96	26.93	20.13	52.97	26.71	24.43	43.06	16.90	23.37	24.85	34.02	28.39 _{±1.39}	
		Neuro-GPT	38.80	16.39	11.96	34.18	12.45	26.12	11.20	47.33	23.35	8.98	41.75	6.98	36.51	28.34	25.72	24.67 _{±0.59}	
		EEGPT	39.91	16.25	11.92	31.87	20.58	25.35	5.19	37.53	13.40	11.00	25.57	21.27	30.93	36.92	18.25	23.06 _{±5.22}	
		CBraMod	28.61	23.79	36.38	53.53	31.40	39.86	10.03	38.13	28.13	32.22	40.81	24.40	31.96	31.95	3.99	30.35 _{±0.94}	
		TFM	1.19	3.53	2.11	8.67	5.54	4.25	1.34	7.73	9.79	4.40	12.69	6.35	1.69	0.80	1.76	4.79 _{±0.42}	
		BrainOmni-Tiny	9.71	4.15	3.05	3.72	2.60	9.77	4.76	6.43	3.95	5.97	7.48	3.69	5.82	10.26	8.10	5.96 _{±0.06}	
		BrainOmni-Base	4.48	8.65	9.05	8.21	11.12	34.75	13.56	24.20	17.94	24.23	26.95	23.37	21.48	21.13	5.47	16.97 _{±0.31}	
		EEGMamba	-0.47	16.86	18.52	39.55	29.61	30.02	21.19	55.73	19.48	40.78	49.07	31.73	3.09	46.91	19.50	28.11 _{±0.06}	
		SingLEM	45.57	17.32	14.62	26.04	15.71	25.28	11.46	26.15	8.37	32.52	40.33	31.32	13.33	29.70	36.57	24.95 _{±2.30}	
		LUNA-Base	6.64	24.37	21.21	39.86	18.53	33.08	24.69	24.70	28.88	42.69	32.41	20.56	39.81	15.82	-0.77	24.83 _{±0.51}	
	Linear Probing	BENDR	0.96	0.29	1.54	1.39	0.28	1.64	1.88	2.77	3.43	2.85	1.31	0.97	1.31	1.11	4.50	1.75 _{±0.16}	
		BIOT-1D	0.85	11.02	22.75	45.07	10.65	35.31	27.37	40.38	25.76	24.03	43.13	23.61	35.55	46.18	-1.43	26.02 _{±0.20}	
		BIOT-2D	-2.55	8.03	30.83	45.86	12.79	45.44	17.25	41.72	28.46	21.35	44.65	24.94	36.78	46.43	-0.12	26.79 _{±1.14}	
		BIOT-6D	1.04	14.44	21.12	37.67	15.25	43.09	21.87	44.82	25.43	26.05	38.15	26.29	34.50	49.56	-0.47	26.59 _{±1.18}	
		LaBraM	33.89	22.52	20.11	36.57	18.24	33.63	12.63	49.14	10.73	30.02	46.07	28.10	7.26	42.55	38.03	28.63 _{±0.07}	
		Neuro-GPT	-0.84	19.02	7.50	35.43	14.13	30.43	41.71	32.39	18.86	33.14	31.57	27.65	39.07	44.52	-2.28	24.82 _{±1.18}	
		EEGPT	5.45	13.60	15.58	34.33	14.59	13.40	22.13	52.89	25.30	29.52	49.40	34.29	25.31	35.69	2.49	24.93 _{±1.58}	
		CBraMod	0.73	7.93	22.37	47.98	11.90	44.81	15.14	56.32	26.26	46.42	40.30	33.19	15.32	38.58	12.28	27.97 _{±0.11}	
		TFM	-0.81	2.85	0.61	4.11	2.50	3.72	0.43	3.40	5.12	1.72	5.44	1.65	1.26	0.59	2.15	2.32 _{±0.13}	
		BrainOmni-Tiny	6.19	8.01	10.54	6.42	11.14	28.90	13.68	23.14	21.65	18.74	27.27	20.76	22.04	22.60	3.41	16.30 _{±0.16}	
		BrainOmni-Base	5.00	8.85	10.75	7.33	11.12	28.15	14.49	19.85	19.03	20.90	26.84	19.53	19.37	24.53	3.49	15.95 _{±0.40}	
		EEGMamba	-4.37	9.48	23.64	38.29	28.32	34.24	23.19	56.74	16.67	37.52	44.27	30.04	11.08	42.67	25.85	27.84 _{±0.22}	
		SingLEM	0.74	0.71	1.73	2.34	1.97	7.17	2.82	5.48	1.92	3.91	5.06	7.78	2.52	0.57	4.07	3.25 _{±0.06}	
		LUNA-Base	0.78	0.86	15.05	0.72	12.05	44.75	6.02	19.68	17.50	23.64	21.02	18.29	1.66	25.80	2.60	14.03 _{±0.34}	
Within-subject (Few-shot)	Full Fine-tuning	Specialist Models	CSP+LDA	45.65	30.60	11.64	37.66	23.21	40.53	31.45	44.11	52.58	41.78	51.43	11.19	0.00	31.21	-0.05	30.20
			EEGNet	1.81	17.04	6.49	42.15	15.55	0.33	31.90	52.78	63.07	34.83	44.69	30.36	41.56	29.98	8.49	28.07 _{±0.71}
			ShallowConv	-0.09	44.12	4.17	57.39	17.03	8.88	23.26	50.20	51.15	42.56	46.92	32.22	0.12	32.98	9.68	28.04 _{±0.39}
			LMDA	4.91	36.14	19.52	62.19	14.33	7.77	19.18	48.74	41.32	38.20	47.45	26.68	14.38	34.10	34.99	29.99 _{±2.00}
			CNN-T	5.86	37.68	18.25	67.04	18.38	4.17	5.76	45.78	50.76	37.22	45.50	32.16	1.42	24.17	27.99	28.14 _{±1.58}
			Deformer	-0.04	42.24	11.61	64.09	21.92	10.92	8.36	55.28	49.01	45.52	46.11	30.99	2.60	30.20	7.40	28.41 _{±0.45}
			Conformer	33.43	25.04	11.66	31.78	15.54	49.68	34.78	48.42	42.74	37.94	52.45	26.91	1.53	34.40	58.68	33.67 _{±2.45}
			TSception	-0.05	31.02	18.26	66.89	17.12	3.45	5.36	51.32	55.34	22.15	43.20	29.45	42.80	18.31	32.40	29.13 _{±1.79}
	Foundation Models	BENDR	-0.19	5.62	8.27	12.56	3.47	3.09	1.91	31.37	54.10	13.27	11.79	17.20	-2.02	6.89	8.26	11.71 _{±0.81}	
		BIOT-1D	-2.97	30.60	17.37	38.48	1.25	9.61	4.14	44.42	54.75	31.05	42.40	31.19	10.18	16.22	11.17	22.66 _{±1.73}	
		BIOT-2D	-2.43	27.77	21.72	33.36	8.33	16.24	4.21	48.62	53.03	30.41	43.76	29.87	7.30	14.83	5.90	22.86 _{±0.97}	
		BIOT-6D	-1.31	24.56	12.04	46.12	6.51	10.81	0.83	38.09	54.07	34.16	52.98	27.82	0.44	13.35	21.63	22.81 _{±0.90}	
		LaBraM	-1.62	13.30	12.59	21.81	17.62	9.01	15.30	44.71	32.41	38.05	41.84	27.94	1.10	30.59	4.22	20.59 _{±1.34}	
		Neuro-GPT	37.28	33.44	9.54	13.64	15.60	39.68	40.17	47.60	43.31	37.45	50.27	27.85	7.72	44.17	62.50	34.01 _{±0.13}	
		EEGPT	3.17	0.05	5.40	1.61	4.13	10.77	6.54	32.72	30.65	7.03	27.16	8.02	8.64	10.51	4.71	10.74 _{±0.66}	
		CBraMod	-1.24	54.04	16.20	60.97	24.49	21.52	22.53	47.75	46.43	32.73	47.91	31.34	-2.82	28.23	79.29	33.96 _{±0.59}	
		TFM	8.10	2.14	2.86	3.01	4.42	0.80	1.45	6.10	7.22	0.39	4.87	3.59	-0.72	0.94	2.95	3.21 _{±0.20}	
		BrainOmni-Tiny	21.25	5.11	8.74	8.30	13.21	31.96	16.30	34.49	29.84	14.21	26.16	23.31	10.12	22.37	37.49	20.19 _{±0.58}	
		BrainOmni-Base	-1.48	8.86	11.32	7.97	15.04	4.37	11.81	33.68	31.68	17.51	29.96	25.13	16.63	21.57	15.49	16.64 _{±0.19}	
		EEGMamba	-8.43	38.30	15.29	54.37	25.61	28.25	29.47	45.32	42.20	31.66	42.93	29.68	-4.17	26.34	10.56	27.16 _{±0.41}	
		SingLEM	12.45	2.14	10.97	11.70	8.87	21.27	18.30	12.59	35.25	26.05	36.74	23.64	14.63	15.01	59.39	20.60 _{±3.09}	
		LUNA-Base	0.13	16.04	11.51	25.38	11.98	3.40	10.42	35.99	33.39	28.52	45.18	18.28	-1.83	33.98	14.87	19.15 _{±0.27}	
	Linear Probing	BENDR	-1.19	-1.46	1.71	1.80	3.19	-0.40	0.64	1.35	1.84	0.30	3.85	-0.21	1.19	0.86	1.50	1.00 _{±0.25}	
		BIOT-1D	0.99	22.65	21.98	31.97	0.59	11.81	18.20	41.31	51.42	27.79	43.67	29.96	-5.81	22.26	16.96	22.38 _{±0.82}	
		BIOT-2D	-3.76	10.44	22.66	38.50	-0.08	25.99	0.46	39.14	51.57	33.09	41.80	26.30	-1.22	12.45	19.92	21.15 _{±0.62}	
		BIOT-6D	-1.45	28.00	15.50	42.77	-1.47	12.99	1.08	49.41	50.41	32.19	48.69	28.68	-8.03	13.92	24.71	22.23 _{±0.92}	
		LaBraM	8.53	30.47	18.44	29.53	11.00	14.12	3.10	44.60	43.49	22.27	46.96	32.82	2.55	17.55	39.95	24.36 _{±0.22}	
		Neuro-GPT	1.96	34.03	2.35	39.23	26.26	27.23	24.11	53.19	12.44	29.88	49.02	34.42	35.02	43.35	3.76	27.82 _{±0.14}	
		EEGPT	0.11	9.79	8.71	20.94	17.82	1.28	4.46	27.06	42.54	16.44	40.05	18.68	0.00	3.23	6.84	14.48 _{±0.63}	
		CBraMod	3.87	39.64	22.48	61.46	17.63	36.53	5.59	47.90	46.24	25.31	46.27						

Table 40. Classification accuracies (%) on Nakanishi2015. The best accuracies are marked in bold, and the second best by an underline.

Scenario	Tuning	Model Type	Approach	S0	S1	S2	S3	S4	S5	S6	S7	S8	Avg.
Cross-subject (LOSO)	Specialist Models	Specialist Models	TRCA	91.67	72.78	85.00	100.00	99.44	100.00	99.44	98.89	99.44	94.07
			EEGNet	<u>92.78</u>	<u>78.70</u>	<u>92.96</u>	100.00	99.63	100.00	<u>99.44</u>	100.00	<u>99.44</u>	<u>95.88</u> _{±0.18}
			ShallowConv	56.85	29.63	71.11	83.89	36.48	92.96	80.37	90.93	84.26	69.61 _{±0.86}
			LMDA	62.59	42.59	81.48	98.89	87.59	<u>99.63</u>	95.56	100.00	97.78	85.12 _{±0.94}
			CNN-T	29.26	19.63	42.59	55.00	41.67	63.15	54.07	61.30	50.37	46.34 _{±0.34}
			Deformer	93.33	86.85	94.81	100.00	99.63	100.00	100.00	100.00	100.00	97.18 _{±0.13}
			Conformer	82.41	65.56	86.48	100.00	92.96	99.44	96.48	<u>99.26</u>	<u>99.44</u>	91.34 _{±1.34}
			TSception	40.19	20.56	47.04	91.11	35.74	88.33	80.93	90.74	50.00	60.51 _{±0.88}
	Full Fine-tuning	Foundation Models	BENDR	85.56	62.96	90.37	100.00	99.44	100.00	99.44	<u>99.26</u>	<u>99.44</u>	92.94 _{±0.30}
			BIOT-1D	69.81	29.81	84.26	88.89	63.52	88.89	83.15	95.00	82.22	76.17 _{±1.59}
			BIOT-2D	73.33	47.04	81.85	62.22	75.00	85.37	55.93	93.89	58.70	70.37 _{±6.88}
			BIOT-6D	53.33	40.37	79.44	87.78	45.00	87.04	83.15	92.41	82.59	72.35 _{±3.04}
			LaBraM	48.33	37.22	83.15	98.52	73.70	100.00	91.48	97.96	82.22	79.18 _{±0.73}
			Neuro-GPT	72.59	61.67	77.04	98.33	90.74	97.96	93.70	96.30	97.78	87.35 _{±0.40}
			EEGPT	64.63	57.96	92.78	100.00	87.78	98.52	97.78	99.07	96.30	88.31 _{±3.30}
			CBraMod	73.33	47.59	84.44	<u>99.63</u>	81.11	<u>99.63</u>	95.56	96.85	90.37	85.39 _{±0.60}
			TFM	8.70	7.41	12.59	14.44	10.00	19.26	16.11	16.11	10.93	12.84 _{±0.68}
			BrainOmni-Tiny	45.56	39.63	82.59	98.89	69.81	<u>99.63</u>	86.11	96.48	86.30	78.33 _{±0.23}
			BrainOmni-Base	27.41	31.67	27.04	67.04	59.63	64.81	59.07	55.19	66.11	50.88 _{±1.99}
			EEGMamba	66.30	44.07	68.52	76.11	59.81	78.52	68.15	87.22	82.04	70.08 _{±0.94}
			SingLEM	13.15	14.26	43.15	35.93	22.22	56.48	42.78	55.19	18.70	33.54 _{±1.23}
			LUNA-Base	7.96	7.04	8.89	8.33	9.81	8.70	8.52	9.07	8.33	8.52 _{±0.22}
	Linear Probing	Foundation Models	BENDR	29.63	18.70	50.93	70.37	59.26	84.26	61.11	79.07	65.19	57.61 _{±4.18}
			BIOT-1D	70.19	29.07	81.30	61.48	28.52	84.07	82.78	95.19	85.93	68.72 _{±0.99}
			BIOT-2D	35.74	29.44	77.96	80.74	56.30	79.44	77.22	89.44	74.44	66.75 _{±2.27}
			BIOT-6D	49.44	26.48	80.19	82.22	60.37	88.15	76.67	92.59	58.70	68.31 _{±3.34}
			LaBraM	22.96	22.41	53.15	65.74	61.11	82.96	61.30	74.07	35.74	53.27 _{±0.83}
			Neuro-GPT	51.67	42.96	57.41	88.33	65.74	92.22	76.30	83.33	93.52	72.39 _{±1.75}
			EEGPT	74.63	59.44	76.85	99.44	82.59	100.00	96.67	87.22	92.41	85.47 _{±4.26}
			CBraMod	13.70	11.48	12.59	17.59	8.70	27.41	20.00	27.96	17.22	17.41 _{±0.28}
			TFM	10.19	9.26	10.37	12.96	10.19	15.93	11.67	7.04	10.37	10.88 _{±0.58}
			BrainOmni-Tiny	60.19	42.78	59.81	93.89	74.81	91.11	81.30	70.19	89.81	73.77 _{±0.66}
			BrainOmni-Base	15.74	15.37	17.59	27.96	32.59	30.74	26.85	27.96	29.63	24.94 _{±0.15}
			EEGMamba	18.89	12.96	9.81	22.04	14.81	22.04	15.37	15.37	29.07	17.82 _{±0.03}
			SingLEM	11.85	13.89	17.04	29.26	14.81	20.74	21.30	21.11	11.11	17.90 _{±0.22}
			LUNA-Base	8.89	7.96	8.33	9.81	10.56	9.44	8.33	8.89	11.85	9.34 _{±0.34}
Within-subject (Few-shot)	Specialist Models	Specialist Models	TRCA	97.22	94.44	100.00	100.00	100.00	100.00	100.00	100.00	<u>97.22</u>	98.77
			EEGNet	15.74	27.78	12.04	84.26	95.37	89.81	85.19	100.00	89.81	66.67 _{±4.00}
			ShallowConv	33.33	11.11	38.89	43.52	82.41	87.04	58.33	65.74	40.74	51.23 _{±2.02}
			LMDA	5.56	9.26	11.11	52.78	95.37	94.44	72.22	85.19	19.44	49.49 _{±1.96}
			CNN-T	31.48	10.19	27.78	76.85	94.44	83.33	85.19	87.96	57.41	61.63 _{±1.19}
			Deformer	17.59	17.59	54.63	<u>96.30</u>	97.22	100.00	92.59	100.00	68.52	71.60 _{±1.90}
			Conformer	40.74	8.33	35.19	49.07	55.56	44.44	41.67	60.19	37.04	41.36 _{±3.07}
			TSception	61.11	8.33	54.63	86.11	84.26	100.00	91.67	<u>99.07</u>	69.44	72.74 _{±2.79}
	Full Fine-tuning	Foundation Models	BENDR	73.15	16.67	86.11	100.00	98.15	100.00	99.07	100.00	100.00	85.91 _{±0.29}
			BIOT-1D	72.22	47.22	77.78	87.04	85.19	94.44	90.74	96.30	79.63	81.17 _{±2.67}
			BIOT-2D	52.78	28.70	77.78	88.89	92.59	92.59	84.26	93.52	87.96	77.67 _{±1.62}
			BIOT-6D	69.44	50.93	<u>92.59</u>	87.96	90.74	95.37	85.19	97.22	92.59	84.67 _{±1.48}
			LaBraM	61.11	26.85	74.07	92.59	96.30	98.15	92.59	100.00	59.26	77.88 _{±2.14}
			Neuro-GPT	40.74	21.30	54.63	90.74	94.44	98.15	95.37	100.00	92.59	76.44 _{±1.48}
			EEGPT	55.56	16.67	71.30	92.59	81.48	96.30	95.37	<u>99.07</u>	73.15	75.72 _{±8.11}
			CBraMod	49.07	12.04	38.89	80.56	77.78	87.96	81.48	93.52	39.81	62.35 _{±1.15}
			TFM	12.96	7.41	12.04	17.59	8.33	14.81	12.96	12.04	7.41	11.73 _{±1.26}
			BrainOmni-Tiny	60.19	24.07	76.85	<u>96.30</u>	93.52	<u>99.07</u>	96.30	100.00	91.67	82.00 _{±0.52}
			BrainOmni-Base	25.00	12.04	16.67	75.00	88.89	86.11	35.19	71.30	65.74	52.88 _{±1.29}
			EEGMamba	30.56	19.44	25.93	44.44	47.22	70.37	33.33	47.22	77.78	44.03 _{±0.15}
			SingLEM	25.93	10.19	26.85	44.44	17.59	48.15	48.15	57.41	18.52	33.02 _{±2.63}
			LUNA-Base	7.41	11.11	9.26	9.26	11.11	10.19	6.48	10.19	5.56	8.95 _{±0.50}
	Linear Probing	Foundation Models	BENDR	7.41	8.33	15.74	28.70	21.30	48.15	30.56	60.19	41.67	29.12 _{±2.45}
			BIOT-1D	65.74	<u>59.26</u>	82.41	91.67	92.59	97.22	91.67	94.44	<u>97.22</u>	85.80 _{±1.01}
			BIOT-2D	55.56	11.11	63.89	76.85	77.78	94.44	82.41	87.96	77.78	69.75 _{±2.15}
			BIOT-6D	65.74	51.85	79.63	83.33	88.89	67.59	82.41	93.52	90.74	78.19 _{±5.09}
			LaBraM	57.41	19.44	50.00	69.44	78.70	89.81	89.81	93.52	54.63	66.98 _{±0.50}
			Neuro-GPT	42.59	19.44	26.85	64.81	50.93	88.89	52.78	43.52	84.26	52.67 _{±1.24}
			EEGPT	<u>86.11</u>	45.37	71.30	95.37	44.44	96.30	66.67	<u>99.07</u>	95.37	77.78 _{±3.81}
			CBraMod	17.59	17.59	17.59	20.37	25.93	33.33	38.89	34.26	12.96	24.28 _{±1.77}
			TFM	13.89	12.96	10.19	12.04	9.26	8.33	7.41	11.11	5.56	10.08 _{±1.72}
			BrainOmni-Tiny	35.19	22.22	40.74	61.11	74.07	70.37	42.59	52.78	68.52	51.95 _{±1.39}
			BrainOmni-Base	19.44	8.33	12.96	31.48	40.74	35.19	17.59	25.93	20.37	23.56 _{±2.52}
			EEGMamba	12.96	15.74	16.67	17.59	17.59	34.26	18.52	15.74	39.81	20.99 _{±0.50}
			SingLEM	12.96	16.67	13.89	20.37	19.44	32.41	23.15	17.59	15.74	19.14 _{±0.91}
			LUNA-Base	12.04	10.19	7.41	11.11	10.19	9.26	10.19	5.56	9.26	9.47 _{±1.72}

Table 41. Cohen's Kappa (%) on Nakanishi2015. The best metrics are marked in bold, and the second best by an underline.

Scenario	Tuning	Model Type	Approach	S0	S1	S2	S3	S4	S5	S6	S7	S8	Avg.
Cross-subject (LOSO)		Specialist Models	TRCA	90.91	70.30	83.64	100.00	<u>99.39</u>	100.00	<u>99.39</u>	98.79	<u>99.39</u>	93.54
			EEGNet	<u>92.12</u>	<u>76.77</u>	<u>92.32</u>	100.00	99.60	100.00	<u>99.39</u>	100.00	<u>99.39</u>	<u>95.51</u> _{±0.19}
			ShallowConv	52.93	23.23	68.48	82.42	30.71	92.32	78.59	90.10	82.83	66.85 _{±0.94}
			LMDA	59.19	37.37	79.80	98.79	86.46	<u>99.60</u>	95.15	100.00	97.58	83.77 _{±1.03}
			CNN-T	22.83	12.32	37.37	50.91	36.36	59.80	49.90	57.78	45.86	41.46 _{±0.37}
			Deformer	92.73	85.66	94.34	100.00	99.60	100.00	100.00	100.00	100.00	96.92 _{±0.14}
			Conformer	80.81	62.42	85.25	100.00	92.32	99.39	96.16	<u>99.19</u>	<u>99.39</u>	90.55 _{±1.47}
			TSception	34.75	13.33	42.22	90.30	29.90	87.27	79.19	89.90	45.45	56.92 _{±0.96}
	Full Fine-tuning	Foundation Models	BENDR	84.24	59.60	89.49	100.00	<u>99.39</u>	100.00	<u>99.39</u>	<u>99.19</u>	<u>99.39</u>	92.30 _{±0.32}
			BIOT-1D	67.07	23.43	82.83	87.88	60.20	87.88	81.62	94.55	80.61	74.01 _{±1.74}
			BIOT-2D	70.91	42.22	80.20	58.79	72.73	84.04	51.92	93.33	54.95	67.68 _{±7.51}
			BIOT-6D	49.09	34.95	77.58	86.67	40.00	85.86	81.62	91.72	81.01	69.83 _{±3.32}
			LaBraM	43.64	31.52	81.62	98.38	71.31	100.00	90.71	97.78	80.61	77.28 _{±0.80}
			Neuro-GPT	70.10	58.18	74.95	98.18	89.90	97.78	93.13	95.96	97.58	86.20 _{±0.44}
			EEGPT	61.41	54.14	92.12	100.00	86.67	98.38	97.58	98.99	95.96	87.25 _{±3.60}
			CBraMod	70.91	42.83	83.03	<u>99.60</u>	79.39	<u>99.60</u>	95.15	96.57	89.49	84.06 _{±0.66}
			TFM	0.40	-1.01	4.65	6.67	1.82	11.92	8.48	8.48	2.83	4.92 _{±0.74}
			BrainOmni-Tiny	40.61	34.14	81.01	98.79	67.07	99.60	84.85	96.16	85.05	76.36 _{±0.25}
			BrainOmni-Base	20.81	25.45	20.40	64.04	55.96	61.62	55.35	51.11	63.03	46.42 _{±2.17}
			EEGMamba	63.23	38.99	65.66	73.94	56.16	76.57	65.25	86.06	80.40	67.36 _{±1.02}
			SingLEM	5.25	6.46	37.98	30.10	15.15	52.53	37.58	51.11	11.31	27.50 _{±1.34}
			LUNA-Base	-0.40	-1.41	0.61	0.00	1.62	0.40	0.20	0.81	0.00	0.20 _{±0.24}
	Linear Probing	Foundation Models	BENDR	23.23	11.31	46.46	67.68	55.56	82.83	57.58	77.17	62.02	53.76 _{±4.57}
			BIOT-1D	67.47	22.63	79.60	57.98	22.02	82.63	81.21	94.75	84.65	65.88 _{±1.08}
			BIOT-2D	29.90	23.03	75.96	78.99	52.32	77.58	75.15	88.48	72.12	63.73 _{±2.47}
			BIOT-6D	44.85	19.80	78.38	80.61	56.77	87.07	74.55	91.92	54.95	65.43 _{±3.64}
			LaBraM	15.96	15.35	48.89	62.63	57.58	81.41	57.78	71.72	29.90	49.02 _{±0.91}
			Neuro-GPT	47.27	37.78	53.54	87.27	62.63	91.52	74.14	81.82	92.93	69.88 _{±1.91}
			EEGPT	72.32	55.76	74.75	99.39	81.01	100.00	96.36	86.06	91.72	84.15 _{±4.65}
			CBraMod	5.86	3.43	4.65	10.10	0.40	20.81	12.73	21.41	9.70	9.90 _{±0.31}
			TFM	2.02	1.01	2.22	5.05	2.02	8.28	3.64	-1.41	2.22	2.78 _{±0.63}
			BrainOmni-Tiny	56.57	37.58	56.16	93.33	72.53	90.30	79.60	67.47	88.89	71.38 _{±0.72}
			BrainOmni-Base	8.08	7.68	10.10	21.41	26.46	24.44	20.20	21.41	23.23	18.11 _{±0.16}
			EEGMamba	11.52	5.05	1.62	14.95	7.07	14.95	7.68	7.68	22.63	10.35 _{±0.03}
			SingLEM	3.84	6.06	9.49	22.83	7.07	13.54	14.14	13.94	3.03	10.44 _{±0.24}
			LUNA-Base	0.61	-0.40	0.00	1.62	2.42	1.21	0.00	0.61	3.84	1.10 _{±0.37}
Within-subject (Few-shot)		Specialist Models	TRCA	96.97	93.94	100.00	100.00	100.00	100.00	100.00	100.00	<u>96.97</u>	98.65
			EEGNet	8.08	21.21	4.04	82.83	94.95	88.89	83.84	100.00	88.89	63.64 _{±4.36}
			ShallowConv	27.27	3.03	33.33	38.38	80.81	85.86	54.55	62.63	35.35	46.80 _{±2.20}
			LMDA	-3.03	1.01	3.03	48.48	94.95	93.94	69.70	83.84	12.12	44.89 _{±2.14}
			CNN-T	25.25	2.02	21.21	74.75	93.94	81.82	83.84	86.87	53.54	58.14 _{±1.30}
			Deformer	10.10	10.10	50.51	<u>95.96</u>	96.97	100.00	91.92	100.00	65.66	69.02 _{±0.08}
			Conformer	35.35	0.00	29.29	44.44	51.52	39.39	36.36	56.57	31.31	36.03 _{±3.34}
			TSception	57.58	0.00	50.51	84.85	82.83	100.00	90.91	<u>98.99</u>	66.67	70.26 _{±3.04}
	Full Fine-tuning	Foundation Models	BENDR	70.71	9.09	84.85	100.00	<u>97.98</u>	100.00	<u>98.99</u>	100.00	100.00	84.62 _{±0.32}
			BIOT-1D	69.70	42.42	75.76	85.86	83.84	93.94	89.90	95.96	77.78	79.46 _{±2.91}
			BIOT-2D	48.48	22.22	75.76	87.88	91.92	91.92	82.83	92.93	86.87	75.65 _{±1.77}
			BIOT-6D	66.67	46.46	<u>91.92</u>	86.87	89.90	94.95	83.84	96.97	91.92	83.28 _{±1.61}
			LaBraM	57.58	20.20	71.72	91.92	95.96	97.98	91.92	100.00	55.56	75.87 _{±2.34}
			Neuro-GPT	35.35	14.14	50.51	89.90	93.94	97.98	94.95	100.00	91.92	74.30 _{±1.61}
			EEGPT	51.52	9.09	68.69	91.92	79.80	95.96	94.95	<u>98.99</u>	70.71	73.51 _{±8.85}
			CBraMod	44.44	4.04	33.33	78.79	75.76	86.87	79.80	92.93	34.34	58.92 _{±1.26}
			TFM	5.05	-1.01	4.04	10.10	0.00	7.07	5.05	4.04	-1.01	3.70 _{±1.37}
			BrainOmni-Tiny	56.57	17.17	74.75	<u>95.96</u>	92.93	<u>98.99</u>	95.96	100.00	90.91	80.36 _{±0.57}
			BrainOmni-Base	18.18	4.04	9.09	72.73	87.88	84.85	29.29	68.69	62.63	48.60 _{±1.41}
			EEGMamba	24.24	12.12	19.19	39.39	42.42	67.68	27.27	42.42	75.76	38.95 _{±0.16}
			SingLEM	19.19	2.02	20.20	39.39	10.10	43.43	43.43	53.54	11.11	26.94 _{±2.87}
			LUNA-Base	-1.01	3.03	1.01	1.01	3.03	2.02	-2.02	2.02	-3.03	0.67 _{±0.55}
	Linear Probing	Foundation Models	BENDR	-1.01	0.00	8.08	22.22	14.14	43.43	24.24	56.57	36.36	22.67 _{±2.67}
			BIOT-1D	62.63	<u>55.56</u>	80.81	90.91	91.92	96.97	90.91	93.94	<u>96.97</u>	84.51 _{±1.10}
			BIOT-2D	51.52	3.03	60.61	74.75	75.76	93.94	80.81	86.87	75.76	67.00 _{±2.35}
			BIOT-6D	62.63	47.47	77.78	81.82	87.88	64.65	80.81	92.93	89.90	76.21 _{±5.56}
			LaBraM	53.54	12.12	45.45	66.67	76.77	88.89	88.89	92.93	50.51	63.97 _{±0.55}
			Neuro-GPT	37.37	12.12	20.20	61.62	46.46	87.88	48.48	38.38	82.83	48.37 _{±1.36}
			EEGPT	<u>84.85</u>	40.40	68.69	94.95	39.39	95.96	63.64	<u>98.99</u>	94.95	75.76 _{±4.16}
			CBraMod	10.10	10.10	10.10	13.13	19.19	27.27	33.33	28.28	5.05	17.40 _{±1.93}
			TFM	6.06	5.05	2.02	4.04	1.01	0.00	-1.01	3.03	-3.03	1.91 _{±1.87}
			BrainOmni-Tiny	29.29	15.15	35.35	57.58	71.72	67.68	37.37	48.48	65.66	47.59 _{±1.51}
			BrainOmni-Base	12.12	0.00	5.05	25.25	35.35	29.29	10.10	19.19	13.13	16.61 _{±2.75}
			EEGMamba	5.05	8.08	9.09	10.10	10.10	28.28	11.11	8.08	34.34	13.80 _{±0.55}
			SingLEM	5.05	9.09	6.06	13.13	12.12	26.26	16.16	10.10	8.08	11.78 _{±0.99}
			LUNA-Base	4.04	2.02	-1.01	3.03	2.02	1.01	2.02	-3.03	1.01	1.23 _{±1.87}

Table 42. Classification accuracies (%) on EEGMat. The best accuracies are marked in bold, and the second best by an underline.

Scenario	Tuning	Model Type	Approach	S0	S1	S2	S3	S4	S5	S6	S7	S8	S9	S10	S11	S12	S13	S14	S15	S16	S17	S18	
Cross-subject (LOSO)		Specialist Models	CSP+LDA	<u>83.33</u>	50.00	50.00	50.00	100.00	53.33	93.33	<u>80.00</u>	56.67	76.67	73.33	46.67	50.00	50.00	50.00	76.67	70.00	50.00	93.33	
			EEGNet	58.89	51.11	50.00	94.44	90.00	54.44	63.33	61.11	70.00	83.33	85.56	57.78	53.33	40.00	48.89	65.56	46.67	63.33	84.44	
			ShallowConv	68.89	58.89	61.11	75.56	86.67	57.78	86.67	66.67	65.56	<u>91.11</u>	94.44	83.33	<u>56.67</u>	47.78	53.33	63.33	74.44	70.00	97.78	
			LMDA	68.89	58.89	47.78	<u>91.11</u>	82.22	81.11	63.33	55.56	71.11	85.56	75.56	58.89	38.89	61.11	55.56	70.00	50.00	75.56	75.56	
			CNN-T	68.89	67.78	53.33	57.78	74.44	73.33	70.00	62.22	81.11	74.44	84.44	73.33	36.67	42.22	48.89	62.22	<u>83.33</u>	70.00	65.56	
			Deformer	77.78	57.78	54.44	82.22	<u>94.44</u>	50.00	87.78	73.33	65.56	85.56	92.22	76.67	45.56	40.00	61.11	65.56	71.11	92.22	64.44	
			Conformer	78.89	64.44	55.56	71.11	88.89	55.56	<u>90.00</u>	64.44	65.56	76.67	100.00	88.89	51.11	47.78	53.33	61.11	74.44	64.44	92.22	
			TSception	82.22	52.22	67.78	<u>91.11</u>	80.00	53.33	85.56	58.89	68.89	92.22	<u>96.67</u>	71.11	37.78	47.78	56.67	56.67	78.89	56.67	93.33	
	Full Fine-tuning	Foundation Models	BENDR	48.89	52.22	61.11	52.22	46.67	50.00	43.33	45.56	66.67	62.22	41.11	68.89	48.89	47.78	58.89	56.67	53.33	56.67	62.22	
			BIOT-1D	<u>83.33</u>	55.56	53.33	82.22	74.44	52.22	76.67	72.22	72.22	74.44	75.56	74.44	43.33	<u>64.44</u>	50.00	73.33	82.22	80.00	80.00	
			BIOT-2D	68.89	50.00	53.33	82.22	67.78	57.78	62.22	78.89	83.33	72.22	77.78	65.56	48.89	47.78	53.33	63.33	84.44	<u>82.22</u>	85.56	
			BIOT-6D	84.44	54.44	54.44	80.00	76.67	54.44	60.00	62.22	75.56	78.89	81.11	71.11	48.89	<u>64.44</u>	55.56	70.00	77.78	92.22	86.67	
			LaBraM	67.78	68.89	<u>70.00</u>	74.44	81.11	62.22	63.33	60.00	64.44	77.78	66.67	65.56	41.11	57.78	63.33	61.11	62.22	53.33	53.33	
			Neuro-GPT	82.22	82.22	51.11	76.67	77.78	58.89	88.89	70.00	78.89	85.56	93.33	81.11	43.33	42.22	54.44	63.33	81.11	81.11	<u>94.44</u>	
			EEGPT	54.44	62.22	53.33	68.89	43.33	51.11	61.11	48.89	64.44	70.00	53.33	54.44	37.78	50.00	58.89	65.56	60.00	61.11	61.11	
			CBraMod	75.56	56.67	65.56	56.67	64.44	64.44	58.89	60.00	66.67	77.78	75.56	64.44	31.11	55.56	52.22	55.56	73.33	81.11	50.00	
			TFM	67.78	40.00	55.56	78.89	78.89	57.78	56.67	53.33	70.00	78.89	70.00	58.89	32.22	36.67	43.33	82.22	61.11	62.22	81.11	
			BrainOmni-Tiny	60.00	51.11	<u>70.00</u>	70.00	54.44	56.67	50.00	55.56	55.56	65.56	47.78	60.00	36.67	47.78	63.33	61.11	57.78	48.89	57.78	
			BrainOmni-Base	60.00	53.33	51.11	54.44	51.11	55.56	57.78	50.00	52.22	43.33	54.44	53.33	57.78	46.67	51.11	46.67	46.67	56.67	46.67	
			EEGMamba	50.00	50.00	50.00	50.00	50.00	50.00	50.00	50.00	50.00	50.00	50.00	50.00	50.00	50.00	50.00	50.00	50.00	50.00	50.00	
			SingLEM	50.00	48.89	50.00	51.11	50.00	50.00	50.00	50.00	50.00	50.00	50.00	50.00	50.00	50.00	50.00	51.11	50.00	50.00	50.00	50.00
			LUNA-Base	50.00	50.00	50.00	50.00	50.00	50.00	50.00	50.00	50.00	50.00	50.00	50.00	50.00	50.00	50.00	50.00	50.00	50.00	50.00	50.00
	Linear Probing	Foundation Models	BENDR	42.22	55.56	43.33	51.11	50.00	45.56	53.33	48.89	54.44	60.00	48.89	48.89	57.78	45.56	48.89	58.89	62.22	50.00	56.67	
			BIOT-1D	63.33	51.11	33.33	<u>91.11</u>	77.78	68.89	64.44	75.56	76.67	<u>91.11</u>	82.22	81.11	43.33	37.78	50.00	<u>84.44</u>	75.56	81.11	90.00	
			BIOT-2D	84.44	52.22	53.33	84.44	74.44	68.89	61.11	86.67	70.00	84.44	83.33	78.89	51.11	44.44	48.89	87.78	64.44	<u>82.22</u>	86.67	
			BIOT-6D	55.56	57.78	61.11	74.44	63.33	46.67	82.22	63.33	76.67	64.44	74.44	<u>84.44</u>	41.11	44.44	<u>64.44</u>	58.89	63.33	70.00	75.56	
			LaBraM	71.11	64.44	72.22	58.89	65.56	55.56	66.67	68.89	71.11	68.89	62.22	65.56	48.89	51.11	67.78	67.78	64.44	61.11	68.89	
			Neuro-GPT	75.56	<u>72.22</u>	51.11	75.56	70.00	<u>74.44</u>	87.78	70.00	72.22	84.44	80.00	60.00	51.11	32.22	55.56	58.89	73.33	71.11	87.78	
			EEGPT	60.00	61.11	53.33	73.33	55.56	51.11	55.56	50.00	71.11	80.00	43.33	56.67	33.33	48.89	58.89	66.67	53.33	70.00	70.00	
			CBraMod	58.89	63.33	63.33	58.89	54.44	48.89	63.33	52.22	<u>82.22</u>	62.22	43.33	54.44	36.67	67.78	58.89	54.44	71.11	76.67	72.22	
			TFM	63.33	51.11	54.44	80.00	70.00	63.33	55.56	54.44	75.56	72.22	41.11	57.78	22.22	47.78	57.78	78.89	63.33	50.00	87.78	
			BrainOmni-Tiny	55.56	53.33	60.00	74.44	64.44	56.67	52.22	55.56	68.89	67.78	71.11	71.11	44.44	45.56	<u>64.44</u>	64.44	61.11	60.00	74.44	
			BrainOmni-Base	50.00	54.44	48.89	52.22	50.00	54.44	52.22	54.44	52.22	43.33	55.56	47.78	48.89	53.33	44.44	50.00	52.22	51.11	48.89	
			EEGMamba	50.00	50.00	50.00	50.00	50.00	50.00	50.00	50.00	50.00	50.00	50.00	50.00	50.00	50.00	50.00	50.00	50.00	50.00	51.11	
			SingLEM	40.00	53.33	55.56	42.22	45.56	48.89	44.44	66.67	50.00	44.44	37.78	48.89	55.56	42.22	54.44	58.89	37.78	48.89	47.78	
			LUNA-Base	50.00	51.11	50.00	51.11	50.00	55.56	50.00	47.78	50.00	48.89	50.00	50.00	50.00	50.00	50.00	50.00	55.56	48.89	50.00	50.00
Within-subject (Few-shot)		Specialist Models	CSP+LDA	91.67	100.00	91.67	100.00	91.67	100.00	100.00	100.00	100.00	75.00	91.67	91.67	100.00	91.67	100.00	100.00	100.00	<u>91.67</u>		
			EEGNet	55.56	50.00	61.11	75.00	61.11	72.22	55.56	33.33	75.00	50.00	75.00	52.78	86.11	52.78	63.89	88.89	80.56	55.56	75.00	
			ShallowConv	41.67	52.78	61.11	63.89	61.11	69.44	75.00	61.11	61.11	52.78	55.56	66.67	83.33	83.33	66.67	83.33	58.33	69.44	77.78	
			LMDA	44.44	61.11	52.78	80.56	27.78	30.56	52.78	33.33	44.44	44.44	44.44	77.78	58.33	50.00	63.89	44.44	61.11	55.56	55.56	
			CNN-T	50.00	50.00	50.00	61.11	50.00	50.00	50.00	50.00	50.00	50.00	50.00	50.00	50.00	50.00	50.00	50.00	50.00	50.00	50.00	
			Deformer	58.33	58.33	50.00	55.56	36.11	30.56	61.11	41.67	41.67	61.11	47.22	55.56	47.22	47.22	50.00	30.56	58.33	55.56	66.67	
			Conformer	63.89	<u>97.22</u>	<u>86.11</u>	91.67	77.78	50.00	61.11	72.22	63.89	75.00	69.44	61.11	72.22	63.89	63.89	58.33	77.78	55.56	77.78	
			TSception	77.78	80.56	91.67	86.11	88.89	100.00	<u>97.22</u>	<u>86.11</u>	83.33	<u>94.44</u>	<u>88.89</u>	<u>97.22</u>	100.00	88.89	<u>94.44</u>	<u>91.67</u>	100.00	<u>97.22</u>	100.00	
	Full Fine-tuning	Foundation Models	BENDR	47.22	58.33	47.22	58.33	55.56	52.78	58.33	47.22	61.11	61.11	36.11	66.67	55.56	63.89	55.56	55.56	52.78	50.00	52.78	
			BIOT-1D	66.67	69.44	61.11	94.44	83.33	61.11	<u>97.22</u>	75.00	86.11	88.89	83.33	94.44	63.89	<u>97.22</u>	80.56	86.11	100.00	83.33	<u>91.67</u>	
			BIOT-2D	69.44	63.89	55.56	86.11	69.44	58.33	86.11	61.11	<u>91.67</u>	72.22	77.78	91.67	66.67	100.00	61.11	88.89	<u>94.44</u>	83.33	<u>91.67</u>	
			BIOT-6D	58.33	77.78	72.22	<u>97.22</u>	88.89	52.78	88.89	<u>86.11</u>	100.00	91.67	80.56	88.89	58.33	100.00	88.89	100.00	91.67	100.00	88.89	
			LaBraM	66.67	77.78	47.22	63.89	55.56	61.11	58.33	27.78	75.00	58.33	75.00	69.44	63.89	75.00	61.11	80.56	86.11	52.78	75.00	
			Neuro-GPT	58.33	63.89	83.33	91.67	69.44	55.56	80.56	30.56	63.89	63.89	69.44	86.11	80.56	80.56	66.67	55.56	86.11	69.44	<u>91.67</u>	
			EEGPT	44.44	47.22	61.11	66.67	55.56	52.78	61.11	61.11	77.78	58.33	75.00	61.11	72.22	80.56	77.78	86.11	72.22	58.33	77.78	
			CBraMod	47.22	86.11	80.56	66.67	55.56	<u>91.67</u>	80.56	83.33	86.11	100.00	72.22	83.33	<u>94.44</u>	<u>97.22</u>	69.44	80.56	100.00	94.44	86.11	
			TFM	63.89	80.56	50.00	86.11	<u>97.22</u>	83.33	75.00	75.00	83.33	58.33	66.67	86.11	72.22	83.33	86.11	88.89	94.44	77.78	83.33	
			BrainOmni-Tiny	44.44	77.78	52.78	58.33	58.33	50.00	52.78	25.00	55.56	75.00	52.78	63.89	55.56	61.11	58.33	52.78	66.67	47.22	77.78	
			BrainOmni-Base	55.56	55.56	50.00	47.22	50.00	50.00	50.00													

Table 43. Classification accuracies (%) on EEGMat. The best accuracies are marked in bold, and the second best by an underline.

Scenario	Tuning	Model Type	Approach	S19	S20	S21	S22	S23	S24	S25	S26	S27	S28	S29	S30	S31	S32	S33	S34	S35	Avg.
Cross-subject (LOSO)		Specialist Models	CSP+LDA	96.67	90.00	70.00	70.00	36.67	56.67	50.00	83.33	60.00	93.33	50.00	50.00	63.33	70.00	56.67	80.00	96.67	67.41
			EEGNet	92.22	85.56	64.44	64.44	65.56	67.78	68.89	61.11	71.11	62.22	65.56	56.67	60.00	67.78	62.22	78.89	81.11	66.60 _{±0.63}
			ShallowConv	94.44	73.33	77.78	76.67	81.11	<u>84.44</u>	86.67	70.00	60.00	<u>92.22</u>	52.22	57.78	56.67	62.22	63.33	72.22	78.89	72.22 _{±1.24}
			LMDA	82.22	65.56	64.44	91.11	71.11	50.00	73.33	71.11	77.78	62.22	60.00	73.33	43.33	51.11	70.00	87.78	67.78	67.47 _{±1.21}
			CNN-T	90.00	88.89	66.67	93.33	75.56	73.33	68.83	71.11	71.11	62.22	51.11	83.33	72.22	84.44	87.78	90.00	67.78	70.77 _{±2.49}
			Deformer	97.78	75.56	82.22	77.78	83.33	83.33	82.22	70.00	58.89	83.33	50.00	62.22	62.22	68.89	56.67	74.44	75.56	71.73 _{±0.37}
			Conformer	94.44	61.11	75.56	61.11	<u>85.53</u>	76.67	<u>85.56</u>	71.11	58.89	90.00	50.00	52.22	53.33	65.53	64.44	70.00	77.78	70.49 _{±0.93}
			TSception	95.56	87.78	74.44	85.56	76.67	87.78	84.44	71.11	<u>81.11</u>	86.67	48.89	81.11	61.11	72.22	<u>82.22</u>	86.67	76.67	74.10 _{±0.04}
	Full Fine-tuning	Foundation Models	BENDR	55.56	56.67	51.11	56.67	46.67	60.00	56.67	47.78	65.56	50.00	61.11	53.33	54.44	48.89	52.22	61.11	54.44	54.32 _{±0.71}
			BIOT-1D	96.67	67.78	77.78	86.67	57.78	71.11	68.89	68.89	70.00	67.78	51.11	65.56	45.56	71.11	64.44	78.89	71.11	69.48 _{±2.63}
			BIOT-2D	98.89	84.44	71.11	73.33	57.78	82.22	76.67	78.89	72.22	51.11	55.56	86.67	42.22	52.22	58.89	81.11	77.78	69.07 _{±1.74}
			BIOT-6D	96.67	76.67	76.67	78.89	72.22	77.78	77.78	65.56	67.78	74.44	55.56	80.00	38.89	70.00	65.55	77.78	66.67	70.77 _{±1.49}
			LaBraM	64.44	83.33	58.89	83.33	75.56	68.89	78.83	65.56	72.22	61.11	56.67	60.00	40.00	60.00	67.78	77.78	77.78	65.74 _{±1.61}
			Neuro-GPT	98.89	70.00	76.67	71.11	80.00	78.89	71.11	67.78	63.33	86.67	52.22	74.44	48.89	56.67	77.78	76.67	76.67	72.62 _{±1.35}
			EEGPT	82.22	63.33	62.22	63.33	63.33	56.67	60.00	61.11	68.89	54.44	54.44	47.78	42.22	50.00	56.67	70.00	52.22	58.02 _{±0.102}
			CBraMod	82.22	65.56	73.33	98.89	86.67	62.22	81.11	60.00	63.33	91.11	54.44	90.00	70.00	64.44	68.89	92.22	73.33	68.43 _{±0.72}
			TFM	87.78	71.11	67.78	75.56	64.44	64.44	62.22	63.33	66.67	64.44	52.22	78.89	44.44	54.44	67.78	68.89	48.89	63.02 _{±1.95}
			BrainOmni-Tiny	60.00	75.56	47.78	67.78	58.89	61.11	56.67	51.11	73.33	51.11	48.89	48.89	47.78	60.00	53.33	64.44	73.33	57.50 _{±0.80}
			BrainOmni-Base	44.44	42.22	52.22	54.44	51.11	48.89	64.44	41.11	56.67	53.33	61.11	46.67	44.44	50.00	51.11	52.22	51.11	51.51 _{±0.76}
			EEGMamba	48.89	50.00	50.00	50.00	50.00	50.00	50.00	50.00	50.00	50.00	50.00	50.00	50.00	50.00	50.00	50.00	50.00	49.97 _{±0.04}
			SingLEM	50.00	50.00	48.89	50.00	50.00	46.67	46.67	50.00	50.00	52.22	50.00	48.89	50.00	50.00	50.00	50.00	50.00	49.85 _{±0.16}
			LUNA-Base	50.00	50.00	50.00	50.00	50.00	50.00	50.00	50.00	50.00	50.00	50.00	50.00	50.00	50.00	50.00	50.00	50.00	50.00 _{±0.00}
	Linear Probing	Foundation Models	BENDR	52.22	53.33	57.78	63.33	40.00	56.67	52.22	50.00	53.33	52.22	46.67	50.00	53.33	42.22	55.56	52.22	41.11	51.51 _{±1.57}
			BIOT-1D	100.00	63.33	86.67	<u>95.56</u>	74.44	67.78	70.00	65.56	85.56	93.33	50.00	86.67	50.00	66.67	72.22	<u>93.33</u>	63.33	72.31 _{±0.97}
			BIOT-2D	87.78	67.78	<u>84.44</u>	93.33	74.44	62.22	78.89	67.78	78.89	91.11	57.78	66.67	54.44	78.89	65.56	95.56	62.22	<u>72.84</u> _{±1.86}
			BIOT-6D	97.78	78.89	66.67	87.73	70.00	74.44	65.56	68.89	65.56	78.89	47.78	71.11	44.44	64.44	51.11	78.89	57.78	66.45 _{±1.24}
			LaBraM	88.89	74.44	62.22	77.78	54.44	54.44	57.78	66.67	72.22	46.67	48.89	57.78	51.11	52.22	70.00	86.67	77.78	64.48 _{±0.56}
			Neuro-GPT	<u>98.89</u>	82.22	64.44	63.33	77.78	70.00	77.78	57.78	51.11	87.78	46.67	<u>87.78</u>	57.78	58.89	64.44	75.56	43.33	68.58 _{±1.89}
			EEGPT	82.22	62.22	66.67	66.67	54.44	50.00	58.89	64.44	70.00	55.56	56.67	58.89	41.11	52.22	54.44	73.33	57.78	59.38 _{±0.53}
			CBraMod	70.00	55.56	45.56	67.78	58.89	68.89	52.22	63.33	67.78	64.44	50.00	58.89	44.44	57.78	72.22	70.00	61.11	60.34 _{±0.27}
			TFM	93.33	66.67	52.22	80.00	71.11	71.11	53.33	64.44	70.00	64.44	48.89	70.00	44.44	48.89	76.67	74.44	51.11	62.44 _{±0.09}
			BrainOmni-Tiny	72.22	73.33	61.11	80.00	66.67	73.33	56.67	55.56	61.11	54.44	<u>62.22</u>	57.78	42.22	62.22	62.22	68.89	74.44	62.50 _{±0.50}
			BrainOmni-Base	48.89	56.67	50.00	43.33	51.11	47.78	48.89	41.11	48.89	52.22	48.89	51.11	57.78	53.33	48.89	52.22	48.89	50.40 _{±0.69}
			EEGMamba	50.00	50.00	50.00	50.00	50.00	50.00	50.00	50.00	50.00	50.00	51.11	50.00	50.00	50.00	50.00	50.00	50.00	50.06 _{±0.09}
			SingLEM	42.22	58.89	46.67	73.33	56.67	54.44	55.56	54.44	66.67	47.78	55.56	41.11	54.44	45.56	48.89	43.33	45.56	50.40 _{±0.43}
			LUNA-Base	55.56	44.44	51.11	43.33	50.00	50.00	50.00	43.33	42.22	50.00	50.00	60.00	50.00	50.00	50.00	50.00	50.00	49.97 _{±0.39}
Within-subject (Few-shot)		Specialist Models	CSP+LDA	100.00	100.00	100.00	91.67	100.00	100.00	66.67	100.00	100.00	100.00	83.33	100.00	100.00	100.00	100.00	91.67	91.67	95.60
			EEGNet	91.67	61.11	55.56	72.22	61.11	55.56	47.22	52.78	66.67	55.56	47.22	36.11	30.56	55.56	58.33	55.56	58.33	60.57 _{±1.29}
			ShallowConv	<u>97.22</u>	75.00	88.83	69.44	77.78	77.78	44.44	61.11	77.78	91.67	47.22	69.44	75.00	83.33	58.33	61.11	<u>97.22</u>	69.37 _{±1.43}
			LMDA	72.22	61.11	61.11	69.44	66.67	55.56	44.44	33.33	75.00	61.11	44.44	66.67	33.33	72.22	58.33	61.11	52.78	54.78 _{±0.22}
			CNN-T	50.00	50.00	50.00	50.00	50.00	58.33	50.00	50.00	50.00	52.78	50.00	50.00	50.00	66.67	50.00	50.00	50.00	51.08 _{±0.95}
			Deformer	63.89	61.11	55.56	58.33	63.89	50.00	55.56	58.33	55.56	50.00	27.78	55.56	44.44	61.11	44.44	55.56	58.33	52.01 _{±0.39}
			Conformer	91.67	86.11	80.56	63.89	66.67	60.00	50.00	50.00	72.22	63.89	38.89	86.11	58.33	47.22	61.11	61.11	77.78	68.33 _{±1.71}
			TSception	100.00	<u>91.67</u>	<u>97.22</u>	91.67	<u>94.44</u>	100.00	50.00	75.00	<u>88.89</u>	<u>94.44</u>	55.56	100.00	<u>91.67</u>	69.44	80.56	83.33	100.00	<u>89.12</u> _{±0.95}
	Full Fine-tuning	Foundation Models	BENDR	30.56	55.56	41.67	61.11	44.44	50.00	55.56	52.78	50.00	52.78	38.89	50.00	47.22	55.56	66.67	41.67	47.22	52.16 _{±0.22}
			BIOT-1D	100.00	75.00	63.89	83.33	80.56	83.33	61.11	69.44	77.78	100.00	94.44	80.56	83.33	100.00	77.78	83.33	91.67	82.48 _{±1.71}
			BIOT-2D	91.67	<u>91.67</u>	58.33	83.33	66.67	69.44	75.00	61.11	75.00	77.78	88.89	77.78	58.33	<u>94.44</u>	86.11	75.00	97.22	77.70 _{±0.48}
			BIOT-6D	100.00	<u>91.67</u>	77.78	91.67	75.00	<u>91.67</u>	69.44	<u>91.67</u>	75.00	<u>94.44</u>	94.44	86.11	75.00	100.00	91.67	97.22	86.11	86.11 _{±3.16}
			LaBraM	91.67	80.56	58.33	83.33	69.44	61.11	58.33	44.44	72.22	63.89	58.33	86.11	50.00	63.89	63.89	63.89	66.67	65.74 _{±1.50}
			Neuro-GPT	<u>97.22</u>	<u>91.67</u>	86.11	75.00	72.22	77.78	58.33	52.78	72.22	88.89	44.44	<u>97.22</u>	38.89	58.33	72.22	69.44	63.89	71.22 _{±3.69}
			EEGPT	91.67	77.78	61.11	75.00	83.33	69.44	55.56	55.56	75.00	58.33	61.11	63.89	25.00	52.78	72.22	72.22	72.22	65.59 _{±2.51}
			CBraMod	100.00	100.00	75.00	100.00	77.78	80.56	69.44	38.89	<u>88.89</u>	<u>94.44</u>	38.89	100.00	72.22	47.22	75.00	80.56	61.11	79.32 _{±1.15}
			TFM	100.00	<u>91.67</u>	88.89	83.33	72.22	69.44	<u>72.22</u>	55.56	77.78	83.33	47.22	86.11	52.78	69.44	86.11	80.56	83.33	77.55 _{±0.93}
			BrainOmni-Tiny	86.11	69.44	75.00	47.22	61.11	68.89	52.78	33.33	63.89	58.33	72.22	69.44	27.78	69.44	66.67	72.22	63.89	58.26 _{±0.79}
			BrainOmni-Base	55.56	47.22	58.33	44.44	50.00	66.67	50.00	50.00	58.33	47.22	47							

Table 44. Cohen's Kappa (%) on EEGMat. The best metrics are marked in bold, and the second best by an underline.

Scenario	Tuning	Model Type	Approach	S0	S1	S2	S3	S4	S5	S6	S7	S8	S9	S10	S11	S12	S13	S14	S15	S16	S17	S18
Cross-subject (LOSO)	Specialist Models	Specialist Models	CSP+LDA	66.67	0.00	0.00	0.00	100.00	6.67	86.67	60.00	13.33	53.33	46.67	-6.67	0.00	0.00	0.00	53.33	40.00	0.00	86.67
			EEGNet	17.78	2.22	0.00	88.89	80.00	8.89	26.67	22.22	40.00	66.67	71.11	15.56	6.67	-20.00	-2.22	31.11	-6.67	26.67	68.89
			ShallowConv	37.78	17.78	22.22	51.11	73.33	15.56	73.33	33.33	31.11	82.22	88.89	66.67	13.33	-4.44	6.67	26.67	48.89	40.00	95.56
			LMDA	37.78	17.78	-4.44	82.22	64.44	62.22	26.67	11.11	42.22	71.11	51.11	17.78	-22.22	22.22	11.11	40.00	0.00	51.11	51.11
			CNN-T	37.78	35.56	6.67	15.56	48.89	46.67	40.00	24.44	62.22	48.89	68.89	46.67	-26.67	-15.56	-2.22	24.44	66.67	40.00	31.11
			Deformer	55.56	15.56	8.89	64.44	88.89	0.00	75.56	46.67	31.11	71.11	84.44	53.33	-8.89	-20.00	22.22	31.11	42.22	84.44	28.89
			Conformer	57.78	28.89	11.11	42.22	77.78	11.11	80.00	28.89	31.11	53.33	100.00	77.78	2.22	-4.44	6.67	22.22	48.89	28.89	84.44
			TSception	64.44	4.44	35.56	82.22	60.00	6.67	71.11	17.78	37.78	84.44	93.33	42.22	-24.44	-4.44	13.33	13.33	57.78	13.33	86.67
	Full Fine-tuning	Foundation Models	BENDR	-2.22	4.44	22.22	4.44	-6.67	0.00	-13.33	-8.89	33.33	24.44	-17.78	37.78	-2.22	-4.44	17.78	13.33	6.67	13.33	24.44
			BIOT-1D	66.67	11.11	6.67	64.44	48.89	4.44	53.33	44.44	44.44	48.89	51.11	48.89	-13.33	28.89	0.00	46.67	64.44	60.00	60.00
			BIOT-2D	37.78	0.00	6.67	64.44	35.56	15.56	24.44	57.78	66.67	44.44	55.56	31.11	-2.22	-4.44	6.67	26.67	68.89	64.44	71.11
			BIOT-6D	68.89	8.89	8.89	60.00	53.33	8.89	20.00	24.44	51.11	57.78	62.22	42.22	-2.22	28.89	11.11	40.00	55.56	84.44	73.33
			LaBraM	35.56	37.78	40.00	48.89	62.22	24.44	26.67	20.00	28.89	55.56	33.33	31.11	-17.78	15.56	26.67	22.22	24.44	6.67	6.67
			Neuro-GPT	64.44	64.44	2.22	53.33	55.56	17.78	77.78	40.00	57.78	71.11	86.67	62.22	-13.33	-15.56	8.89	26.67	62.22	62.22	88.89
			EEGPT	8.89	24.44	6.67	37.78	-13.33	2.22	22.22	-2.22	28.89	40.00	6.67	8.89	-24.44	0.00	17.78	31.11	20.00	22.22	22.22
			CBraMod	51.11	13.33	31.11	13.33	28.89	28.89	17.78	20.00	33.33	55.56	51.11	28.89	-37.78	11.11	4.44	11.11	46.67	62.22	0.00
			TFM	35.56	-20.00	11.11	57.78	57.78	15.56	13.33	6.67	40.00	57.78	40.00	17.78	-35.56	-26.67	-13.33	64.44	22.22	24.44	62.22
			BrainOmni-Tiny	20.00	2.22	40.00	40.00	8.89	13.33	0.00	11.11	11.11	31.11	-4.44	20.00	-26.67	-4.44	26.67	22.22	15.56	-2.22	15.56
	Linear Probing	Foundation Models	BrainOmni-Base	20.00	6.67	2.22	8.89	2.22	11.11	15.56	0.00	4.44	-13.33	8.89	6.67	15.56	-6.67	2.22	-6.67	-6.67	13.33	-6.67
			EEGMamba	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
			SingLEM	0.00	-2.22	0.00	2.22	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	2.22	0.00	0.00	0.00
			LUNA-Base	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
			BENDR	-15.56	11.11	-13.33	2.22	0.00	-8.89	6.67	-2.22	8.89	20.00	-2.22	-2.22	15.56	-8.89	-2.22	17.78	24.44	0.00	13.33
			BIOT-1D	26.67	2.22	-33.33	82.22	55.56	37.78	28.89	51.11	53.33	82.22	64.44	62.22	-13.33	-24.44	0.00	68.89	51.11	62.22	80.00
			BIOT-2D	68.89	4.44	6.67	68.89	48.89	37.78	22.22	73.33	40.00	68.89	66.67	57.78	2.22	-11.11	-2.22	75.56	28.89	64.44	73.33
			BIOT-6D	11.11	15.56	22.22	48.89	26.67	-6.67	64.44	26.67	53.33	28.89	48.89	68.89	-17.78	-11.11	28.89	17.78	26.67	40.00	51.11
	Full Fine-tuning	Foundation Models	LaBraM	42.22	28.89	44.44	17.78	31.11	11.11	33.33	37.78	42.22	37.78	24.44	31.11	-2.22	2.22	35.56	35.56	28.89	22.22	37.78
			Neuro-GPT	51.11	44.44	2.22	51.11	40.00	48.89	75.56	40.00	44.44	68.89	60.00	20.00	2.22	-35.56	11.11	17.78	46.67	42.22	75.56
			EEGPT	20.00	22.22	6.67	46.67	11.11	2.22	11.11	0.00	42.22	60.00	-13.33	13.33	-33.33	-2.22	17.78	33.33	6.67	40.00	40.00
			CBraMod	17.78	26.67	26.67	17.78	8.89	-2.22	26.67	4.44	64.44	24.44	-13.33	8.89	-26.67	35.56	17.78	8.89	42.22	53.33	44.44
			TFM	26.67	2.22	8.89	60.00	40.00	26.67	11.11	8.89	51.11	44.44	-17.78	15.56	-55.56	-4.44	15.56	57.78	26.67	0.00	75.56
			BrainOmni-Tiny	11.11	6.67	20.00	48.89	28.89	13.33	4.44	11.11	37.78	35.56	42.22	42.22	-11.11	-8.89	28.89	28.89	22.22	20.00	48.89
			BrainOmni-Base	0.00	8.89	-2.22	4.44	0.00	8.89	4.44	8.89	4.44	-13.33	11.11	-4.44	-2.22	6.67	-11.11	0.00	4.44	2.22	-2.22
			EEGMamba	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	2.22
			SingLEM	-20.00	6.67	11.11	-15.56	-8.89	-2.22	-11.11	33.33	0.00	-11.11	-24.44	-2.22	11.11	-15.56	8.89	17.78	-24.44	-2.22	-4.44
			LUNA-Base	0.00	2.22	0.00	2.22	0.00	11.11	0.00	-4.44	0.00	-2.22	0.00	0.00	0.00	0.00	0.00	0.00	0.00	11.11	-2.22
Within-subject (Few-shot)	Specialist Models	Specialist Models	CSP+LDA	83.33	100.00	83.33	100.00	83.33	100.00	100.00	100.00	100.00	50.00	83.33	83.33	100.00	83.33	100.00	100.00	100.00	100.00	83.33
			EEGNet	11.11	0.00	22.22	50.00	22.22	44.44	11.11	-33.33	50.00	0.00	50.00	5.56	72.22	5.56	27.78	77.78	61.11	11.11	50.00
			ShallowConv	-16.67	5.56	22.22	27.78	22.22	38.89	50.00	22.22	22.22	5.56	11.11	33.33	66.67	66.67	33.33	66.67	16.67	38.89	55.56
			LMDA	-11.11	22.22	5.56	61.11	-44.44	-38.89	5.56	-33.33	-11.11	-11.11	-11.11	55.56	16.67	0.00	27.78	-11.11	22.22	11.11	11.11
			CNN-T	0.00	0.00	0.00	22.22	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
			Deformer	16.67	16.67	0.00	11.11	-27.78	-38.89	22.22	-16.67	-16.67	22.22	-5.56	11.11	-5.56	-5.56	0.00	-38.89	16.67	11.11	33.33
			Conformer	27.78	94.44	72.22	83.33	55.56	0.00	22.22	44.44	27.78	50.00	38.89	22.22	44.44	27.78	27.78	16.67	55.56	11.11	55.56
			TSception	55.56	61.11	83.33	72.22	77.78	100.00	94.44	72.22	66.67	88.89	77.78	94.44	100.00	77.78	88.89	83.33	100.00	94.44	100.00
	Full Fine-tuning	Foundation Models	BENDR	-5.56	16.67	-5.56	16.67	11.11	5.56	16.67	-5.56	22.22	22.22	-27.78	33.33	11.11	27.78	11.11	11.11	5.56	0.00	5.56
			BIOT-1D	33.33	38.89	22.22	88.89	66.67	22.22	94.44	50.00	72.22	77.78	66.67	88.89	27.78	94.44	61.11	72.22	100.00	66.67	83.33
			BIOT-2D	38.89	27.78	11.11	72.22	38.89	16.67	72.22	22.22	83.33	44.44	55.56	83.33	33.33	100.00	22.22	77.78	88.89	66.67	83.33
			BIOT-6D	16.67	55.56	44.44	94.44	77.78	5.56	77.78	72.22	100.00	83.33	61.11	77.78	16.67	100.00	77.78	100.00	83.33	100.00	77.78
			LaBraM	33.33	55.56	-5.56	27.78	11.11	22.22	16.67	-44.44	50.00	16.67	50.00	38.89	27.78	50.00	22.22	61.11	72.22	5.56	50.00
			Neuro-GPT	16.67	27.78	66.67	83.33	38.89	11.11	61.11	-38.89	27.78	27.78	38.89	72.22	61.11	61.11	33.33	11.11	72.22	38.89	83.33
			EEGPT	-11.11	-5.56	22.22	33.33	11.11	5.56	22.22	22.22	55.56	16.67	50.00	22.22	44.44	61.11	55.56	72.22	44.44	16.67	55.56
			CBraMod	-5.56	72.22	61.11	33.33	11.11	83.33	61.11	66.67	72.22	100.00	44.44	66.67	88.89	94.44	38.89	61.11	100.00	88.89	72.22
Linear Probing	Foundation Models	TFM	27.78	61.11	0.00	72.22	94.44	66.67	50.00	50.00	66.67	16.67	33.33	72.22	44.44	66.67	72.22	77.78	88.89	55.56	66.67	
		BrainOmni-Tiny	-11.11	55.56	5.56	16.67	16.67	0.00	5.56	-50.00	11.11	50.00	5.56	27.78	11.11	22.22	16.67	5.56	33.33	-5.56	55.56	
		BrainOmni-Base	11.11	11.11	0.00	-5.56	0.00	0.00	0.00	5.56	16.67	-11.11	0.00	33.33	5.56	-11.11	5.56	16.67	-16.67	5.56	22.22	
		EEGMamba	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.56	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.56	0.00	0.00	0.00	
		SingLEM	0.00	11.11	0.00	0.00	0.00	16.67	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.56	0.00	0.00	0.00	0.00	
		LUNA-Base	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	22.22	0.00	0.00	0.00	0.00	22.22	0.00	0.00	
		BENDR	0.0																			

Table 45. Cohen's Kappa (%) on EEGMat. The best metrics are marked in bold, and the second best by an underline.

Scenario	Tuning	Model Type	Approach	S19	S20	S21	S22	S23	S24	S25	S26	S27	S28	S29	S30	S31	S32	S33	S34	S35	Avg.
Cross-subject (LOSO)	Specialist Models	Specialist Models	CSP+LDA	93.33	80.00	40.00	40.00	-26.67	13.33	0.00	66.67	20.00	86.67	0.00	0.00	26.67	40.00	13.33	60.00	93.33	34.81
			EEGNet	84.44	71.11	28.89	28.89	31.11	35.56	37.78	22.22	42.22	24.44	31.11	13.33	20.00	35.56	24.44	57.78	62.22	33.21 _{a1.26}
			ShallowConv	88.89	46.67	55.56	53.33	62.22	68.89	73.33	40.00	20.00	<u>84.44</u>	4.44	15.56	13.33	24.44	26.67	44.44	57.73	44.44 _{a2.49}
			LMDA	64.44	31.11	28.89	82.22	42.22	0.00	46.67	42.22	55.56	24.44	20.00	46.67	-13.33	2.22	40.00	75.56	35.56	34.94 _{a2.42}
			CNN-T	80.00	<u>77.78</u>	33.33	86.67	51.11	46.67	37.78	42.22	42.22	24.44	2.22	66.67	44.44	68.89	75.56	80.00	35.56	41.54 _{a4.08}
			Deformer	95.56	51.11	64.44	55.56	66.67	66.67	64.44	40.00	17.78	66.67	0.00	24.44	24.44	37.78	13.33	48.89	51.11	43.46 _{a0.75}
			Conformer	88.89	22.22	51.11	22.22	<u>71.11</u>	53.33	<u>71.11</u>	42.22	17.78	80.00	0.00	4.44	6.67	31.11	28.89	40.00	55.56	40.99 _{a1.87}
			TSception	91.11	75.56	48.89	71.11	53.33	75.56	68.89	42.22	<u>62.22</u>	73.33	-2.22	62.22	22.22	44.44	<u>64.44</u>	73.33	53.33	48.21 _{a0.09}
	Full Fine-tuning	Foundation Models	BENDR	11.11	13.33	2.22	13.33	-6.67	20.00	13.33	-4.44	31.11	0.00	22.22	6.67	8.89	-2.22	4.44	22.22	8.89	8.64 _{a1.43}
			BIOT-1D	93.33	35.56	55.56	73.33	15.56	42.22	37.78	37.78	40.00	35.56	2.22	31.11	-8.89	42.22	28.89	57.78	42.22	38.95 _{a5.26}
			BIOT-2D	<u>97.78</u>	68.89	42.22	46.67	15.56	64.44	53.33	<u>57.78</u>	44.44	2.22	11.11	73.33	-15.56	4.44	17.78	62.22	55.56	38.15 _{a3.49}
			BIOT-6D	93.33	53.33	53.33	57.78	44.44	55.56	55.56	31.11	35.56	48.89	11.11	60.00	-22.22	40.00	31.11	55.56	33.33	41.54 _{a2.99}
			LaBraM	28.89	66.67	17.78	66.67	51.11	37.78	57.78	31.11	44.44	22.22	13.33	20.00	-20.00	20.00	35.56	55.56	55.55	31.43 _{a3.22}
			Neuro-GPT	<u>97.78</u>	40.00	53.33	42.22	60.00	57.78	42.22	35.56	26.67	73.33	4.44	48.89	-2.22	13.33	55.56	53.33	53.33	45.25 _{a2.71}
			EEGPT	64.44	26.67	24.44	26.67	26.67	13.33	20.00	22.22	37.78	8.89	8.89	-4.44	-15.56	0.00	13.33	40.00	4.44	16.05 _{a2.04}
			CBraMod	64.44	31.11	46.67	97.78	73.33	24.44	62.22	20.00	26.67	82.22	8.89	80.00	40.00	28.89	37.78	84.44	46.67	36.85 _{a1.44}
			TFM	75.56	42.22	35.56	51.11	28.89	28.89	24.44	26.67	33.33	28.89	4.44	57.78	-11.11	8.89	35.56	37.78	-2.22	26.05 _{a3.90}
			BrainOmni-Tiny	20.00	51.11	-4.44	35.56	17.78	22.22	13.33	2.22	46.67	2.22	-2.22	-2.22	-4.44	20.00	6.67	28.89	46.67	15.00 _{a1.60}
			BrainOmni-Base	-11.11	-15.56	4.44	8.89	2.22	-2.22	28.89	-17.78	13.33	6.67	22.22	-6.67	-11.11	0.00	2.22	4.44	2.22	3.02 _{a1.52}
			EEGMamba	-2.22	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	-0.06 _{a0.09}
			SingLEM	0.00	0.00	-2.22	0.00	0.00	-6.67	-6.67	0.00	0.00	4.44	0.00	-2.22	0.00	0.00	0.00	0.00	0.00	-0.31 _{a0.31}
			LUNA-Base	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00 _{a0.00}
	Linear Probing	Foundation Models	BENDR	4.44	6.67	15.56	26.67	-20.00	13.33	4.44	0.00	6.67	4.44	-6.67	0.00	6.67	-15.56	11.11	4.44	-17.78	3.02 _{a3.14}
			BIOT-1D	100.00	26.67	73.33	<u>91.11</u>	48.89	35.56	40.00	31.11	71.11	86.67	0.00	<u>73.33</u>	0.00	33.33	44.44	<u>86.67</u>	26.67	44.63 _{a1.93}
			BIOT-2D	75.56	35.56	68.89	86.67	62.22	24.44	57.78	35.56	57.78	82.22	15.56	33.33	8.89	<u>57.78</u>	31.11	91.11	24.44	45.68 _{a3.72}
			BIOT-6D	95.56	57.78	33.33	75.56	40.00	48.89	31.11	37.78	31.11	57.78	-4.44	42.22	-11.11	28.89	2.22	57.78	15.56	32.90 _{a2.49}
			LaBraM	77.78	48.89	24.44	55.56	8.89	8.89	15.56	33.33	44.44	-6.67	-2.22	15.56	2.22	4.44	20.00	73.33	55.56	28.95 _{a1.11}
			Neuro-GPT	<u>97.78</u>	64.44	28.89	26.67	55.56	40.00	55.56	15.56	2.22	75.56	-6.67	<u>75.56</u>	15.56	17.78	28.89	51.11	-13.33	37.16 _{a3.78}
			EEGPT	64.44	24.44	33.33	33.33	8.89	0.00	17.78	28.89	40.00	11.11	13.33	17.78	-17.78	4.44	8.89	46.67	15.56	18.77 _{a1.06}
			CBraMod	40.00	11.11	-8.89	35.56	17.78	37.78	4.44	26.67	35.56	28.83	0.00	17.78	-11.11	15.53	44.44	40.00	22.22	20.68 _{a0.53}
			TFM	86.67	33.33	4.44	60.00	42.22	42.22	6.67	28.89	40.00	28.89	-2.22	40.00	-11.11	-2.22	53.33	48.89	2.22	24.88 _{a0.17}
			BrainOmni-Tiny	44.44	46.67	22.22	60.00	33.33	46.67	13.33	11.11	22.22	8.89	<u>24.44</u>	15.56	-15.56	24.44	24.44	37.78	48.89	25.00 _{a0.99}
			BrainOmni-Base	-2.22	13.33	0.00	-13.33	2.22	-4.44	-2.22	-17.78	-2.22	4.44	-2.22	2.22	-15.56	6.67	-2.22	4.44	-2.22	0.80 _{a1.37}
			EEGMamba	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	2.22	0.00	0.00	0.00	0.00	0.00	0.00	0.12 _{a0.17}
			SingLEM	-15.56	17.78	-6.67	46.67	13.33	8.89	11.11	8.89	33.33	-4.44	11.11	-17.78	8.89	-8.89	-2.22	-13.33	-8.89	0.80 _{a0.86}
			LUNA-Base	11.11	-11.11	2.22	-13.33	0.00	0.00	0.00	-13.33	-15.56	0.00	0.00	20.00	0.00	0.00	0.00	0.00	0.00	-0.06 _{a0.78}
Within-subject (Few-shot)	Specialist Models	Specialist Models	CSP+LDA	100.00	100.00	100.00	83.33	100.00	100.00	33.33	100.00	100.00	100.00	66.67	100.00	100.00	100.00	100.00	83.33	83.33	91.20
			EEGNet	83.33	22.22	11.11	44.44	22.22	11.11	-5.56	5.56	33.33	11.11	-5.56	-27.78	-38.89	11.11	16.67	11.11	16.67	21.14 _{a2.57}
			ShallowConv	<u>94.44</u>	50.00	77.78	38.89	55.56	55.56	-11.11	22.22	55.56	83.33	-5.56	38.89	50.00	66.67	16.67	22.22	<u>94.44</u>	38.73 _{a2.86}
			LMDA	44.44	22.22	22.22	38.89	33.33	11.11	-11.11	-33.33	50.00	22.22	-11.11	33.33	-33.33	44.44	16.67	22.22	5.56	9.57 _{a0.44}
			CNN-T	0.00	0.00	0.00	0.00	0.00	16.67	0.00	0.00	0.00	5.56	0.00	0.00	0.00	33.33	0.00	0.00	0.00	2.16 _{a1.90}
			Deformer	27.78	22.22	11.11	16.67	27.78	0.00	11.11	16.67	11.11	0.00	-44.44	11.11	-11.11	22.22	-11.11	11.11	16.67	4.01 _{a0.79}
			Conformer	83.33	72.22	61.11	27.78	33.33	33.33	0.00	0.00	44.44	27.78	-22.22	72.22	16.67	-5.56	22.22	22.22	55.53	36.73 _{a3.43}
			TSception	100.00	<u>83.33</u>	<u>94.44</u>	83.33	88.89	100.00	0.00	50.00	<u>77.78</u>	88.89	11.11	100.00	83.33	38.89	61.11	66.67	100.00	<u>78.24</u> _{a1.89}
	Full Fine-tuning	Foundation Models	BENDR	-38.89	11.11	-16.67	22.22	-11.11	0.00	11.11	5.56	0.00	5.56	-22.22	0.00	-5.56	11.11	33.33	-16.67	-5.56	4.32 _{a0.44}
			BIOT-1D	100.00	50.00	27.78	66.67	61.11	66.67	22.22	38.89	55.56	100.00	88.89	61.11	66.67	100.00	55.56	66.67	83.33	64.97 _{a3.43}
			BIOT-2D	83.33	<u>83.33</u>	16.67	66.67	33.33	38.89	50.00	22.22	50.00	55.56	77.78	55.56	16.67	88.89	72.22	50.00	<u>94.44</u>	55.40 _{a0.95}
			BIOT-6D	100.00	<u>83.33</u>	55.56	83.33	50.00	<u>83.33</u>	38.89	<u>83.33</u>	50.00	88.89	<u>88.83</u>	72.22	50.00	100.00	83.33	94.44	72.22	72.22 _{a6.31}
			LaBraM	83.33	61.11	16.67	66.67	38.89	22.22	16.67	-11.11	44.44	27.78	16.67	72.22	0.00	27.78	27.78	27.78	33.33	31.48 _{a3.00}
			Neuro-GPT	<u>94.44</u>	<u>83.33</u>	72.22	50.00	44.44	55.56	16.67	5.56	44.44	77.78	-11.11	<u>94.44</u>	-22.22	16.67	44.44	38.89	27.78	42.44 _{a7.37}
			EEGPT	83.33	55.56	22.22	50.00	66.67	38.89	11.11	11.11	50.00	16.67	22.22	27.78	-50.00	5.56	44.44	44.44	27.78	31.17 _{a5.02}
			CBraMod	100.00	100.00	50.00	100.00	55.53	61.11	38.89	-22.22	<u>77.78</u>	88.89	-22.22	100.00	44.44	-5.56	50.00	61.11	22.22	58.64 _{a2.3}

Table 46. Classification Top-5 accuracies (%) on Things-EEG2. The best accuracies are marked in bold, and the second best by an underline.

Scenario	Tuning	Model Type	Approach	S0	S1	S2	S3	S4	S5	S6	S7	S8	S9	Avg.
Cross-subject (LOSO)		Specialist Models	EEGNet	7.33	12.17	12.67	8.17	11.83	9.00	8.67	10.83	9.83	13.00	10.35 _{±1.25}
			ShallowConv	10.00	12.67	9.67	10.67	11.17	9.83	12.00	10.50	9.83	10.33	10.67 _{±0.45}
			LMDA	11.33	16.33	<u>15.83</u>	15.50	15.67	13.67	15.67	15.67	15.83	14.83	15.03 _{±1.27}
			CNN-T	3.33	6.50	4.33	4.33	4.83	4.67	4.67	4.50	5.00	4.33	4.65 _{±0.50}
			Deformer	14.00	17.33	16.50	<u>17.33</u>	21.17	19.33	19.67	19.67	<u>17.33</u>	15.17	17.75 _{±0.74}
			Conformer	5.67	8.67	8.50	6.33	5.83	8.00	5.50	6.17	6.83	6.83	6.83 _{±0.44}
			TSception	5.50	6.50	6.67	6.33	6.33	8.33	5.67	8.00	6.33	6.50	6.62 _{±0.17}
	Full Fine-tuning	Foundation Models	BENDR	9.67	10.00	10.50	11.67	11.50	9.00	9.00	9.83	10.83	11.33	10.33 _{±0.37}
			BIOT-1D	2.25	2.50	2.58	2.67	2.50	2.75	2.17	2.75	2.08	2.75	2.50 _{±0.00}
			BIOT-2D	2.50	2.67	2.67	3.00	1.83	2.83	2.00	2.50	2.33	2.50	2.48 _{±0.00}
			BIOT-6D	2.67	2.67	2.17	2.67	2.50	3.00	3.00	2.83	2.50	2.83	2.68 _{±0.22}
			LaBraM	15.67	<u>18.83</u>	13.50	18.67	12.67	10.33	14.67	<u>18.50</u>	18.17	14.17	15.52 _{±0.96}
			Neuro-GPT	<u>15.17</u>	17.33	14.83	15.83	<u>16.33</u>	<u>16.50</u>	<u>16.83</u>	17.00	15.67	16.17	<u>16.17</u> _{±0.09}
			EEGPT	11.67	15.67	14.17	14.00	13.67	12.67	14.67	13.50	15.00	13.17	13.82 _{±0.13}
			CBraMod	12.33	19.17	<u>15.83</u>	14.67	15.83	12.50	15.17	13.67	15.83	<u>15.67</u>	15.07 _{±0.18}
			TFM	2.50	2.33	2.50	2.83	3.17	2.50	2.50	2.67	3.00	2.50	2.65 _{±0.04}
			BrainOmni-Tiny	10.00	7.17	10.17	7.83	6.67	10.00	9.67	6.17	10.50	9.50	8.77 _{±0.48}
			BrainOmni-Base	8.67	7.67	10.50	8.50	8.00	7.67	9.00	10.17	8.50	8.00	8.67 _{±0.08}
			EEGMamba	12.33	17.83	13.83	12.83	15.83	14.00	16.17	14.67	13.67	14.33	14.55 _{±0.25}
			SingLEM	9.00	4.83	4.33	6.83	8.83	4.67	8.50	6.67	8.00	3.83	6.55 _{±1.26}
			LUNA-Base	5.67	5.67	5.33	5.50	6.83	5.00	6.33	6.17	6.33	5.50	5.83 _{±0.25}
	Linear Probing	Foundation Models	BENDR	3.17	4.83	3.00	4.50	3.17	3.83	4.50	3.17	5.00	2.67	3.78 _{±0.42}
			BIOT-1D	2.33	2.83	2.50	2.67	2.17	2.67	2.33	2.50	3.00	2.67	2.57 _{±0.00}
			BIOT-2D	3.00	3.17	3.17	2.00	1.67	2.83	2.17	2.00	2.17	2.17	2.43 _{±0.00}
			BIOT-6D	2.83	2.33	2.83	2.67	2.67	2.83	2.67	2.83	2.67	2.33	2.67 _{±0.12}
			LaBraM	4.00	6.17	4.67	4.83	4.83	4.67	6.83	4.83	5.50	5.67	5.20 _{±0.39}
			Neuro-GPT	8.67	8.50	8.00	7.83	9.17	7.50	7.17	8.00	6.67	6.83	7.83 _{±0.06}
			EEGPT	7.83	8.33	5.50	9.50	8.17	9.17	9.33	7.33	7.33	8.83	8.13 _{±0.15}
			CBraMod	9.50	15.50	12.17	13.67	12.50	11.33	14.17	10.17	13.17	14.83	12.70 _{±0.11}
			TFM	2.50	2.67	2.67	1.67	3.17	2.67	3.33	3.83	2.17	3.00	2.77 _{±0.34}
			BrainOmni-Tiny	5.50	5.50	5.50	6.00	5.33	4.67	6.17	5.33	5.33	6.50	5.58 _{±0.45}
			BrainOmni-Base	5.50	7.17	8.33	5.33	5.83	7.83	5.33	6.83	6.50	8.83	6.75 _{±0.57}
			EEGMamba	8.00	16.00	12.17	12.17	13.83	10.83	14.50	13.17	13.67	14.50	12.88 _{±0.48}
			SingLEM	6.67	8.17	5.00	7.00	5.50	6.00	7.17	6.17	8.33	7.33	6.73 _{±0.64}
			LUNA-Base	2.50	2.50	2.50	2.50	2.50	2.50	2.50	2.50	2.50	2.50	2.20 _{±0.00}
Within-subject (Few-shot)		Specialist Models	EEGNet	<u>21.67</u>	36.33	35.67	43.67	32.33	36.50	35.83	45.67	36.67	43.00	36.73 _{±0.55}
			ShallowConv	5.83	6.67	5.17	7.50	5.00	9.50	8.17	7.00	4.67	10.67	7.02 _{±0.42}
			LMDA	22.50	22.50	<u>28.17</u>	32.67	19.17	<u>28.83</u>	23.50	32.67	23.83	32.50	<u>26.63</u> _{±0.89}
			CNN-T	5.50	5.33	4.50	4.33	6.17	4.67	5.17	5.67	4.00	5.83	5.12 _{±0.14}
			Deformer	15.33	22.33	24.17	30.50	18.50	24.67	18.83	29.83	21.17	28.00	23.33 _{±1.57}
			Conformer	5.17	4.33	3.00	4.67	4.50	5.00	4.67	5.33	4.33	6.33	4.73 _{±0.26}
			TSception	5.33	4.83	5.67	4.67	3.50	4.50	6.17	7.17	5.33	11.83	5.90 _{±0.39}
	Full Fine-tuning	Foundation Models	BENDR	5.50	7.17	7.33	8.83	7.83	5.50	8.67	11.00	5.50	15.17	8.25 _{±0.21}
			BIOT-1D	2.67	2.83	3.00	2.50	2.67	2.67	2.83	1.83	2.83	2.67	2.65 _{±0.00}
			BIOT-2D	3.33	2.17	2.83	2.17	2.67	2.67	2.17	1.83	2.50	3.17	2.55 _{±0.00}
			BIOT-6D	2.83	2.50	2.67	2.50	2.50	2.67	2.33	3.17	2.67	2.33	2.62 _{±0.19}
			LaBraM	17.67	21.33	23.67	37.17	18.33	23.00	21.50	31.17	19.17	27.33	24.03 _{±0.78}
			Neuro-GPT	11.83	15.50	14.67	22.17	17.67	20.67	12.00	22.67	15.50	22.83	17.55 _{±0.49}
			EEGPT	5.83	7.50	8.00	8.67	6.83	9.67	5.00	9.83	5.33	9.50	7.62 _{±0.97}
			CBraMod	17.67	13.67	13.50	<u>42.00</u>	17.83	22.50	<u>27.50</u>	<u>40.00</u>	18.67	<u>36.17</u>	24.95 _{±0.18}
			TFM	3.00	2.67	2.33	2.33	2.67	2.67	3.00	2.83	2.17	3.33	2.70 _{±0.40}
			BrainOmni-Tiny	5.83	6.33	8.67	11.33	11.00	7.67	7.67	13.50	6.83	9.00	8.78 _{±0.19}
			BrainOmni-Base	5.67	8.83	8.83	10.50	8.67	7.50	8.83	10.50	5.83	11.33	8.65 _{±0.64}
			EEGMamba	18.17	<u>24.83</u>	21.50	34.33	16.00	26.50	26.67	32.50	<u>24.17</u>	34.50	25.92 _{±0.48}
			SingLEM	4.00	2.33	2.83	3.00	2.67	3.50	2.50	2.50	2.67	2.67	2.87 _{±0.14}
			LUNA-Base	2.83	2.67	3.00	2.67	2.83	2.50	2.33	2.67	2.50	3.67	2.77 _{±0.06}
	Linear Probing	Foundation Models	BENDR	4.33	4.50	4.50	4.50	3.67	4.00	3.50	4.50	3.67	6.50	4.37 _{±0.51}
			BIOT-1D	2.83	2.33	2.67	2.83	3.00	2.50	2.50	2.17	2.17	2.50	2.58 _{±0.00}
			BIOT-2D	3.00	2.67	3.17	2.17	2.83	3.00	2.67	3.17	3.00	2.67	2.83 _{±0.00}
			BIOT-6D	1.83	3.00	2.33	2.50	2.50	1.83	3.00	3.00	2.83	2.17	2.50 _{±0.07}
			LaBraM	5.83	5.00	4.83	5.83	7.00	8.17	5.83	6.50	5.00	7.50	6.15 _{±0.57}
			Neuro-GPT	3.83	7.17	7.17	9.50	9.50	9.50	9.00	11.83	4.33	10.17	8.20 _{±0.47}
			EEGPT	7.17	6.00	9.33	11.17	8.00	11.50	5.67	11.50	5.83	11.67	8.78 _{±0.73}
			CBraMod	12.67	14.83	22.00	29.17	<u>19.83</u>	18.33	19.00	25.50	17.67	26.17	20.52 _{±0.17}
			TFM	4.00	2.50	1.83	3.00	2.83	3.00	2.33	2.83	3.50	3.00	2.88 _{±0.13}
			BrainOmni-Tiny	3.00	6.83	7.83	7.50	7.00	8.17	7.33	10.00	6.00	9.17	7.28 _{±0.18}
			BrainOmni-Base	3.67	4.83	6.67	7.83	6.50	7.17	6.67	10.17	4.33	10.00	6.78 _{±0.28}
			EEGMamba	12.33	14.50	16.67	21.17	13.00	15.33	16.00	23.50	17.50	22.00	17.20 _{±0.49}
			SingLEM	11.17	9.17	16.67	21.17	18.33	12.67	14.50	23.50	13.67	18.50	15.93 _{±0.31}
			LUNA-Base	2.33	2.50	2.83	2.83	2.50	2.17	2.17	2.67	2.50	2.33	2.48 _{±0.10}

Table 47. Classification 2-Way accuracies (%) on Things-EEG2. The best accuracies are marked in bold, and the second best by an underline.

Scenario	Tuning	Model Type	Approach	S0	S1	S2	S3	S4	S5	S6	S7	S8	S9	Avg.
Cross-subject (LOSO)		Specialist Models	EEGNet	65.33	<u>80.17</u>	78.17	66.00	76.50	77.17	74.00	76.00	74.00	76.83	74.42 _{+3.67}
			ShallowConv	69.83	75.83	70.83	70.67	72.83	75.00	74.33	69.83	71.50	69.67	72.03 _{+0.38}
			LMDA	76.50	78.67	74.50	<u>79.50</u>	80.67	80.33	<u>79.33</u>	<u>81.00</u>	79.33	77.33	<u>78.72</u> _{+0.86}
			CNN-T	60.33	62.67	59.33	56.83	60.00	60.17	58.50	60.00	59.00	53.67	59.05 _{+2.13}
			Deformer	<u>79.00</u>	<u>80.17</u>	<u>78.67</u>	79.17	78.67	78.50	76.83	77.67	78.50	<u>77.50</u>	78.47 _{+0.65}
			Conformer	63.00	66.17	63.17	64.00	63.83	63.17	62.33	65.33	64.83	64.17	64.00 _{+0.91}
			TSception	62.17	63.83	64.17	67.33	64.83	68.50	59.00	64.67	58.67	60.17	63.33 _{+1.47}
	Full Fine-tuning	Foundation Models	BENDR	70.33	73.17	69.83	71.33	73.33	71.17	70.17	73.17	69.67	72.50	71.47 _{+0.39}
			BIOT-1D	49.67	50.08	47.58	51.42	49.08	50.83	48.25	50.25	50.75	50.00	49.79 _{+0.01}
			BIOT-2D	48.33	49.83	49.83	53.17	43.50	51.83	50.67	51.33	48.67	47.33	49.45 _{+0.00}
			BIOT-6D	50.33	53.00	49.67	52.33	51.67	50.67	49.17	50.33	49.33	49.17	50.57 _{+0.22}
			LaBraM	76.33	78.17	68.83	78.50	71.83	70.50	73.00	81.67	<u>80.67</u>	72.00	75.15 _{+0.93}
			Neuro-GPT	82.17	82.33	80.67	79.83	<u>79.17</u>	<u>78.67</u>	81.50	78.33	81.00	79.17	80.28 _{+1.08}
			EEGPT	74.17	76.33	72.17	74.67	77.17	73.50	78.33	75.33	73.50	73.83	74.90 _{+1.35}
			CBraMod	73.67	78.83	73.50	76.67	76.33	76.33	78.50	73.67	77.83	73.50	75.88 _{+0.89}
			TFM	50.83	51.50	51.83	53.33	49.00	48.33	49.50	51.00	50.33	49.17	50.48 _{+0.45}
			BrainOmni-Tiny	69.00	63.00	67.17	63.33	67.00	66.00	70.17	66.67	68.50	67.50	66.83 _{+1.16}
			BrainOmni-Base	67.50	68.83	68.00	67.83	62.67	66.83	69.33	67.50	66.50	66.33	67.13 _{+0.39}
			EEGMamba	74.83	74.67	72.67	74.00	74.50	74.83	73.33	75.50	75.17	75.33	74.48 _{+0.12}
			SingLEM	73.00	58.33	58.83	61.67	66.67	59.00	62.33	61.00	61.33	57.50	61.97 _{+4.80}
			LUNA-Base	62.00	63.00	63.33	62.83	68.50	59.67	64.50	63.00	65.00	64.50	63.68 _{+0.55}
	Linear Probing	Foundation Models	BENDR	52.83	58.00	55.00	58.33	50.83	55.83	54.67	49.67	58.17	55.33	54.87 _{+0.75}
			BIOT-1D	47.83	52.00	50.50	49.33	48.50	51.00	52.50	50.33	51.17	51.17	50.43 _{+0.00}
			BIOT-2D	50.67	48.50	51.67	53.17	48.83	52.33	49.00	50.00	50.17	50.17	50.45 _{+0.01}
			BIOT-6D	50.17	34.67	51.33	48.50	48.17	51.33	50.00	51.17	49.50	48.67	48.35 _{+3.15}
			LaBraM	59.50	64.33	56.17	59.83	57.83	60.00	60.33	56.67	61.33	60.50	59.65 _{+1.31}
			Neuro-GPT	67.50	69.33	63.00	66.50	68.00	66.33	64.00	67.33	65.00	65.00	66.20 _{+0.99}
			EEGPT	68.83	64.83	68.00	66.33	66.33	64.33	65.83	69.50	63.50	66.67	66.42 _{+0.49}
			CBraMod	75.00	78.33	72.83	76.67	75.67	75.50	77.50	73.50	76.00	77.00	75.80 _{+1.19}
			TFM	49.67	47.50	50.33	47.50	49.17	46.67	50.17	51.17	50.50	51.67	49.43 _{+0.95}
			BrainOmni-Tiny	62.17	60.17	62.50	57.67	60.17	63.50	61.17	57.83	63.33	59.00	60.75 _{+0.25}
			BrainOmni-Base	64.33	64.83	64.17	62.17	59.00	65.67	67.00	65.50	63.83	62.00	63.85 _{+0.95}
			EEGMamba	70.00	75.83	74.33	73.17	71.50	72.33	74.83	74.17	72.83	75.00	73.40 _{+0.29}
			SingLEM	61.33	66.33	62.67	62.67	62.00	59.83	65.67	64.17	64.50	62.00	63.12 _{+0.48}
			LUNA-Base	48.50	48.17	49.67	48.83	47.33	49.50	50.33	52.00	48.67	49.00	49.20 _{+0.32}
Within-subject (Few-shot)		Specialist Models	EEGNet	84.50	90.00	91.17	<u>90.50</u>	87.50	87.50	88.83	<u>91.00</u>	89.00	92.50	89.25 _{+0.60}
			ShallowConv	59.67	65.83	63.50	64.50	62.00	67.67	62.33	65.50	60.00	74.17	64.52 _{+0.56}
			LMDA	81.50	82.17	<u>89.17</u>	89.33	75.67	83.50	83.83	90.17	81.67	88.33	84.53 _{+0.94}
			CNN-T	59.17	57.67	57.67	58.17	54.83	57.67	58.00	59.00	54.00	58.83	57.50 _{+0.94}
			Deformer	79.00	81.50	87.17	85.50	78.17	82.67	81.00	86.83	82.00	85.67	82.95 _{+0.72}
			Conformer	54.00	63.17	52.00	59.83	59.67	58.50	61.17	68.00	55.83	66.83	59.90 _{+0.19}
			TSception	65.00	65.33	63.67	60.67	52.33	61.17	66.83	64.50	60.17	72.67	63.23 _{+0.86}
	Full Fine-tuning	Foundation Models	BENDR	57.67	66.17	63.83	68.33	63.67	63.83	68.83	75.17	58.83	72.17	65.85 _{+0.49}
			BIOT-1D	51.50	50.17	49.17	50.00	49.67	51.33	51.17	48.67	48.50	51.17	50.13 _{+0.01}
			BIOT-2D	53.50	50.67	48.00	49.83	51.17	48.67	49.17	53.67	49.67	49.33	50.37 _{+0.01}
			BIOT-6D	49.50	53.50	32.50	48.83	50.17	52.67	52.00	51.00	49.50	52.50	49.22 _{+2.46}
			LaBraM	83.17	79.67	84.83	89.67	76.50	83.00	82.83	90.00	81.17	86.67	83.75 _{+0.74}
			Neuro-GPT	76.00	84.00	77.67	82.33	77.00	80.00	78.33	88.83	78.00	88.00	81.02 _{+1.10}
			EEGPT	64.00	64.50	67.83	67.83	64.83	63.33	59.17	65.67	60.00	66.33	64.35 _{+1.28}
			CBraMod	81.50	80.17	73.83	92.17	81.67	86.17	87.00	92.50	81.17	92.17	84.83 _{+0.43}
			TFM	50.67	47.17	49.33	51.83	52.17	49.50	51.67	53.67	50.17	47.83	50.40 _{+0.55}
			BrainOmni-Tiny	61.00	64.17	67.33	69.83	71.33	71.00	62.83	74.83	65.50	71.67	67.95 _{+0.25}
			BrainOmni-Base	63.17	67.00	65.83	71.33	69.00	69.67	65.33	69.50	63.00	73.17	67.70 _{+0.21}
			EEGMamba	<u>84.00</u>	<u>86.83</u>	87.17	87.00	80.17	85.83	86.67	89.33	83.83	90.33	86.12 _{+0.65}
			SingLEM	56.00	51.33	52.67	49.50	49.50	52.00	52.17	54.83	50.67	51.33	52.00 _{+1.27}
			LUNA-Base	51.50	54.17	50.33	53.67	49.33	50.33	53.00	50.50	51.67	54.17	51.87 _{+0.80}
	Linear Probing	Foundation Models	BENDR	55.67	56.17	60.33	61.17	55.50	55.50	54.83	55.67	57.00	56.83	56.87 _{+1.17}
			BIOT-1D	49.50	48.33	48.50	51.83	50.17	48.67	51.50	49.83	51.67	47.00	49.70 _{+0.01}
			BIOT-2D	50.67	52.00	48.00	50.33	49.33	52.33	48.17	47.83	50.50	49.67	49.88 _{+0.01}
			BIOT-6D	51.17	54.17	50.83	52.67	55.00	49.67	52.17	51.00	52.50	49.17	51.83 _{+0.53}
			LaBraM	58.00	57.33	58.00	61.00	62.83	65.33	61.67	61.50	58.50	62.50	60.67 _{+0.47}
			Neuro-GPT	59.67	64.33	62.67	66.17	63.83	66.50	63.67	73.00	63.17	70.67	65.37 _{+0.33}
			EEGPT	63.17	62.00	68.50	68.50	68.17	67.50	61.00	74.00	61.67	68.50	66.30 _{+1.97}
			CBraMod	76.00	79.33	85.50	86.17	76.00	83.17	81.83	86.33	78.17	84.17	81.67 _{+0.38}
			TFM	52.83	47.50	48.00	47.67	47.67	48.00	53.50	49.33	53.50	49.67	49.77 _{+1.25}
			BrainOmni-Tiny	57.67	60.50	61.33	65.33	67.33	65.83	60.33	69.17	62.83	67.33	63.77 _{+0.50}
			BrainOmni-Base	58.50	58.67	62.00	67.00	65.83	64.50	61.50	67.67	62.50	64.50	63.27 _{+0.72}
			EEGMamba	71.17	80.67	79.00	80.67	75.83	78.00	77.33	81.67	78.00	82.67	78.50 _{+0.33}
			SingLEM	70.67	71.17	78.67	79.50	72.33	75.00	76.50	82.33	73.67	79.17	75.90 _{+0.53}
			LUNA-Base	51.50	53.50	50.50	51.67	48.50	50.00	48.17	51.00	48.83	50.67	50.43 _{+0.56}

Table 48. Root Mean Square Error (RMSE) in SEED-VIG. The best RMSEs are marked in bold, and the second best by an underline.

Scenario	Tuning	Model Type	Approach	S0	S1	S2	S3	S4	S5	S6	S7	S8	S9	S10
Cross-subject (LOSO)	Specialist Models	Specialist Models	PSD+Ridge	0.2986	0.1886	0.3192	0.2643	0.2820	0.1505	0.1423	0.2458	0.1408	0.2209	0.1552
			EEGNet	<u>0.1518</u>	0.3864	0.4450	<u>0.1933</u>	0.2665	0.1276	0.1443	0.3423	0.1173	0.1643	0.2006
			ShallowConv	0.1809	0.3228	0.3684	0.2217	0.2833	0.1342	0.1150	0.2960	0.1537	0.2130	0.1367
			LMDA	0.1395	0.1609	0.3758	0.2198	0.2790	0.1324	0.1491	0.2789	<u>0.1188</u>	<u>0.1484</u>	0.1425
			CNN-T	0.2862	0.1654	0.2435	0.2600	0.2610	0.1525	0.1669	0.3070	0.2060	0.2359	0.1886
			Deformer	0.1740	0.2165	0.2887	0.3162	0.2636	0.1755	0.1410	0.3724	0.1508	0.2188	0.1707
			Conformer	0.2218	0.2354	0.2721	0.3173	0.2703	0.2135	0.1308	0.2798	0.1731	0.3430	0.2548
			TSception	0.2361	0.1814	0.2484	0.3169	<u>0.2516</u>	0.1395	0.1244	0.2997	0.1482	0.1962	0.1589
	Full Fine-tuning	Foundation Models	BENDR	0.2117	0.1851	0.2493	0.2338	0.2546	0.1715	0.1692	0.3345	0.1841	0.2553	0.1578
			BIOT-1D	0.1626	0.1792	0.2335	0.3788	0.2576	0.1556	<u>0.1111</u>	0.2805	0.1612	0.1926	0.1477
			BIOT-2D	0.1718	0.1760	0.2292	0.3594	0.2594	0.1523	0.1154	0.2974	0.1550	0.2146	0.1541
			BIOT-6D	0.1866	0.1795	0.2485	0.3312	0.2614	0.1414	0.1327	0.3724	0.1552	0.1878	0.1423
			LaBraM	0.1643	0.1822	0.2085	0.4225	0.2412	0.1304	0.1261	0.3143	0.1278	0.2668	0.1723
			Neuro-GPT	0.2076	0.2549	0.3934	0.2025	0.2836	0.1477	0.1776	0.2609	0.1672	0.2670	0.2270
			EEGPT	0.2789	0.1723	0.2340	0.2033	0.2731	0.1468	0.1403	0.2626	0.1397	0.2060	0.1536
			CBraMod	0.1631	0.2182	0.2880	0.5829	0.3064	0.1893	0.1611	0.3579	0.1504	0.2909	0.1702
			TFM	0.2077	0.1986	0.2074	0.3092	0.2825	0.1351	0.1367	0.2804	0.1679	0.2041	0.1505
			BrainOmni-Tiny	0.1887	0.1751	0.2482	0.2396	0.2726	0.1615	0.1708	0.2350	0.1390	0.1935	0.1608
			BrainOmni-Base	0.1728	0.1857	0.2904	0.2537	0.2718	0.1696	0.1857	0.2637	0.1725	0.2072	0.1980
			EEGMamba	0.1883	0.1670	0.2697	0.2412	0.2758	0.1504	0.1239	0.2641	0.1297	0.1489	0.1644
			SingLEM	0.1901	0.1834	0.2327	0.3071	0.2886	0.1436	0.1817	0.2614	0.1389	0.2046	0.1406
			LUNA-Base	0.1633	0.2063	0.2221	0.2810	0.2732	0.1316	0.1362	0.2643	0.1493	0.2144	0.1328
	Linear Probing	Foundation Models	BENDR	0.3846	0.3026	0.2419	0.2996	0.3585	0.2184	0.2048	0.3323	0.2218	0.2449	0.2216
			BIOT-1D	0.2133	0.1852	0.2283	0.3056	0.2832	0.1411	0.1170	0.2871	0.1735	0.1749	0.1369
			BIOT-2D	0.2078	0.1918	0.2244	0.3213	0.2864	0.1387	0.1067	0.3062	0.1487	0.1801	0.1372
			BIOT-6D	0.1988	0.1956	0.2259	0.3572	0.2635	<u>0.1285</u>	0.1127	0.2900	0.1517	0.2227	0.1648
			LaBraM	0.2181	0.2067	<u>0.1953</u>	0.2590	0.2770	0.1348	0.1195	0.3450	0.1440	0.2301	0.1538
			Neuro-GPT	0.3119	0.1735	0.2305	0.2422	0.3033	0.1397	0.1579	0.2883	0.1366	0.1696	0.1397
			EEGPT	0.2553	<u>0.1626</u>	0.2561	0.1728	0.2861	0.1394	0.1506	0.1767	0.1449	0.1648	0.1568
			CBraMod	0.1809	0.2647	0.3394	0.2431	0.2832	0.2828	0.1463	0.3635	0.1589	0.2387	0.1423
			TFM	0.2930	0.2212	0.1935	0.2522	0.3049	0.1396	0.1332	0.2881	0.1476	0.1470	0.1402
			BrainOmni-Tiny	0.3006	0.1706	0.2204	0.2317	0.2904	0.1392	0.1483	<u>0.2200</u>	0.1214	0.1876	0.1280
			BrainOmni-Base	0.1781	0.1919	0.2366	0.2356	0.2758	0.1498	0.1262	0.2892	0.1317	0.1746	0.1321
			EEGMamba	0.1714	0.1911	0.2558	0.2293	0.2946	0.1512	0.1535	0.3214	0.1526	0.1673	<u>0.1290</u>
			SingLEM	0.3629	0.2122	0.1981	0.2464	0.3016	0.1754	0.1483	0.2979	0.1575	0.1928	0.1522
			LUNA-Base	0.2770	0.1728	0.2292	0.2684	0.2916	0.1336	0.1311	0.2832	0.1297	0.1503	0.1329
Within-subject (Few-shot)	Specialist Models	Specialist Models	PSD+Ridge	0.1936	<u>0.1462</u>	<u>0.1170</u>	0.2600	0.1441	0.1994	0.1450	0.2417	0.1127	0.2583	0.1064
			EEGNet	0.2419	0.1726	0.1316	0.1888	0.2967	0.1395	0.1166	0.2400	0.1163	0.1552	0.1347
			ShallowConv	0.4239	0.3460	0.2623	0.4605	0.3985	0.4335	0.3603	0.5759	0.3616	0.5374	0.2940
			LMDA	0.1569	0.1824	0.1295	0.1725	0.2593	0.1299	0.1179	0.1740	0.1008	0.1438	0.1293
			CNN-T	0.1174	0.1497	0.1144	0.1681	0.2188	0.1591	0.1004	0.2268	<u>0.1003</u>	<u>0.1351</u>	0.1002
			Deformer	0.2196	0.2886	0.2848	0.3235	0.3751	0.2957	0.2510	0.3823	0.3235	0.2508	0.1971
			Conformer	0.1231	0.1320	0.1192	0.1634	0.2116	0.1292	0.1217	0.1212	0.0989	0.1319	0.1131
			TSception	0.1517	0.1468	0.1345	0.1720	0.2959	0.1510	0.1347	0.1835	0.1491	0.1540	0.1456
	Full Fine-tuning	Foundation Models	BENDR	0.3212	0.1768	0.1830	0.2100	0.3537	0.1378	0.1257	0.2774	0.1391	0.1587	0.1469
			BIOT-1D	0.1864	0.1682	0.1502	0.1670	0.2828	0.4321	0.1355	1.4617	0.1409	0.2123	0.1510
			BIOT-2D	0.1623	0.1663	0.1557	0.1515	0.2930	0.2724	0.1194	1.4215	0.1270	0.1720	0.1389
			BIOT-6D	0.1387	0.1619	0.1340	0.1966	<u>0.1291</u>	0.2570	0.1517	0.1890	0.1135	0.1886	<u>0.0898</u>
			LaBraM	0.2022	0.1704	0.1480	0.1917	0.3057	0.1349	0.1222	0.1889	0.1248	0.1571	0.1363
			Neuro-GPT	0.1502	0.1746	0.1619	0.1900	0.2753	0.1384	0.1301	0.1733	0.1293	0.1611	0.1347
			EEGPT	0.1869	0.1651	0.1620	0.1833	0.2911	0.1288	0.1184	0.1870	0.1223	0.1549	0.1314
			CBraMod	0.1672	0.1821	0.1440	0.1969	0.3016	0.1873	0.1203	0.5143	0.1579	0.1922	0.1457
			TFM	0.2552	0.1788	0.1837	0.1995	0.3158	0.1550	0.1262	0.2204	0.1285	0.1652	0.1408
			BrainOmni-Tiny	0.1574	0.2051	0.1803	0.2129	0.2776	0.1765	0.1633	0.2058	0.1362	0.1728	0.1669
			BrainOmni-Base	0.1522	0.1798	0.2028	0.1812	0.2600	0.1405	0.1780	0.1194	0.1196	0.1796	0.1834
			EEGMamba	0.1487	0.1619	0.1466	0.1714	0.2683	0.1457	0.1286	0.1342	0.1241	0.1631	0.1274
			SingLEM	0.2019	0.1918	0.1633	0.2139	0.3423	0.1294	0.1225	0.2189	0.1294	0.1553	0.1391
			LUNA-Base	0.1108	0.1665	0.1530	0.1696	0.2765	<u>0.1277</u>	0.1151	0.0999	0.1182	0.1450	0.1305
	Linear Probing	Foundation Models	BENDR	0.2886	0.1591	0.1342	0.3921	0.2058	0.2329	0.1682	0.3348	0.2021	0.3342	0.1182
			BIOT-1D	0.1487	0.1683	0.1732	0.1618	0.3521	0.1615	0.1581	0.5968	0.1216	0.1675	0.1383
			BIOT-2D	0.1843	0.1675	0.1543	<u>0.1566</u>	0.4053	0.1789	0.1140	1.0112	0.1570	0.2390	0.1378
			BIOT-6D	0.1359	0.1475	0.1303	0.1944	0.1209	0.2230	0.1485	0.2066	0.1163	0.2232	0.0860
			LaBraM	0.1737	0.1631	0.1258	0.2809	0.1611	0.2195	0.1563	0.2738	0.1603	0.2258	0.1154
			Neuro-GPT	0.1619	0.1700	0.1695	0.1936	0.2856	0.1489	0.1403	0.1850	0.1287	0.1768	0.1383
			EEGPT	0.1878	0.1679	0.1613	0.1956	0.2906	0.1281	0.1177	0.1923	0.1342	0.1641	0.1374
			CBraMod	0.1527	0.1952	0.1231	0.2375	0.1338	0.2099	0.1501	0.2599	0.1359	0.2148	0.1044
			TFM	0.2552	0.1790	0.1838	0.1995	0.3159	0.1556	0.1263	0.2206	0.1287	0.1654	0.1409
			BrainOmni-Tiny	0.2064	0.1915	0.1623	0.2336	0.2744	0.1651	0.1755	0.1807	0.1356	0.1989	0.1461
			BrainOmni-Base	0.1335	0.1797	0.1812	0.2008	0.2714	0.1404	0.1291	0.1342	0.1130	0.1541	0.1423
			EEGMamba	0.1361	0.1596	0.1347	0.1706	0.2615	0.1385	0.1220	0.1437	0.1166	0.1536	0.1210
			SingLEM	0.3449	0.1972	0.2214	0.2328	0.3555	0.1673	0.1825	0.2580	0.1876	0.1960	0.1869
			LUNA-Base	<u>0.1165</u>	0.1657	0.1477	0.1670	0.2603	0.1268	<u>0.1120</u>	<u>0.1155</u>	0.1138	0.1432	0.1286

Table 49. Root Mean Square Error (RMSE) in SEED-VIG. The best RMSEs are marked in bold, and the second best by an underline (continued).

Scenario	Tuning	Model Type	Approach	S11	S12	S13	S14	S15	S16	S17	S18	S19	S20	Avg.
Cross-subject (LOSO)		Specialist Models	PSD+Ridge	0.3325	0.2546	0.2191	0.1703	0.3184	0.2466	0.2548	0.1535	0.4711	0.3976	0.2489
			EEGNet	0.3956	0.3087	0.2731	0.2051	0.3506	0.2449	0.2563	<u>0.1148</u>	0.3752	0.3146	0.2561 _{±0.0092}
			ShallowConv	0.3256	0.2457	0.2382	0.1856	<u>0.2009</u>	0.1842	0.1952	0.1263	<u>0.3681</u>	0.3136	0.2290 _{±0.0029}
			LMDA	0.4162	0.3128	0.2343	0.1865	0.3086	0.2624	0.2570	0.1272	0.4555	0.3119	0.2389 _{±0.0029}
			CNN-T	0.4165	0.3413	0.2450	0.2343	0.2644	0.2416	0.2806	0.1600	0.3743	0.3374	0.2556 _{±0.0140}
			Deformer	0.3223	0.3495	0.2493	0.2954	0.2303	0.1736	0.2007	0.1941	0.4119	0.3607	0.2512 _{±0.0053}
			Conformer	0.3417	<u>0.2076</u>	<u>0.2125</u>	0.1788	0.2395	<u>0.1628</u>	0.1537	0.1170	0.4354	<u>0.2889</u>	0.2405 _{±0.0044}
			TSception	0.3909	0.3081	0.1840	0.1580	0.2049	0.2282	0.2903	0.1198	0.3931	0.2551	0.2302 _{±0.0019}
		Foundation Models	BENDR	0.3027	0.2982	0.2440	0.2298	0.2617	0.2614	0.2091	0.1834	0.3702	0.2981	0.2412 _{±0.0025}
			BIOT-1D	<u>0.2932</u>	0.2507	0.2374	0.1884	0.3049	0.2356	0.2129	0.2144	0.4161	0.3164	0.2348 _{±0.0094}
			BIOT-2D	0.3103	0.2546	0.2282	0.1936	0.3203	0.2290	0.2092	0.2209	0.4217	0.3093	0.2372 _{±0.0032}
			BIOT-6D	0.2945	0.2304	0.2160	0.2109	0.3020	0.2548	0.2150	0.1759	0.4289	0.3183	0.2374 _{±0.0033}
			LaBraM	0.3970	0.2042	0.2028	0.1703	0.1869	0.1847	0.1977	0.1669	0.4418	0.2807	0.2281 _{±0.0035}
			Neuro-GPT	0.2801	0.3105	0.2266	0.1687	0.2201	0.2782	0.1928	0.2230	0.4635	0.3151	0.2509 _{±0.0055}
			EEGPT	0.3370	0.2994	0.2371	0.1711	0.3126	0.2261	0.2882	0.1671	0.4336	0.3611	0.2402 _{±0.0025}
			CBraMod	0.4254	0.2917	0.2203	0.2485	0.2672	0.1623	<u>0.1680</u>	0.2296	0.4765	0.3390	0.2718 _{±0.0019}
			TFM	0.3259	0.2366	0.2136	0.1852	0.2867	0.1743	0.2219	0.1948	0.3753	0.3002	<u>0.2283</u> _{±0.0026}
			BrainOmni-Tiny	0.3531	0.3049	0.2214	0.2122	0.2855	0.2659	0.2439	0.1469	0.5252	0.3673	0.2434 _{±0.0003}
			BrainOmni-Base	<u>0.5224</u>	0.2575	0.2156	0.2155	0.4240	0.1959	0.2162	0.1776	0.4315	0.3263	0.2549 _{±0.0055}
			EEGMamba	0.3218	0.2420	0.2076	0.2735	0.2779	0.2106	0.2312	0.1407	0.4382	0.3567	0.2297 _{±0.0010}
			SingLEM	0.3653	0.2183	<u>0.2016</u>	0.2022	0.3014	0.2196	0.2013	0.1845	0.4124	0.3536	0.2349 _{±0.0004}
			LUNA-Base	0.3974	0.2627	0.2194	0.2061	0.3113	0.2118	0.1816	0.1661	0.4319	0.3546	0.2342 _{±0.0043}
		Foundation Models	BENDR	0.4328	0.3788	0.2865	0.2331	0.4012	0.2653	0.3836	0.2072	0.4328	0.3897	0.3068 _{±0.0017}
			BIOT-1D	0.3008	0.2311	0.2142	0.2132	0.3146	0.2092	0.2032	0.1926	0.4186	0.3256	0.2319 _{±0.0006}
			BIOT-2D	0.3158	0.2321	0.2089	0.2026	0.3176	0.2074	0.2023	0.1801	0.4430	0.3397	0.2333 _{±0.0033}
			BIOT-6D	0.3177	0.2284	0.2128	0.2330	0.2618	0.2048	0.2049	0.1695	0.4567	0.3328	0.2349 _{±0.0011}
			LaBraM	0.3289	0.2365	0.2119	0.2406	0.2318	0.2065	0.2421	0.2215	0.3660	0.3327	0.2334 _{±0.0010}
			Neuro-GPT	0.3804	0.2879	0.2262	0.1788	0.3378	0.1862	0.3147	0.1175	0.4251	0.3820	0.2443 _{±0.0035}
			EEGPT	0.4045	0.3240	0.2265	0.1661	0.3490	0.2387	0.2810	0.1226	0.3908	0.3342	0.2335 _{±0.0043}
			CBraMod	0.5335	0.2452	0.2201	0.2762	0.3189	0.2845	0.2028	0.1570	0.5323	0.3921	0.2765 _{±0.0020}
			TFM	0.3779	0.2946	0.2381	0.1938	0.3302	0.1962	0.3120	0.1284	0.4100	0.3347	0.2417 _{±0.0004}
			BrainOmni-Tiny	0.3772	0.3308	0.2548	<u>0.1607</u>	0.3559	0.2391	0.3363	0.1178	0.4642	0.3450	0.2447 _{±0.0012}
			BrainOmni-Base	0.4268	0.2840	0.2024	0.2181	0.3448	0.1805	0.2235	0.1388	0.4731	0.3416	0.2360 _{±0.0032}
			EEGMamba	0.3608	0.2461	0.2148	0.2219	0.3114	0.2265	0.2209	0.1403	0.4349	0.3253	0.2343 _{±0.0002}
			SingLEM	0.3737	0.3547	0.2444	0.1995	0.3797	0.2495	0.3364	0.1571	0.4218	0.3640	0.2632 _{±0.0006}
			LUNA-Base	0.3661	0.2817	0.2188	0.1688	0.3263	0.2102	0.2817	0.1143	0.4635	0.3391	0.2367 _{±0.0013}
Within-subject (Few-shot)		Specialist Models	PSD+Ridge	0.1551	0.2992	0.2731	<u>0.1323</u>	0.1699	0.2425	0.1232	0.1029	0.1772	0.1044	0.1764 _{±0.0013}
			EEGNet	0.3122	0.1603	0.2282	0.1440	0.3110	0.1347	0.3596	0.1173	0.3719	0.2997	0.2082 _{±0.0045}
			ShallowConv	0.3172	0.3825	0.4472	0.3398	0.2967	0.4746	0.3768	0.3002	0.3114	0.3624	0.3839 _{±0.0063}
			LMDA	0.2795	0.1416	0.3108	0.1487	0.4559	0.1362	0.3642	0.1144	0.3479	0.2621	0.2027 _{±0.0110}
			CNN-T	0.1446	0.1114	0.2042	<u>0.1323</u>	0.2438	0.1097	0.1948	0.0785	<u>0.1537</u>	0.2654	0.1538 _{±0.0100}
			Deformer	0.3766	0.1687	0.2609	0.2288	0.5325	0.1688	0.3306	0.2544	0.3026	0.2781	0.2902 _{±0.0167}
			Conformer	<u>0.1455</u>	0.1229	0.1855	0.1273	0.2125	0.1194	0.1448	<u>0.0874</u>	0.1281	0.2461	0.1421 _{±0.0034}
			TSception	0.2254	0.2250	0.2639	0.1553	0.2321	0.1505	0.2798	0.1242	0.1978	0.2144	0.1851 _{±0.0063}
		Foundation Models	BENDR	0.4155	0.2036	0.2529	0.1872	0.3665	0.2220	0.3583	0.1143	0.4058	0.3605	0.2436 _{±0.0023}
			BIOT-1D	0.1927	0.1456	0.2334	0.1886	0.1804	<u>0.1105</u>	0.2754	0.1049	0.3086	0.2245	0.2597 _{±0.0170}
			BIOT-2D	0.1963	0.1358	0.2915	0.1614	0.1819	0.1152	0.1982	0.1049	0.2909	0.2311	0.2423 _{±0.0410}
			BIOT-6D	0.1716	0.2337	0.2228	0.1500	<u>0.1614</u>	0.2767	0.1718	0.1181	1.3133	<u>0.1126</u>	0.2230 _{±0.0414}
			LaBraM	0.2550	0.1740	0.2344	0.1653	0.2809	0.1458	0.2348	0.1232	0.2960	0.3151	0.1956 _{±0.0017}
			Neuro-GPT	0.2819	0.1680	0.2237	0.1728	0.2285	0.1281	0.2171	0.1213	0.3289	0.2585	0.1880 _{±0.0035}
			EEGPT	0.3240	0.1757	0.2210	0.1634	0.2726	0.1446	0.2750	0.1220	0.3642	0.2847	0.1990 _{±0.0007}
			CBraMod	0.2116	0.1278	0.2385	0.1581	0.2596	0.1389	0.2215	0.1348	0.2383	0.2686	0.2051 _{±0.0036}
			TFM	0.3439	0.1940	0.2545	0.1776	0.2917	0.2034	0.3094	0.1189	0.3571	0.3174	0.2208 _{±0.0058}
			BrainOmni-Tiny	0.3204	0.1711	0.2489	0.1565	0.2667	0.1792	0.2781	0.1460	0.3914	0.2934	0.2146 _{±0.0089}
			BrainOmni-Base	0.2822	0.1794	0.2084	0.2023	0.2345	0.1363	0.2088	0.1247	0.2941	0.2718	0.1923 _{±0.0062}
			EEGMamba	0.2462	0.1588	0.2208	0.1667	0.2485	0.1385	0.1833	0.1228	0.2689	0.2518	0.1774 _{±0.0018}
			SingLEM	0.3316	0.1764	0.2453	0.1791	0.3469	0.2049	0.3572	0.1194	0.4225	0.3785	0.2271 _{±0.0021}
			LUNA-Base	0.2073	0.1593	0.2083	0.1605	0.1976	0.1559	0.1444	0.1044	0.3294	0.2392	0.1676 _{±0.0034}
		Foundation Models	BENDR	0.1692	0.3690	0.3362	0.1691	0.2027	0.3310	<u>0.1329</u>	0.1203	0.2435	0.1244	0.2271 _{±0.0015}
			BIOT-1D	0.2032	0.1307	0.2805	0.1629	0.1949	0.1290	0.2314	0.1074	0.3081	0.2334	0.2062 _{±0.0022}
			BIOT-2D	0.2046	<u>0.1158</u>	0.2738	0.1625	0.1890	0.1145	0.1842	0.1137	0.2620	0.2295	0.2264 _{±0.0096}
			BIOT-6D	0.1753	0.2752	0.2257	0.1374	0.1609	0.2850	0.3807	0.0990	2.5759	0.1191	0.2937 _{±0.0419}
			LaBraM	0.1741	0.3135	0.3128	0.1562	0.1878	0.2867	0.1398	0.1187	0.1985	0.1183	0.1935 _{±0.0020}
			Neuro-GPT	0.3035	0.1981	0.2398	0.1757	0.2535	0.1513	0.2473	0.1312	0.3379	0.2943	0.2015 _{±0.0051}
			EEGPT	0.3201	0.1789	0.2218	0.1675	0.2616	0.1448	0.2675	0.1304	0.3423	0.2971	0.2004 _{±0.0062}
			CBraMod	0.1763	0.2235	0.2628	0.1341	0.1643	0.2728	0.2247	0.1168	0.3644	0.1222	0.1895 _{±0.0075}
			TFM	0.3440	0.1941	0.2547	0.1778	0.2919	0.2037	0.3097	0.1191	0.3572	0.3136	0.2208 _{±0.0060}
			BrainOmni-Tiny	0.3425	0.2059	0.2260	0.1686	0.2752	0.1937	0.2750	0.1653	0.3608	0.3422	0.2202 _{±0.0035}
			BrainOmni-Base	0.3042	0.1733	0.2111	0.1766	0.2294	0.1386	0.2099	0.1182	0.3000	0.2678	0.1861 _{±0.0031}

Table 50. Correlation Coefficient (CC) in SEED-VIG. The best CCs are marked in bold, and the second best by an underline.

Scenario	Tuning	Model Type	Approach	S0	S1	S2	S3	S4	S5	S6	S7	S8	S9	S10
Cross-subject (LOSO)		Specialist Models	PSD+Ridge	0.5198	0.2097	0.6967	0.4877	0.5218	0.1122	0.0000	0.6737	0.1956	0.1524	0.3600
			EEGNet	<u>0.9086</u>	0.0000	0.7628	0.5818	0.5747	0.2428	0.3931	0.7300	0.5399	0.3688	0.2927
			ShallowConv	0.9244	0.0000	<u>0.7713</u>	0.4348	0.4839	0.2310	0.4500	0.8661	0.2337	0.4032	<u>0.4544</u>
			LMDA	0.8850	0.3588	0.6897	0.3258	0.6412	0.1284	0.3926	0.8624	0.4727	0.2447	0.3763
			CNN-T	0.2768	0.0000	0.4097	0.2972	0.6544	0.1490	0.1744	0.5294	0.4185	0.2513	0.1913
			Deformer	0.8462	0.0965	0.6380	0.2934	0.6298	0.1677	0.4134	0.7577	0.1776	0.3351	0.2401
			Conformer	0.7444	0.0000	0.7264	0.2970	0.5771	0.2283	0.4708	0.8690	0.3256	<u>0.4261</u>	0.0699
			TSception	0.6892	-0.1142	0.7760	0.4700	<u>0.7069</u>	0.1953	0.4857	<u>0.8867</u>	0.1980	0.3793	0.3221
	Full Fine-tuning	Foundation Models	BENDR	0.7231	0.1267	0.5133	0.3951	0.6449	0.1286	0.2706	0.6890	0.3737	0.3748	0.4128
			BIOT-1D	0.8794	0.1529	0.4565	0.5388	0.6548	-0.0531	0.4121	0.7249	0.2535	0.3076	0.4227
			BIOT-2D	0.8940	0.0998	0.4486	0.5352	0.6210	-0.0929	<u>0.4898</u>	0.7006	0.2608	0.2981	0.4241
			BIOT-6D	0.8822	0.0992	0.3347	0.5095	0.6072	0.0342	0.3251	0.6071	0.2164	0.3157	0.3947
			LaBraM	0.8946	0.1274	0.7181	0.2799	0.7558	0.3703	0.2166	0.8592	<u>0.4848</u>	0.4271	0.4221
			Neuro-GPT	0.7564	0.0500	0.7611	0.4091	0.6607	0.1901	0.4666	0.8350	0.1370	0.4118	0.3324
			EEGPT	0.7147	0.1150	0.6268	0.4923	0.6051	0.2064	0.3091	0.7409	0.0906	0.2785	0.3649
			CBraMod	0.8478	0.0973	0.6755	0.3276	0.4693	0.1232	0.0376	0.7804	0.3272	0.3733	0.4253
			TFM	0.7696	0.2279	0.3340	0.3263	0.5118	0.0446	0.3293	0.8360	0.3295	0.2699	0.2325
			BrainOmni-Tiny	0.8270	0.2189	0.5358	0.4230	0.5837	-0.0371	0.1734	0.8219	0.2381	0.2102	0.2788
			BrainOmni-Base	0.8758	0.2336	0.5388	0.5875	0.6151	-0.0376	0.2247	0.8688	0.3408	0.2690	0.2923
			EEGMamba	0.8899	0.1830	0.6663	0.5991	0.6160	0.0550	0.1444	0.8679	0.2221	0.3132	0.3158
			SingLEM	0.7883	0.0650	0.6399	0.4890	0.4754	<u>0.3078</u>	0.0822	0.8787	0.4233	0.3122	0.3132
			LUNA-Base	0.8849	0.1471	0.5793	0.5445	0.6064	0.0143	-0.0212	0.8963	0.4111	0.3194	0.2134
	Linear Probing	Foundation Models	BENDR	0.0646	0.0420	0.0149	0.0264	0.0465	0.0000	0.0068	0.1697	0.0000	0.0000	0.0215
			BIOT-1D	0.8785	0.1003	0.3514	0.5340	0.5317	0.0844	0.3029	0.7911	0.1330	0.2946	0.3200
			BIOT-2D	0.8840	0.0652	0.2804	0.5351	0.5161	0.0891	0.4054	0.8216	0.2359	0.3396	0.3336
			BIOT-6D	0.8613	0.0101	0.3812	0.3693	0.5985	0.1590	0.5362	0.7399	0.2214	0.3637	0.3926
			LaBraM	0.8143	0.0658	0.6642	0.4647	0.6547	0.3032	0.2936	0.8391	0.3024	0.3411	0.4738
			Neuro-GPT	0.7031	0.0921	0.4517	0.4238	0.4578	0.0957	0.2240	0.7239	0.1739	0.1302	0.3000
			EEGPT	0.8923	0.1802	0.5139	<u>0.6042</u>	0.6122	0.1252	0.2722	0.8710	0.1014	0.2414	0.3519
			CBraMod	0.7952	0.0403	0.6007	0.4324	0.5398	0.1146	0.2297	0.7313	0.1861	0.2329	0.2124
			TFM	0.6200	0.0437	0.2327	0.2654	0.4322	-0.0319	0.0056	0.7035	0.1726	0.1485	0.1819
			BrainOmni-Tiny	0.8540	0.1980	0.5035	0.4359	0.6462	-0.0596	0.3463	0.8572	0.2764	0.1543	0.3233
			BrainOmni-Base	0.8914	<u>0.3108</u>	0.6191	0.6568	0.5860	-0.1028	0.1658	0.8603	0.2713	0.2769	0.3190
			EEGMamba	0.8837	0.1842	0.5305	0.5438	0.5281	0.0428	0.2429	0.8070	0.2255	0.2518	0.3330
			SingLEM	0.2197	0.1981	0.3002	0.0923	0.3905	0.0615	0.1505	0.3070	0.1617	0.0964	0.2343
			LUNA-Base	0.8774	0.2164	0.5241	0.5591	0.6094	-0.1136	0.3078	0.8815	0.2070	0.2705	0.2960
Within-subject (Few-shot)		Specialist Models	PSD+Ridge	0.7536	0.2109	0.4806	0.7401	0.7063	0.5018	0.4671	0.7035	<u>0.8335</u>	0.6832	0.4440
			EEGNet	0.7476	0.1740	0.6263	0.4365	0.4916	0.0376	0.1450	0.2980	0.4820	0.1535	0.2769
			ShallowConv	0.2673	0.0019	0.1119	0.4057	0.1718	0.0362	0.0002	0.1538	0.1758	0.1688	0.0025
			LMDA	0.8435	0.0233	0.6016	0.4799	0.6406	0.0591	0.1181	0.7738	0.5925	0.1871	0.2097
			CNN-T	0.9309	<u>0.4047</u>	<u>0.7817</u>	0.6009	0.7924	0.2742	0.6333	0.5343	0.6335	0.3082	0.6664
			Deformer	0.8860	0.2823	0.6592	0.5088	0.7292	0.0085	0.3118	0.7431	0.5048	0.1648	0.3015
			Conformer	0.9398	0.6335	0.8333	0.6405	0.8331	0.2106	<u>0.5435</u>	0.8849	0.6352	0.4733	0.6051
			TSception	0.9049	0.5549	0.7198	0.5935	0.6038	0.4522	0.4987	0.7230	0.3873	0.4068	0.5203
	Full Fine-tuning	Foundation Models	BENDR	0.1382	0.0407	0.0323	0.0986	0.1354	0.0317	0.0415	0.0646	0.0000	0.0100	0.0192
			BIOT-1D	0.8281	0.1219	0.5194	0.5633	0.5702	-0.0181	0.3492	0.1379	0.3701	0.1327	0.1399
			BIOT-2D	0.8498	0.1099	0.4561	0.6281	0.5126	0.0086	0.4027	0.0823	0.4290	0.1029	0.1551
			BIOT-6D	0.8889	0.1230	0.2868	<u>0.8558</u>	0.8015	0.2550	0.4342	0.8232	0.8456	0.8410	<u>0.6683</u>
			LaBraM	0.7485	0.0365	0.4122	0.2559	0.4316	0.1092	0.1519	0.6661	0.3049	0.0892	0.1040
			Neuro-GPT	0.8655	0.0787	0.4544	0.3109	0.5694	0.0962	0.1511	0.7082	0.2282	0.0690	0.2069
			EEGPT	0.8351	0.1137	0.3437	0.4294	0.5130	0.1133	0.1299	0.7743	0.3906	0.0914	0.1606
			CBraMod	0.8878	0.1933	0.6013	0.5462	0.6542	0.1298	0.3835	0.3146	0.4261	0.2454	0.4629
			TFM	0.5065	0.0281	0.1265	0.2376	0.4234	0.0731	0.1653	0.5505	0.1062	0.0438	0.1741
			BrainOmni-Tiny	0.8750	0.1253	0.4272	0.3631	0.5909	0.1880	0.1270	0.7675	0.2186	0.0413	0.2259
			BrainOmni-Base	0.9131	0.1149	0.5710	0.5189	0.6450	0.1284	0.2684	0.8826	0.5929	0.1915	0.3720
			EEGMamba	0.8745	0.2324	0.4908	0.5202	0.6018	0.0932	0.1286	0.8304	0.4573	0.2250	0.3841
			SingLEM	0.6939	-0.0088	0.2152	0.1245	0.2443	-0.0031	0.0288	0.4286	0.0592	0.0293	0.0345
			LUNA-Base	<u>0.9357</u>	0.1222	0.4205	0.6053	0.6738	0.0671	0.2208	0.9154	0.4920	0.2315	0.2842
	Linear Probing	Foundation Models	BENDR	0.1344	0.0000	0.0137	0.0110	0.0509	0.0143	0.0297	0.0214	0.0498	0.2143	0.2644
			BIOT-1D	0.8671	0.1437	0.4315	0.5858	0.3940	0.0761	0.4132	0.2203	0.4827	0.1593	0.2127
			BIOT-2D	0.8383	0.1410	0.5436	0.6232	0.4641	0.0668	0.5330	0.2583	0.4285	0.1564	0.2160
			BIOT-6D	0.8874	0.3082	0.3110	0.8612	<u>0.8200</u>	0.4030	0.5079	0.8022	0.8325	0.7988	0.7038
			LaBraM	0.8737	0.1142	0.3301	0.7309	0.7113	0.3391	0.2958	0.5972	0.6732	<u>0.8330</u>	0.4118
			Neuro-GPT	0.8401	0.1459	0.3909	0.3225	0.5521	0.0524	0.1176	0.6340	0.2287	0.0274	0.2270
			EEGPT	0.8417	0.0755	0.3781	0.4421	0.6197	0.0850	0.2036	0.8543	0.4434	-0.0114	0.2119
			CBraMod	0.8692	0.1512	0.4405	0.7802	0.7564	<u>0.4587</u>	0.4020	0.6655	0.7983	0.8314	0.4807
			TFM	0.4886	0.0015	0.0163	0.2409	0.3574	-0.0204	0.0900	0.5261	0.2136	-0.0011	0.1528
			BrainOmni-Tiny	0.8442	0.1310	0.3157	0.2634	0.6188	0.2200	0.1248	0.7672	0.3198	0.0812	0.2719
			BrainOmni-Base	0.9083	0.1156	0.4890	0.5477	0.6330	0.1366	0.1679	<u>0.8988</u>	0.5979	0.1862	0.3970
			EEGMamba	0.8977	0.2681	0.6017	0.5187	0.6247	0.1561	0.2069	0.8234	0.4560	0.2591	0.4439
			SingLEM	0.1562	-0.0082	0.0703	0.0588	0.1310	0.0038	-0.0029	0.1066	-0.0348	0.0063	0.0295
			LUNA-Base	0.9161	0.1181	0.4375	0.5891	0.6285	0.1496	0.2369	0.8856	0.4727	0.1851	0.3334

Table 51. Correlation Coefficient (CC) in SEED-VIG. The best CCs are marked in bold, and the second best by an underline (continued).

Scenario	Tuning	Model Type	Approach	S11	S12	S13	S14	S15	S16	S17	S18	S19	S20	Avg.
Cross-subject (LOSO)	Specialist Models	Specialist Models	PSD+Ridge	0.7425	0.5121	0.4216	0.4431	0.5063	0.6157	0.7274	0.3163	0.0832	0.5362	0.4185
			EEGNet	0.7512	0.7363	0.4771	0.4507	0.7983	0.6666	0.8440	0.5182	0.5550	0.7253	0.5646 _{±0.0009}
			ShallowConv	0.8175	0.8256	0.5585	0.4405	0.8090	0.7443	0.8694	0.4415	<u>0.5858</u>	0.7832	<u>0.5712</u> _{±0.0076}
			LMDA	0.5047	0.6273	0.4702	0.4962	0.6890	0.5058	0.8048	0.3974	0.2411	0.7105	0.5155 _{±0.0047}
			CNN-T	0.4985	0.2257	0.4278	0.3998	0.5263	0.6796	0.5103	0.2959	0.5074	0.2711	0.3648 _{±0.1177}
			Deformer	<u>0.8158</u>	0.8056	0.5001	0.2724	0.7546	0.7037	0.8634	0.4499	0.5214	0.7117	0.5235 _{±0.0135}
			Conformer	0.6501	0.8492	0.5833	0.4974	0.7426	0.7995	0.9072	0.4777	0.5022	0.7820	0.5418 _{±0.0147}
			TSception	0.6748	<u>0.8280</u>	0.6699	0.4340	0.8563	0.7040	0.8417	0.4847	0.6191	0.8140	0.5677 _{±0.0072}
	Full Fine-tuning	Foundation Models	BENDR	0.7381	0.5333	0.4009	0.2671	0.6846	0.5670	0.8131	0.3256	0.5704	0.6248	0.4847 _{±0.0114}
			BIOT-1D	0.7478	0.7332	0.4047	0.3820	0.4961	0.7033	0.7804	0.4142	0.0653	0.7751	0.4882 _{±0.0166}
			BIOT-2D	0.7089	0.7543	0.4404	0.2782	0.5215	0.6182	0.7951	0.4596	0.1433	<u>0.7967</u>	0.4855 _{±0.0084}
			BIOT-6D	0.7072	0.7253	0.4639	0.2793	0.5311	0.4806	0.7866	0.4623	-0.0267	0.7818	0.4532 _{±0.0069}
			LaBraM	0.6880	0.7985	0.5829	<u>0.5464</u>	<u>0.8268</u>	0.6739	0.8779	0.4035	0.4935	0.7655	0.5816 _{±0.0074}
			Neuro-GPT	0.7999	0.8056	<u>0.6253</u>	0.4680	0.7976	0.7320	0.8640	0.4544	0.4803	0.7509	0.5613 _{±0.0095}
			EEGPT	0.6823	0.6947	0.4032	0.4159	0.6868	0.4240	0.7389	0.4207	0.2973	0.7095	0.4770 _{±0.0022}
			CBraMod	0.5184	0.5132	0.4776	0.5624	0.6705	0.6806	<u>0.8901</u>	<u>0.5050</u>	0.4180	0.7151	0.4960 _{±0.0170}
			TFM	0.6470	0.7463	0.5048	0.4206	0.5448	0.7479	0.7738	0.2662	0.2932	0.6924	0.4690 _{±0.0037}
			BrainOmni-Tiny	0.5799	0.6605	0.4509	0.3717	0.6134	0.6233	0.7727	0.3273	0.2743	0.6595	0.4575 _{±0.0048}
			BrainOmni-Base	0.5563	0.7365	0.5753	0.4265	0.5679	0.7642	0.8145	0.2746	0.3298	0.7129	0.5032 _{±0.0094}
			EEGMamba	0.7072	0.7119	0.5562	0.4197	0.7255	0.5669	0.8273	0.3450	0.3072	0.7218	0.5124 _{±0.0022}
			SingLEM	0.6576	0.7121	0.5483	0.4981	0.5736	0.6915	0.8177	0.4599	0.3784	0.6842	0.5141 _{±0.0033}
			LUNA-Base	0.5066	0.8114	0.5387	0.2750	0.4227	<u>0.7779</u>	0.8613	0.2370	0.2055	0.7493	0.4753 _{±0.0149}
	Linear Probing	Foundation Models	BENDR	0.0467	0.0945	0.0550	0.0228	0.0000	0.0413	0.0349	0.0173	0.0236	0.0876	0.0364 _{±0.0001}
			BIOT-1D	0.7036	0.7536	0.4847	0.2950	0.5572	0.5928	0.8137	0.3563	0.1329	0.7703	0.4658 _{±0.0054}
			BIOT-2D	0.6702	0.7465	0.5016	0.2494	0.5237	0.5324	0.8087	0.4606	0.0385	0.7389	0.4656 _{±0.0018}
			BIOT-6D	0.6823	0.7486	0.4553	0.3129	0.6689	0.5535	0.8182	0.4504	0.0472	0.7383	0.4814 _{±0.0032}
			LaBraM	0.7218	0.7174	0.5229	0.4506	0.7062	0.6261	0.7860	0.3652	0.5581	0.7063	0.5418 _{±0.0018}
			Neuro-GPT	0.6705	0.5040	0.4089	0.3394	0.5651	0.6105	0.7238	0.3873	0.0618	0.6192	0.4127 _{±0.0205}
			EEGPT	0.6788	0.6007	0.4259	0.3922	0.6915	0.6352	0.8319	0.3585	0.2493	0.7030	0.4920 _{±0.0018}
			CBraMod	0.6434	0.7606	0.5867	0.3329	0.7095	0.4571	0.8430	0.4032	0.3096	0.7209	0.4706 _{±0.0021}
			TFM	0.5331	0.5479	0.2712	0.0127	0.4305	0.6168	0.5776	0.3828	0.1461	0.5387	0.3253 _{±0.0072}
			BrainOmni-Tiny	0.6011	0.7149	0.4741	0.4196	0.5756	0.6832	0.7918	0.4223	0.1335	0.6736	0.4774 _{±0.0028}
			BrainOmni-Base	0.6250	0.7391	0.5853	0.3928	0.6399	0.7155	0.8633	0.2739	0.2305	0.7453	0.5079 _{±0.0018}
			EEGMamba	0.7328	0.5963	0.5054	0.4090	0.4956	0.6582	0.8307	0.4037	0.2276	0.6821	0.4816 _{±0.0008}
Within-subject (Few-shot)	Specialist Models	Specialist Models	PSD+Ridge	0.3635	0.6049	0.5820	0.6068	0.5026	0.6850	0.2461	0.4745	0.7124	0.5326	0.5636 _{±0.0038}
			EEGNet	0.6045	0.7190	0.3789	0.5291	0.5761	0.7771	0.5053	0.3264	0.2618	0.4636	0.4291 _{±0.0192}
			ShallowConv	0.0195	0.3899	0.0760	0.1505	0.1447	0.1633	0.0205	0.0977	0.2767	0.0180	0.1233 _{±0.0168}
			LMDA	0.6699	0.7318	0.2876	0.4468	0.4407	0.7784	0.5598	0.2919	0.4270	0.6011	0.4650 _{±0.0154}
			CNN-T	0.9346	0.8787	0.6432	<u>0.6376</u>	0.8132	<u>0.8775</u>	0.8602	0.7783	<u>0.9189</u>	0.7281	<u>0.6960</u> _{±0.0199}
			Deformer	0.8340	0.7959	0.4658	0.3871	0.7877	0.8011	0.8312	0.2805	0.6768	0.6832	0.5541 _{±0.0219}
			Conformer	<u>0.9342</u>	<u>0.8628</u>	0.7132	0.6815	0.8682	0.8868	0.9219	<u>0.6951</u>	0.9464	0.7931	0.7398 _{±0.0235}
			TSception	0.8545	0.6963	0.4007	0.5500	0.8113	0.7569	0.7719	0.4667	0.8475	<u>0.7721</u>	0.6330 _{±0.0089}
	Full Fine-tuning	Foundation Models	BENDR	0.0721	0.1978	0.0515	0.0000	0.0539	0.0164	0.2014	0.3627	0.0214	0.1117	0.0749 _{±0.0109}
			BIOT-1D	0.8669	0.7513	0.3192	0.2213	<u>0.8404</u>	0.8418	0.7296	0.5669	0.6497	0.7376	0.4876 _{±0.0115}
			BIOT-2D	0.8597	0.7782	0.2247	0.3215	0.8309	0.8278	0.8255	0.5910	0.6809	0.7363	0.4959 _{±0.0127}
			BIOT-6D	0.1265	0.7786	0.7383	0.4936	0.6114	0.5787	0.0453	0.4592	0.1701	0.4384	0.5364 _{±0.0117}
			LaBraM	0.7507	0.6323	0.1511	0.2321	0.5646	0.7605	0.7838	0.3170	0.6145	0.2879	0.4002 _{±0.0052}
			Neuro-GPT	0.6955	0.6011	0.3567	0.2533	0.7063	0.7785	0.7729	0.2913	0.4168	0.6063	0.4389 _{±0.0171}
			EEGPT	0.5574	0.5482	0.2984	0.2618	0.6280	0.7682	0.7124	0.2248	0.2743	0.5410	0.4147 _{±0.0118}
			CBraMod	0.8386	0.8039	0.4414	0.4419	0.7561	0.7821	0.8479	0.5050	0.7750	0.6481	0.5564 _{±0.0010}
			TFM	0.5228	0.4890	0.1809	0.2161	0.4284	0.4311	0.4831	0.2158	0.1745	0.3031	0.2800 _{±0.0063}
			BrainOmni-Tiny	0.5774	0.6744	0.3430	0.3869	0.6405	0.7381	0.7563	0.3500	0.3166	0.4617	0.4378 _{±0.0094}
			BrainOmni-Base	0.7404	0.6854	0.5041	0.3396	0.7903	0.8283	0.8727	0.5282	0.6759	0.7119	0.5655 _{±0.0052}
			EEGMamba	0.7663	0.6851	0.4254	0.3480	0.6494	0.7503	0.8486	0.3644	0.6822	0.6499	0.5242 _{±0.0069}
			SingLEM	0.4961	0.4971	0.0998	0.1222	0.2625	0.1811	0.1911	0.0636	0.0112	0.0461	0.1818 _{±0.0115}
			LUNA-Base	0.8367	0.7316	0.5136	0.2809	0.8393	0.7476	<u>0.9193</u>	0.5665	0.5398	0.7606	0.5574 _{±0.0112}
	Linear Probing	Foundation Models	BENDR	0.0159	0.0079	0.0198	0.0350	0.0103	0.0593	0.0127	0.0000	0.0247	0.0449	0.0405 _{±0.0056}
			BIOT-1D	0.8645	0.8253	0.2393	0.3811	0.8096	0.8190	0.7899	0.5596	0.6426	0.7235	0.5067 _{±0.0067}
			BIOT-2D	0.8462	0.8261	0.3237	0.4608	0.8251	0.8340	0.8542	0.6212	0.7171	0.7155	0.5378 _{±0.0125}
			BIOT-6D	0.1166	0.7075	<u>0.7334</u>	0.5989	0.6161	0.6306	0.0367	0.5919	0.0651	0.3955	0.5583 _{±0.0175}
			LaBraM	0.0989	0.5975	0.4672	0.4859	0.3684	0.5422	0.2299	0.1789	0.7319	0.2997	0.4720 _{±0.0188}
			Neuro-GPT	0.6396	0.5385	0.3419	0.2363	0.6394	0.7598	0.7115	0.2889	0.3351	0.5161	0.4069 _{±0.0237}
			EEGPT	0.6426	0.6249	0.4294	0.2918	0.7176	0.8151	0.8083	0.2904	0.4037	0.5501	0.4627 _{±0.0140}
			CBraMod	0.0959	0.7922	0.6629	0.6080	0.5612	0.6185	0.0892	0.2571	0.3850	0.4353	0.5305 _{±0.0078}
			TFM	0.4418	0.3966	0.0832	0.0926	0.4619	0.3493	0.4840	0.3287	0.2176	0.2696	0.2472 _{±0.0204}
			BrainOmni-Tiny	0.5604	0.7264	0.2806	0.3758	0.5715	0.7862	0.7418	0.3196	0.2518	0.5155	0.4327 _{±0.0133}
			BrainOmni-Base	0.6623	0.6773	0.5152	0.2131	0.7607	0.8351	0.8582	0.4841	0.6636	0.7239	0.5463 _{±0.0028}
			EEGMamba	0.7850	0.7458	0.4320	0.4263	0.6989	0.7836	0.8529	0.4086	0.6992	0.6291	0.5580 _{±0.0052}
			SingLEM	0.0484	0.1077	0.0307	0.0280	0.0543	-0.0106	0.0576	-0.0118	0.0263	0.0379	0.0

Table 52. Overall ranking of EEG FMs and specialist models across 13 datasets under LOSO and few-shot scenarios.

Total Rank	Model	Avg. Rank	Model Size
1	CBraMod (FM)	6.56	4.0M
2	EEGNet	7.40	2K
3	Conformer	7.40	0.16M
4	ShallowConv	7.60	36K
5	Deformer	7.72	0.8M
6	LMDA	7.76	3K
7	Neuro-GPT (FM)	7.84	0.16M
8	TSception	9.24	9K
9	LaBraM (FM)	9.36	5.8M
10	EEGMamba (FM)	9.36	3.3M
11	Traditional ML	9.91	–
12	BENDR (FM)	10.48	4.0M
13	BIOT-6D (FM)	11.44	3.2M
14	CNN-T	11.96	2.8M
15	EEGPT (FM)	12.08	25M
16	BrainOmni-Tiny (FM)	12.33	8.4M
17	BrainOmni-Base (FM)	12.76	33M
18	LUNA-Base (FM)	14.44	7.0M
19	SingLEM (FM)	15.12	3.3M
20	TFM (FM)	16.20	1.9M

Table 53. Overall ranking of evaluated EEG foundation models across 13 datasets under LOSO and few-shot scenarios, grouped by pre-training objective.

Overall Rank	Model	Pre-training Objective	Avg. Rank	Model Size
1	CBraMod	Raw Signals	6.56	4.0M
7	Neuro-GPT	Embedded Tokens	7.84	0.16M
9	LaBraM	EEG Codebook Index	9.36	5.8M
10	EEGMamba	Raw Signals	9.36	3.3M
12	BENDR	Embedded Tokens	10.48	4.0M
13	BIOT-6D	Embedded Tokens	11.44	3.2M
15	EEGPT	Raw Signals	12.08	25M
16	BrainOmni-Tiny	Codebook Index	12.33	8.4M
17	BrainOmni-Base	Codebook Index	12.76	33M
18	LUNA-Base	Embedded Tokens	14.44	7.0M
19	SingLEM	Raw Signals + Frequency	15.12	3.3M
20	TFM	Codebook Index	16.20	1.9M

Table 52 reports the overall ranking, average rank, and parameter size of the evaluated EEG decoding models across the 13 datasets under both the LOSO and within-subject few-shot scenarios. Table 53 further groups the 12 evaluated EEG FMs by their pre-training objectives, and reports the corresponding average ranks and model sizes. We additionally summarized a leakage-aware ranking in Table 54.

Computational Efficiency

Tables 55 and 56 compare the fine-tuning computational cost of all evaluated models under the *flatten* and *pooling* feature aggregation strategies. Each cell reports *Full / Linear*, corresponding to full-parameter fine-tuning and head-only fine-tuning, respectively; specialist models have

Table 54. Leakage-aware overall ranking of EEG FMs and specialist models across 13 datasets under LOSO and few-shot scenarios. For each model, rankings on downstream datasets that overlap with its pretraining data are excluded before computing the average rank.

Total Rank	Model	Avg. Rank	Model Size
1	CBraMod (FM)	6.63	4.0M
2	EEGNet	7.08	2K
3	Conformer	7.08	0.16M
4	LMDA	7.44	3K
5	ShallowConv	7.48	36K
6	Deformer	7.64	0.8M
7	Neuro-GPT (FM)	7.75	0.16M
8	TSception	8.96	9K
9	LaBraM (FM)	9.17	5.8M
10	Traditional ML	9.43	–
11	EEGMamba (FM)	9.58	3.3M
12	BENDR (FM)	10.38	4.0M
13	BIOT-6D (FM)	11.41	3.2M
14	EEGPT (FM)	11.57	25M
15	CNN-T	11.60	2.8M
16	BrainOmni-Tiny (FM)	11.90	8.4M
17	BrainOmni-Base (FM)	12.52	33M
18	SingLEM (FM)	14.60	3.3M
19	LUNA-Base (FM)	14.83	7.0M
20	TFM (FM)	16.00	1.9M

no linear-probing counterpart and are marked as “–”. Here, # Trainable Params denotes the parameters actually updated during optimization, Feature Dim is the dimensionality of the aggregated feature fed into the classification head, and FLOPs (G) measures the floating-point operations of a single forward pass per sample. All measurements were obtained with PyTorch 2.6.0 (CUDA 12.4) on a single NVIDIA A100-SXM4-80GB GPU with a fixed batch size of 32.

Parameter-Efficient Fine-tuning

To assess the feasibility of parameter-efficient fine-tuning (PEFT) for EEG FMs, we conducted a preliminary experiment using Low-Rank Adaptation (LoRA) [114] on two representative FMs, CBraMod and Neuro-GPT, evaluated on the BNCI2014001 and BNCI2014009 datasets under both the LOSO and within-subject few-shot scenarios. The results are reported in Table 57.

Statistical Significance Analysis

For each EEG FM, we conducted a scenario-level paired Wilcoxon signed-rank test between full fine-tuning and linear probing over dataset–scenario pairs. Each valid pair corresponds to the full and linear results of the same model on the same dataset–scenario. A positive Δ indicates that full fine-tuning outperforms linear probing;

for SEED-VIG, where a lower RMSE is better, the sign was reversed accordingly. “Full wins”, “Linear wins”, and “Ties” denote the numbers of dataset–scenario pairs favoring full fine-tuning, linear probing, and neither strategy, respectively. The rank-biserial correlation r is reported as the effect size. The results are summarized in Table 58.

In addition, we assessed the statistical reliability of the overall model rankings using the Friedman test followed by the Nemenyi post-hoc test. The 25 dataset–scenario combinations were used as evaluation blocks. To satisfy the complete-block assumption of the Friedman/Nemenyi procedure, models with incomplete results, including Traditional ML, BrainOmni-Tiny, and BrainOmni-Base, were excluded, and ranks were recomputed within each scenario for the remaining 17 models. The Friedman test rejected the null hypothesis that all models have equivalent ranks ($\chi^2_F = 98.19$, $p = 7.56 \times 10^{-14}$), and the Iman-Davenport corrected statistic was also significant ($F_F = 7.81$, $p = 4.34 \times 10^{-16}$). The Nemenyi post-hoc test yielded a critical difference of 4.94 at $\alpha = 0.05$. The resulting critical difference diagram is shown in Fig. 8, where models connected by a horizontal bar are not significantly different at the 0.05 level.

Comparison of different fine-tuning ratios

We conducted an analysis on different fine-tuning data ratios, varying from 10% to 90% in increments of 20%, across all specialist and FMs. The results are presented in Figs. 9 - 14.

Most models exhibited consistent performance improvements as the fine-tuning data ratio increased from 10% to 90%, which aligns with intuitive expectations. Notably, the relative ranking among models remained largely stable across different fine-tuning ratios. For instance, TRCA consistently achieved the highest accuracy on the Nakanishi2015 dataset regardless of the data ratio, while EEGNet maintained competitive performance across all settings on the BNCI2014008 dataset. This observation suggests that model superiority is relatively independent of fine-tuning data availability, and that a well-performing model under low-data conditions tends to preserve its advantage as more data becomes available. However, minimal calibration or even calibration-free adaptation remains a critical requirement for practical BCI deployment. Developing models capable of rapid adaptation to downstream tasks with limited calibration data continues to be an important and open challenge.

A5. Representation Visualization

Figs. 15–17 provide the complete t -SNE visualizations of the learned representations on the SEED, TUAB, and BNCI2014008 datasets, respectively. For each dataset, we visualize seven specialist models and nine EEG FMs; for every FM, both full-parameter fine-tuning (full) and linear probing (linear) are shown.

A6. EXISTING SURVEYS AND BENCHMARKS

Table 59 compares our work with representative surveys, reviews, and benchmarks on EEG and brain foundation models published over the past two years. The comparison covers six aspects: the type of contribution, the number of EEG FMs included, the scope of benchmark experiments, the evaluation protocols, whether specialist baselines are included, and the main distinction of each work. Our work differs from prior efforts in two main respects. First, it surveys 55 EEG FMs, substantially more than any existing review or survey. Second, it provides a unified empirical benchmark of 12 open-source EEG FMs against 7 specialist baselines on 13 datasets covering 9 BCI paradigms, evaluated under both the LOSO and within-subject few-shot scenarios with both full-parameter fine-tuning and linear probing. To our knowledge, this is the largest EEG FM survey and the most comprehensive benchmark to date that systematically contrasts FMs with specialist baselines under unified evaluation protocols.

A7. SYMBOLS AND NOTATION

Tables 60 and 61 summarize the principal mathematical symbols and loss functions used throughout this paper. Model-specific auxiliary losses that appear only in individual models are not exhaustively enumerated here; the complete list is provided in Table 5.

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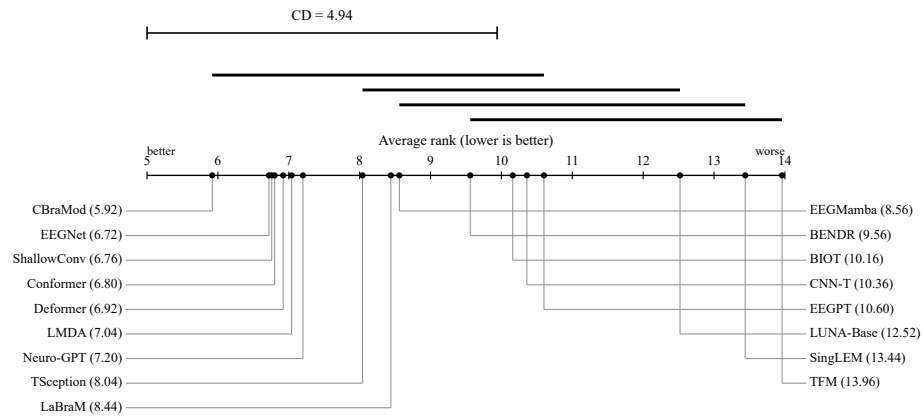


Figure 8. Critical difference (CD) diagram of the average ranks of all evaluated models across the 25 dataset–scenario combinations, based on the Nemenyi post-hoc test. A lower rank indicates better performance, and models connected by a horizontal bar are not significantly different at the 0.05 level.

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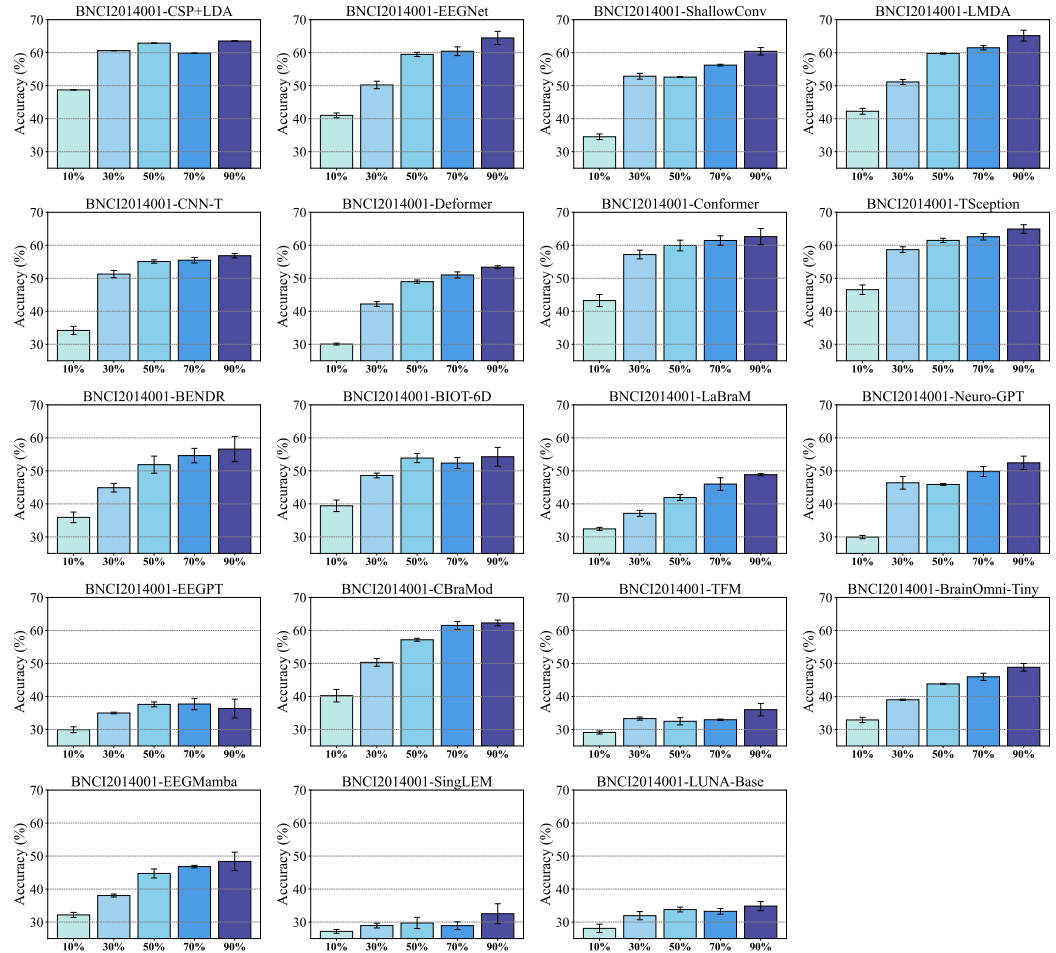


Figure 9. Accuracy comparison on the BNCI2014001 dataset across different fine-tuning ratios.

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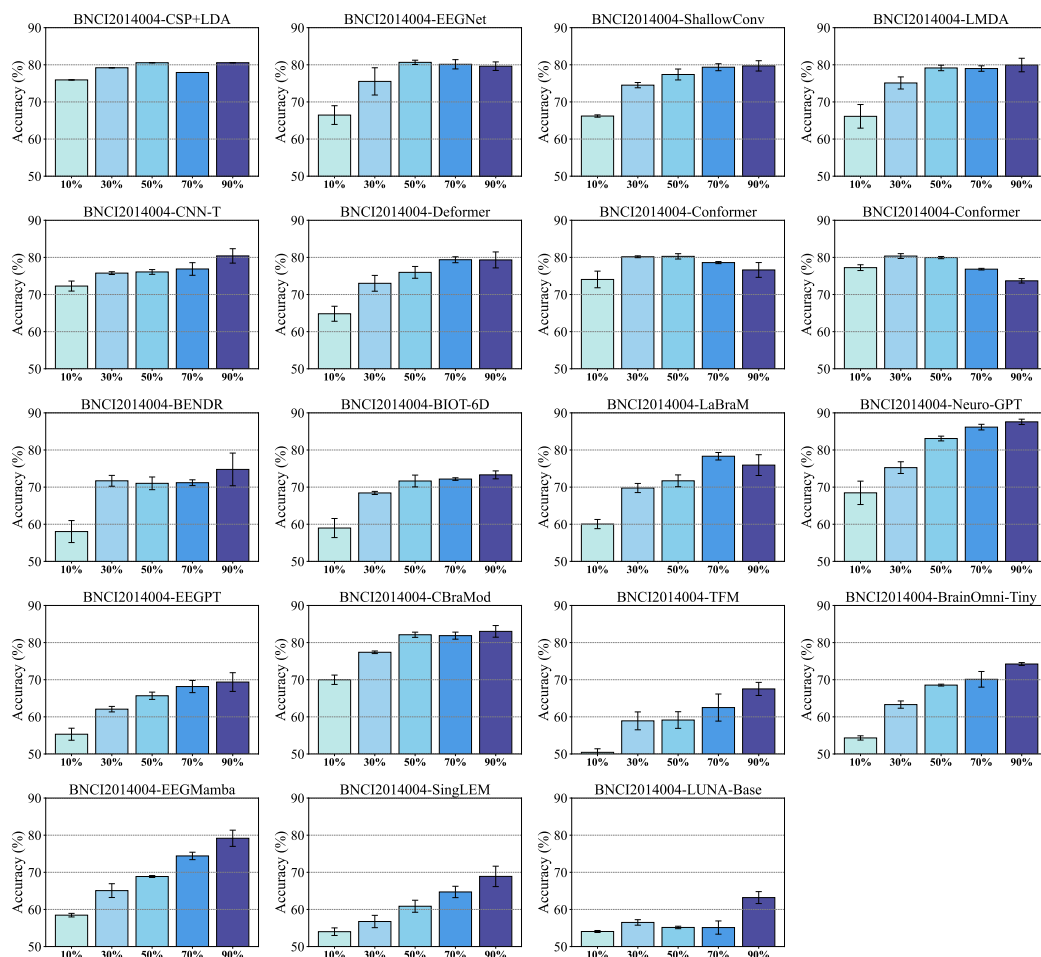


Figure 10. Accuracy comparison on the BNCI2014004 dataset across different fine-tuning ratios.

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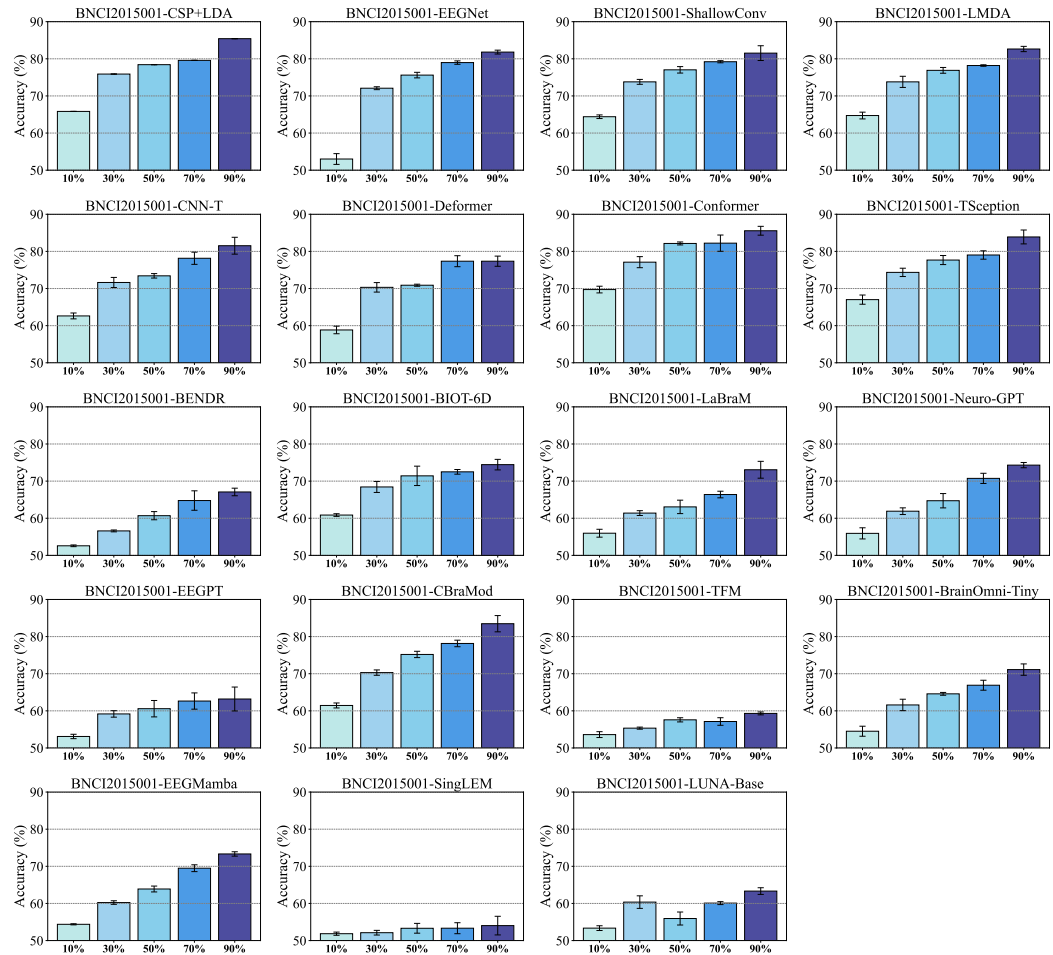


Figure 11. Accuracy comparison on the BNCI2015001 dataset across different fine-tuning ratios.

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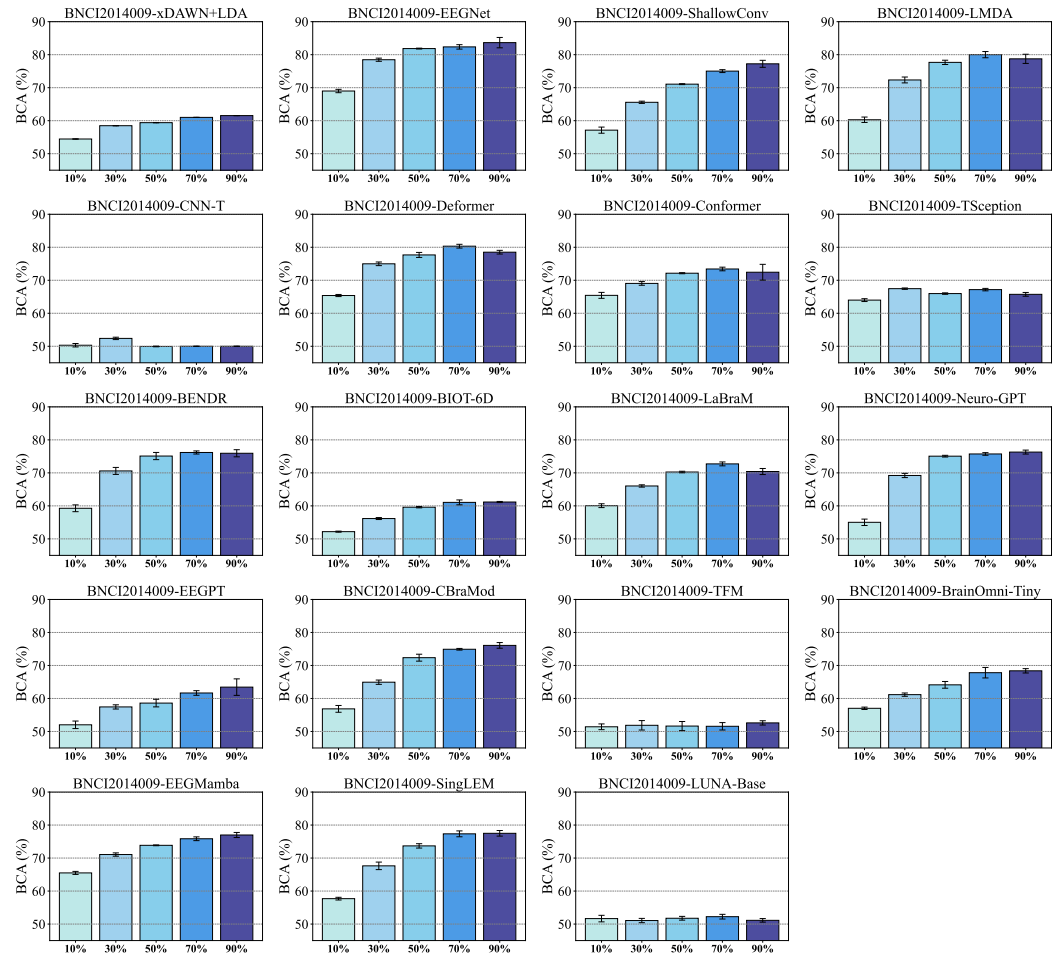


Figure 12. Balanced classification accuracy comparison on the BNCI2014009 dataset across different fine-tuning ratios.

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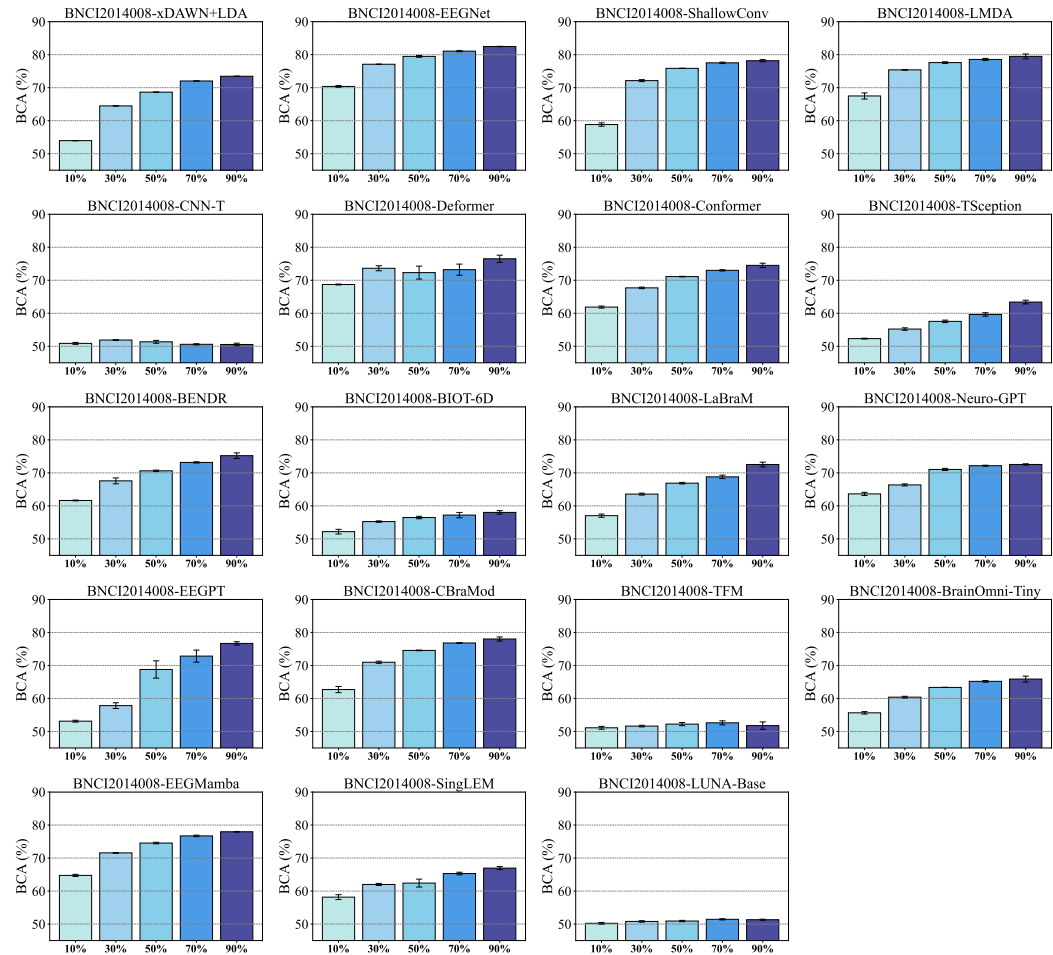


Figure 13. Balanced classification accuracy comparison on the BNCI2014008 dataset across different fine-tuning ratios.

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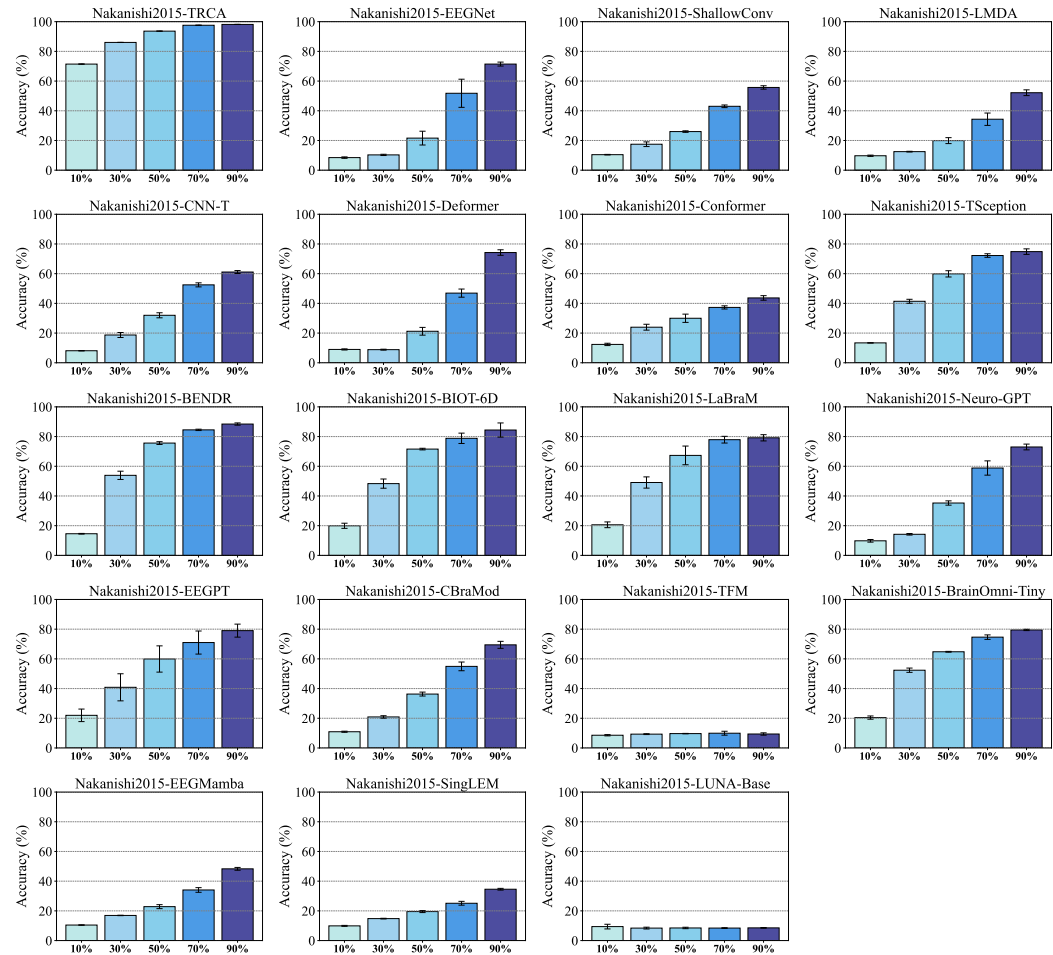


Figure 14. Accuracy comparison on the Nakanishi2015 dataset across different fine-tuning ratios.

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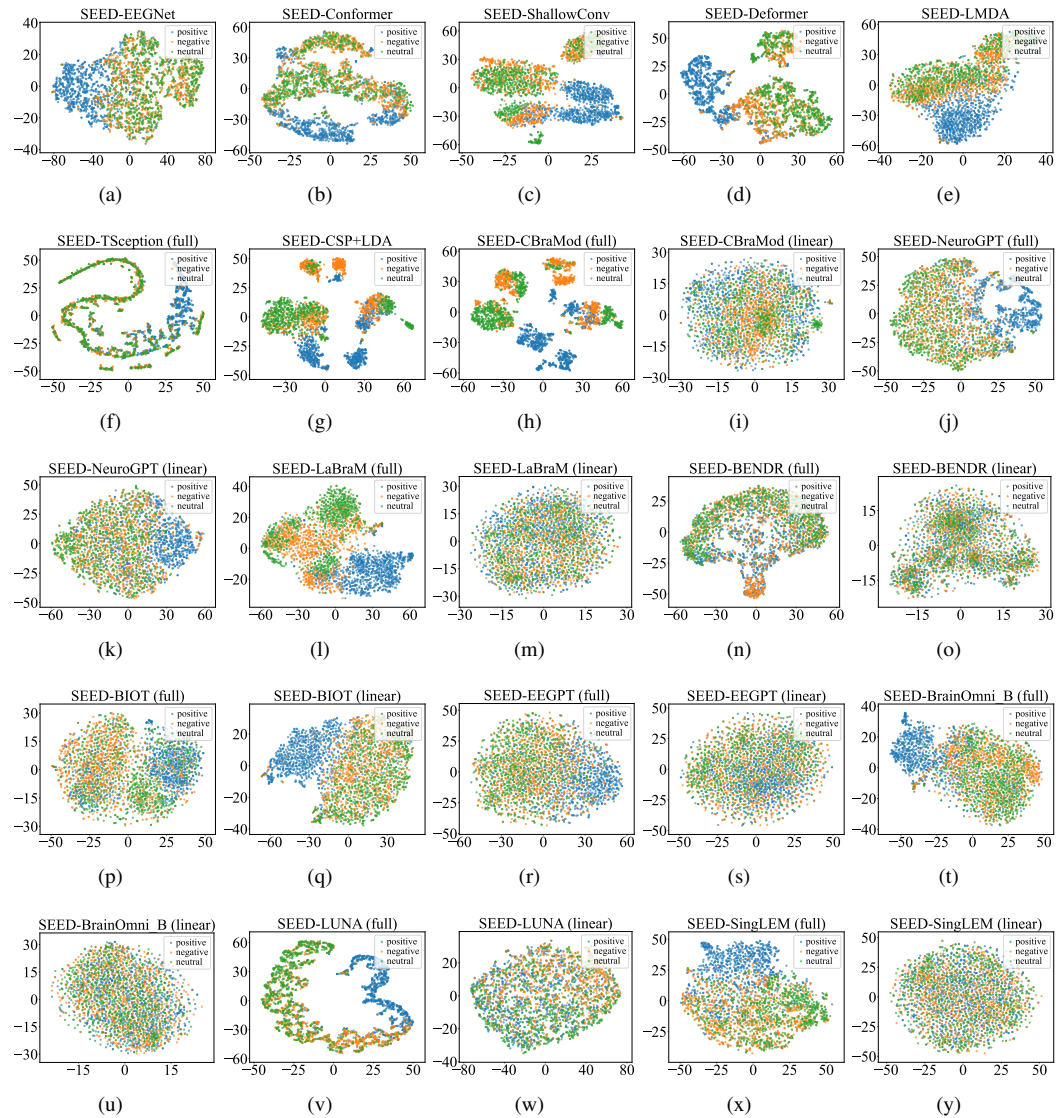


Figure 15. *t*-SNE visualization of the SEED datasets.

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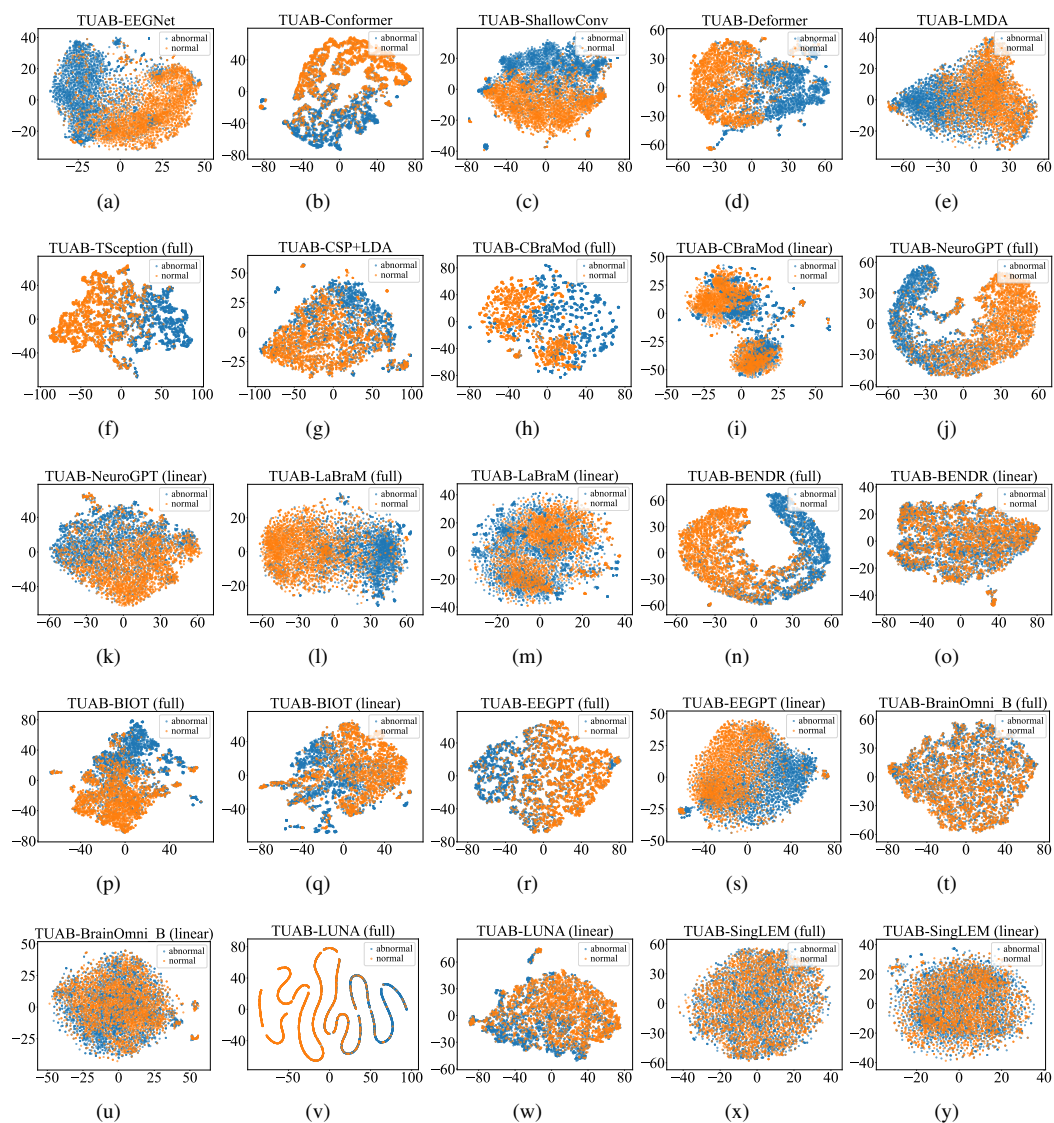


Figure 16. *t*-SNE visualization of the TUAB datasets.

Table 55. Fine-tuning computational cost under the flatten feature aggregation strategy. Values are reported as *Full* / *Linear*, where *Full* denotes full-parameter fine-tuning and *Linear* denotes head-only fine-tuning. Specialist models do not have a linear-probing counterpart, and unavailable entries are denoted by “–”.

Channel	Duration	Category	Model	Time Pts	# Backbone Params	# MLP Params	# Trainable Params	Feature Dim	FLOPs (G)	Peak Mem. (MB)	FT Time (ms / batch)	Latency (ms / sample)	Throughput (sample / s)
		Specialist Models	EEGNet	250 / -	1.3K / -	452 / -	1.7K / -	- / -	0.09 / -	25.0 / -	2.2 / -	0.016 / -	61,566 / -
			ShallowConv	250 / -	17.2K / -	1.8K / -	18.9K / -	- / -	0.38 / -	59.1 / -	1.7 / -	0.011 / -	94,679 / -
			LMDA	250 / -	2.5K / -	256 / -	2.8K / -	- / -	0.26 / -	67.1 / -	3.0 / -	0.027 / -	36,585 / -
			CNN-T	250 / -	2.75M / -	1.0K / -	2.76M / -	- / -	1.50 / -	88.2 / -	25.8 / -	0.297 / -	3,370 / -
			Deformer	250 / -	352.9K / -	2.8K / -	355.7K / -	- / -	1.81 / -	153.1 / -	13.2 / -	0.120 / -	8,299 / -
			Conformer	250 / -	137.1K / -	244 / -	137.4K / -	- / -	0.46 / -	60.9 / -	17.7 / -	0.140 / -	7,119 / -
			TSception	128 / -	5.5K / -	644 / -	6.1K / -	- / -	0.09 / -	29.2 / -	3.7 / -	0.032 / -	30,801 / -
10	1s	Foundation Models	BENDR	256 / 256	3.97M / 3.97M	6.1K / 6.1K	3.98M / 6.4K	1,536 / 1,536	0.0004 / 0.0004	113.3 / 68.4	4.5 / 3.1	0.037 / 0.036	27,335 / 27,631
			BIOT-6D	200 / 200	3.19M / 3.19M	18.4K / 18.4K	3.21M / 18.6K	4,608 / 4,608	3.73 / 3.73	102.8 / 64.0	29.5 / 24.9	0.272 / 0.266	3,676 / 3,756
			LaBraM	200 / 200	5.82M / 5.82M	8.8K / 8.8K	5.83M / 8.8K	2,200 / 2,200	4.10 / 4.10	159.4 / 42.7	33.7 / 11.4	0.280 / 0.281	3,565 / 3,557
			Neuro-GPT	250 / 250	156.6K / 156.6K	1.8K / 1.8K	158.3K / 2.0K	440 / 440	0.91 / 0.91	106.1 / 93.6	18.6 / 14.0	0.153 / 0.149	6,545 / 6,701
			EEGPT	256 / 256	25.29M / 25.29M	32.8K / 32.8K	25.32M / 32.9K	8,192 / 8,192	90.30 / 90.30	793.4 / 428.9	24.2 / 19.7	0.240 / 0.240	4,174 / 4,158
			CBraMod	200 / 200	4.88M / 4.88M	8.0K / 8.0K	4.89M / 8.0K	2,000 / 2,000	3.15 / 3.15	135.7 / 38.5	53.2 / 16.6	0.287 / 0.288	3,479 / 3,474
			TFM	200 / 200	1.89M / 1.89M	4.4K / 4.4K	1.90M / 1.17M	1,088 / 1,088	2.92 / 2.92	97.7 / 92.2	23.1 / 11.2	0.296 / 0.295	3,383 / 3,388
			BrainOmni-Tiny	256 / 256	13.48M / 13.48M	131.1K / 131.1K	13.61M / 131.1K	32,768 / 32,768	70.38 / 70.38	1126.8 / 201.4	50.7 / 19.4	0.571 / 0.572	1,752 / 1,748
			BrainOmni-Base	256 / 256	37.76M / 37.76M	262.1K / 262.1K	38.02M / 262.1K	65,536 / 65,536	264.80 / 264.80	2394.1 / 297.7	79.0 / 30.7	0.882 / 0.883	1,133 / 1,133
			SingLEM	128 / 128	3.27M / 3.27M	644 / 644	3.27M / 644	160 / 160	4.60 / 4.60	226.0 / 54.3	27.1 / 7.6	0.208 / 0.210	4,810 / 4,767
LUNA-Base	256 / 256	7.36M / 7.36M	1.0K / 1.0K	7.36M / 1.0K	256 / 256	3.35 / 3.35	178.2 / 58.6	42.2 / 16.0	0.462 / 0.451	2,163 / 2,219			
		Specialist Models	EEGNet	1,000 / -	1.3K / -	2.0K / -	3.3K / -	- / -	0.35 / -	52.5 / -	2.8 / -	0.017 / -	59,844 / -
			ShallowConv	1,000 / -	17.2K / -	9.8K / -	26.9K / -	- / -	1.62 / -	176.7 / -	2.6 / -	0.012 / -	86,717 / -
			LMDA	1,000 / -	2.5K / -	1.3K / -	3.9K / -	- / -	1.35 / -	267.8 / -	7.8 / -	0.047 / -	21,250 / -
			CNN-T	1,000 / -	2.75M / -	1.0K / -	2.76M / -	- / -	4.19 / -	153.4 / -	25.9 / -	0.298 / -	3,358 / -
			Deformer	1,000 / -	751.1K / -	9.0K / -	760.0K / -	- / -	5.64 / -	292.8 / -	14.9 / -	0.122 / -	8,175 / -
			Conformer	1,000 / -	137.1K / -	244 / -	137.4K / -	- / -	2.19 / -	183.7 / -	17.9 / -	0.150 / -	6,671 / -
			TSception	512 / -	5.5K / -	644 / -	6.1K / -	- / -	0.50 / -	80.9 / -	3.9 / -	0.037 / -	27,203 / -
4s		Foundation Models	BENDR	1,024 / 1,024	3.97M / 3.97M	22.5K / 22.5K	3.99M / 22.7K	5,632 / 5,632	0.0014 / 0.0014	224.0 / 169.0	4.4 / 3.0	0.037 / 0.036	27,016 / 27,644
			BIOT-6D	800 / 800	3.19M / 3.19M	129.0K / 129.0K	3.32M / 129.2K	32,256 / 32,256	26.13 / 26.13	373.2 / 244.4	31.1 / 25.9	0.275 / 0.275	3,632 / 3,630
			LaBraM	800 / 800	5.82M / 5.82M	32.8K / 32.8K	5.86M / 32.8K	8,200 / 8,200	15.67 / 15.67	393.6 / 55.6	34.5 / 11.1	0.275 / 0.278	3,639 / 3,591
			Neuro-GPT	1,000 / 1,000	156.6K / 156.6K	9.8K / 9.8K	166.3K / 10.0K	2,440 / 2,440	4.14 / 4.14	358.6 / 244.9	23.9 / 21.2	0.150 / 0.146	6,671 / 6,831
			EEGPT	1,024 / 1,024	25.29M / 25.29M	131.1K / 131.1K	25.42M / 131.2K	32,768 / 32,768	361.18 / 361.18	2212.0 / 1371.1	79.4 / 54.2	0.769 / 0.771	1,300 / 1,296
			CBraMod	800 / 800	4.88M / 4.88M	32.0K / 32.0K	4.92M / 32.0K	8,000 / 8,000	12.60 / 12.60	337.8 / 49.6	55.4 / 17.0	0.295 / 0.289	3,395 / 3,460
			TFM	800 / 800	1.89M / 1.89M	28.9K / 28.9K	1.92M / 1.20M	7,232 / 7,232	20.38 / 20.38	492.9 / 488.6	24.0 / 11.5	0.305 / 0.302	3,274 / 3,308
			BrainOmni-Tiny	1,024 / 1,024	13.48M / 13.48M	393.2K / 393.2K	13.87M / 393.2K	98,304 / 98,304	210.44 / 210.44	3094.5 / 452.1	74.9 / 32.7	0.923 / 0.917	1,084 / 1,091
			BrainOmni-Base	1,024 / 1,024	37.76M / 37.76M	786.4K / 786.4K	38.54M / 786.4K	196,608 / 196,608	793.67 / 793.67	6335.2 / 548.8	204.0 / 70.0	2.136 / 2.137	468 / 468
			SingLEM	512 / 512	3.27M / 3.27M	2.6K / 2.6K	3.27M / 2.6K	640 / 640	18.38 / 18.38	744.7 / 130.3	34.0 / 11.2	0.339 / 0.339	2,949 / 2,950
LUNA-Base	1,024 / 1,024	7.36M / 7.36M	1.0K / 1.0K	7.36M / 1.0K	256 / 256	12.43 / 12.43	360.8 / 77.1	42.6 / 16.7	0.459 / 0.458	2,178 / 2,182			
		Specialist Models	EEGNet	250 / -	1.6K / -	452 / -	2.1K / -	- / -	0.27 / -	42.8 / -	2.6 / -	0.016 / -	61,418 / -
			ShallowConv	250 / -	52.4K / -	1.8K / -	54.1K / -	- / -	1.20 / -	149.4 / -	2.5 / -	0.011 / -	94,754 / -
			LMDA	250 / -	2.9K / -	256 / -	3.2K / -	- / -	0.84 / -	175.1 / -	5.9 / -	0.033 / -	30,011 / -
			CNN-T	250 / -	2.77M / -	1.0K / -	2.77M / -	- / -	2.07 / -	99.3 / -	32.5 / -	0.501 / -	1,994 / -
			Deformer	250 / -	443.0K / -	2.8K / -	445.8K / -	- / -	3.50 / -	251.5 / -	14.0 / -	0.125 / -	8,008 / -
			Conformer	250 / -	172.3K / -	244 / -	172.6K / -	- / -	1.29 / -	151.2 / -	17.7 / -	0.150 / -	6,645 / -
			TSception	128 / -	12.0K / -	644 / -	12.6K / -	- / -	0.27 / -	55.7 / -	3.8 / -	0.036 / -	27,850 / -
32	1s	Foundation Models	BENDR	256 / 256	3.97M / 3.97M	6.1K / 6.1K	3.98M / 6.8K	1,536 / 1,536	0.0004 / 0.0004	114.9 / 69.1	4.5 / 3.0	0.037 / 0.036	27,123 / 27,639
			BIOT-6D	200 / 200	3.19M / 3.19M	18.4K / 18.4K	3.21M / 19.0K	4,608 / 4,608	3.73 / 3.73	103.3 / 64.5	29.4 / 24.8	0.268 / 0.268	3,731 / 3,734
			LaBraM	200 / 200	5.82M / 5.82M	26.4K / 26.4K	5.85M / 26.4K	6,600 / 6,600	12.53 / 12.53	324.9 / 50.8	34.5 / 11.1	0.277 / 0.279	3,610 / 3,584
			Neuro-GPT	250 / 250	157.0K / 157.0K	1.8K / 1.8K	158.8K / 2.5K	440 / 440	0.91 / 0.91	106.8 / 94.3	18.1 / 14.0	0.153 / 0.146	6,555 / 6,848
			EEGPT	256 / 256	25.29M / 25.29M	32.8K / 32.8K	25.32M / 33.8K	8,192 / 8,192	232.24 / 232.24	1544.1 / 922.2	51.7 / 34.6	0.498 / 0.498	2,009 / 2,010
			CBraMod	200 / 200	4.88M / 4.88M	25.6K / 25.6K	4.91M / 25.6K	6,400 / 6,400	10.18 / 10.18	292.6 / 46.2	52.7 / 16.4	0.288 / 0.287	3,467 / 3,485
			TFM	200 / 200	1.89M / 1.89M	4.4K / 4.4K	1.90M / 1.17M	1,088 / 1,088	2.92 / 2.92	98.3 / 92.7	23.3 / 11.2	0.298 / 0.294	3,354 / 3,396
			BrainOmni-Tiny	256 / 256	13.48M / 13.48M	131.1K / 131.1K	13.61M / 131.1K	32,768 / 32,768	72.68 / 72.68	1127.7 / 469.4	54.1 / 23.2	0.648 / 0.670	1,542 / 1,492
			BrainOmni-Base	256 / 256	37.76M / 37.76M	262.1K / 262.1K	38.02M / 262.1K	65,536 / 65,536	267.09 / 267.09	2394.8 / 564.9	81.2 / 37.0	0.989 / 0.990	1,011 / 1,010
			SingLEM	128 / 128	3.27M / 3.27M	2.1K / 2.1K	3.27M / 2.1K	512 / 512	14.71 / 14.71	606.7 / 110.0	28.6 / 9.2	0.277 / 0.276	3,616 / 3,617
LUNA-Base	256 / 256	7.36M / 7.36M	1.0K / 1.0K	7.36M / 1.0K	256 / 256	3.71 / 3.71	206.4 / 69.7	42.9 / 16.6	0.458 / 0.449	2,182 / 2,229			
		Specialist Models	EEGNet	1,000 / -	1.6K / -	2.0K / -	3.6K / -	- / -	1.09 / -	122.9 / -	6.1 / -	0.024 / -	42,424 / -
			ShallowConv	1,000 / -	52.4K / -	9.8K / -	62.1K / -	- / -	5.20 / -	512.2 / -	4.5 / -	0.031 / -	32,458 / -
			LMDA	1,000 / -	2.9K / -	1.3K / -	4.3K / -	- / -	4.30 / -	776.6 / -	25.4 / -	0.118 / -	8,503 / -
			CNN-T	1,000 / -	2.77M / -	1.0K / -	2.77M / -	- / -	6.34 / -	225.0 / -	33.0 / -	0.510 / -	1,962 / -
			Deformer	1,000 / -	841.2K / -	9.0K / -	850.1K / -	- / -	12.40 / -	835.6 / -	16.6 / -	0.128 / -	7,815 / -
			Conformer	1,000 / -	172.3K / -	244 / -	172.6K / -	- / -	5.77 / -	514.0 / -	19.5 / -	0.146 / -	6,859 / -
			TSception	512 / -	12.0K / -	644 / -	12.6K / -	- / -	1.55 / -	223.9 / -	5.0 / -	0.034 / -	29,742 / -
4s		Foundation Models	BENDR	1,024 / 1,024	3.97M / 3.97M	22.5K / 22.5K	4.00M / 23.2K	5,632 / 5,632	0.0014 / 0.0014	226.5 / 172.4	4.4 / 3.1	0.037 / 0.037	27,298 / 27,364
			BIOT-6D	800 / 800	3.19M / 3.19M	129.0K / 129.0K	3.32M / 129.6K	32,256 / 32,256	26.13 / 26.13	375.3 / 245.4	31.0 / 26.0	0.278 / 0.279	3,603 / 3,583
			LaBraM	800 / 800	5.82M / 5.82M	103.2K / 103.2K	5.93M / 103.2K	25,800 / 25,800	52.81 / 52.81	1183.1 / 111.8	42.9 / 15.0	0.364 / 0.364	2,750 / 2,750
			Neuro-GPT	1,000 / 1,000	157.0K / 157.0K	9.8K / 9.8K	166.8K / 10.5K	2,440 / 2,440	4.14 / 4.14	361.4 / 247.8	24.9 / 19.7	0.148 / 0.153	6,761 / 6,520
			EEGPT	1,024 / 1,024	25.29M / 25.29M	131.1K / 131.1K	25.42M / 132.1K	32,768 / 32,768	928.95 / 928.95	5195.3 / 3339.4	188.1 / 126.5	1.765 / 1.765	566 / 567
			CBraMod	800 / 800	4.88M / 4.88M	102.4K / 102.4K	4.99M / 102.4K	25,600 / 25,600	40.77 / 40.77	932.8 / 79.9	56.0 / 17.6	0.295 / 0.293	3,385 / 3,414
			TFM	800 / 800	1.89M / 1.89M	28.9K / 28.9K	1.92M / 1.20M	7,232 / 7,232	20.38 / 20.38	495.9 / 490.4	24.0 / 11.5	0.302 / 0.302	3,307 / 3,306
			BrainOmni-Tiny	1,024 / 1,024	13.48M / 13.48M	393.2K / 393.2K	13.87M / 393.2K	98,304 / 98,304	215.75 / 215.75	3097.8 / 1255.8	83.9 / 41.2	1.225 / 1.225	816 / 816
			BrainOmni-Base	1,024 / 1,024	37.76M / 37.76M	786.4K / 786.4K	38.54M / 786.4K	196,608 / 196,608	798.99 / 798.99	6336.6 / 1353.6	213.7 / 79.8	2.445 / 2.447	409 / 409
			SingLEM	512 / 512	3.27M / 3.27M	8.2K / 8.2K	3.28M / 8.2K	2,048 / 2,048	58.83 / 58.83	2227.0 / 353.3	46.7 / 32.3	0.992 / 0.993	1,008 / 1,007
LUNA-Base	1,024 / 1,024	7.36M / 7.36M	1.0K / 1.0K	7.36M / 1.0K	256 / 256	10.70 / 13.70	40.6 / 16.8	118.1	42.6 / 16.8	0.460 / 0.476	2,172 / 2,099		

Table 56. Fine-tuning computational cost under the pooling feature aggregation strategy. Values are reported as *Full* / *Linear*, where *Full* denotes full-parameter fine-tuning and *Linear* denotes head-only fine-tuning. Specialist models do not have a linear-probing counterpart, and unavailable entries are denoted by “–”.

Channel	Duration	Category	Model	Time Pts	# Backbone Params	# MLP Params	# Trainable Params	Feature Dim	FLOPs (G)	Peak Mem. (MB)	FT Time (ms / batch)	Latency (ms / sample)	Throughput (sample / s)
	Specialist Models	EEGNet	250 / -	1.3K / -	452 / -	1.7K / -	- / -	0.09 / -	25.0 / -	2.2 / -	0.016 / -	61,566 / -	
		ShallowConv	250 / -	17.2K / -	1.8K / -	18.9K / -	- / -	0.38 / -	59.1 / -	1.7 / -	0.011 / -	94,679 / -	
		LMDA	250 / -	2.5K / -	256 / -	2.8K / -	- / -	0.26 / -	67.1 / -	3.0 / -	0.027 / -	36,585 / -	
		CNN-T	250 / -	2.75M / -	1.0K / -	2.76M / -	- / -	1.50 / -	88.2 / -	25.8 / -	0.297 / -	3,370 / -	
		Deformer	250 / -	352.9K / -	2.8K / -	355.7K / -	- / -	1.81 / -	153.1 / -	13.2 / -	0.120 / -	8,299 / -	
		Conformer	250 / -	137.1K / -	244 / -	137.4K / -	- / -	0.46 / -	60.9 / -	17.7 / -	0.140 / -	7,119 / -	
		TSception	128 / -	5.5K / -	644 / -	6.1K / -	- / -	0.09 / -	29.2 / -	3.7 / -	0.032 / -	30,801 / -	
	Foundation Models	BENDR	256 / 256	3.97M / 3.97M	2.1K / 2.1K	3.97M / 2.3K	512 / 512	0.0001 / 0.0001	114.2 / 68.4	4.6 / 3.1	0.037 / 0.037	26,909 / 27,120	
		BIOT-6D	200 / 200	3.19M / 3.19M	1.0K / 1.0K	3.19M / 1.2K	256 / 256	3.73 / 3.73	102.5 / 63.7	29.5 / 25.4	0.265 / 0.273	3,773 / 3,659	
		LaBraM	200 / 200	5.82M / 5.82M	804 / 804	5.83M / 804	200 / 200	4.10 / 4.10	159.2 / 42.6	34.6 / 11.5	0.287 / 0.287	3,490 / 3,488	
Neuro-GPT		250 / 250	156.6K / 156.6K	164 / 164	156.7K / 406	40 / 40	0.91 / 0.91	106.1 / 93.6	18.4 / 14.3	0.155 / 0.152	6,463 / 6,559		
EEGPT		256 / 256	25.29M / 25.29M	2.1K / 2.1K	25.29M / 2.2K	512 / 512	90.29 / 90.29	792.9 / 428.5	26.6 / 16.3	0.240 / 0.240	4,175 / 4,166		
CBraMod		200 / 200	4.88M / 4.88M	804 / 804	4.88M / 804	200 / 200	3.15 / 3.15	135.6 / 38.4	52.7 / 16.5	0.288 / 0.289	3,477 / 3,463		
TFM		200 / 200	1.89M / 1.89M	260 / 260	1.89M / 1.17M	64 / 64	2.92 / 2.92	97.7 / 92.1	23.2 / 11.4	0.296 / 0.301	3,379 / 3,323		
10	Foundation Models	BrainOmni-Tiny	256 / 256	13.48M / 13.48M	1.0K / 1.0K	13.48M / 1.0K	256 / 256	70.38 / 70.38	1124.8 / 199.9	51.1 / 19.5	0.575 / 0.573	1,739 / 1,747	
		BrainOmni-Base	256 / 256	37.76M / 37.76M	2.1K / 2.1K	37.76M / 2.1K	512 / 512	264.78 / 264.78	2390.1 / 294.7	79.2 / 32.9	0.885 / 0.883	1,130 / 1,133	
		SingLEM	128 / 128	3.27M / 3.27M	68 / 68	3.27M / 68	16 / 16	4.60 / 4.60	226.0 / 54.3	27.0 / 7.5	0.211 / 0.206	4,745 / 4,850	
		LUNA-Base	256 / 256	7.36M / 7.36M	1.0K / 1.0K	7.36M / 1.0K	256 / 256	3.35 / 3.35	178.2 / 58.6	43.1 / 16.6	0.469 / 0.465	2,132 / 2,149	
		Specialist Models	EEGNet	1,000 / -	1.3K / -	2.0K / -	3.3K / -	- / -	0.35 / -	52.5 / -	2.8 / -	0.017 / -	59,844 / -
			ShallowConv	1,000 / -	17.2K / -	9.8K / -	26.9K / -	- / -	1.62 / -	176.7 / -	2.6 / -	0.012 / -	86,717 / -
			LMDA	1,000 / -	2.5K / -	1.3K / -	3.9K / -	- / -	1.35 / -	267.8 / -	7.8 / -	0.047 / -	21,250 / -
	Foundation Models	CNN-T	1,000 / -	2.75M / -	1.0K / -	2.76M / -	- / -	4.19 / -	153.4 / -	25.9 / -	0.298 / -	3,358 / -	
		Deformer	1,000 / -	751.1K / -	9.0K / -	760.0K / -	- / -	5.64 / -	292.8 / -	14.9 / -	0.122 / -	8,175 / -	
		Conformer	1,000 / -	137.1K / -	244 / -	137.4K / -	- / -	2.19 / -	183.7 / -	17.9 / -	0.150 / -	6,671 / -	
TSception		512 / -	5.5K / -	644 / -	6.1K / -	- / -	0.50 / -	80.9 / -	3.9 / -	0.037 / -	27,203 / -		
4s		Foundation Models	BENDR	1,024 / 1,024	3.97M / 3.97M	2.1K / 2.1K	3.97M / 2.3K	512 / 512	0.0001 / 0.0001	223.7 / 168.7	4.5 / 3.1	0.037 / 0.037	27,030 / 26,894
			BIOT-6D	800 / 800	3.19M / 3.19M	1.0K / 1.0K	3.19M / 1.2K	256 / 256	26.12 / 26.12	371.3 / 242.1	30.8 / 26.0	0.274 / 0.277	3,654 / 3,613
			LaBraM	800 / 800	5.82M / 5.82M	804 / 804	5.83M / 804	200 / 200	15.67 / 15.67	393.1 / 55.2	35.4 / 11.3	0.284 / 0.284	3,522 / 3,517
	Neuro-GPT		1,000 / 1,000	156.6K / 156.6K	164 / 164	156.7K / 406	40 / 40	4.14 / 4.14	358.4 / 244.7	23.8 / 19.1	0.155 / 0.145	6,445 / 6,889	
	EEGPT		1,024 / 1,024	25.29M / 25.29M	2.1K / 2.1K	25.29M / 2.2K	512 / 512	361.17 / 361.17	2210.1 / 1369.1	79.3 / 54.1	0.769 / 0.770	1,300 / 1,299	
	CBraMod		800 / 800	4.88M / 4.88M	804 / 804	4.88M / 804	200 / 200	12.60 / 12.60	337.3 / 49.2	56.3 / 17.1	0.299 / 0.290	3,347 / 3,448	
	TFM		800 / 800	1.89M / 1.89M	260 / 260	1.89M / 1.17M	64 / 64	20.38 / 20.38	492.6 / 488.3	24.1 / 11.8	0.304 / 0.310	3,294 / 3,222	
	BrainOmni-Tiny	1,024 / 1,024	13.48M / 13.48M	1.0K / 1.0K	13.48M / 1.0K	256 / 256	210.41 / 210.41	3087.8 / 444.8	73.8 / 32.6	0.913 / 0.913	1,096 / 1,096		
	BrainOmni-Base	1,024 / 1,024	37.76M / 37.76M	2.1K / 2.1K	37.76M / 2.1K	512 / 512	793.62 / 793.62	6323.0 / 539.9	203.7 / 69.6	2.135 / 2.013	468 / 468		
	SingLEM	512 / 512	3.27M / 3.27M	68 / 68	3.27M / 68	16 / 16	18.38 / 18.38	744.6 / 130.2	34.0 / 11.1	0.338 / 0.338	2,956 / 2,960		
LUNA-Base	1,024 / 1,024	7.36M / 7.36M	1.0K / 1.0K	7.36M / 1.0K	256 / 256	12.43 / 12.43	360.8 / 77.1	43.2 / 16.8	0.468 / 0.466	2,136 / 2,145			
	Specialist Models	EEGNet	250 / -	1.6K / -	452 / -	2.1K / -	- / -	0.27 / -	42.8 / -	2.6 / -	0.016 / -	61,418 / -	
		ShallowConv	250 / -	52.4K / -	1.8K / -	54.1K / -	- / -	1.20 / -	149.4 / -	2.5 / -	0.011 / -	94,754 / -	
		LMDA	250 / -	2.9K / -	256 / -	3.2K / -	- / -	0.84 / -	175.1 / -	5.9 / -	0.033 / -	30,011 / -	
		CNN-T	250 / -	2.77M / -	1.0K / -	2.77M / -	- / -	2.07 / -	99.3 / -	32.5 / -	0.501 / -	1,994 / -	
		Deformer	250 / -	443.0K / -	2.8K / -	445.8K / -	- / -	3.50 / -	251.5 / -	14.0 / -	0.125 / -	8,008 / -	
		Conformer	250 / -	172.3K / -	244 / -	172.6K / -	- / -	1.29 / -	151.2 / -	17.7 / -	0.150 / -	6,645 / -	
		TSception	128 / -	12.0K / -	644 / -	12.6K / -	- / -	0.27 / -	55.7 / -	3.8 / -	0.036 / -	27,850 / -	
	Foundation Models	BENDR	256 / 256	3.97M / 3.97M	2.1K / 2.1K	3.97M / 2.7K	512 / 512	0.0001 / 0.0001	114.8 / 69.1	4.6 / 3.1	0.037 / 0.037	26,753 / 26,927	
		BIOT-6D	200 / 200	3.19M / 3.19M	1.0K / 1.0K	3.19M / 1.6K	256 / 256	3.73 / 3.73	103.1 / 64.2	29.3 / 24.8	0.264 / 0.268	3,795 / 3,725	
		LaBraM	200 / 200	5.82M / 5.82M	804 / 804	5.83M / 804	200 / 200	12.53 / 12.53	324.5 / 50.5	35.0 / 12.0	0.286 / 0.298	3,495 / 3,359	
Neuro-GPT		250 / 250	157.0K / 157.0K	164 / 164	157.2K / 890	40 / 40	0.91 / 0.91	106.8 / 94.2	18.4 / 14.4	0.151 / 0.149	6,608 / 6,698		
EEGPT		256 / 256	25.29M / 25.29M	2.1K / 2.1K	25.29M / 3.1K	512 / 512	232.24 / 232.24	1543.6 / 921.7	51.7 / 34.4	0.497 / 0.498	2,011 / 2,010		
CBraMod		200 / 200	4.88M / 4.88M	804 / 804	4.88M / 804	200 / 200	10.18 / 10.18	292.2 / 45.9	53.8 / 16.3	0.293 / 0.286	3,417 / 3,496		
TFM		200 / 200	1.89M / 1.89M	260 / 260	1.89M / 1.17M	64 / 64	2.92 / 2.92	98.2 / 92.7	24.2 / 11.5	0.308 / 0.304	3,249 / 3,287		
32	Foundation Models	BrainOmni-Tiny	256 / 256	13.48M / 13.48M	1.0K / 1.0K	13.48M / 1.0K	256 / 256	72.67 / 72.67	1125.8 / 467.9	54.1 / 22.3	0.656 / 0.646	1,523 / 1,548	
		BrainOmni-Base	256 / 256	37.76M / 37.76M	2.1K / 2.1K	37.76M / 2.1K	512 / 512	267.07 / 267.07	2390.8 / 561.9	81.2 / 36.0	0.989 / 0.990	1,011 / 1,010	
		SingLEM	128 / 128	3.27M / 3.27M	68 / 68	3.27M / 68	16 / 16	14.71 / 14.71	600.6 / 110.0	28.8 / 9.2	0.277 / 0.277	3,616 / 3,616	
		LUNA-Base	256 / 256	7.36M / 7.36M	1.0K / 1.0K	7.36M / 1.0K	256 / 256	3.71 / 3.71	206.4 / 69.7	43.0 / 16.8	0.466 / 0.467	2,144 / 2,143	
		Specialist Models	EEGNet	1,000 / -	1.6K / -	2.0K / -	3.6K / -	- / -	1.09 / -	122.9 / -	6.1 / -	0.024 / -	42,424 / -
			ShallowConv	1,000 / -	52.4K / -	9.8K / -	62.1K / -	- / -	5.20 / -	512.2 / -	4.5 / -	0.031 / -	32,458 / -
			LMDA	1,000 / -	2.9K / -	1.3K / -	4.3K / -	- / -	4.30 / -	776.6 / -	25.4 / -	0.118 / -	8,503 / -
	Foundation Models	CNN-T	1,000 / -	2.77M / -	1.0K / -	2.77M / -	- / -	6.34 / -	225.0 / -	33.0 / -	0.510 / -	1,962 / -	
		Deformer	1,000 / -	841.2K / -	9.0K / -	850.1K / -	- / -	12.40 / -	835.6 / -	16.6 / -	0.128 / -	7,815 / -	
		Conformer	1,000 / -	172.3K / -	244 / -	172.6K / -	- / -	5.77 / -	514.0 / -	19.5 / -	0.146 / -	6,859 / -	
TSception	512 / -	12.0K / -	644 / -	12.6K / -	- / -	1.55 / -	223.9 / -	5.0 / -	0.034 / -	29,742 / -			
4s	Foundation Models	BENDR	1,024 / 1,024	3.97M / 3.97M	2.1K / 2.1K	3.97M / 2.7K	512 / 512	0.0001 / 0.0001	226.2 / 172.1	4.1 / 3.1	0.037 / 0.037	26,823 / 26,775	
		BIOT-6D	800 / 800	3.19M / 3.19M	1.0K / 1.0K	3.19M / 1.6K	256 / 256	26.12 / 26.12	373.4 / 243.4	31.2 / 25.8	0.279 / 0.274	3,590 / 3,647	
		LaBraM	800 / 800	5.82M / 5.82M	804 / 804	5.83M / 804	200 / 200	52.80 / 52.80	1181.5 / 110.6	41.2 / 12.5	0.364 / 0.365	2,745 / 2,737	
		Neuro-GPT	1,000 / 1,000	157.0K / 157.0K	164 / 164	157.2K / 890	40 / 40	4.14 / 4.14	361.3 / 247.6	24.9 / 20.4	0.152 / 0.145	6,561 / 6,900	
		EEGPT	1,024 / 1,024	25.29M / 25.29M	2.1K / 2.1K	25.29M / 3.1K	512 / 512	928.95 / 928.95	5193.3 / 3337.4	188.2 / 126.3	1.764 / 1.764	567 / 567	
		CBraMod	800 / 800	4.88M / 4.88M	804 / 804	4.88M / 804	200 / 200	40.76 / 40.76	931.2 / 78.7	56.4 / 17.0	0.295 / 0.291	3,385 / 3,439	
		TFM	800 / 800	1.89M / 1.89M	260 / 260	1.89M / 1.17M	64 / 64	20.38 / 20.38	495.6 / 490.0	24.9 / 12.4	0.317 / 0.322	3,156 / 3,106	
	BrainOmni-Tiny	1,024 / 1,024	13.48M / 13.48M	1.0K / 1.0K	13.48M / 1.0K	256 / 256	215.72 / 215.72	3090.8 / 1250.9	83.9 / 41.9	1.225 / 1.224	816 / 817		
	BrainOmni-Base	1,024 / 1,024	37.76M / 37.76M	2.1K / 2.1K	37.76M / 2.1K	512 / 512	798.93 / 798.93	6324.6 / 1344.6	213.8 / 79.6	2.447 / 2.447	409 / 409		
	SingLEM	512 / 512	3.27M / 3.27M	68 / 68	3.27M / 68	16 / 16	58.83 / 58.83	2226.9 / 335.2	96.7 / 32.3	0.991 / 0.993	1,009 / 1,007		
LUNA-Base	1,024 / 1,024	7.36M / 7.36M	1.0K / 1.0K	7.36M / 1.0K	256 / 256	13.70 / 13.70	461.8 / 118.1	43.3 / 16.8	0.470 / 0.469	2,129 / 2,132			

Table 57. Balanced classification accuracies (%) of Full Fine-tuning, Linear Probing, and LoRA fine-tuning on EEG FMs on BNCI2014001 and BNCI2014009.

Scenario	Dataset	Tuning	Model	S0	S1	S2	S3	S4	S5	S6	S7	S8	S9	Avg.
Cross-subject (LOSO)	BNCI2014001	Full Fine-tuning	CBraMod	54.51	46.99	63.19	47.92	43.29	44.91	54.75	60.19	61.57	–	53.03 _{±0.22}
			Neuro-GPT	59.72	32.18	62.62	34.49	31.71	37.27	39.81	65.62	59.26	–	46.97 _{±0.71}
		Linear Probing	CBraMod	49.31	31.37	55.67	34.61	29.28	29.17	32.75	56.60	54.28	–	41.45 _{±0.50}
			Neuro-GPT	54.63	38.43	62.85	47.45	32.87	33.33	39.81	64.12	60.65	–	48.24 _{±1.04}
	BNCI2014009	LoRA	CBraMod	46.30	27.08	52.66	31.02	25.69	33.56	28.24	59.03	55.44	–	39.89 _{±0.38}
			Neuro-GPT	53.36	33.80	66.44	39.24	37.50	37.27	41.55	63.54	58.45	–	47.90 _{±1.44}
		Full Fine-tuning	CBraMod	66.42	74.76	67.81	81.98	85.76	69.83	72.22	77.88	91.67	84.65	77.30 _{±0.28}
			Neuro-GPT	72.05	80.52	66.42	72.33	83.89	69.62	72.40	73.85	86.74	81.91	75.97 _{±0.53}
		Linear Probing	CBraMod	57.15	55.52	58.09	56.98	65.87	55.21	54.93	53.40	67.53	61.63	58.63 _{±0.54}
			Neuro-GPT	55.45	54.48	56.42	59.17	67.05	54.06	51.91	54.41	65.83	68.23	58.70 _{±0.26}
		LoRA	CBraMod	53.16	50.28	54.90	60.69	67.95	51.22	53.75	62.15	65.45	56.74	57.63 _{±0.31}
			Neuro-GPT	68.19	78.99	66.81	71.88	82.78	70.24	72.26	71.35	83.96	83.19	74.97 _{±0.14}
Within-subject (Few-shot)	BNCI2014001	Full Fine-tuning	CBraMod	56.37	46.57	69.61	38.40	39.38	30.07	56.54	61.93	54.25	–	50.34 _{±1.18}
			Neuro-GPT	44.77	40.20	62.42	37.91	35.29	39.54	38.40	64.54	54.41	–	46.39 _{±1.92}
		Linear Probing	CBraMod	34.97	37.42	43.95	35.78	27.94	32.52	34.15	60.62	59.31	–	40.74 _{±1.06}
			Neuro-GPT	55.23	41.01	60.29	42.81	36.11	41.18	46.08	71.24	54.41	–	49.82 _{±1.55}
	BNCI2014009	LoRA	CBraMod	35.95	44.61	67.65	40.52	30.39	33.66	41.50	66.18	72.55	–	48.11 _{±0.33}
			Neuro-GPT	48.04	39.05	48.86	40.52	32.84	35.29	39.54	56.70	51.96	–	43.65 _{±1.22}
		Full Fine-tuning	CBraMod	57.18	53.56	53.63	55.73	60.83	54.67	53.08	55.10	64.30	60.47	56.85 _{±1.03}
			Neuro-GPT	52.39	55.36	50.00	54.17	62.09	60.17	51.19	51.76	52.26	60.83	55.02 _{±0.97}
		Linear Probing	CBraMod	50.00	50.22	49.96	50.00	51.10	49.92	49.65	49.81	50.15	50.95	50.18 _{±0.15}
			Neuro-GPT	50.00	50.00	49.96	50.00	50.34	50.46	49.96	50.38	51.34	54.02	50.65 _{±0.31}
		LoRA	CBraMod	49.88	53.88	49.56	54.12	56.03	52.69	52.69	50.11	55.75	52.89	52.76 _{±0.42}
			Neuro-GPT	56.42	66.40	54.04	63.59	69.44	58.50	55.62	63.77	64.81	68.53	62.11 _{±0.88}

Table 58. Scenario-level paired Wilcoxon signed-rank tests comparing full fine-tuning and linear probing for EEG FMs.

Model	N valid pairs	Full wins	Linear wins	Ties	Median Δ	Wilcoxon W	p -value	Sig.	Rank-biserial r
BENDR	25	23	2	0	12.2200	4.0000	4.17×10^{-7}	****	0.9754
BIOT-6D	25	21	4	0	2.2200	34.0000	2.17×10^{-4}	***	0.7908
LaBraM	25	19	6	0	2.0300	64.0000	0.0067	**	0.6062
Neuro-GPT	25	18	7	0	2.8200	52.5000	0.0031	**	0.6769
EEGPT	25	7	18	0	-1.3600	80.0000	0.0255	*	-0.5077
CBraMod	25	23	2	0	6.6700	7.0000	1.13×10^{-6}	****	0.9569
TFM	25	24	0	1	1.6500	0.0000	1.82×10^{-5}	****	1.0000
BrainOmni-Tiny	21	15	6	0	1.8000	64.0000	0.0760	ns	0.4459
BrainOmni-Base	21	16	5	0	1.1700	27.5000	0.0022	**	0.7619
EEGMamba	25	19	6	0	4.1100	34.0000	2.17×10^{-4}	***	0.7908
SingLEM	25	15	10	0	0.2400	117.0000	0.2304	ns	0.2800
LUNA-Base	25	17	8	0	0.5400	71.0000	0.0125	*	0.5631

Note: ****: $p < 0.0001$; ***: $p < 0.001$; **: $p < 0.01$; *: $p < 0.05$; ns: not significant.

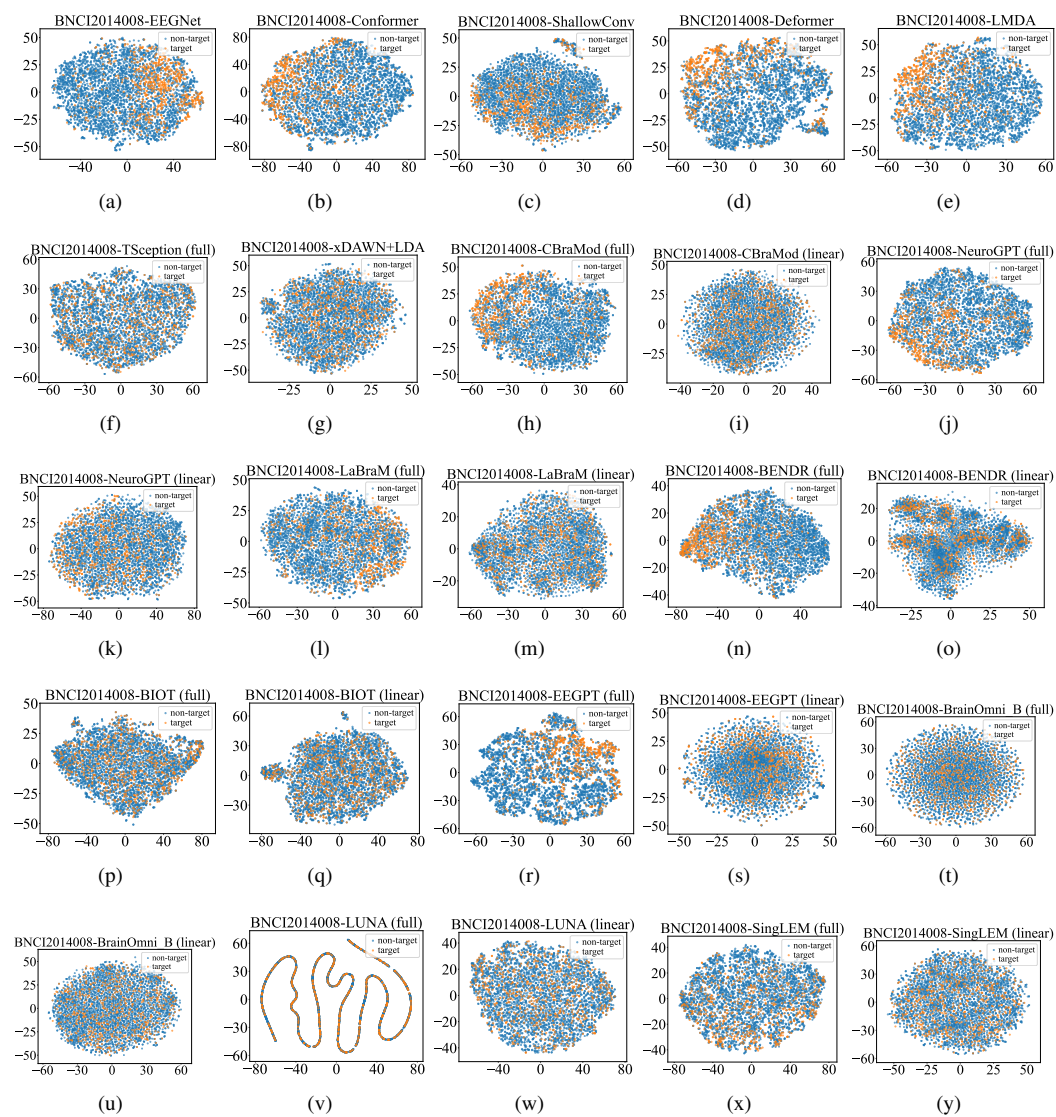


Figure 17. t -SNE visualization of the BNCI2014008 datasets.

Table 59. Comparison of our work with representative surveys, reviews, and benchmarks on EEG and brain foundation models. “–” indicates that the corresponding work does not provide benchmark experiments or the relevant component is not applicable.

Work	Type	# EEG FMs	Benchmark coverage	Evaluation protocols	Specialist baselines	Main distinction
Lai et al. [115]	Review	13	–	–	–	Concise EEG-FM review
Altaheri et al. [116]	Survey	27	–	–	–	SSL-focused brain-signal survey
Wu et al. [117]	Benchmark	4	13 datasets; 7 BCI tasks	cross-subj.; multi-subj.; few-shot; full / linear	Yes	BCI transfer benchmark
Zhou et al. [118]	Survey	15	–	–	–	Broad BFM framework
Kuruppu et al. [119]	Review	10	–	–	–	Early EEG-FM critical review
Gu et al. [120]	Survey	8	–	–	–	Biosignal-FM roadmap
Xiong et al. [121]	Benchmark	7	14 datasets; 10 paradigms	full / frozen / LoRA; single-/multi-task	Partial	Systematic EEG-FM benchmark
Yao et al. [27]	Review	17	6 datasets; 4 task categories	Unified comparison	Yes	EEG-FM progress review + comparison
Lee et al. [122]	Survey	29	–	–	–	Comprehensive biosignal-FM review
Kwon et al. [123]	Review	3	–	–	–	EEG-centered perspective
Shen et al. [124]	Benchmark	15	18 EEG/iEEG datasets; 11 tasks	cross-subj.; group-wise CV	Partial	EEG / iEEG SSL taxonomy + benchmark
Ours	Survey + benchmark	55	12 open-source FMs; 13 datasets; 9 paradigms	LOSO; few-shot; full / linear	Yes	Largest EEG-FM survey + benchmark

Table 60. Main mathematical symbols used in the manuscript.

Symbol	Meaning	Usage
$X \in \mathbb{R}^{C \times T}$	Raw EEG trial with C channels and T samples	EEG input representation
$\tilde{X} = \mathcal{G}(X)$	Preprocessed EEG signal	Data standardization
$\mathcal{G}(\cdot)$	Preprocessing operator (e.g., z -score, CAR, EA, EMA)	Preprocessing pipeline
C	Number of EEG channels	Input shape
T	Number of temporal samples per trial	Input shape
f_s	Sampling rate	Preprocessing
M	Temporal patch length	Patch construction
S	Patch stride	Patch construction
$O = M - S$	Patch overlap	Patch construction
$N_p = \lfloor (T - M) / S \rfloor + 1$	Number of temporal patches	Patch construction
$\mathcal{S}(\cdot)$	Patching operator	Tokenization
$P = \mathcal{S}(\tilde{X})$	Patch tensor of shape $N_p \times C \times M$	Pre-training input
$p_{k,c}$	k -th patch of channel c	Masked reconstruction
Ω	Set of masked patch indices	Masked modeling
$\mathcal{M}_x(\cdot)$	Masking operator on raw-signal patches	Raw-signal reconstruction
$\mathcal{M}_z(\cdot)$	Masking operator on embedded tokens	Token reconstruction
P_{msk}	Masked raw-signal patches	Masked reconstruction
$\mathcal{E}_\theta(\cdot)$	Encoder with parameters θ	FM backbone
$\mathcal{D}_\phi(\cdot)$	Decoder or reconstruction head with parameters ϕ	Pre-training head
$\mathcal{T}_\psi(\cdot)$	Tokenizer or patch-embedding module with parameters ψ	Tokenization
$Z = \mathcal{T}_\psi(P) \in \mathbb{R}^{N_p \times d}$	Embedded EEG tokens	Token reconstruction / contrastive learning
$z_{k,c}$	Embedded token corresponding to patch (k, c)	Token-level objectives
d	Token embedding dimension	Model architecture
$\hat{X}, \hat{P}, \hat{Z}$	Reconstructions of X , P , and Z	Reconstruction objectives
h_i, h_i^+	Anchor and positive representations	Contrastive learning
$\mathcal{N}_i$	Negative set for anchor h_i	Contrastive learning
$\text{sim}(\cdot, \cdot)$	Similarity function (e.g., cosine similarity)	Contrastive learning
τ	Temperature parameter	Contrastive learning
$\mathcal{F}_r(\cdot)$	Spectral transform yielding target type r	Frequency-domain reconstruction
$S^r = \mathcal{F}_r(P)$	Spectral target of EEG patches	Frequency-domain reconstruction
r	Spectral target type, $r \in \{\text{spec}, \text{sa}, \text{bp}, \text{freq}\}$	Frequency-domain reconstruction
$\mathcal{Q}(\cdot)$	Quantizer mapping embeddings to codebook indices	Codebook objectives
K	Codebook size	Codebook objectives
$I \in \{1, \dots, K\}^L$	Sequence of discrete codebook indices	Codebook prediction
$\hat{\pi}_{k,c}$	Predicted categorical distribution over codebook entries or pseudo-labels	Codebook / pseudo-label prediction
$u_{1:L}$	Ordered token sequence for autoregressive modeling	Autoregressive pre-training
L	Length of token or index sequence	Sequence modeling
f_Θ	Pre-trained EEG FM with parameters Θ	Model definition
Θ^*, Θ_j^*	Pre-training optimum and task- j fine-tuning optimum	Model optimization
$\mathcal{D}_{pre}$	Pre-training corpus	Pre-training data
$\mathcal{T} = \{\tau_j\}_{j=1}^J$	Set of downstream tasks; τ_j denotes task j	Downstream evaluation
$\mathcal{D}_{task}^{(j)}, \mathcal{D}_{ft}^{(j)}, \mathcal{D}_{te}^{(j)}$	Full, fine-tuning, and test splits of task τ_j	Downstream evaluation
N_j, n_j	Total trials and fine-tuning trials in task τ_j	Downstream evaluation
C_j	Number of classes for task τ_j	Downstream classification
$y_i, \hat{y}_i$	Ground-truth label and predicted distribution for sample i	Classification
$\mathbf{1}(\cdot)$	Indicator function	Evaluation metric
B	Mini-batch size	Optimization
λ	Weighting coefficient for multi-term losses	Hybrid objectives

Table 61. Common loss symbols used in the manuscript. Model-specific losses that occur only in individual models are summarized in Table 5.

Symbol	Meaning	Usage
$\mathcal{L}_{pre}$	Overall pre-training objective, typically a weighted combination of reconstruction, prediction, contrastive, or auxiliary terms	General pre-training
$\mathcal{L}_{mse}^{rs}$	Mean squared error for raw-signal reconstruction	Raw EEG waveform
$\mathcal{L}_{mse}^{nrs}$	Mean squared error for denoised raw-signal reconstruction	Denoised EEG waveform
$\mathcal{L}_{s-l1}^{rs}, \mathcal{L}_{huber}^{rs}$	Smooth- ℓ_1 or Huber-style robust reconstruction loss for raw signals	Robust waveform reconstruction
$\mathcal{L}_{mae}^{os}$	Masked-autoencoding reconstruction loss on original-signal patches	Used by SAMBA
$\mathcal{L}_{mse}^{et}$	Mean squared error for embedded-token reconstruction	Embedded EEG tokens
$\mathcal{L}_{mse}^{emb}$	Mean squared error for latent-embedding reconstruction (equivalent to $\mathcal{L}_{mse}^{et}$ in some models, e.g., EEGPT)	Latent embeddings
$\mathcal{L}_{ce}$	Standard cross-entropy loss	Building block for $\mathcal{L}_{cls}^{ci}$, $\mathcal{L}_{cls}^{skl}$, etc.
$\mathcal{L}_{cl}$	Generic contrastive loss, typically InfoNCE-style	Representation alignment
$\mathcal{L}_{cl}^{et}$	Contrastive loss on embedded EEG tokens	Token-level contrastive
$\mathcal{L}_{cl}^{emb}$	Contrastive loss on latent embeddings	Embedding-level contrastive
$\mathcal{L}_{cl}^{glob}$	Global contrastive loss for discriminative structure	Global representation
$\mathcal{L}_{mse}^{spec}$	Mean squared error for spectrogram (time-frequency) reconstruction	Spectrogram target
$\mathcal{L}_{mse}^{sa}$	Mean squared error for spectral-amplitude reconstruction	Spectral amplitude
$\mathcal{L}_{mse}^{bp}$	Mean squared error for band-power reconstruction	Band-power features
$\mathcal{L}_{mse}^{freq}$	Generic frequency-domain reconstruction loss	Fourier-domain target
$\mathcal{L}_{\ell_1}^{stft}, \mathcal{L}_{\ell_2}^{stft}$	Multi-scale STFT ℓ_1 / squared- ℓ_2 reconstruction loss	Time-frequency codec
$\mathcal{L}_{cls}^{skl}$	Cross-entropy loss for spectrogram-derived k -means pseudo-label prediction	Spectral pseudo-labels
$\mathcal{L}_{cls}^{ci}$	Cross-entropy loss for masked codebook-index prediction	Discrete neural codes
$\mathcal{L}_{nll}^{ci}$	Negative log-likelihood for autoregressive codebook-index prediction	Autoregressive neural codes
$\mathcal{L}_{mse}^{cbe}$	Mean squared error for codebook-embedding reconstruction	Codebook embeddings
$\mathcal{L}_{cls}$	Supervised classification loss; task-specific instances denoted $\mathcal{L}_{cls}^{(j)}$	Downstream classification
$\mathcal{L}_{aux}$	Model-specific auxiliary regularization (alignment, MoE balancing, decomposition, or consistency-style terms)	Auxiliary objectives